SMITHIAN FOLD THEORY V3

From One Law to a Working World

An Exact, Zero-Parameter and Machine-Closed Foundation for Engineering Translation from Smithian Fold Theory

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Contents

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> ***Publication record.*** *Maria Smith authorised this status-only correction.*
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> *Zenodo's new-version route after 10.5281/zenodo.21760449*
> *in the same concept DOI lineage. The correction changes no scientific claim,*
> *evidence classification, mathematical value, source identity or machine record.*
> ***Lean verification integration.*** *The current SFT ordered census passed the*
> *independent Lean 4 verification layer on 2 August 2026: 2,777 accepted claims,*
> *898,902 candidate records and decisions, 11,108 controls, seventeen branches*
> *and no reported issue. Lean natively proves the two-class operational root and*
> *its unique survivor and constructs proof-bearing acceptance-gate certificates.*
> *The remaining claim content is checked as the complete registered artifact*
> *graph; that distinction is retained throughout this successor.*

Abstract

This successor presents the complete current `engineering_translation` paper surface: 80 live model-admitted claims, 20,480 generated candidates, 80 unique survivors and 320 passed controls. It integrates 8 claim section(s) not present in the preceding abstract's dated surface and remains complete only to this versioned registered census, with lawful extension open. The independent Lean 4 layer returned PASS for this surface as part of all 2,777 current claims.

Formal admission, implementation validation, observation, measurement, empirical support, confirmation, adverse evidence, unresolved evidence and publication status remain separate classifications. Lean does not reclassify an empirical result, repair an adverse observation or convert a later comparison into an earlier prediction. Lawful criticism, falsification and versioned extension remain open.

Successor headline findings

1. **Independent whole-model result.** Lean 4 returned `PASS` for 2,777/2,777 current claims, 898,902 candidates and decisions, 11,108 controls and all 17 branches, with no issue.
2. **Operational root.** The exact registered two-class grammar is formalised natively, and `presentedOccurrence` is proved to be its unique survivor without an imported or user-declared axiom in the exported theorem's axiom audit.
3. **Typed validation.** The root and generic acceptance implications are native Lean theorems; the remaining scientific claims are checked as complete registered artifacts. Empirical truth is not inferred from proof-assistant acceptance alone.
4. **Fail-closed custody.** A six-file byte-identity mismatch halted the first run. Exact registered bytes were restored, all 385 source bindings were re-audited, and the unchanged verifier then passed.
5. **Current paper surface.** This successor accounts for all 80 current claim(s) in its declared branch ownership boundary; 8 later claim section(s) are integrated in the successor supplement.
6. **Newly integrated results.** `SFT-ENG-TESLA-RESONANT-TRANSFER-PROTOCOL-002` - Tesla resonant-transfer engineering protocol; `SFT-ENG-VACUUM-INERTIA-RESPONSE-PROTOCOL-002` - Driven vacuum/inertial-response engineering protocol; `SFT-ENG-VACUUM-BEAT-RESTORATION-PROTOCOL-002` - Vacuum-beat transfer and restoration protocol; `SFT-ENG-SECTOR-FIVE-SEVEN-DETECTION-PROTOCOL-002` - Sector-five/seven blind detection protocol; `SFT-ENG-SMITHIUM-SYNTHESIS-IDENTIFICATION-PROTOCOL-002` - Smithium synthesis and joint-identification protocol; `SFT-ENG-NOVEL-TRANSLATIONS-COMPLETE-FAMILY-002` - Complete prior-return engineering protocol family; `SFT-ENG-CONSCIOUSNESS-PLACEBO-CROSS-BINDING-PROTOCOL-002` - Open consciousness, placebo and cross-binding protocol; `SFT-ENG-NOVEL-TRANSLATIONS-NO-OMISSION-ADDENDUM-002` - Engineering novel-translations no-omission addendum.

These findings supplement rather than erase the branch-specific headline results retained below. Any adverse, corrected, unresolved, unavailable or chronology-bound result in the preceding scientific record remains governed by its current claim package.

Current status, evidence language and reader map

Status field	Current position
Paper and completion boundary	80 current claims in the declared branch surface. Completion is dated to the current registered census and remains open to lawful versioned extension.
Formal status	The Lean root theorem is kernel checked. Every live model claim has a current admitted receipt and passed the Lean artifact gates. Formal admission does not imply empirical confirmation.
Empirical status	Claim specific. Blind, non-blind, development-observed, holdout, favourable, adverse, null, missing, unavailable, disputed and unresolved distinctions remain exactly as recorded.
Chronology	Claim-specific registration, sealing, custody and observation order controls. Later evidence never becomes an earlier prediction without the registered chronology.
Publication status	Version 1.1.1 is published open access at 10.5281/zenodo.21761664 in the existing Zenodo concept DOI lineage.
Ownership	This paper retains its declared scientific boundary. Lean verification creates no new branch ownership and no application may select a law.
What is not claimed	Permanent closure of science, universal empirical confirmation, conversion of compatibility into confirmation, or superiority to every rival model.

Corpus-wide terminology and evidence key

Term	Reserved meaning in this paper
Theorem	A formally closed proposition at its declared grammar and dependency boundary.
Law	An admitted branch relation with its declared carrier, boundary, dependencies and extension rule.
Claim	The registered unit judged by the engine and bound to one immutable receipt.
Constitution	The rules governing admissible objects, derivations, evidence and publication; not an empirical result.
Derivation	Generation and elimination of the declared candidate space to its surviving structure.
Prediction	A consequence sealed before the matching target is released under the registered custody protocol.
Observation	A source-bound external record; it does not enter candidate generation.
Measurement	An instrument-, method-, condition- and uncertainty-bound observed value.
Reconstruction	A separately implemented regeneration or an explicitly identified inference of a retained state or history.
Exact numerical correspondence	Equality or registered interval relation between exact numerical objects at the declared boundary.
Structural correspondence	A post-derivation relation between forms without a claim of exact numerical prediction.
Boundary correspondence	Agreement restricted to the named interface, limit or ownership boundary.
Compatibility	Non-adverse but non-discriminating evidence.
Support	Relevant evidence that is not unique confirmation.
Confirmation	Used only when the current registered evidence protocol warrants that classification.
Validation	Successful execution of the named formal, computational or empirical test; its kind must be stated.
Adverse result	A registered test result that conflicts with the tested claim at its declared boundary.
Unresolved result	Required evidence remains unavailable, incomplete, disputed or insufficiently classified.
Implementation identity	The hash-bound identity of executable material; implementation success is not empirical confirmation.
Formal, empirical and publication status	Three independent classifications; none substitutes for another.
Foundational closure	Completion of the registered foundation only; not field-wide closure.
Field-wide closure	Completion of the named frozen or registered dated field census.
Current-evidence closure	Closure only to the evidence surface presently registered and preserved.
Extension openness	New questions may enter a later version without rewriting receipts or history.

Proof language is reserved for formal closure; derivation for generated structure; implementation for executable demonstration; prediction for a correctly sealed prospective consequence; observation and measurement for source-bound external records; reconstruction for separately regenerated or inferred states; correspondence for post-derivation relations; compatibility and support for non-unique evidence; confirmation only where the registered protocol authorises it; adverse for conflict; and unresolved where required evidence remains incomplete.

Three reading levels

1. **Conceptual paper.** The abstract, headline findings, branch narrative, major derivations, evidence synthesis, limitations and conclusion provide the readable scientific argument.
2. **Scientific audit layer.** Family and claim sections preserve exact status, dependencies, candidate and survivor counts, controls, chronology, sources, corrections, adverse evidence and reconciliation.
3. **Machine archive.** The repository packages preserve complete candidates, decisions, hashes, receipts, executable traces, source snapshots and certificates. They remain authoritative where prose abbreviates display.

Editorial change control

This successor improves currency, expression, typography and navigation. It does not change scientific meaning, chronology, evidence class, branch ownership, claim status or machine identity. Any conflict between authoritative records must be resolved by explicit supersession or reported to Maria Smith; prose alone cannot manufacture agreement.

Authorship, access and the public scientific mission

Maria Smith produced this work outside credentialed institutional access. That fact is not offered as personal exceptionalism. It is an indictment of all the minds and contributions society loses when financial gatekeeping, credentials, paywalls, prestige, institutional permission and access to funded infrastructure are dressed as rigour. The appropriate response is not to ask whether an excluded author was somehow uniquely permitted to think. It is to ask how many minds, observations and corrections were never heard because the gate was mistaken for the method.

Ernos Labs is an open-source science movement and a standards designation. The repository, derivations, source failures, controls, papers and machine receipts are public so that any person can inspect, challenge, reproduce or extend them. Transparent access is not a relaxation of rigour: it makes rigour inspectable. Paywalled claims, opaque oracles, credential votes and funding-driven restrictions cannot substitute for a complete derivation, a sealed prediction, an unfavourable control or an independently reproducible result.

The paper is licensed **CC BY 4.0** and the platform code **Apache-2.0**. Maria Smith retains authorship, copyright and creative rights. Anyone may use and criticize the work under those licenses. The **Ernos Labs** designation may be used only when the root theorem and constraints remain visible, the protected engine and verification authority remain unchanged, every candidate is enumerated, outcomes do not select laws, adverse and missing evidence is preserved, authorship and licenses remain intact, and extensions pass the same admission route. A weakened fork remains open-source software but is not an Ernos Labs scientific admission.

Open criticism requires no credential. Scientific admission requires the common evidence protocol. Contact: **Maria.Smith.Sftoe@gmail.com**. Public submissions and review: **https://discord.gg/ucwGryVxGr**. GitHub: **https://github.com/MettaMazza**.

Mathematical constitution

Every law traces to **There Is No Nothing** and to already admitted Fold dependencies. No additional axiom is introduced. No free or fitted parameter can select a survivor. Proof-state arithmetic is exact and finite. Conventional numerical zero is not a proof carrier; the glyph `0` may be retained as a typed marker for absence, non-response or a source field, but it does not import ordinary zero semantics. Negative, irrational and imaginary quantities may appear as held source labels where an external dataset uses them, but never as proof scalars. Completed infinities and ungenerated continua are excluded.

For each claim the generator builds eight binary dimensions: carrier, boundary, relation, record, evidence, provenance, generality and extension. Their literal Cartesian product contains 2^8 = 256 candidates. One choice on each dimension alone preserves its required distinction. The program decides all 256 candidates; one survives and 255 are eliminated. Across 72 claims this is exactly 18,432 decided forms. A separate implementation rebuilds the product and survivor. Positive-finite induction proves that adding one lawful component, interface, platform, requirement, test, failure, lifecycle stage or anomaly preserves all earlier versions, adverse rows, boundaries and receipts while appending its trace.

This closure is exact but bounded. It establishes the foundational forms necessary to make Engineering Translation reconstructible and falsifiable. It does not prohibit new technologies, users, platforms, hazards, tests, standards or lawful discoveries. The branch is current-evidence closed and extension-open, never permanently locked.

Empirical constitution and blind order

The complete inventory, candidate grammar, predicted structural labels and falsification conditions were frozen before any external standard, handbook, product, test or outcome was opened. Primary custodians were selected afterward for requirements, systems, measurement, safety, accessibility, reproducibility and lifecycle records-not because their status could prove a law. A capability-closed prediction process cannot read a file, network, clock, environment variable, process or target. Only after its derivation seal exists may the external evaluator open the registered record.

Every comparison retains successful, adverse, absent, missing, disputed, unresolved, access-denied and transport-failed rows. A handbook is not an implementation. A standard is not automatic conformity. A model is not an observation. A prototype is not a production system. A simulation, demonstration, verification, validation and acceptance decision remain different evidence classes. These distinctions are part of the empirical result, not administrative bookkeeping.

The immutable engine seal is `sha256:4f4cdd7986808e6a6102d650c85e6093d6425e49f14a5f05d70fa05e6031d46a`. The verification-authority seal is `sha256:bf810a190b504f0f874a778a52e23251904b17b40a7364135e74b34e8ba0c3b8`. A mismatch halts. Those identities secure a shared admission route; they never replace the readable derivations.

Dependency and ownership boundary

Each categorical science retains ownership of its laws and evidence. Mathematics owns exact structure; computation owns formal semantics; Physics, Chemistry and Materials own physical relations; Biology, Consciousness and Medicine own organism, experience and clinical state; Earth and Astronomy own their observed systems; Social Science owns collective context, access and legitimacy. Engineering Translation owns versioned requirements, designs, artifacts, tests, deployments and lifecycle evidence. It may discover an anomaly and return a question, but it may never directly rewrite the upstream law.

Registered primary-source surface

- **NASA-SE-HANDBOOK-001 - National Aeronautics and Space Administration.** NASA Systems Engineering Handbook Rev 2. Registered families: requirements_function_boundary, components_interfaces_architecture, verification_validation_traceability. Locator: https://www.nasa.gov/wp-content/uploads/2018/09/nasa_systems_engineering_handbook_0.pdf. Its authority here is source custody and documented provenance, not prestige or consensus.
- **NASA-SE-APPENDIX-001 - National Aeronautics and Space Administration.** Systems Engineering Handbook Appendix. Registered families: measurement_calibration_acceptance, verification_validation_traceability, science_design_evidence_boundary. Locator: https://www.nasa.gov/reference/system-engineering-handbook-appendix/. Its authority here is source custody and documented provenance, not prestige or consensus.
- **NASA-SWE-VALIDATION-001 - National Aeronautics and Space Administration.** Software Requirements Validation. Registered families: requirements_function_boundary, verification_validation_traceability, cross_platform_accessibility. Locator: https://swehb.nasa.gov/spaces/SWEHBVB/pages/32604513/SWE-055%2B-%2BRequirements%2BValidation. Its authority here is source custody and documented provenance, not prestige or consensus.
- **NIST-TRACEABILITY-001 - National Institute of Standards and Technology.** Metrological Traceability Policy and FAQ. Registered families: measurement_calibration_acceptance, verification_validation_traceability, science_design_evidence_boundary. Locator: https://www.nist.gov/metrology/metrological-traceability. Its authority here is source custody and documented provenance, not prestige or consensus.
- **NIST-SSE-001 - National Institute of Standards and Technology.** Engineering Trustworthy Secure Systems SP 800-160 Vol 1 Rev 1. Registered families: components_interfaces_architecture, safety_hazard_resilience, lifecycle_maintenance_sustainability. Locator: https://csrc.nist.gov/pubs/sp/800/160/v1/r1/final. Its authority here is source custody and documented provenance, not prestige or consensus.
- **NIST-SSDF-001 - National Institute of Standards and Technology.** Secure Software Development Framework SP 800-218. Registered families: verification_validation_traceability, cross_platform_accessibility, lifecycle_maintenance_sustainability. Locator: https://csrc.nist.gov/pubs/sp/800/218/final. Its authority here is source custody and documented provenance, not prestige or consensus.
- **HSE-COMAH-001 - UK Health and Safety Executive.** Control of Major Accident Hazards 2015. Registered families: safety_hazard_resilience, alternatives_tradeoffs_optimization, lifecycle_maintenance_sustainability. Locator: https://www.hse.gov.uk/comah/comah15.htm. Its authority here is source custody and documented provenance, not prestige or consensus.
- **W3C-WCAG22-001 - World Wide Web Consortium.** Web Content Accessibility Guidelines 2.2. Registered families: cross_platform_accessibility, requirements_function_boundary, verification_validation_traceability. Locator: https://www.w3.org/TR/WCAG22/. Its authority here is source custody and documented provenance, not prestige or consensus.
- **FDA-DESIGN-CONTROLS-001 - US Food and Drug Administration.** Design Control Guidance for Medical Device Manufacturers. Registered families: requirements_function_boundary, verification_validation_traceability, ownership_anomaly_handoffs. Locator: https://www.fda.gov/media/116573/download. Its authority here is source custody and documented provenance, not prestige or consensus.

- **REPRODUCIBLE-BUILDS-001 - Reproducible Builds Project.** Reproducible Builds Documentation. Registered families: verification_validation_traceability, cross_platform_accessibility, lifecycle_maintenance_sustainability. Locator: https://reproducible-builds.org/docs/. Its authority here is source custody and documented provenance, not prestige or consensus.
- **EPA-LIFECYCLE-001 - US Environmental Protection Agency.** Life Cycle Engineering Guidelines. Registered families: resources_efficiency_reliability, alternatives_tradeoffs_optimization, lifecycle_maintenance_sustainability. Locator: https://nepis.epa.gov/Exe/ZyPURL.cgi?Dockey=P10071L2.TXT. Its authority here is source custody and documented provenance, not prestige or consensus.
- **CISA-SECURE-BY-DESIGN-001 - US Cybersecurity and Infrastructure Security Agency.** Secure by Design. Registered families: safety_hazard_resilience, science_design_evidence_boundary, ownership_anomaly_handoffs. Locator: https://www.cisa.gov/securebydesign. Its authority here is source custody and documented provenance, not prestige or consensus.
- **PYTHON-PLATFORM-001 - Python Software Foundation.** Python platform module documentation. Registered families: cross_platform_accessibility, components_interfaces_architecture, science_design_evidence_boundary. Locator: https://docs.python.org/3/library/platform.html. Its authority here is source custody and documented provenance, not prestige or consensus.

Family results

Requirements, function, constraints and operating boundaries

This family contains 6 separately admitted obligations. Its closure preserves artifact, requirement, version, environment, operating boundary, failure and source record; it does not universalise one implementation or successful run.

- **Engineering requirement** (`SFT-ENG-REQUIREMENT-001`): A requirement is a source-custodied, purpose-bound and verifiable condition with conflicts, revisions and acceptance retained.
- **Required function** (`SFT-ENG-FUNCTION-001`): A function is a declared input-to-output transformation whose state, loss, context and failure modes remain explicit.
- **Engineering constraint** (`SFT-ENG-CONSTRAINT-001`): A constraint is a source- and context-bound limit on generated designs, never an unexplained fitted preference.
- **Operating boundary** (`SFT-ENG-OPERATING-BOUNDARY-001`): An operating boundary retains environment, loads, duration, transients, exceptions and observed failures.
- **Use and misuse case** (`SFT-ENG-USE-CASE-001`): A use case retains actor, goal, preconditions, path and outcome; foreseeable misuse and harm remain co-registered.
- **Requirement acceptance boundary** (`SFT-ENG-ACCEPTANCE-BOUNDARY-001`): Acceptance requires a preregistered criterion, method, result, uncertainty and preserved failure record.

Components, interfaces, architecture and integration

This family contains 6 separately admitted obligations. Its closure preserves artifact, requirement, version, environment, operating boundary, failure and source record; it does not universalise one implementation or successful run.

- **Engineering component** (`SFT-ENG-COMPONENT-001`): A component is a versioned bounded part with declared state, capability, interfaces and failures.
- **Engineering interface** (`SFT-ENG-INTERFACE-001`): An interface is a typed exchange contract retaining endpoints, units, timing, errors and version.
- **System architecture** (`SFT-ENG-ARCHITECTURE-001`): Architecture is a versioned composition of components, interfaces and allocated responsibilities.
- **Engineered system** (`SFT-ENG-SYSTEM-001`): An engineered system composes parts and relations around a declared purpose while retaining environment, users and failures.
- **Modular decomposition** (`SFT-ENG-MODULARITY-001`): A modular decomposition is lawful only when interface contracts permit exact recomposition of required behavior.

- **System integration** (`SFT-ENG-INTEGRATION-001`): Integration retains component versions, assembly order, interface tests, failures and rollback state.

Resources, capacity, efficiency, reliability and failure

This family contains 6 separately admitted obligations. Its closure preserves artifact, requirement, version, environment, operating boundary, failure and source record; it does not universalise one implementation or successful run.

- **Engineering resource** (`SFT-ENG-RESOURCE-001`): Resource accounting retains stock, flow, units, time, loss and source for each operation.
- **Capacity and load** (`SFT-ENG-CAPACITY-001`): Capacity is a load-profile-bound relation retaining duration, queuing, degradation and failure.
- **Engineering efficiency** (`SFT-ENG-EFFICIENCY-001`): Efficiency compares useful output with all registered input and loss inside an explicit boundary.
- **Reliability** (`SFT-ENG-RELIABILITY-001`): Reliability is a demand-, duration- and population-bound record of successful and failed service.
- **Availability** (`SFT-ENG-AVAILABILITY-001`): Availability retains serviceable and unavailable intervals, causes, maintenance and demand windows.
- **Failure mode** (`SFT-ENG-FAILURE-MODE-001`): A failure mode is an observed or generated loss-of-function path retaining cause, detectability, effect and recovery.

Measurement, calibration, tolerance and acceptance

This family contains 6 separately admitted obligations. Its closure preserves artifact, requirement, version, environment, operating boundary, failure and source record; it does not universalise one implementation or successful run.

- **Engineering measurement** (`SFT-ENG-MEASUREMENT-001`): Engineering measurement retains measurand, unit, instrument, conditions, uncertainty and record custody.
- **Calibration** (`SFT-ENG-CALIBRATION-001`): Calibration is a dated comparison to a traceable reference across a declared range and condition.
- **Tolerance** (`SFT-ENG-TOLERANCE-001`): A tolerance is a purpose- and uncertainty-bound allowed region registered before acceptance data.
- **Engineering test** (`SFT-ENG-TEST-001`): A test retains article version, setup, procedure, raw observations, deviations and unfavorable outcomes.
- **Acceptance test** (`SFT-ENG-ACCEPTANCE-TEST-001`): An acceptance test compares a frozen requirement and threshold to a source-custodied result.
- **Measurement uncertainty** (`SFT-ENG-MEASUREMENT-UNCERTAINTY-001`): Measurement uncertainty retains declared sources, correlations, method, coverage and revision.

Alternatives, trade-offs and bounded optimisation

This family contains 6 separately admitted obligations. Its closure preserves artifact, requirement, version, environment, operating boundary, failure and source record; it does not universalise one implementation or successful run.

- **Design alternative** (`SFT-ENG-ALTERNATIVE-001`): A design alternative must be generated from the frozen requirement grammar with every rejected option retained.
- **Engineering trade-off** (`SFT-ENG-TRADEOFF-001`): A trade-off retains distinct criteria, evidence, conflicts, stakeholders and losses without collapsing them into a hidden scalar.
- **Bounded engineering optimization** (`SFT-ENG-OPTIMIZATION-001`): Engineering optimization enumerates a declared finite design set and preserves ties, failures and constraint violations.
- **Multi-criteria decision record** (`SFT-ENG-MULTI-CRITERIA-001`): Multi-criteria choice keeps measured evidence separate from declared values, priorities and dissent.

- **Robust design** (`SFT-ENG-ROBUST-DESIGN-001`): A robust design retains performance across the preregistered condition family, including adverse corners.
- **Design freeze and change control** (`SFT-ENG-DESIGN-FREEZE-001`): A design freeze is a versioned baseline whose later changes require explicit impact, retest and provenance.

Control, feedback, stability, observability and fault response

This family contains 6 separately admitted obligations. Its closure preserves artifact, requirement, version, environment, operating boundary, failure and source record; it does not universalise one implementation or successful run.

- **Control action** (`SFT-ENG-CONTROL-001`): Control maps an observed state to a bounded action with target, latency, limits and effects retained.
- **Feedback loop** (`SFT-ENG-FEEDBACK-001`): A feedback loop retains plant, sensor, controller, actuator, delay, bounds, noise and failure paths.
- **Operational stability** (`SFT-ENG-STABILITY-001`): Stability is a finite operating-region claim retaining perturbation, return bound, overshoot and failures.
- **Engineering observability** (`SFT-ENG-OBSERVABILITY-001`): Observability requires declared output histories to distinguish the registered internal state set.
- **Engineering controllability** (`SFT-ENG-CONTROLLABILITY-001`): Controllability is reachability under declared actions, limits and duration with failed paths retained.
- **Fault detection and response** (`SFT-ENG-FAULT-RESPONSE-001`): Fault response retains detection, misses, latency, bounded action, residual state and recovery.

Hazard, risk, safety, fail-safe operation and resilience

This family contains 6 separately admitted obligations. Its closure preserves artifact, requirement, version, environment, operating boundary, failure and source record; it does not universalise one implementation or successful run.

- **Engineering hazard** (`SFT-ENG-HAZARD-001`): A hazard is a source-to-harm path retaining exposure, severity, likelihood and uncertainty.
- **Engineering risk record** (`SFT-ENG-RISK-001`): Risk retains scenario evidence, exposure, consequence, uncertainty and declared value judgments separately.
- **Safety case** (`SFT-ENG-SAFETY-001`): A safety case is a scoped claim supported by traceable evidence, controls, gaps and review.
- **Fail-safe operation** (`SFT-ENG-FAIL-SAFE-001`): Fail-safe operation requires every declared fault to reach a tested bounded safe state or preserve a visible halt.
- **Engineering resilience** (`SFT-ENG-RESILIENCE-001`): Resilience retains service loss, adaptation resources, recovery time, residual degradation and failed recovery.
- **Defence in depth** (`SFT-ENG-DEFENCE-IN-DEPTH-001`): Defence in depth requires separately testable barriers with dependencies, bypasses and common causes retained.

Verification, validation, traceability and end-to-end proof

This family contains 6 separately admitted obligations. Its closure preserves artifact, requirement, version, environment, operating boundary, failure and source record; it does not universalise one implementation or successful run.

- **Engineering verification** (`SFT-ENG-VERIFICATION-001`): Verification tests whether a versioned artifact satisfies its frozen specification.
- **Engineering validation** (`SFT-ENG-VALIDATION-001`): Validation tests whether the verified system serves its declared use in its declared context.
- **Bidirectional traceability** (`SFT-ENG-TRACEABILITY-001`): Traceability connects need, requirement, design, implementation, test and result in both directions.

- **Engineering reproducibility** (`SFT-ENG-REPRODUCIBILITY-001`): Reproducibility retains source, dependencies, environment, commands, outputs, failures and identities.
- **Implementation-distinct check** (`SFT-ENG-INDEPENDENT-CHECK-001`): An independent check reconstructs the result through a separately identified implementation path.
- **End-to-end verification** (`SFT-ENG-E2E-001`): End-to-end verification exercises the declared public entry through every stage to its final receipt.

Cross-platform, portable, accessible and learnable operation

This family contains 6 separately admitted obligations. Its closure preserves artifact, requirement, version, environment, operating boundary, failure and source record; it does not universalise one implementation or successful run.

- **Cross-platform behavior** (`SFT-ENG-CROSS-PLATFORM-001`): Cross-platform evidence preserves macOS, Windows and Linux environments, results and platform-specific differences.
- **Portable data and record format** (`SFT-ENG-PORTABLE-DATA-001`): A portable record has an open schema, explicit encoding, version, units, errors and migration path.
- **Accessible human interface** (`SFT-ENG-ACCESSIBLE-INTERFACE-001`): An accessible interface retains user diversity, input and output modes, errors, guidance and direct testing.
- **Lightweight deployment** (`SFT-ENG-LIGHTWEIGHT-DEPLOYMENT-001`): Lightweight deployment declares and minimizes runtime dependencies, permissions, downloads and infrastructure.
- **One-command public verification** (`SFT-ENG-ONE-COMMAND-001`): A single documented command reaches the complete verification route and reports progress, duration, failures and receipt.
- **Learnable public operation** (`SFT-ENG-LEARNABLE-USE-001`): Learnable operation retains audience, prerequisites, progressive guidance, errors, examples and user tests.

Configuration, maintenance, lifecycle and sustainability

This family contains 6 separately admitted obligations. Its closure preserves artifact, requirement, version, environment, operating boundary, failure and source record; it does not universalise one implementation or successful run.

- **Configuration identity** (`SFT-ENG-CONFIGURATION-001`): Configuration identity binds source, dependencies, settings, platform and hashes to a dated behavior.
- **Maintenance action** (`SFT-ENG-MAINTENANCE-001`): Maintenance retains diagnosis, parts, action, test, downtime and resulting configuration.
- **Change impact** (`SFT-ENG-CHANGE-IMPACT-001`): Change impact traces a proposed modification through affected requirements, interfaces, hazards, tests and releases.
- **Decommissioning** (`SFT-ENG-DECOMMISSION-001`): Decommissioning retains authority, users, data, materials, hazards, recovery and archive custody.
- **Full lifecycle record** (`SFT-ENG-LIFECYCLE-001`): A lifecycle record connects need, design, build, use, maintenance and retirement with resources and impacts retained.
- **Bounded sustainability claim** (`SFT-ENG-SUSTAINABILITY-001`): A sustainability claim retains lifecycle boundary, resources, emissions, labour, distribution, uncertainty and trade-offs.

Scientific law, design, performance, simulation and demonstration

This family contains 6 separately admitted obligations. Its closure preserves artifact, requirement, version, environment, operating boundary, failure and source record; it does not universalise one implementation or successful run.

- **Scientific-law translation** (`SFT-ENG-LAW-TRANSLATION-001`): Engineering translation consumes an admitted law and records every added design choice, assumption, boundary and test.
- **Design performance** (`SFT-ENG-PERFORMANCE-001`): Performance is a version- and condition-bound implementation result, not a fundamental scientific law.

- **Implementation failure boundary** (`SFT-ENG-IMPLEMENTATION-FAILURE-001`): A failed implementation can falsify its operating claim while leaving the upstream law unchanged unless a valid anomaly bridge is established.
- **Prototype evidence** (`SFT-ENG-PROTOTYPE-001`): Prototype evidence retains omissions, fidelity, setup, failures and the bounded question it answers.
- **Engineering simulation boundary** (`SFT-ENG-SIMULATION-001`): A simulation retains its model, discretization, inputs, assumptions, error and comparison; it is not an observation.
- **Engineering demonstration** (`SFT-ENG-DEMONSTRATION-001`): A demonstration shows bounded behavior in a retained scenario and remains distinct from verification, validation and science admission.

Anomaly return, disciplinary handoffs and the future V4 boundary

This family contains 6 separately admitted obligations. Its closure preserves artifact, requirement, version, environment, operating boundary, failure and source record; it does not universalise one implementation or successful run.

- **Engineering anomaly** (`SFT-ENG-ANOMALY-001`): An anomaly is a source-custodied deviation from a preregistered engineering expectation with controls and repeats retained.
- **Return of anomaly to science** (`SFT-ENG-SCIENCE-RETURN-001`): A validated anomaly returns as a question to its categorical science owner; it does not enter the model through Engineering.
- **Software and computation handoff** (`SFT-ENG-SOFTWARE-HANDOFF-001`): Computation owns formal semantics; Engineering owns the versioned executable translation and deployment evidence.
- **Physical sciences handoff** (`SFT-ENG-PHYSICAL-HANDOFF-001`): Physics, Chemistry, Materials, Earth and Astronomy own their laws; Engineering owns bounded artifacts using them.
- **Biological, medical and social handoff** (`SFT-ENG-BIO-MED-SOCIAL-HANDOFF-001`): Biology, Medicine, Consciousness and Social Science retain their evidence while Engineering translates a bounded intervention.
- **SFT-derived self-hosted V4 handoff** (`SFT-ENG-V4-HANDOFF-001`): A future V4 self-hosted rebuild must reproduce V3 semantics, receipts and constraints through an independently auditable bootstrap.

Complete claim-by-claim derivation record

The following seventy-two sections are the human-readable chain. The machine packages preserve all 256 candidate rows, 256 decisions, hostile controls, source manifest, independent output, empirical receipt and engine receipt for each claim.

1. Engineering requirement

Claim: `SFT-ENG-REQUIREMENT-001`

Family: `requirements_function_boundary`

Exact statement: A requirement is a source-custodied, purpose-bound and verifiable condition with conflicts, revisions and acceptance retained.

Forward chain and exact carrier

The generated carrier is `stakeholder-need-purpose-context-carrier`. Its required relation is `need-to-verifiable-condition-relation`. The reconstruction must retain `source-purpose-priority-version-conflict-and-acceptance-held`. Its declared empirical boundary is `requirement-scope-bounded`. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is `SFT-FOUNDATION-ONE-001`, `SFT-FOUNDATION-FOLD-001`, `SFT-FOUNDATION-FOLD-DYNAMICS-001`, `SFT-MATH-EXACT-ARITHMETIC-001`, `SFT-MATH-GRAPH-NETWORK-001`, `SFT-MATH-OPTIMIZATION-001`, `SFT-MATH-DYNAMICAL-SYSTEMS-001`, `SFT-MATH-LOGIC-PROOF-001`, `SFT-INFO-QUANTITY-001`, `SFT-INFO-CHANNEL-CAPACITY-001`, `SFT-COMP-FORM-STATE-TRANSITION-001`, `SFT-COMP-SEM-CORRECTNESS-001`, `SFT-COMP-SEM-COMPILATION-001`, `SFT-COMP-DIST-CAUSALITY-001`, `SFT-PHYS-MEAS-VALUE-RECORD-001`, `SFT-PHYS-MEAS-CALIBRATION-001`, `SFT-CHEM-MEAS-TRACEABILITY-001`, `SFT-MAT-SUST-LIFECYCLE-001`, `SFT-BIO-BIO-HANDOFF-001`,

`SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001`, `SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001`, `SFT-EARTH-HAZARD-RISK-HANDOFF-001`, `SFT-ASTRO-ASTRO-HANDOFF-001`, `SFT-SOCIAL-ENGINEERING-HANDOFF-001`. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external standard, application, performance target, product or prior implementation to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external standards, products, tests or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite components, versions, platforms, tests and lifecycle records; performance remains artifact, environment, user, method and operating-boundary limited.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - which requirement, artifact, component, user, environment or test bears the claim: `stakeholder-need-purpose-context-carrier`.
- **boundary** - which operating envelope, version, platform, user and lifecycle stage bound the result: `requirement-scope-bounded`.
- **relation** - which generated transformation, interface, comparison or handoff connects the carriers: `need-to-verifiable-condition-relation`.
- **record** - which requirements, versions, alternatives, failures, uncertainty and rollback remain reconstructible: `source-purpose-priority-version-conflict-and-acceptance-held`.
- **evidence** - whether law, requirement, design, simulation, test, demonstration, performance and anomaly remain distinct: `science-design-test-and-decision-classes-explicit`.
- **provenance** - whether the result traces to the One and names every upstream receipt and design choice: `root-bound-forward-translation`.
- **generality** - whether finite extension preserves platforms, versions, failures and operating limits: `positive-finite-extension-preserves-platforms-failures-and-versions`.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: `no-extra-rule`.

Their ordered product gives the unique survivor:

```
stakeholder-need-purpose-context-carrier__requirement-scope-bounded__need-to-verifiab
le-condition-relation__source-purpose-priority-version-conflict-and-acceptance-held__
science-design-test-and-decision-classes-explicit__root-bound-forward-translation__po
sitive-finite-extension-preserves-platforms-failures-and-versions__no-extra-rule
```

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses the requirement, artifact, user or environment. Erasing the boundary universalises a version, platform or test. Importing the relation lets a target or preferred application choose the result. Retaining only successful output removes requirements, alternatives, failures and rollback. Conflating evidence classes turns simulation, demonstration or performance into observation or law. Breaking provenance removes the trace to the One and upstream receipts. Erasing finite context makes one implementation universal. Adding a silent dependency or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite requirement or admitted law, versioned artifact, boundary, testable relation and complete record. The successor adds one lawful component, interface, platform, requirement, test, failure, lifecycle stage or anomaly while preserving all earlier identities, adverse rows and receipts.

Falsification and deliberately unfavourable controls

The registered falsification condition is: Reject if an unstated preference or implementation detail is presented as a scientific or stakeholder requirement.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: `source_bound_engineering_requirement_design_test_lifecycle_or_access_record/primary_standard_handbook_guidance_or_project_record_structural_correspondence`. Captured registered sources: 2. Unresolved transports retained: 1. The registered structural consequence corresponded: `True`. External evidence selected the survivor: `false`. Application or performance selected the law: `false`. Successful performance was relabelled scientific proof: `false`. Failed implementation was relabelled scientific falsification: `false`. Simulation or demonstration was relabelled observation: `false`.

- `NASA-SE-HANDBOOK-001` - transport `captured`; captured bytes `3773440`; snapshot `sha256:8eeb4887a4dc57a23049da7dd2ed556833cf98e214b240468d987873164ff688`.
- `NASA-SWE-VALIDATION-001` - transport `captured`; captured bytes `114542`; snapshot `sha256:15ed8ce3e5acbf6c24380016a30ece410cc61b9b31ea16d01b31f00cac52e888`.
- `FDA-DESIGN-CONTROLS-001` - transport `failed_preserved`; captured bytes `0`; snapshot `unresolved and preserved`.

The comparison is structural and source-bound. It checks that a relevant primary handbook, standard, guidance record or open engineering project can represent the declared requirement, artifact, environment, operating boundary, version, test and lifecycle without rewriting the Fold law. It does not claim that a standard proves conformity, that one handbook exhausts Engineering, or that a custodian's reputation proves the result. New records can extend or challenge the surface only with source, version, configuration, method, failures and adverse rows retained.

Scientific meaning

For **engineering requirement**, the consequence is practical and exact: A requirement is a source-custodied, purpose-bound and verifiable condition with conflicts, revisions and acceptance retained. An Engineering claim therefore has to expose the carrier `stakeholder-need-purpose-context-carrier`, the relation `need-to-verifiable-condition-relation`, the record `source-purpose-priority-version-conflict-and-acceptance-held`, and the boundary `requirement-scope-bounded` together. If any one is hidden, the result may remain a concept, prototype, simulation, demonstration or product choice, but it is not this admitted law.

This distinction protects users, maintainers and science. It prevents a successful output from erasing requirements, failed alternatives, hidden resources, inaccessible users, hazards, maintenance burdens or end-of-life costs. It prevents an application from selecting the upstream law and prevents engineering failure from becoming automatic scientific falsification. A replacement must still generate its forms, eliminate alternatives, survive controls, preserve custody and reproduce through the shared engine.

Machine certificate

Candidate count: `256`; survivor count: `1`; derivation seal: `sha256:7ab55fba4215cf16301d1d3d279d23fe99b3c648cb5ab543bc86afaeb9512c84`; independent certificate: `sha256:b3f295ad5ce58c2c908900bbb48d1f223055fe852a92e101933c9c86ee096207`; engine receipt: `sha256:d576e9c15d258d3c64253f4e8f05ed95a275c544a6db286dc23b6c3007b70316`. The independent implementation recomputed the candidate product: `True`. All external rows were preserved: `True`.

2. Required function

Claim: `SFT-ENG-FUNCTION-001`

Family: `requirements_function_boundary`

Exact statement: A function is a declared input-to-output transformation whose state, loss, context and failure modes remain explicit.

Forward chain and exact carrier

The generated carrier is `input-operation-output-purpose-carrier`. Its required relation is `declared-transformation-relation`. The reconstruction must retain `inputs-outputs-state-loss-context-and-failure-held`. Its declared empirical boundary is `function-domain-bounded`. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is `SFT-FOUNDATION-ONE-001`, `SFT-FOUNDATION-FOLD-001`, `SFT-FOUNDATION-FOLD-DYNAMICS-001`, `SFT-MATH-EXACT-ARITHMETIC-001`, `SFT-MATH-GRAPH-NETWORK-001`, `SFT-MATH-OPTIMIZATION-001`, `SFT-MATH-DYNAMICAL-SYSTEMS-001`, `SFT-MATH-LOGIC-PROOF-001`, `SFT-INFO-QUANTITY-001`, `SFT-INFO-CHANNEL-CAPACITY-001`, `SFT-COMP-FORM-STATE-TRANSITION-001`, `SFT-COMP-SEM-CORRECTNESS-001`, `SFT-COMP-SEM-COMPILATION-001`, `SFT-COMP-DIST-CAUSALITY-001`, `SFT-PHYS-MEAS-VALUE-RECORD-001`, `SFT-PHYS-MEAS-CALIBRATION-001`, `SFT-CHEM-MEAS-TRACEABILITY-001`, `SFT-MAT-SUST-LIFECYCLE-001`, `SFT-BIO-BIO-HANDOFF-001`, `SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001`, `SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001`, `SFT-EARTH-HAZARD-RISK-HANDOFF-001`, `SFT-ASTRO-ASTRO-HANDOFF-001`, `SFT-SOCIAL-ENGINEERING-HANDOFF-001`, `SFT-ENG-REQUIREMENT-001`. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external standard, application, performance target, product or prior implementation to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external standards, products, tests or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite components, versions, platforms, tests and lifecycle records; performance remains artifact, environment, user, method and operating-boundary limited.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - which requirement, artifact, component, user, environment or test bears the claim: `input-operation-output-purpose-carrier`.
- **boundary** - which operating envelope, version, platform, user and lifecycle stage bound the result: `function-domain-bounded`.
- **relation** - which generated transformation, interface, comparison or handoff connects the carriers: `declared-transformation-relation`.
- **record** - which requirements, versions, alternatives, failures, uncertainty and rollback remain reconstructible: `inputs-outputs-state-loss-context-and-failure-held`.
- **evidence** - whether law, requirement, design, simulation, test, demonstration, performance and anomaly remain distinct: `science-design-test-and-decision-classes-explicit`.
- **provenance** - whether the result traces to the One and names every upstream receipt and design choice: `root-bound-forward-translation`.
- **generality** - whether finite extension preserves platforms, versions, failures and operating limits: `positive-finite-extension-preserves-platforms-failures-and-versions`.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: `no-extra-rule`.

Their ordered product gives the unique survivor:

`input-operation-output-purpose-carrier__function-domain-bounded__declared-transformation-relation__inputs-outputs-state-loss-context-and-failure-held__science-design-test-and-decision-classes-explicit__root-bound-forward-translation__positive-finite-extension-preserves-platforms-failures-and-versions__no-extra-rule`

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses the requirement, artifact, user or environment. Erasing the boundary universalises a version, platform or test. Importing the relation lets a target or preferred application choose the result. Retaining only successful output removes requirements, alternatives, failures and rollback. Conflating evidence classes turns simulation, demonstration or performance into observation or law. Breaking provenance removes the trace to the One and upstream receipts. Erasing finite context makes one implementation universal. Adding a silent dependency or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite requirement or admitted law, versioned artifact, boundary, testable relation and complete record. The successor adds one lawful component, interface, platform, requirement, test, failure, lifecycle stage or anomaly while preserving all earlier identities, adverse rows and receipts.

Falsification and deliberately unfavourable controls

The registered falsification condition is: Reject if a product label, component name or favourable output substitutes for the transformation.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: `source_bound_engineering_requirement_design_test_lifecycle_or_access_record/primary_standard_handbook_guidance_or_project_record_structural_correspondence`. Captured registered sources: 2. Unresolved transports retained: 1. The registered structural consequence corresponded: `True`. External evidence selected the survivor: `false`. Application or performance selected the law: `false`. Successful performance was relabelled scientific proof: `false`. Failed implementation was relabelled scientific falsification: `false`. Simulation or demonstration was relabelled observation: `false`.

- `NASA-SE-HANDBOOK-001` - transport `captured`; captured bytes `3773440`; snapshot `sha256:8eeb4887a4dc57a23049da7dd2ed556833cf98e214b240468d987873164ff688`.
- `NASA-SWE-VALIDATION-001` - transport `captured`; captured bytes `114542`; snapshot `sha256:15ed8ce3e5acbf6c24380016a30ece410cc61b9b31ea16d01b31f00cac52e888`.
- `FDA-DESIGN-CONTROLS-001` - transport `failed_preserved`; captured bytes `0`; snapshot `unresolved and preserved`.

The comparison is structural and source-bound. It checks that a relevant primary handbook, standard, guidance record or open engineering project can represent the declared requirement, artifact, environment, operating boundary, version, test and lifecycle without rewriting the Fold law. It does not claim that a standard proves conformity, that one handbook exhausts Engineering, or that a custodian's reputation proves the result. New records can extend or challenge the surface only with source, version, configuration, method, failures and adverse rows retained.

Scientific meaning

For **required function**, the consequence is practical and exact: A function is a declared input-to-output transformation whose state, loss, context and failure modes remain explicit. An Engineering claim therefore has to expose the carrier `input-operation-output-purpose-carrier`, the relation `declared-transformation-relation`, the record `inputs-outputs-state-loss-context-and-failure-held`, and the boundary `function-domain-bounded` together. If any one is hidden, the result may remain a concept, prototype, simulation, demonstration or product choice, but it is not this admitted law.

This distinction protects users, maintainers and science. It prevents a successful output from erasing requirements, failed alternatives, hidden resources, inaccessible users, hazards, maintenance burdens or end-of-life costs. It prevents an application from selecting the upstream law and prevents engineering failure from becoming automatic scientific falsification. A replacement must still generate its forms, eliminate alternatives, survive controls, preserve custody and reproduce through the shared engine.

Machine certificate

Candidate count: `256`; survivor count: `1`; derivation seal: `sha256:a12a2ff27de91e678143a68f38a4830932f497cf6c7c54366f85ed66bee4eb67`; independent certificate: `sha256:eb6ad906b003aeb9cde3058aeedc2dea8629bb5163ca7c5585e237b58983d2ba`; engine receipt: `sha256:c3cde1442d2ae6fbed97c59b287da2c790327f965736295951dd68cb1ac50206`. The independent implementation recomputed the candidate product: `True`. All external rows were preserved: `True`.

3. Engineering constraint

Claim: `SFT-ENG-CONSTRAINT-001`

Family: `requirements_function_boundary`

Exact statement: A constraint is a source- and context-bound limit on generated designs, never an unexplained fitted preference.

Forward chain and exact carrier

The generated carrier is `design-variable-limit-source-carrier`. Its required relation is `variable-to-admissible-region-relation`. The reconstruction must retain `source-units-tolerance-context-version-and-conflict-held`. Its declared empirical boundary is `constraint-authority-bounded`. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is `SFT-FOUNDATION-ONE-001`, `SFT-FOUNDATION-FOLD-001`, `SFT-FOUNDATION-FOLD-DYNAMICS-001`, `SFT-MATH-EXACT-ARITHMETIC-001`, `SFT-MATH-GRAPH-NETWORK-001`, `SFT-MATH-OPTIMIZATION-001`, `SFT-MATH-DYNAMICAL-SYSTEMS-001`, `SFT-MATH-LOGIC-PROOF-001`, `SFT-INFO-QUANTITY-001`, `SFT-INFO-CHANNEL-CAPACITY-001`, `SFT-COMP-FORM-STATE-TRANSITION-001`, `SFT-COMP-SEM-CORRECTNESS-001`, `SFT-COMP-SEM-COMPILATION-001`, `SFT-COMP-DIST-CAUSALITY-001`, `SFT-PHYS-MEAS-VALUE-RECORD-001`, `SFT-PHYS-MEAS-CALIBRATION-001`, `SFT-CHEM-MEAS-TRACEABILITY-001`, `SFT-MAT-SUST-LIFECYCLE-001`, `SFT-BIO-BIO-HANDOFF-001`, `SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001`, `SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001`, `SFT-EARTH-HAZARD-RISK-HANDOFF-001`, `SFT-ASTRO-ASTRO-HANDOFF-001`, `SFT-SOCIAL-ENGINEERING-HANDOFF-001`, `SFT-ENG-FUNCTION-001`. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external standard, application, performance target, product or prior implementation to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external standards, products, tests or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite components, versions, platforms, tests and lifecycle records; performance remains artifact, environment, user, method and operating-boundary limited.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - which requirement, artifact, component, user, environment or test bears the claim: `design-variable-limit-source-carrier`.
- **boundary** - which operating envelope, version, platform, user and lifecycle stage bound the result: `constraint-authority-bounded`.
- **relation** - which generated transformation, interface, comparison or handoff connects the carriers: `variable-to-admissible-region-relation`.
- **record** - which requirements, versions, alternatives, failures, uncertainty and rollback remain reconstructible: `source-units-tolerance-context-version-and-conflict-held`.
- **evidence** - whether law, requirement, design, simulation, test, demonstration, performance and anomaly remain distinct: `science-design-test-and-decision-classes-explicit`.
- **provenance** - whether the result traces to the One and names every upstream receipt and design choice: `root-bound-forward-translation`.
- **generality** - whether finite extension preserves platforms, versions, failures and operating limits: `positive-finite-extension-preserves-platforms-failures-and-versions`.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: `no-extra-rule`.

Their ordered product gives the unique survivor:

```
design-variable-limit-source-carrier__constraint-authority-bounded__variable-to-admis
sible-region-relation__source-units-tolerance-context-version-and-conflict-held__scie
nce-design-test-and-decision-classes-explicit__root-bound-forward-translation__positi
ve-finite-extension-preserves-platforms-failures-and-versions__no-extra-rule
```

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses the requirement, artifact, user or environment. Erasing the boundary universalises a version, platform or test. Importing the relation lets a target or preferred application choose the result. Retaining only successful output removes requirements, alternatives, failures and rollback. Conflating evidence classes turns simulation, demonstration or performance into observation or law. Breaking provenance removes the trace to the One and upstream receipts. Erasing finite context makes one implementation universal. Adding a silent dependency or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite requirement or admitted law, versioned artifact, boundary, testable relation and complete record. The successor adds one lawful component, interface, platform, requirement, test, failure, lifecycle stage or anomaly while preserving all earlier identities, adverse rows and receipts.

Falsification and deliberately unfavourable controls

The registered falsification condition is: Reject if a convenient target or hidden parameter is introduced as a necessary limit.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: `source_bound_engineering_requirement_design_test_lifecycle_or_access_record/primary_standard_handbook_guidance_or_project_record_structural_correspondence`. Captured registered sources: 2. Unresolved transports retained: 1. The registered structural consequence corresponded: `True`. External evidence selected the survivor: `false`. Application or performance selected the law: `false`. Successful performance was relabelled scientific proof: `false`. Failed implementation was relabelled scientific falsification: `false`. Simulation or demonstration was relabelled observation: `false`.

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The comparison is structural and source-bound. It checks that a relevant primary handbook, standard, guidance record or open engineering project can represent the declared requirement, artifact, environment, operating boundary, version, test and lifecycle without rewriting the Fold law. It does not claim that a standard proves conformity, that one handbook exhausts Engineering, or that a custodian's reputation proves the result. New records can extend or challenge the surface only with source, version, configuration, method, failures and adverse rows retained.

Scientific meaning

For **engineering constraint**, the consequence is practical and exact: A constraint is a source- and context-bound limit on generated designs, never an unexplained fitted preference. An Engineering claim therefore has to expose the carrier `design-variable-limit-source-carrier`, the relation `variable-to-admissible-region-relation`, the record `source-units-tolerance-context-version-and-conflict-held`, and the boundary `constraint-authority-bounded` together. If any one is hidden, the result may remain a concept, prototype, simulation, demonstration or product choice, but it is not this admitted law.

This distinction protects users, maintainers and science. It prevents a successful output from erasing requirements, failed alternatives, hidden resources, inaccessible users, hazards, maintenance burdens or end-of-life costs. It prevents an application from selecting the upstream law and prevents engineering failure from becoming automatic scientific falsification. A replacement must still generate its forms, eliminate alternatives, survive controls, preserve custody and reproduce through the shared engine.

Machine certificate

Candidate count: `256`; survivor count: `1`; derivation seal: `sha256:415de770658c8a91cab450d930efdc4d6edf8a08ef72c6f19a3a4d4b397bf84a`; independent certificate: `sha256:655d1e169faccbb36281abf187f6c71aebc402c2fb54ea62d0f51ffee6589645`; engine receipt: `sha256:0b201dc8237be12a2290a8d7113ce6f834d1a17d7d0289889b81130e30b89f0b`. The independent implementation recomputed the candidate product: `True`. All external rows were preserved: `True`.

4. Operating boundary

Claim: `SFT-ENG-OPERATING-BOUNDARY-001`

Family: `requirements_function_boundary`

Exact statement: An operating boundary retains environment, loads, duration, transients, exceptions and observed failures.

Forward chain and exact carrier

The generated carrier is `system-environment-state-limit-carrier`. Its required relation is `condition-to-valid-operation-relation`. The reconstruction must retain `conditions-duration-loads-transients-exceptions-and-failures-held`. Its declared empirical boundary is `declared-operation-envelope`. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is `SFT-FOUNDATION-ONE-001`, `SFT-FOUNDATION-FOLD-001`, `SFT-FOUNDATION-FOLD-DYNAMICS-001`, `SFT-MATH-EXACT-ARITHMETIC-001`, `SFT-MATH-GRAPH-NETWORK-001`, `SFT-MATH-OPTIMIZATION-001`, `SFT-MATH-DYNAMICAL-SYSTEMS-001`, `SFT-MATH-LOGIC-PROOF-001`, `SFT-INFO-QUANTITY-001`, `SFT-INFO-CHANNEL-CAPACITY-001`, `SFT-COMP-FORM-STATE-TRANSITION-001`, `SFT-COMP-SEM-CORRECTNESS-001`, `SFT-COMP-SEM-COMPILATION-001`, `SFT-COMP-DIST-CAUSALITY-001`, `SFT-PHYS-MEAS-VALUE-RECORD-001`, `SFT-PHYS-MEAS-CALIBRATION-001`, `SFT-CHEM-MEAS-TRACEABILITY-001`, `SFT-MAT-SUST-LIFECYCLE-001`, `SFT-BIO-BIO-HANDOFF-001`, `SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001`, `SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001`, `SFT-EARTH-HAZARD-RISK-HANDOFF-001`, `SFT-ASTRO-ASTRO-HANDOFF-001`, `SFT-SOCIAL-ENGINEERING-HANDOFF-001`, `SFT-ENG-CONSTRAINT-001`. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external standard, application, performance target, product or prior implementation to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external standards, products, tests or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite components, versions, platforms, tests and lifecycle records; performance remains artifact, environment, user, method and operating-boundary limited.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - which requirement, artifact, component, user, environment or test bears the claim: `system-environment-state-limit-carrier`.
- **boundary** - which operating envelope, version, platform, user and lifecycle stage bound the result: `declared-operation-envelope`.
- **relation** - which generated transformation, interface, comparison or handoff connects the carriers: `condition-to-valid-operation-relation`.
- **record** - which requirements, versions, alternatives, failures, uncertainty and rollback remain reconstructible: `conditions-duration-loads-transients-exceptions-and-failures-held`.
- **evidence** - whether law, requirement, design, simulation, test, demonstration, performance and anomaly remain distinct: `science-design-test-and-decision-classes-explicit`.
- **provenance** - whether the result traces to the One and names every upstream receipt and design choice: `root-bound-forward-translation`.
- **generality** - whether finite extension preserves platforms, versions, failures and operating limits: `positive-finite-extension-preserves-platforms-failures-and-versions`.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: `no-extra-rule`.

Their ordered product gives the unique survivor:

```
system-environment-state-limit-carrier__declared-operation-envelope__condition-to-val
id-operation-relation__conditions-duration-loads-transients-exceptions-and-failures-h
eld__science-design-test-and-decision-classes-explicit__root-bound-forward-translatio
n__positive-finite-extension-preserves-platforms-failures-and-versions__no-extra-rule
```

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses the requirement, artifact, user or environment. Erasing the boundary universalises a version, platform or test. Importing the relation lets a target or preferred application choose the result. Retaining only successful output removes requirements, alternatives, failures and rollback. Conflating evidence classes turns simulation, demonstration or performance into observation or law. Breaking provenance removes the trace to the One and upstream receipts. Erasing finite context makes one implementation universal. Adding a silent dependency or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite requirement or admitted law, versioned artifact, boundary, testable relation and complete record. The successor adds one lawful component, interface, platform, requirement, test, failure, lifecycle stage or anomaly while preserving all earlier identities, adverse rows and receipts.

Falsification and deliberately unfavourable controls

The registered falsification condition is: Reject if success inside one test envelope is universalized beyond its registered boundary.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: `source_bound_engineering_requirement_design_test_lifecycle_or_access_record/primary_standard_handbook_guidance_or_project_record_structural_correspondence`. Captured registered sources: 2. Unresolved transports retained: 1. The registered structural consequence corresponded: `True`. External evidence selected the survivor: `false`. Application or performance selected the law: `false`. Successful performance was relabelled scientific proof: `false`. Failed implementation was relabelled scientific falsification: `false`. Simulation or demonstration was relabelled observation: `false`.

- `NASA-SE-HANDBOOK-001` - transport `captured`; captured bytes `3773440`; snapshot `sha256:8eeb4887a4dc57a23049da7dd2ed556833cf98e214b240468d987873164ff688`.
- `NASA-SWE-VALIDATION-001` - transport `captured`; captured bytes `114542`; snapshot `sha256:15ed8ce3e5acbf6c24380016a30ece410cc61b9b31ea16d01b31f00cac52e888`.
- `FDA-DESIGN-CONTROLS-001` - transport `failed_preserved`; captured bytes `0`; snapshot `unresolved and preserved`.

The comparison is structural and source-bound. It checks that a relevant primary handbook, standard, guidance record or open engineering project can represent the declared requirement, artifact, environment, operating boundary, version, test and lifecycle without rewriting the Fold law. It does not claim that a standard proves conformity, that one handbook exhausts Engineering, or that a custodian's reputation proves the result. New records can extend or challenge the surface only with source, version, configuration, method, failures and adverse rows retained.

Scientific meaning

For **operating boundary**, the consequence is practical and exact: An operating boundary retains environment, loads, duration, transients, exceptions and observed failures. An Engineering claim therefore has to expose the carrier `system-environment-state-limit-carrier`, the relation `condition-to-valid-operation-relation`, the record `conditions-duration-loads-transients-exceptions-and-failures-held`, and the boundary `declared-operation-envelope` together. If any one is hidden, the result may remain a concept, prototype, simulation, demonstration or product choice, but it is not this admitted law.

This distinction protects users, maintainers and science. It prevents a successful output from erasing requirements, failed alternatives, hidden resources, inaccessible users, hazards, maintenance burdens or end-of-life costs. It prevents an application from selecting the upstream law and prevents engineering failure from becoming automatic scientific falsification. A replacement must still generate its forms, eliminate alternatives, survive controls, preserve custody and reproduce through the shared engine.

Machine certificate

Candidate count: `256`; survivor count: `1`; derivation seal: `sha256:1575ba4d03cf91565f594dc9d5b3a2540a0282950b5fdaab73f9d90be4b71b26`; independent certificate: `sha256:733273f4b0d6b643eb78f51a552583808a3f7707a70feca499bdcfff36ba0e93`; engine receipt: `sha256:3447dce9ca5104791f130d5cf2b4612259d9b5ad5e73cd45bce488c29270fca5`. The independent implementation recomputed the candidate product: `True`. All external rows were preserved: `True`.

5. Use and misuse case

Claim: `SFT-ENG-USE-CASE-001`

Family: `requirements_function_boundary`

Exact statement: A use case retains actor, goal, preconditions, path and outcome; foreseeable misuse and harm remain co-registered.

Forward chain and exact carrier

The generated carrier is `actor-goal-system-context-carrier`. Its required relation is `action-to-system-response-relation`. The reconstruction must retain `actors-preconditions-path-outcomes-misuse-and-harm-held`. Its declared empirical boundary is `use-context-bounded`. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is `SFT-FOUNDATION-ONE-001`, `SFT-FOUNDATION-FOLD-001`, `SFT-FOUNDATION-FOLD-DYNAMICS-001`, `SFT-MATH-EXACT-ARITHMETIC-001`, `SFT-MATH-GRAPH-NETWORK-001`, `SFT-MATH-OPTIMIZATION-001`, `SFT-MATH-DYNAMICAL-SYSTEMS-001`, `SFT-MATH-LOGIC-PROOF-001`, `SFT-INFO-QUANTITY-001`, `SFT-INFO-CHANNEL-CAPACITY-001`, `SFT-COMP-FORM-STATE-TRANSITION-001`, `SFT-COMP-SEM-CORRECTNESS-001`, `SFT-COMP-SEM-COMPILATION-001`, `SFT-COMP-DIST-CAUSALITY-001`, `SFT-PHYS-MEAS-VALUE-RECORD-001`, `SFT-PHYS-MEAS-CALIBRATION-001`, `SFT-CHEM-MEAS-TRACEABILITY-001`, `SFT-MAT-SUST-LIFECYCLE-001`, `SFT-BIO-BIO-HANDOFF-001`, `SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001`, `SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001`, `SFT-EARTH-HAZARD-RISK-HANDOFF-001`, `SFT-ASTRO-ASTRO-HANDOFF-001`, `SFT-SOCIAL-ENGINEERING-HANDOFF-001`, `SFT-ENG-OPERATING-BOUNDARY-001`. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external standard, application, performance target, product or prior implementation to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external standards, products, tests or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite components, versions, platforms, tests and lifecycle records; performance remains artifact, environment, user, method and operating-boundary limited.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - which requirement, artifact, component, user, environment or test bears the claim: `actor-goal-system-context-carrier`.
- **boundary** - which operating envelope, version, platform, user and lifecycle stage bound the result: `use-context-bounded`.
- **relation** - which generated transformation, interface, comparison or handoff connects the carriers: `action-to-system-response-relation`.
- **record** - which requirements, versions, alternatives, failures, uncertainty and rollback remain reconstructible: `actors-preconditions-path-outcomes-misuse-and-harm-held`.

- **evidence** - whether law, requirement, design, simulation, test, demonstration, performance and anomaly remain distinct: `science-design-test-and-decision-classes-explicit`.
- **provenance** - whether the result traces to the One and names every upstream receipt and design choice: `root-bound-forward-translation`.
- **generality** - whether finite extension preserves platforms, versions, failures and operating limits: `positive-finite-extension-preserves-platforms-failures-and-versions`.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: `no-extra-rule`.

Their ordered product gives the unique survivor:

```
actor-goal-system-context-carrier__use-context-bounded__action-to-system-response-relation__actors-preconditions-path-outcomes-misuse-and-harm-held__science-design-test-and-decision-classes-explicit__root-bound-forward-translation__positive-finite-extension-preserves-platforms-failures-and-versions__no-extra-rule
```

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses the requirement, artifact, user or environment. Erasing the boundary universalises a version, platform or test. Importing the relation lets a target or preferred application choose the result. Retaining only successful output removes requirements, alternatives, failures and rollback. Conflating evidence classes turns simulation, demonstration or performance into observation or law. Breaking provenance removes the trace to the One and upstream receipts. Erasing finite context makes one implementation universal. Adding a silent dependency or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite requirement or admitted law, versioned artifact, boundary, testable relation and complete record. The successor adds one lawful component, interface, platform, requirement, test, failure, lifecycle stage or anomaly while preserving all earlier identities, adverse rows and receipts.

Falsification and deliberately unfavourable controls

The registered falsification condition is: Reject if intended use erases observed or foreseeable misuse.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: `source_bound_engineering_requirement_design_test_lifecycle_or_access_record/primary_standard_handbook_guidance_or_project_record_structural_correspondence`. Captured registered sources: 2. Unresolved transports retained: 1. The registered structural consequence corresponded: `True`. External evidence selected the survivor: `false`. Application or performance selected the law: `false`. Successful performance was relabelled scientific proof: `false`. Failed implementation was relabelled scientific falsification: `false`. Simulation or demonstration was relabelled observation: `false`.

- `NASA-SE-HANDBOOK-001` - transport `captured`; captured bytes `3773440`; snapshot `sha256:8eeb4887a4dc57a23049da7dd2ed556833cf98e214b240468d987873164ff688`.
- `NASA-SWE-VALIDATION-001` - transport `captured`; captured bytes `114542`; snapshot `sha256:15ed8ce3e5acbf6c24380016a30ece410cc61b9b31ea16d01b31f00cac52e888`.
- `FDA-DESIGN-CONTROLS-001` - transport `failed_preserved`; captured bytes `0`; snapshot `unresolved and preserved`.

The comparison is structural and source-bound. It checks that a relevant primary handbook, standard, guidance record or open engineering project can represent the declared requirement, artifact, environment, operating boundary, version, test and lifecycle without rewriting the Fold law. It does not claim that a standard proves conformity, that one handbook exhausts Engineering, or that a custodian's reputation proves the result. New records can extend or challenge the surface only with source, version, configuration, method, failures and adverse rows retained.

Scientific meaning

For **use and misuse case**, the consequence is practical and exact: A use case retains actor, goal, preconditions, path and outcome; foreseeable misuse and harm remain co-registered. An Engineering claim therefore has to expose the carrier `actor-goal-system-context-carrier`, the relation `action-to-system-response-relation`, the record `actors-preconditions-path-outcomes-misuse-and-harm-held`, and the boundary `use-context-bounded` together. If any one is hidden, the result may remain a concept, prototype, simulation, demonstration or product choice, but it is not this admitted law.

This distinction protects users, maintainers and science. It prevents a successful output from erasing requirements, failed alternatives, hidden resources, inaccessible users, hazards, maintenance burdens or end-of-life costs. It prevents an application from selecting the upstream law and prevents engineering failure from becoming automatic scientific falsification. A replacement must still generate its forms, eliminate alternatives, survive controls, preserve custody and reproduce through the shared engine.

Machine certificate

Candidate count: `256`; survivor count: `1`; derivation seal: `sha256:ebbf3f76baa02a34d97d02ffffdabad925b69b2002ccb59d2ca1206de03a1629`; independent certificate: `sha256:6d0e4dae77777e7d7024aa88d1aabead0776ee1b1634776d11382738ad73f169`; engine receipt: `sha256:e82fd4f8d04b79886ae1b2da4d9ae232d34bc07029efc5a5a9c3dab431ea50f5`. The independent implementation recomputed the candidate product: `True`. All external rows were preserved: `True`.

6. Requirement acceptance boundary

Claim: `SFT-ENG-ACCEPTANCE-BOUNDARY-001`

Family: `requirements_function_boundary`

Exact statement: Acceptance requires a preregistered criterion, method, result, uncertainty and preserved failure record.

Forward chain and exact carrier

The generated carrier is `requirement-test-evidence-decision-carrier`. Its required relation is `condition-to-acceptance-relation`. The reconstruction must retain `criterion-method-result-uncertainty-failure-and-approver-held`. Its declared empirical boundary is `acceptance-protocol-bounded`. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is `SFT-FOUNDATION-ONE-001`, `SFT-FOUNDATION-FOLD-001`, `SFT-FOUNDATION-FOLD-DYNAMICS-001`, `SFT-MATH-EXACT-ARITHMETIC-001`, `SFT-MATH-GRAPH-NETWORK-001`, `SFT-MATH-OPTIMIZATION-001`, `SFT-MATH-DYNAMICAL-SYSTEMS-001`, `SFT-MATH-LOGIC-PROOF-001`, `SFT-INFO-QUANTITY-001`, `SFT-INFO-CHANNEL-CAPACITY-001`, `SFT-COMP-FORM-STATE-TRANSITION-001`, `SFT-COMP-SEM-CORRECTNESS-001`, `SFT-COMP-SEM-COMPILATION-001`, `SFT-COMP-DIST-CAUSALITY-001`, `SFT-PHYS-MEAS-VALUE-RECORD-001`, `SFT-PHYS-MEAS-CALIBRATION-001`, `SFT-CHEM-MEAS-TRACEABILITY-001`, `SFT-MAT-SUST-LIFECYCLE-001`, `SFT-BIO-BIO-HANDOFF-001`, `SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001`, `SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001`, `SFT-EARTH-HAZARD-RISK-HANDOFF-001`, `SFT-ASTRO-ASTRO-HANDOFF-001`, `SFT-SOCIAL-ENGINEERING-HANDOFF-001`, `SFT-ENG-USE-CASE-001`. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external standard, application, performance target, product or prior implementation to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external standards, products, tests or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite components, versions, platforms, tests and lifecycle records; performance remains artifact, environment, user, method and operating-boundary limited.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - which requirement, artifact, component, user, environment or test bears the claim: `requirement-test-evidence-decision-carrier`.
- **boundary** - which operating envelope, version, platform, user and lifecycle stage bound the result: `acceptance-protocol-bounded`.
- **relation** - which generated transformation, interface, comparison or handoff connects the carriers: `condition-to-acceptance-relation`.
- **record** - which requirements, versions, alternatives, failures, uncertainty and rollback remain reconstructible: `criterion-method-result-uncertainty-failure-and-approver-held`.
- **evidence** - whether law, requirement, design, simulation, test, demonstration, performance and anomaly remain distinct: `science-design-test-and-decision-classes-explicit`.
- **provenance** - whether the result traces to the One and names every upstream receipt and design choice: `root-bound-forward-translation`.
- **generality** - whether finite extension preserves platforms, versions, failures and operating limits: `positive-finite-extension-preserves-platforms-failures-and-versions`.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: `no-extra-rule`.

Their ordered product gives the unique survivor:

```
requirement-test-evidence-decision-carrier__acceptance-protocol-bounded__condition-to
-acceptance-relation__criterion-method-result-uncertainty-failure-and-approver-held__
science-design-test-and-decision-classes-explicit__root-bound-forward-translation__po
sitive-finite-extension-preserves-platforms-failures-and-versions__no-extra-rule
```

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses the requirement, artifact, user or environment. Erasing the boundary universalises a version, platform or test. Importing the relation lets a target or preferred application choose the result. Retaining only successful output removes requirements, alternatives, failures and rollback. Conflating evidence classes turns simulation, demonstration or performance into observation or law. Breaking provenance removes the trace to the One and upstream receipts. Erasing finite context makes one implementation universal. Adding a silent dependency or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite requirement or admitted law, versioned artifact, boundary, testable relation and complete record. The successor adds one lawful component, interface, platform, requirement, test, failure, lifecycle stage or anomaly while preserving all earlier identities, adverse rows and receipts.

Falsification and deliberately unfavourable controls

The registered falsification condition is: Reject if the criterion changes after outcome or authority alone declares compliance.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: source_bound_engineering_requirement_design_test_lifecycle_or_access_record/ primary_standard_handbook_guidance_or_project_record_structural_correspondence. Captured registered sources: 2. Unresolved transports retained: 1. The registered structural consequence corresponded: True. External evidence selected the survivor: false. Application or performance selected the law: false. Successful performance was relabelled scientific proof: false. Failed implementation was relabelled scientific falsification: false. Simulation or demonstration was relabelled observation: false.

- NASA-SE-HANDBOOK-001 - transport captured; captured bytes 3773440; snapshot sha256:8eeb4887a4dc57a23049da7dd2ed556833cf98e214b240468d987873164ff688.
- NASA-SWE-VALIDATION-001 - transport captured; captured bytes 114542; snapshot sha256:15ed8ce3e5acbf6c24380016a30ece410cc61b9b31ea16d01b31f00cac52e888.
- FDA-DESIGN-CONTROLS-001 - transport failed_preserved; captured bytes 0; snapshot unresolved and preserved.

The comparison is structural and source-bound. It checks that a relevant primary handbook, standard, guidance record or open engineering project can represent the declared requirement, artifact, environment, operating boundary, version, test and lifecycle without rewriting the Fold law. It does not claim that a standard proves conformity, that one handbook exhausts Engineering, or that a custodian's reputation proves the result. New records can extend or challenge the surface only with source, version, configuration, method, failures and adverse rows retained.

Scientific meaning

For **requirement acceptance boundary**, the consequence is practical and exact: Acceptance requires a preregistered criterion, method, result, uncertainty and preserved failure record. An Engineering claim therefore has to expose the carrier requirement-test-evidence-decision-carrier, the relation condition-to-acceptance-relation, the record criterion-method-result-uncertainty-failure-and-approver-held, and the boundary acceptance-protocol-bounded together. If any one is hidden, the result may remain a concept, prototype, simulation, demonstration or product choice, but it is not this admitted law.

This distinction protects users, maintainers and science. It prevents a successful output from erasing requirements, failed alternatives, hidden resources, inaccessible users, hazards, maintenance burdens or end-of-life costs. It prevents an application from selecting the upstream law and prevents engineering failure from becoming automatic scientific falsification. A replacement must still generate its forms, eliminate alternatives, survive controls, preserve custody and reproduce through the shared engine.

Machine certificate

Candidate count: 256; survivor count: 1; derivation seal: sha256:a38a416dc80fb95b1c4340bdf4b2982b4d49d8111dc571cebe1939aa3be96918; independent certificate: sha256:6d4cb93d0d3ff54cef9008aa3e2d08e74977a0b1cef639f5635ffc66e026cf8a; engine receipt: sha256:cd154a063434c15b09f39717c0853d04b08f9df9a24ebe975c1ad557545d6db4. The independent implementation recomputed the candidate product: True. All external rows were preserved: True.

7. Engineering component

Claim: SFT-ENG-COMPONENT-001

Family: components_interfaces_architecture

Exact statement: A component is a versioned bounded part with declared state, capability, interfaces and failures.

Forward chain and exact carrier

The generated carrier is bounded-part-state-capability-carrier. Its required relation is part-to-declared-function-relation. The reconstruction must retain identity-version-input-output-state-and-failure-held. Its declared empirical boundary is component-boundary-declared. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is `SFT-FOUNDATION-ONE-001`, `SFT-FOUNDATION-FOLD-001`, `SFT-FOUNDATION-FOLD-DYNAMICS-001`, `SFT-MATH-EXACT-ARITHMETIC-001`, `SFT-MATH-GRAPH-NETWORK-001`, `SFT-MATH-OPTIMIZATION-001`, `SFT-MATH-DYNAMICAL-SYSTEMS-001`, `SFT-MATH-LOGIC-PROOF-001`, `SFT-INFO-QUANTITY-001`, `SFT-INFO-CHANNEL-CAPACITY-001`, `SFT-COMP-FORM-STATE-TRANSITION-001`, `SFT-COMP-SEM-CORRECTNESS-001`, `SFT-COMP-SEM-COMPILATION-001`, `SFT-COMP-DIST-CAUSALITY-001`, `SFT-PHYS-MEAS-VALUE-RECORD-001`, `SFT-PHYS-MEAS-CALIBRATION-001`, `SFT-CHEM-MEAS-TRACEABILITY-001`, `SFT-MAT-SUST-LIFECYCLE-001`, `SFT-BIO-BIO-HANDOFF-001`, `SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001`, `SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001`, `SFT-EARTH-HAZARD-RISK-HANDOFF-001`, `SFT-ASTRO-ASTRO-HANDOFF-001`, `SFT-SOCIAL-ENGINEERING-HANDOFF-001`, `SFT-ENG-ACCEPTANCE-BOUNDARY-001`. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external standard, application, performance target, product or prior implementation to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external standards, products, tests or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite components, versions, platforms, tests and lifecycle records; performance remains artifact, environment, user, method and operating-boundary limited.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - which requirement, artifact, component, user, environment or test bears the claim: `bounded-part-state-capability-carrier`.
- **boundary** - which operating envelope, version, platform, user and lifecycle stage bound the result: `component-boundary-declared`.
- **relation** - which generated transformation, interface, comparison or handoff connects the carriers: `part-to-declared-function-relation`.
- **record** - which requirements, versions, alternatives, failures, uncertainty and rollback remain reconstructible: `identity-version-input-output-state-and-failure-held`.
- **evidence** - whether law, requirement, design, simulation, test, demonstration, performance and anomaly remain distinct: `science-design-test-and-decision-classes-explicit`.
- **provenance** - whether the result traces to the One and names every upstream receipt and design choice: `root-bound-forward-translation`.
- **generality** - whether finite extension preserves platforms, versions, failures and operating limits: `positive-finite-extension-preserves-platforms-failures-and-versions`.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: `no-extra-rule`.

Their ordered product gives the unique survivor:

```
bounded-part-state-capability-carrier__component-boundary-declared__part-to-declared-function-relation__identity-version-input-output-state-and-failure-held__science-design-test-and-decision-classes-explicit__root-bound-forward-translation__positive-finite-extension-preserves-platforms-failures-and-versions__no-extra-rule
```

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses the requirement, artifact, user or environment. Erasing the boundary universalises a version, platform or test. Importing the relation lets a target or preferred application choose the result. Retaining only successful output removes requirements, alternatives, failures and rollback. Conflating evidence classes turns simulation, demonstration or performance into observation or law. Breaking provenance removes the trace to the One and upstream receipts. Erasing finite context makes one implementation universal. Adding a silent dependency or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite requirement or admitted law, versioned artifact, boundary, testable relation and complete record. The successor adds one lawful component, interface, platform, requirement, test, failure, lifecycle stage or anomaly while preserving all earlier identities, adverse rows and receipts.

Falsification and deliberately unfavourable controls

The registered falsification condition is: Reject if a name or vendor identity substitutes for measured behaviour.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: `source_bound_engineering_requirement_design_test_lifecycle_or_access_record/primary_standard_handbook_guidance_or_project_record_structural_correspondence`. Captured registered sources: 3. Unresolved transports retained: 0. The registered structural consequence corresponded: `True`. External evidence selected the survivor: `false`. Application or performance selected the law: `false`. Successful performance was relabelled scientific proof: `false`. Failed implementation was relabelled scientific falsification: `false`. Simulation or demonstration was relabelled observation: `false`.

- `NASA-SE-HANDBOOK-001` - transport `captured`; captured bytes `3773440`; snapshot `sha256:8eeb4887a4dc57a23049da7dd2ed556833cf98e214b240468d987873164ff688`.
- `NIST-SSE-001` - transport `captured`; captured bytes `47984`; snapshot `sha256:64c8d41d9a78268d596a0a60aad18eead50fd375e8628486f7202788a207edaa`.
- `PYTHON-PLATFORM-001` - transport `captured`; captured bytes `63923`; snapshot `sha256:6873ff23f5e2e6985050e60024a477265ed779867e32542e133879aae0e01611`.

The comparison is structural and source-bound. It checks that a relevant primary handbook, standard, guidance record or open engineering project can represent the declared requirement, artifact, environment, operating boundary, version, test and lifecycle without rewriting the Fold law. It does not claim that a standard proves conformity, that one handbook exhausts Engineering, or that a custodian's reputation proves the result. New records can extend or challenge the surface only with source, version, configuration, method, failures and adverse rows retained.

Scientific meaning

For **engineering component**, the consequence is practical and exact: A component is a versioned bounded part with declared state, capability, interfaces and failures. An Engineering claim therefore has to expose the carrier `bounded-part-state-capability-carrier`, the relation `part-to-declared-function-relation`, the record `identity-version-input-output-state-and-failure-held`, and the boundary `component-boundary-declared` together. If any one is hidden, the result may remain a concept, prototype, simulation, demonstration or product choice, but it is not this admitted law.

This distinction protects users, maintainers and science. It prevents a successful output from erasing requirements, failed alternatives, hidden resources, inaccessible users, hazards, maintenance burdens or end-of-life costs. It prevents an application from selecting the upstream law and prevents engineering failure from becoming automatic scientific falsification. A replacement must still generate its forms, eliminate alternatives, survive controls, preserve custody and reproduce through the shared engine.

Machine certificate

Candidate count: `256`; survivor count: `1`; derivation seal: `sha256:f537672be611224cad5b31f3ebb6d43cb0a770a6a22f68e8de5cce72f6c9a630`; independent certificate: `sha256:cc23492e177e8a1c81da5b4f4272e8c1599efdebb1399c872af6e8f3804d9445`; engine receipt: `sha256:1a028469b13f45730ec5a6a1c75966b87229f40e7da12df43f434750d1421eac`. The independent implementation recomputed the candidate product: `True`. All external rows were preserved: `True`.

8. Engineering interface

Claim: `SFT-ENG-INTERFACE-001`

Family: `components_interfaces_architecture`

Exact statement: An interface is a typed exchange contract retaining endpoints, units, timing, errors and version.

Forward chain and exact carrier

The generated carrier is `component-pair-exchange-contract-carrier`. Its required relation is `typed-transfer-and-coordination-relation`. The reconstruction must retain `endpoints-signals-units-timing-errors-and-version-held`. Its declared empirical boundary is `interface-contract-bounded`. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is `SFT-FOUNDATION-ONE-001`, `SFT-FOUNDATION-FOLD-001`, `SFT-FOUNDATION-FOLD-DYNAMICS-001`, `SFT-MATH-EXACT-ARITHMETIC-001`, `SFT-MATH-GRAPH-NETWORK-001`, `SFT-MATH-OPTIMIZATION-001`, `SFT-MATH-DYNAMICAL-SYSTEMS-001`, `SFT-MATH-LOGIC-PROOF-001`, `SFT-INFO-QUANTITY-001`, `SFT-INFO-CHANNEL-CAPACITY-001`, `SFT-COMP-FORM-STATE-TRANSITION-001`, `SFT-COMP-SEM-CORRECTNESS-001`, `SFT-COMP-SEM-COMPILATION-001`, `SFT-COMP-DIST-CAUSALITY-001`, `SFT-PHYS-MEAS-VALUE-RECORD-001`, `SFT-PHYS-MEAS-CALIBRATION-001`, `SFT-CHEM-MEAS-TRACEABILITY-001`, `SFT-MAT-SUST-LIFECYCLE-001`, `SFT-BIO-BIO-HANDOFF-001`, `SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001`, `SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001`, `SFT-EARTH-HAZARD-RISK-HANDOFF-001`, `SFT-ASTRO-ASTRO-HANDOFF-001`, `SFT-SOCIAL-ENGINEERING-HANDOFF-001`, `SFT-ENG-ACCEPTANCE-BOUNDARY-001`, `SFT-ENG-COMPONENT-001`. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external standard, application, performance target, product or prior implementation to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external standards, products, tests or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite components, versions, platforms, tests and lifecycle records; performance remains artifact, environment, user, method and operating-boundary limited.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - which requirement, artifact, component, user, environment or test bears the claim: `component-pair-exchange-contract-carrier`.
- **boundary** - which operating envelope, version, platform, user and lifecycle stage bound the result: `interface-contract-bounded`.
- **relation** - which generated transformation, interface, comparison or handoff connects the carriers: `typed-transfer-and-coordination-relation`.
- **record** - which requirements, versions, alternatives, failures, uncertainty and rollback remain reconstructible: `endpoints-signals-units-timing-errors-and-version-held`.

- **evidence** - whether law, requirement, design, simulation, test, demonstration, performance and anomaly remain distinct: `science-design-test-and-decision-classes-explicit`.
- **provenance** - whether the result traces to the One and names every upstream receipt and design choice: `root-bound-forward-translation`.
- **generality** - whether finite extension preserves platforms, versions, failures and operating limits: `positive-finite-extension-preserves-platforms-failures-and-versions`.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: `no-extra-rule`.

Their ordered product gives the unique survivor:

```
component-pair-exchange-contract-carrier__interface-contract-bounded__typed-transfer-
and-coordination-relation__endpoints-signals-units-timing-errors-and-version-held__sc
ience-design-test-and-decision-classes-explicit__root-bound-forward-translation__posi
tive-finite-extension-preserves-platforms-failures-and-versions__no-extra-rule
```

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses the requirement, artifact, user or environment. Erasing the boundary universalises a version, platform or test. Importing the relation lets a target or preferred application choose the result. Retaining only successful output removes requirements, alternatives, failures and rollback. Conflating evidence classes turns simulation, demonstration or performance into observation or law. Breaking provenance removes the trace to the One and upstream receipts. Erasing finite context makes one implementation universal. Adding a silent dependency or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite requirement or admitted law, versioned artifact, boundary, testable relation and complete record. The successor adds one lawful component, interface, platform, requirement, test, failure, lifecycle stage or anomaly while preserving all earlier identities, adverse rows and receipts.

Falsification and deliberately unfavourable controls

The registered falsification condition is: Reject if physical contact or a call name proves compatible exchange.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: `source_bound_engineering_requirement_design_test_lifecycle_or_access_record/primary_standard_handbook_guidance_or_project_record_structural_correspondence`. Captured registered sources: 3. Unresolved transports retained: 0. The registered structural consequence corresponded: `True`. External evidence selected the survivor: `false`. Application or performance selected the law: `false`. Successful performance was relabelled scientific proof: `false`. Failed implementation was relabelled scientific falsification: `false`. Simulation or demonstration was relabelled observation: `false`.

- `NASA-SE-HANDBOOK-001` - transport `captured`; captured bytes `3773440`; snapshot `sha256:8eeb4887a4dc57a23049da7dd2ed556833cf98e214b240468d987873164ff688`.
- `NIST-SSE-001` - transport `captured`; captured bytes `47984`; snapshot `sha256:64c8d41d9a78268d596a0a60aad18eead50fd375e8628486f7202788a207edaa`.
- `PYTHON-PLATFORM-001` - transport `captured`; captured bytes `63923`; snapshot `sha256:6873ff23f5e2e6985050e60024a477265ed779867e32542e133879aae0e01611`.

The comparison is structural and source-bound. It checks that a relevant primary handbook, standard, guidance record or open engineering project can represent the declared requirement, artifact, environment, operating boundary, version, test and lifecycle without rewriting the Fold law. It does not claim that a standard proves conformity, that one handbook exhausts Engineering, or that a custodian's reputation proves the result. New records can extend or challenge the surface only with source, version, configuration, method, failures and adverse rows retained.

Scientific meaning

For **engineering interface**, the consequence is practical and exact: An interface is a typed exchange contract retaining endpoints, units, timing, errors and version. An Engineering claim therefore has to expose the carrier `component-pair-exchange-contract-carrier`, the relation `typed-transfer-and-coordination-relation`, the record `endpoints-signals-units-timing-errors-and-version-held`, and the boundary `interface-contract-bounded` together. If any one is hidden, the result may remain a concept, prototype, simulation, demonstration or product choice, but it is not this admitted law.

This distinction protects users, maintainers and science. It prevents a successful output from erasing requirements, failed alternatives, hidden resources, inaccessible users, hazards, maintenance burdens or end-of-life costs. It prevents an application from selecting the upstream law and prevents engineering failure from becoming automatic scientific falsification. A replacement must still generate its forms, eliminate alternatives, survive controls, preserve custody and reproduce through the shared engine.

Machine certificate

Candidate count: `256`; survivor count: `1`; derivation seal: `sha256:dcc0d066d2d73b35bdb633cef2a5b13d3874578afbe7e13fe0457310e8f72eaf`; independent certificate: `sha256:2fb445f3614a19d40d2cf88c3237ca41f7a71f639a4baae5ad8dc4aee203bd23`; engine receipt: `sha256:37e9f45a48de371bc058941dbc7f3a9d883632e2f78aeea8ff47bb915878cba3`. The independent implementation recomputed the candidate product: `True`. All external rows were preserved: `True`.

9. System architecture

Claim: `SFT-ENG-ARCHITECTURE-001`

Family: `components_interfaces_architecture`

Exact statement: Architecture is a versioned composition of components, interfaces and allocated responsibilities.

Forward chain and exact carrier

The generated carrier is `components-interfaces-allocation-carrier`. Its required relation is `composition-and-dependency-relation`. The reconstruction must retain `parts-links-responsibilities-failures-versions-and-rationale-held`. Its declared empirical boundary is `architecture-scope-bounded`. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is `SFT-FOUNDATION-ONE-001`, `SFT-FOUNDATION-FOLD-001`, `SFT-FOUNDATION-FOLD-DYNAMICS-001`, `SFT-MATH-EXACT-ARITHMETIC-001`, `SFT-MATH-GRAPH-NETWORK-001`, `SFT-MATH-OPTIMIZATION-001`, `SFT-MATH-DYNAMICAL-SYSTEMS-001`, `SFT-MATH-LOGIC-PROOF-001`, `SFT-INFO-QUANTITY-001`, `SFT-INFO-CHANNEL-CAPACITY-001`, `SFT-COMP-FORM-STATE-TRANSITION-001`, `SFT-COMP-SEM-CORRECTNESS-001`, `SFT-COMP-SEM-COMPILATION-001`, `SFT-COMP-DIST-CAUSALITY-001`, `SFT-PHYS-MEAS-VALUE-RECORD-001`, `SFT-PHYS-MEAS-CALIBRATION-001`, `SFT-CHEM-MEAS-TRACEABILITY-001`, `SFT-MAT-SUST-LIFECYCLE-001`, `SFT-BIO-BIO-HANDOFF-001`, `SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001`, `SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001`, `SFT-EARTH-HAZARD-RISK-HANDOFF-001`, `SFT-ASTRO-ASTRO-HANDOFF-001`, `SFT-SOCIAL-ENGINEERING-HANDOFF-001`, `SFT-ENG-ACCEPTANCE-BOUNDARY-001`, `SFT-ENG-INTERFACE-001`. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external standard, application, performance target, product or prior implementation to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external standards, products, tests or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite components, versions, platforms, tests and lifecycle records; performance remains artifact, environment, user, method and operating-boundary limited.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - which requirement, artifact, component, user, environment or test bears the claim: `components-interfaces-allocation-carrier`.
- **boundary** - which operating envelope, version, platform, user and lifecycle stage bound the result: `architecture-scope-bounded`.
- **relation** - which generated transformation, interface, comparison or handoff connects the carriers: `composition-and-dependency-relation`.
- **record** - which requirements, versions, alternatives, failures, uncertainty and rollback remain reconstructible: `parts-links-responsibilities-failures-versions-and-rationale-held`.
- **evidence** - whether law, requirement, design, simulation, test, demonstration, performance and anomaly remain distinct: `science-design-test-and-decision-classes-explicit`.
- **provenance** - whether the result traces to the One and names every upstream receipt and design choice: `root-bound-forward-translation`.
- **generality** - whether finite extension preserves platforms, versions, failures and operating limits: `positive-finite-extension-preserves-platforms-failures-and-versions`.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: `no-extra-rule`.

Their ordered product gives the unique survivor:

```
components-interfaces-allocation-carrier__architecture-scope-bounded__composition-and
-dependency-relation__parts-links-responsibilities-failures-versions-and-rationale-he
ld__science-design-test-and-decision-classes-explicit__root-bound-forward-translation
__positive-finite-extension-preserves-platforms-failures-and-versions__no-extra-rule
```

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses the requirement, artifact, user or environment. Erasing the boundary universalises a version, platform or test. Importing the relation lets a target or preferred application choose the result. Retaining only successful output removes requirements, alternatives, failures and rollback. Conflating evidence classes turns simulation, demonstration or performance into observation or law. Breaking provenance removes the trace to the One and upstream receipts. Erasing finite context makes one implementation universal. Adding a silent dependency or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite requirement or admitted law, versioned artifact, boundary, testable relation and complete record. The successor adds one lawful component, interface, platform, requirement, test, failure, lifecycle stage or anomaly while preserving all earlier identities, adverse rows and receipts.

Falsification and deliberately unfavourable controls

The registered falsification condition is: Reject if a diagram omits dependency, failure propagation or unrepresented components.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: `source_bound_engineering_requirement_design_test_lifecycle_or_access_record/primary_standard_handbook_guidance_or_project_record_structural_correspondence`. Captured registered sources: 3. Unresolved transports retained: 0. The registered structural consequence corresponded: `True`. External evidence selected the survivor: `false`. Application or performance selected the law: `false`. Successful performance was relabelled scientific proof: `false`. Failed implementation was relabelled scientific falsification: `false`. Simulation or demonstration was relabelled observation: `false`.

- `NASA-SE-HANDBOOK-001` - transport `captured`; captured bytes `3773440`; snapshot `sha256:8eeb4887a4dc57a23049da7dd2ed556833cf98e214b240468d987873164ff688`.
- `NIST-SSE-001` - transport `captured`; captured bytes `47984`; snapshot `sha256:64c8d41d9a78268d596a0a60aad18eead50fd375e8628486f7202788a207edaa`.
- `PYTHON-PLATFORM-001` - transport `captured`; captured bytes `63923`; snapshot `sha256:6873ff23f5e2e6985050e60024a477265ed779867e32542e133879aae0e01611`.

The comparison is structural and source-bound. It checks that a relevant primary handbook, standard, guidance record or open engineering project can represent the declared requirement, artifact, environment, operating boundary, version, test and lifecycle without rewriting the Fold law. It does not claim that a standard proves conformity, that one handbook exhausts Engineering, or that a custodian's reputation proves the result. New records can extend or challenge the surface only with source, version, configuration, method, failures and adverse rows retained.

Scientific meaning

For **system architecture**, the consequence is practical and exact: Architecture is a versioned composition of components, interfaces and allocated responsibilities. An Engineering claim therefore has to expose the carrier `components-interfaces-allocation-carrier`, the relation `composition-and-dependency-relation`, the record `parts-links-responsibilities-failures-versions-and-rationale-held`, and the boundary `architecture-scope-bounded` together. If any one is hidden, the result may remain a concept, prototype, simulation, demonstration or product choice, but it is not this admitted law.

This distinction protects users, maintainers and science. It prevents a successful output from erasing requirements, failed alternatives, hidden resources, inaccessible users, hazards, maintenance burdens or end-of-life costs. It prevents an application from selecting the upstream law and prevents engineering failure from becoming automatic scientific falsification. A replacement must still generate its forms, eliminate alternatives, survive controls, preserve custody and reproduce through the shared engine.

Machine certificate

Candidate count: `256`; survivor count: `1`; derivation seal: `sha256:23df1e72e3104bd6334e65727ae9ebf486153e4281ef764a727bda7c40ae9145`; independent certificate: `sha256:ba9e1058a2ef10e8171c3a481381a81d1bb275d3c2cdeb96aeac8ea0031f4a8e`; engine receipt: `sha256:7deffa065dcfc825ec565444d246ff656c6b3f10e75b7bb56efe3682c14aeb9c`. The independent implementation recomputed the candidate product: `True`. All external rows were preserved: `True`.

10. Engineered system

Claim: `SFT-ENG-SYSTEM-001`

Family: `components_interfaces_architecture`

Exact statement: An engineered system composes parts and relations around a declared purpose while retaining environment, users and failures.

Forward chain and exact carrier

The generated carrier is `parts-relations-boundary-purpose-carrier`. Its required relation is `joint-operation-relation`. The reconstruction must retain `parts-environment-state-resources-users-and-failures-held`. Its declared empirical boundary is `system-purpose-bounded`. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is `SFT-FOUNDATION-ONE-001`, `SFT-FOUNDATION-FOLD-001`, `SFT-FOUNDATION-FOLD-DYNAMICS-001`, `SFT-MATH-EXACT-ARITHMETIC-001`, `SFT-MATH-GRAPH-NETWORK-001`, `SFT-MATH-OPTIMIZATION-001`, `SFT-MATH-DYNAMICAL-SYSTEMS-001`, `SFT-MATH-LOGIC-PROOF-001`, `SFT-INFO-QUANTITY-001`, `SFT-INFO-CHANNEL-CAPACITY-001`, `SFT-COMP-FORM-STATE-TRANSITION-001`, `SFT-COMP-SEM-CORRECTNESS-001`, `SFT-COMP-SEM-COMPILATION-001`, `SFT-COMP-DIST-CAUSALITY-001`, `SFT-PHYS-MEAS-VALUE-RECORD-001`, `SFT-PHYS-MEAS-CALIBRATION-001`, `SFT-CHEM-MEAS-TRACEABILITY-001`, `SFT-MAT-SUST-LIFECYCLE-001`, `SFT-BIO-BIO-HANDOFF-001`, `SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001`, `SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001`, `SFT-EARTH-HAZARD-RISK-HANDOFF-001`, `SFT-ASTRO-ASTRO-HANDOFF-001`, `SFT-SOCIAL-ENGINEERING-HANDOFF-001`, `SFT-ENG-ACCEPTANCE-BOUNDARY-001`, `SFT-ENG-ARCHITECTURE-001`. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external standard, application, performance target, product or prior implementation to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external standards, products, tests or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite components, versions, platforms, tests and lifecycle records; performance remains artifact, environment, user, method and operating-boundary limited.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - which requirement, artifact, component, user, environment or test bears the claim: `parts-relations-boundary-purpose-carrier`.
- **boundary** - which operating envelope, version, platform, user and lifecycle stage bound the result: `system-purpose-bounded`.
- **relation** - which generated transformation, interface, comparison or handoff connects the carriers: `joint-operation-relation`.
- **record** - which requirements, versions, alternatives, failures, uncertainty and rollback remain reconstructible: `parts-environment-state-resources-users-and-failures-held`.
- **evidence** - whether law, requirement, design, simulation, test, demonstration, performance and anomaly remain distinct: `science-design-test-and-decision-classes-explicit`.
- **provenance** - whether the result traces to the One and names every upstream receipt and design choice: `root-bound-forward-translation`.
- **generality** - whether finite extension preserves platforms, versions, failures and operating limits: `positive-finite-extension-preserves-platforms-failures-and-versions`.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: `no-extra-rule`.

Their ordered product gives the unique survivor:

```
parts-relations-boundary-purpose-carrier__system-purpose-bounded__joint-operation-rel
ation__parts-environment-state-resources-users-and-failures-held__science-design-test
-and-decision-classes-explicit__root-bound-forward-translation__positive-finite-exten
sion-preserves-platforms-failures-and-versions__no-extra-rule
```

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses the requirement, artifact, user or environment. Erasing the boundary universalises a version, platform or test. Importing the relation lets a target or preferred application choose the result. Retaining only successful output removes requirements, alternatives, failures and rollback. Conflating evidence classes turns simulation, demonstration or performance into observation or law. Breaking provenance removes the trace to the One and upstream receipts. Erasing finite context makes one implementation universal. Adding a silent dependency or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite requirement or admitted law, versioned artifact, boundary, testable relation and complete record. The successor adds one lawful component, interface, platform, requirement, test, failure, lifecycle stage or anomaly while preserving all earlier identities, adverse rows and receipts.

Falsification and deliberately unfavourable controls

The registered falsification condition is: Reject if system-level success is inferred from isolated component success.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: `source_bound_engineering_requirement_design_test_lifecycle_or_access_record/primary_standard_handbook_guidance_or_project_record_structural_correspondence`. Captured registered sources: 3. Unresolved transports retained: 0. The registered structural consequence corresponded: `True`. External evidence selected the survivor: `false`. Application or performance selected the law: `false`. Successful performance was relabelled scientific proof: `false`. Failed implementation was relabelled scientific falsification: `false`. Simulation or demonstration was relabelled observation: `false`.

- `NASA-SE-HANDBOOK-001` - transport `captured`; captured bytes `3773440`; snapshot `sha256:8eeb4887a4dc57a23049da7dd2ed556833cf98e214b240468d987873164ff688`.
- `NIST-SSE-001` - transport `captured`; captured bytes `47984`; snapshot `sha256:64c8d41d9a78268d596a0a60aad18eead50fd375e8628486f7202788a207edaa`.
- `PYTHON-PLATFORM-001` - transport `captured`; captured bytes `63923`; snapshot `sha256:6873ff23f5e2e6985050e60024a477265ed779867e32542e133879aae0e01611`.

The comparison is structural and source-bound. It checks that a relevant primary handbook, standard, guidance record or open engineering project can represent the declared requirement, artifact, environment, operating boundary, version, test and lifecycle without rewriting the Fold law. It does not claim that a standard proves conformity, that one handbook exhausts Engineering, or that a custodian's reputation proves the result. New records can extend or challenge the surface only with source, version, configuration, method, failures and adverse rows retained.

Scientific meaning

For **engineered system**, the consequence is practical and exact: An engineered system composes parts and relations around a declared purpose while retaining environment, users and failures. An Engineering claim therefore has to expose the carrier `parts-relations-boundary-purpose-carrier`, the relation `joint-operation-relation`, the record `parts-environment-state-resources-users-and-failures-held`, and the boundary `system-purpose-bounded` together. If any one is hidden, the result may remain a concept, prototype, simulation, demonstration or product choice, but it is not this admitted law.

This distinction protects users, maintainers and science. It prevents a successful output from erasing requirements, failed alternatives, hidden resources, inaccessible users, hazards, maintenance burdens or end-of-life costs. It prevents an application

from selecting the upstream law and prevents engineering failure from becoming automatic scientific falsification. A replacement must still generate its forms, eliminate alternatives, survive controls, preserve custody and reproduce through the shared engine.

Machine certificate

Candidate count: `256`; survivor count: `1`; derivation seal: `sha256:717d81f897e1edb326d7da30a8995b7fbe5cc121353b4964371fcdd2d578c8fd`; independent certificate: `sha256:a8f18af8146670f0342139e99ed415d3c1950fd64f951d7da1509c835355fab8`; engine receipt: `sha256:5cd0b2eb968f28707d85b6d60ce3977816de74319cc342f3a5c18776a1ec497a`. The independent implementation recomputed the candidate product: `True`. All external rows were preserved: `True`.

11. Modular decomposition

Claim: `SFT-ENG-MODULARITY-001`

Family: `components_interfaces_architecture`

Exact statement: A modular decomposition is lawful only when interface contracts permit exact recomposition of required behavior.

Forward chain and exact carrier

The generated carrier is `system-partition-interface-carrier`. Its required relation is `decomposition-recomposition-relation`. The reconstruction must retain `partition-contract-coupling-state-loss-and-tests-held`. Its declared empirical boundary is `decomposition-boundary-declared`. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is `SFT-FOUNDATION-ONE-001`, `SFT-FOUNDATION-FOLD-001`, `SFT-FOUNDATION-FOLD-DYNAMICS-001`, `SFT-MATH-EXACT-ARITHMETIC-001`, `SFT-MATH-GRAPH-NETWORK-001`, `SFT-MATH-OPTIMIZATION-001`, `SFT-MATH-DYNAMICAL-SYSTEMS-001`, `SFT-MATH-LOGIC-PROOF-001`, `SFT-INFO-QUANTITY-001`, `SFT-INFO-CHANNEL-CAPACITY-001`, `SFT-COMP-FORM-STATE-TRANSITION-001`, `SFT-COMP-SEM-CORRECTNESS-001`, `SFT-COMP-SEM-COMPILATION-001`, `SFT-COMP-DIST-CAUSALITY-001`, `SFT-PHYS-MEAS-VALUE-RECORD-001`, `SFT-PHYS-MEAS-CALIBRATION-001`, `SFT-CHEM-MEAS-TRACEABILITY-001`, `SFT-MAT-SUST-LIFECYCLE-001`, `SFT-BIO-BIO-HANDOFF-001`, `SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001`, `SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001`, `SFT-EARTH-HAZARD-RISK-HANDOFF-001`, `SFT-ASTRO-ASTRO-HANDOFF-001`, `SFT-SOCIAL-ENGINEERING-HANDOFF-001`, `SFT-ENG-ACCEPTANCE-BOUNDARY-001`, `SFT-ENG-SYSTEM-001`. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external standard, application, performance target, product or prior implementation to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external standards, products, tests or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite components, versions, platforms, tests and lifecycle records; performance remains artifact, environment, user, method and operating-boundary limited.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - which requirement, artifact, component, user, environment or test bears the claim: `system-partition-interface-carrier`.
- **boundary** - which operating envelope, version, platform, user and lifecycle stage bound the result: `decomposition-boundary-declared`.
- **relation** - which generated transformation, interface, comparison or handoff connects the carriers: `decomposition-recomposition-relation`.

- **record** - which requirements, versions, alternatives, failures, uncertainty and rollback remain reconstructible: `partition-contract-coupling-state-loss-and-tests-held`.
- **evidence** - whether law, requirement, design, simulation, test, demonstration, performance and anomaly remain distinct: `science-design-test-and-decision-classes-explicit`.
- **provenance** - whether the result traces to the One and names every upstream receipt and design choice: `root-bound-forward-translation`.
- **generality** - whether finite extension preserves platforms, versions, failures and operating limits: `positive-finite-extension-preserves-platforms-failures-and-versions`.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: `no-extra-rule`.

Their ordered product gives the unique survivor:

```
system-partition-interface-carrier__decomposition-boundary-declared__decomposition-re
composition-relation__partition-contract-coupling-state-loss-and-tests-held__science-
design-test-and-decision-classes-explicit__root-bound-forward-translation__positive-f
inite-extension-preserves-platforms-failures-and-versions__no-extra-rule
```

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses the requirement, artifact, user or environment. Erasing the boundary universalises a version, platform or test. Importing the relation lets a target or preferred application choose the result. Retaining only successful output removes requirements, alternatives, failures and rollback. Conflating evidence classes turns simulation, demonstration or performance into observation or law. Breaking provenance removes the trace to the One and upstream receipts. Erasing finite context makes one implementation universal. Adding a silent dependency or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite requirement or admitted law, versioned artifact, boundary, testable relation and complete record. The successor adds one lawful component, interface, platform, requirement, test, failure, lifecycle stage or anomaly while preserving all earlier identities, adverse rows and receipts.

Falsification and deliberately unfavourable controls

The registered falsification condition is: Reject if decomposition hides cross-module state or failure coupling.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: `source_bound_engineering_requirement_design_test_lifecycle_or_access_record/ primary_standard_handbook_guidance_or_project_record_structural_correspondence`. Captured registered sources: 3. Unresolved transports retained: 0. The registered structural consequence corresponded: `True`. External evidence selected the survivor: `false`. Application or performance selected the law: `false`. Successful performance was relabelled scientific proof: `false`. Failed implementation was relabelled scientific falsification: `false`. Simulation or demonstration was relabelled observation: `false`.

- `NASA-SE-HANDBOOK-001` - transport `captured`; captured bytes `3773440`; snapshot `sha256:8eeb4887a4dc57a23049da7dd2ed556833cf98e214b240468d987873164ff688`.
- `NIST-SSE-001` - transport `captured`; captured bytes `47984`; snapshot `sha256:64c8d41d9a78268d596a0a60aad18eead50fd375e8628486f7202788a207edaa`.

- PYTHON-PLATFORM-001 - transport captured; captured bytes 63923; snapshot sha256:6873ff23f5e2e6985050e60024a477265ed779867e32542e133879aae0e01611.

The comparison is structural and source-bound. It checks that a relevant primary handbook, standard, guidance record or open engineering project can represent the declared requirement, artifact, environment, operating boundary, version, test and lifecycle without rewriting the Fold law. It does not claim that a standard proves conformity, that one handbook exhausts Engineering, or that a custodian's reputation proves the result. New records can extend or challenge the surface only with source, version, configuration, method, failures and adverse rows retained.

Scientific meaning

For **modular decomposition**, the consequence is practical and exact: A modular decomposition is lawful only when interface contracts permit exact recomposition of required behaviour. An Engineering claim therefore has to expose the carrier system-partition-interface-carrier, the relation decomposition-recomposition-relation, the record partition-contract-coupling-state-loss-and-tests-held, and the boundary decomposition-boundary-declared together. If any one is hidden, the result may remain a concept, prototype, simulation, demonstration or product choice, but it is not this admitted law.

This distinction protects users, maintainers and science. It prevents a successful output from erasing requirements, failed alternatives, hidden resources, inaccessible users, hazards, maintenance burdens or end-of-life costs. It prevents an application from selecting the upstream law and prevents engineering failure from becoming automatic scientific falsification. A replacement must still generate its forms, eliminate alternatives, survive controls, preserve custody and reproduce through the shared engine.

Machine certificate

Candidate count: 256; survivor count: 1; derivation seal: sha256:99aab7daea7e20b09c0ea1437131d11843a488429637173f1d371fabe84172ab; independent certificate: sha256:97ae01b3feab2aa103c3315d4259205dd4ccb1d42c05cfb791aed18d90b07844; engine receipt: sha256:4927d9b6e8f6fdda5ec599f2badaf1090ad1baa9c03a8110d338d96873c3fa54. The independent implementation recomputed the candidate product: True. All external rows were preserved: True.

12. System integration

Claim: SFT-ENG-INTEGRATION-001

Family: components_interfaces_architecture

Exact statement: Integration retains component versions, assembly order, interface tests, failures and rollback state.

Forward chain and exact carrier

The generated carrier is components-build-sequence-system-carrier. Its required relation is assembly-to-joint-behaviour-relation. The reconstruction must retain versions-order-interfaces-tests-failures-and-rollbacks-held. Its declared empirical boundary is integration-build-bounded. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-OPTIMIZATION-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-CHANNEL-CAPACITY-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-COMP-SEM-CORRECTNESS-001, SFT-COMP-SEM-COMPILATION-001, SFT-COMP-DIST-CAUSALITY-001, SFT-PHYS-MEAS-VALUE-RECORD-001, SFT-PHYS-MEAS-CALIBRATION-001, SFT-CHEM-MEAS-TRACEABILITY-001, SFT-MAT-SUST-LIFECYCLE-001, SFT-BIO-BIO-HANDOFF-001, SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001, SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-ASTRO-HANDOFF-001, SFT-SOCIAL-ENGINEERING-HANDOFF-001, SFT-ENG-ACCEPTANCE-BOUNDARY-001, SFT-ENG-MODULARITY-001. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external standard, application, performance target, product or prior implementation to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external standards, products, tests or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite components, versions, platforms, tests and lifecycle records; performance remains artifact, environment, user, method and operating-boundary limited.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - which requirement, artifact, component, user, environment or test bears the claim: `components-build-sequence-system-carrier`.
- **boundary** - which operating envelope, version, platform, user and lifecycle stage bound the result: `integration-build-bounded`.
- **relation** - which generated transformation, interface, comparison or handoff connects the carriers: `assembly-to-joint-behaviour-relation`.
- **record** - which requirements, versions, alternatives, failures, uncertainty and rollback remain reconstructible: `versions-order-interfaces-tests-failures-and-rollbacks-held`.
- **evidence** - whether law, requirement, design, simulation, test, demonstration, performance and anomaly remain distinct: `science-design-test-and-decision-classes-explicit`.
- **provenance** - whether the result traces to the One and names every upstream receipt and design choice: `root-bound-forward-translation`.
- **generality** - whether finite extension preserves platforms, versions, failures and operating limits: `positive-finite-extension-preserves-platforms-failures-and-versions`.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: `no-extra-rule`.

Their ordered product gives the unique survivor:

```
components-build-sequence-system-carrier__integration-build-bounded__assembly-to-join
t-behaviour-relation__versions-order-interfaces-tests-failures-and-rollbacks-held__sc
ience-design-test-and-decision-classes-explicit__root-bound-forward-translation__posi
tive-finite-extension-preserves-platforms-failures-and-versions__no-extra-rule
```

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses the requirement, artifact, user or environment. Erasing the boundary universalises a version, platform or test. Importing the relation lets a target or preferred application choose the result. Retaining only successful output removes requirements, alternatives, failures and rollback. Conflating evidence classes turns simulation, demonstration or performance into observation or law. Breaking provenance removes the trace to the One and upstream receipts. Erasing finite context makes one implementation universal. Adding a silent dependency or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite requirement or admitted law, versioned artifact, boundary, testable relation and complete record. The successor adds one lawful component, interface, platform, requirement, test, failure, lifecycle stage or anomaly while preserving all earlier identities, adverse rows and receipts.

Falsification and deliberately unfavourable controls

The registered falsification condition is: Reject if a final demonstration erases failed intermediate integrations.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: `source_bound_engineering_requirement_design_test_lifecycle_or_access_record/primary_standard_handbook_guidance_or_project_record_structural_correspondence`. Captured registered sources: 3. Unresolved transports retained: 0. The registered structural consequence corresponded: `True`. External evidence selected the survivor: `false`. Application or performance selected the law: `false`. Successful performance was relabelled scientific proof: `false`. Failed implementation was relabelled scientific falsification: `false`. Simulation or demonstration was relabelled observation: `false`.

- `NASA-SE-HANDBOOK-001` - transport `captured`; captured bytes `3773440`; snapshot `sha256:8eeb4887a4dc57a23049da7dd2ed556833cf98e214b240468d987873164ff688`.
- `NIST-SSE-001` - transport `captured`; captured bytes `47984`; snapshot `sha256:64c8d41d9a78268d596a0a60aad18eead50fd375e8628486f7202788a207edaa`.
- `PYTHON-PLATFORM-001` - transport `captured`; captured bytes `63923`; snapshot `sha256:6873ff23f5e2e6985050e60024a477265ed779867e32542e133879aae0e01611`.

The comparison is structural and source-bound. It checks that a relevant primary handbook, standard, guidance record or open engineering project can represent the declared requirement, artifact, environment, operating boundary, version, test and lifecycle without rewriting the Fold law. It does not claim that a standard proves conformity, that one handbook exhausts Engineering, or that a custodian's reputation proves the result. New records can extend or challenge the surface only with source, version, configuration, method, failures and adverse rows retained.

Scientific meaning

For **system integration**, the consequence is practical and exact: Integration retains component versions, assembly order, interface tests, failures and rollback state. An Engineering claim therefore has to expose the carrier `components-build-sequence-system-carrier`, the relation `assembly-to-joint-behaviour-relation`, the record `versions-order-interfaces-tests-failures-and-rollbacks-held`, and the boundary `integration-build-bounded` together. If any one is hidden, the result may remain a concept, prototype, simulation, demonstration or product choice, but it is not this admitted law.

This distinction protects users, maintainers and science. It prevents a successful output from erasing requirements, failed alternatives, hidden resources, inaccessible users, hazards, maintenance burdens or end-of-life costs. It prevents an application from selecting the upstream law and prevents engineering failure from becoming automatic scientific falsification. A replacement must still generate its forms, eliminate alternatives, survive controls, preserve custody and reproduce through the shared engine.

Machine certificate

Candidate count: `256`; survivor count: `1`; derivation seal: `sha256:8f015f533f7452682be032fc6d5cee3a9c38620603dead0168011b8a22f57c26`; independent certificate: `sha256:21e5f9a8114e8dbe0f8dcbe12214923b8667d3f8fbeafa9c44c49aed73829a9b`; engine receipt: `sha256:d2b8103d27c920f149a191509764f1e24ad891b58516fc00bed2c59c8872cd41`. The independent implementation recomputed the candidate product: `True`. All external rows were preserved: `True`.

13. Engineering resource

Claim: `SFT-ENG-RESOURCE-001`

Family: `resources_efficiency_reliability`

Exact statement: Resource accounting retains stock, flow, units, time, loss and source for each operation.

Forward chain and exact carrier

The generated carrier is `system-operation-resource-carrier`. Its required relation is `operation-to-consumption-relation`. The reconstruction must retain `resource-stock-flow-units-time-loss-and-source-held`. Its declared empirical boundary is `resource-accounting-bounded`. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is `SFT-FOUNDATION-ONE-001`, `SFT-FOUNDATION-FOLD-001`, `SFT-FOUNDATION-FOLD-DYNAMICS-001`, `SFT-MATH-EXACT-ARITHMETIC-001`, `SFT-MATH-GRAPH-NETWORK-001`, `SFT-MATH-OPTIMIZATION-001`, `SFT-MATH-DYNAMICAL-SYSTEMS-001`, `SFT-MATH-LOGIC-PROOF-001`, `SFT-INFO-QUANTITY-001`, `SFT-INFO-CHANNEL-CAPACITY-001`, `SFT-COMP-FORM-STATE-TRANSITION-001`, `SFT-COMP-SEM-CORRECTNESS-001`, `SFT-COMP-SEM-COMPILATION-001`, `SFT-COMP-DIST-CAUSALITY-001`, `SFT-PHYS-MEAS-VALUE-RECORD-001`, `SFT-PHYS-MEAS-CALIBRATION-001`, `SFT-CHEM-MEAS-TRACEABILITY-001`, `SFT-MAT-SUST-LIFECYCLE-001`, `SFT-BIO-BIO-HANDOFF-001`, `SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001`, `SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001`, `SFT-EARTH-HAZARD-RISK-HANDOFF-001`, `SFT-ASTRO-ASTRO-HANDOFF-001`, `SFT-SOCIAL-ENGINEERING-HANDOFF-001`, `SFT-ENG-INTEGRATION-001`. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external standard, application, performance target, product or prior implementation to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external standards, products, tests or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite components, versions, platforms, tests and lifecycle records; performance remains artifact, environment, user, method and operating-boundary limited.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - which requirement, artifact, component, user, environment or test bears the claim: `system-operation-resource-carrier`.
- **boundary** - which operating envelope, version, platform, user and lifecycle stage bound the result: `resource-accounting-bounded`.
- **relation** - which generated transformation, interface, comparison or handoff connects the carriers: `operation-to-consumption-relation`.
- **record** - which requirements, versions, alternatives, failures, uncertainty and rollback remain reconstructible: `resource-stock-flow-units-time-loss-and-source-held`.
- **evidence** - whether law, requirement, design, simulation, test, demonstration, performance and anomaly remain distinct: `science-design-test-and-decision-classes-explicit`.
- **provenance** - whether the result traces to the One and names every upstream receipt and design choice: `root-bound-forward-translation`.
- **generality** - whether finite extension preserves platforms, versions, failures and operating limits: `positive-finite-extension-preserves-platforms-failures-and-versions`.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: `no-extra-rule`.

Their ordered product gives the unique survivor:

```
system-operation-resource-carrier__resource-accounting-bounded__operation-to-consumpt
ion-relation__resource-stock-flow-units-time-loss-and-source-held__science-design-tes
t-and-decision-classes-explicit__root-bound-forward-translation__positive-finite-exte
nsion-preserves-platforms-failures-and-versions__no-extra-rule
```

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses the requirement, artifact, user or environment. Erasing the boundary universalises a version, platform or test. Importing the relation lets a target or preferred application choose the result. Retaining only successful output removes requirements, alternatives, failures and rollback. Conflating evidence classes turns simulation, demonstration or performance into observation or law. Breaking provenance removes the trace to the One and upstream receipts. Erasing finite context makes one implementation universal. Adding a silent dependency or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite requirement or admitted law, versioned artifact, boundary, testable relation and complete record. The successor adds one lawful component, interface, platform, requirement, test, failure, lifecycle stage or anomaly while preserving all earlier identities, adverse rows and receipts.

Falsification and deliberately unfavourable controls

The registered falsification condition is: Reject if an uncounted external supply or hidden computational service is omitted.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: `source_bound_engineering_requirement_design_test_lifecycle_or_access_record/primary_standard_handbook_guidance_or_project_record_structural_correspondence`. Captured registered sources: 3. Unresolved transports retained: 0. The registered structural consequence corresponded: `True`. External evidence selected the survivor: `false`. Application or performance selected the law: `false`. Successful performance was relabelled scientific proof: `false`. Failed implementation was relabelled scientific falsification: `false`. Simulation or demonstration was relabelled observation: `false`.

- `EPA-LIFECYCLE-001` - transport `captured`; captured bytes `980`; snapshot `sha256:1d271583636a679563a046e767d0575a92d23eeafbf99f0da3fef2582e102c4e`.
- `NASA-SE-HANDBOOK-001` - transport `captured`; captured bytes `3773440`; snapshot `sha256:8eeb4887a4dc57a23049da7dd2ed556833cf98e214b240468d987873164ff688`.
- `NIST-SSE-001` - transport `captured`; captured bytes `47984`; snapshot `sha256:64c8d41d9a78268d596a0a60aad18eead50fd375e8628486f7202788a207edaa`.

The comparison is structural and source-bound. It checks that a relevant primary handbook, standard, guidance record or open engineering project can represent the declared requirement, artifact, environment, operating boundary, version, test and lifecycle without rewriting the Fold law. It does not claim that a standard proves conformity, that one handbook exhausts Engineering, or that a custodian's reputation proves the result. New records can extend or challenge the surface only with source, version, configuration, method, failures and adverse rows retained.

Scientific meaning

For **engineering resource**, the consequence is practical and exact: Resource accounting retains stock, flow, units, time, loss and source for each operation. An Engineering claim therefore has to expose the carrier `system-operation-resource-carrier`, the relation `operation-to-consumption-relation`, the record `resource-stock-flow-units-time-loss-and-source-held`, and the boundary `resource-accounting-bounded` together. If any one is hidden, the result may remain a concept, prototype, simulation, demonstration or product choice, but it is not this admitted law.

This distinction protects users, maintainers and science. It prevents a successful output from erasing requirements, failed alternatives, hidden resources, inaccessible users, hazards, maintenance burdens or end-of-life costs. It prevents an application

from selecting the upstream law and prevents engineering failure from becoming automatic scientific falsification. A replacement must still generate its forms, eliminate alternatives, survive controls, preserve custody and reproduce through the shared engine.

Machine certificate

Candidate count: `256`; survivor count: `1`; derivation seal: `sha256:1144f7e7138e0b8f382f5832ae8e44432370d49ad65f6e4001328716a9b48e33`; independent certificate: `sha256:f081b5a86d11fb8c501d8f9c9090b937dc8861507f6f079fe608600c9c74acdd`; engine receipt: `sha256:264d3cd9efaaceea723aae48a58392418af0a8a11c6fcdc6bdf65fec0e0ec1e3`. The independent implementation recomputed the candidate product: `True`. All external rows were preserved: `True`.

14. Capacity and load

Claim: `SFT-ENG-CAPACITY-001`

Family: `resources_efficiency_reliability`

Exact statement: Capacity is a load-profile-bound relation retaining duration, queuing, degradation and failure.

Forward chain and exact carrier

The generated carrier is `system-demand-capability-carrier`. Its required relation is `load-to-service-relation`. The reconstruction must retain `demand-capacity-duration-queue-degradation-and-failure-held`. Its declared empirical boundary is `load-profile-bounded`. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is `SFT-FOUNDATION-ONE-001`, `SFT-FOUNDATION-FOLD-001`, `SFT-FOUNDATION-FOLD-DYNAMICS-001`, `SFT-MATH-EXACT-ARITHMETIC-001`, `SFT-MATH-GRAPH-NETWORK-001`, `SFT-MATH-OPTIMIZATION-001`, `SFT-MATH-DYNAMICAL-SYSTEMS-001`, `SFT-MATH-LOGIC-PROOF-001`, `SFT-INFO-QUANTITY-001`, `SFT-INFO-CHANNEL-CAPACITY-001`, `SFT-COMP-FORM-STATE-TRANSITION-001`, `SFT-COMP-SEM-CORRECTNESS-001`, `SFT-COMP-SEM-COMPILATION-001`, `SFT-COMP-DIST-CAUSALITY-001`, `SFT-PHYS-MEAS-VALUE-RECORD-001`, `SFT-PHYS-MEAS-CALIBRATION-001`, `SFT-CHEM-MEAS-TRACEABILITY-001`, `SFT-MAT-SUST-LIFECYCLE-001`, `SFT-BIO-BIO-HANDOFF-001`, `SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001`, `SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001`, `SFT-EARTH-HAZARD-RISK-HANDOFF-001`, `SFT-ASTRO-ASTRO-HANDOFF-001`, `SFT-SOCIAL-ENGINEERING-HANDOFF-001`, `SFT-ENG-INTEGRATION-001`, `SFT-ENG-RESOURCE-001`. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external standard, application, performance target, product or prior implementation to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external standards, products, tests or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite components, versions, platforms, tests and lifecycle records; performance remains artifact, environment, user, method and operating-boundary limited.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - which requirement, artifact, component, user, environment or test bears the claim: `system-demand-capability-carrier`.
- **boundary** - which operating envelope, version, platform, user and lifecycle stage bound the result: `load-profile-bounded`.
- **relation** - which generated transformation, interface, comparison or handoff connects the carriers: `load-to-service-relation`.

- **record** - which requirements, versions, alternatives, failures, uncertainty and rollback remain reconstructible: `demand-capacity-duration-queue-degradation-and-failure-held`.
- **evidence** - whether law, requirement, design, simulation, test, demonstration, performance and anomaly remain distinct: `science-design-test-and-decision-classes-explicit`.
- **provenance** - whether the result traces to the One and names every upstream receipt and design choice: `root-bound-forward-translation`.
- **generality** - whether finite extension preserves platforms, versions, failures and operating limits: `positive-finite-extension-preserves-platforms-failures-and-versions`.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: `no-extra-rule`.

Their ordered product gives the unique survivor:

```
system-demand-capability-carrier__load-profile-bounded__load-to-service-relation__dem
and-capacity-duration-queue-degradation-and-failure-held__science-design-test-and-dec
ision-classes-explicit__root-bound-forward-translation__positive-finite-extension-pre
serves-platforms-failures-and-versions__no-extra-rule
```

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses the requirement, artifact, user or environment. Erasing the boundary universalises a version, platform or test. Importing the relation lets a target or preferred application choose the result. Retaining only successful output removes requirements, alternatives, failures and rollback. Conflating evidence classes turns simulation, demonstration or performance into observation or law. Breaking provenance removes the trace to the One and upstream receipts. Erasing finite context makes one implementation universal. Adding a silent dependency or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite requirement or admitted law, versioned artifact, boundary, testable relation and complete record. The successor adds one lawful component, interface, platform, requirement, test, failure, lifecycle stage or anomaly while preserving all earlier identities, adverse rows and receipts.

Falsification and deliberately unfavourable controls

The registered falsification condition is: Reject if peak, average or one successful load is universalized.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: `source_bound_engineering_requirement_design_test_lifecycle_or_access_record/ primary_standard_handbook_guidance_or_project_record_structural_correspondence`. Captured registered sources: 3. Unresolved transports retained: 0. The registered structural consequence corresponded: `True`. External evidence selected the survivor: `false`. Application or performance selected the law: `false`. Successful performance was relabelled scientific proof: `false`. Failed implementation was relabelled scientific falsification: `false`. Simulation or demonstration was relabelled observation: `false`.

- `EPA-LIFECYCLE-001` - transport `captured`; captured bytes `980`; snapshot `sha256:1d271583636a679563a046e767d0575a92d23eeafbf99f0da3fef2582e102c4e`.
- `NASA-SE-HANDBOOK-001` - transport `captured`; captured bytes `3773440`; snapshot `sha256:8eeb4887a4dc57a23049da7dd2ed556833cf98e214b240468d987873164ff688`.

- `NIST-SSE-001` - transport `captured`; captured bytes `47984`; snapshot `sha256:64c8d41d9a78268d596a0a60aad18eead50fd375e8628486f7202788a207edaa`.

The comparison is structural and source-bound. It checks that a relevant primary handbook, standard, guidance record or open engineering project can represent the declared requirement, artifact, environment, operating boundary, version, test and lifecycle without rewriting the Fold law. It does not claim that a standard proves conformity, that one handbook exhausts Engineering, or that a custodian's reputation proves the result. New records can extend or challenge the surface only with source, version, configuration, method, failures and adverse rows retained.

Scientific meaning

For **capacity and load**, the consequence is practical and exact: Capacity is a load-profile-bound relation retaining duration, queuing, degradation and failure. An Engineering claim therefore has to expose the carrier `system-demand-capability-carrier`, the relation `load-to-service-relation`, the record `demand-capacity-duration-queue-degradation-and-failure-held`, and the boundary `load-profile-bounded` together. If any one is hidden, the result may remain a concept, prototype, simulation, demonstration or product choice, but it is not this admitted law.

This distinction protects users, maintainers and science. It prevents a successful output from erasing requirements, failed alternatives, hidden resources, inaccessible users, hazards, maintenance burdens or end-of-life costs. It prevents an application from selecting the upstream law and prevents engineering failure from becoming automatic scientific falsification. A replacement must still generate its forms, eliminate alternatives, survive controls, preserve custody and reproduce through the shared engine.

Machine certificate

Candidate count: `256`; survivor count: `1`; derivation seal: `sha256:e40e1a26fc3b5836762ac7e2559bd8de458883f6b5158f9b8126a046208f5176`; independent certificate: `sha256:d497222fc51ead74dc583a358e75c82c30f0b592d0c072398e941ce356047fa7`; engine receipt: `sha256:3963d278a9dde0a0c4f1c137720a3335d73c5557527eb06a8c99e84cccc16796`. The independent implementation recomputed the candidate product: `True`. All external rows were preserved: `True`.

15. Engineering efficiency

Claim: `SFT-ENG-EFFICIENCY-001`

Family: `resources_efficiency_reliability`

Exact statement: Efficiency compares useful output with all registered input and loss inside an explicit boundary.

Forward chain and exact carrier

The generated carrier is `input-useful-output-loss-carrier`. Its required relation is `accounted-output-to-input-relation`. The reconstruction must retain `boundary-input-output-loss-time-and-uncertainty-held`. Its declared empirical boundary is `efficiency-boundary-declared`. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is `SFT-FOUNDATION-ONE-001`, `SFT-FOUNDATION-FOLD-001`, `SFT-FOUNDATION-FOLD-DYNAMICS-001`, `SFT-MATH-EXACT-ARITHMETIC-001`, `SFT-MATH-GRAPH-NETWORK-001`, `SFT-MATH-OPTIMIZATION-001`, `SFT-MATH-DYNAMICAL-SYSTEMS-001`, `SFT-MATH-LOGIC-PROOF-001`, `SFT-INFO-QUANTITY-001`, `SFT-INFO-CHANNEL-CAPACITY-001`, `SFT-COMP-FORM-STATE-TRANSITION-001`, `SFT-COMP-SEM-CORRECTNESS-001`, `SFT-COMP-SEM-COMPILATION-001`, `SFT-COMP-DIST-CAUSALITY-001`, `SFT-PHYS-MEAS-VALUE-RECORD-001`, `SFT-PHYS-MEAS-CALIBRATION-001`, `SFT-CHEM-MEAS-TRACEABILITY-001`, `SFT-MAT-SUST-LIFECYCLE-001`, `SFT-BIO-BIO-HANDOFF-001`, `SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001`, `SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001`, `SFT-EARTH-HAZARD-RISK-HANDOFF-001`, `SFT-ASTRO-ASTRO-HANDOFF-001`, `SFT-SOCIAL-ENGINEERING-HANDOFF-001`, `SFT-ENG-INTEGRATION-001`, `SFT-ENG-CAPACITY-001`. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external standard, application, performance target, product or prior implementation to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external standards, products, tests or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite components, versions, platforms, tests and lifecycle records; performance remains artifact, environment, user, method and operating-boundary limited.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - which requirement, artifact, component, user, environment or test bears the claim: `input-useful-output-loss-carrier`.
- **boundary** - which operating envelope, version, platform, user and lifecycle stage bound the result: `efficiency-boundary-declared`.
- **relation** - which generated transformation, interface, comparison or handoff connects the carriers: `accounted-output-to-input-relation`.
- **record** - which requirements, versions, alternatives, failures, uncertainty and rollback remain reconstructible: `boundary-input-output-loss-time-and-uncertainty-held`.
- **evidence** - whether law, requirement, design, simulation, test, demonstration, performance and anomaly remain distinct: `science-design-test-and-decision-classes-explicit`.
- **provenance** - whether the result traces to the One and names every upstream receipt and design choice: `root-bound-forward-translation`.
- **generality** - whether finite extension preserves platforms, versions, failures and operating limits: `positive-finite-extension-preserves-platforms-failures-and-versions`.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: `no-extra-rule`.

Their ordered product gives the unique survivor:

```
input-useful-output-loss-carrier__efficiency-boundary-declared__accounted-output-to-input-relation__boundary-input-output-loss-time-and-uncertainty-held__science-design-test-and-decision-classes-explicit__root-bound-forward-translation__positive-finite-extension-preserves-platforms-failures-and-versions__no-extra-rule
```

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses the requirement, artifact, user or environment. Erasing the boundary universalises a version, platform or test. Importing the relation lets a target or preferred application choose the result. Retaining only successful output removes requirements, alternatives, failures and rollback. Conflating evidence classes turns simulation, demonstration or performance into observation or law. Breaking provenance removes the trace to the One and upstream receipts. Erasing finite context makes one implementation universal. Adding a silent dependency or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite requirement or admitted law, versioned artifact, boundary, testable relation and complete record. The successor adds one lawful component, interface, platform, requirement, test, failure, lifecycle stage or anomaly while preserving all earlier identities, adverse rows and receipts.

Falsification and deliberately unfavourable controls

The registered falsification condition is: Reject efficiency above the closed resource account or omit externalized loss.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: `source_bound_engineering_requirement_design_test_lifecycle_or_access_record/primary_standard_handbook_guidance_or_project_record_structural_correspondence`. Captured registered sources: 3. Unresolved transports retained: 0. The registered structural consequence corresponded: `True`. External evidence selected the survivor: `false`. Application or performance selected the law: `false`. Successful performance was relabelled scientific proof: `false`. Failed implementation was relabelled scientific falsification: `false`. Simulation or demonstration was relabelled observation: `false`.

- `EPA-LIFECYCLE-001` - transport `captured`; captured bytes `980`; snapshot `sha256:1d271583636a679563a046e767d0575a92d23eeafbf99f0da3fef2582e102c4e`.
- `NASA-SE-HANDBOOK-001` - transport `captured`; captured bytes `3773440`; snapshot `sha256:8eeb4887a4dc57a23049da7dd2ed556833cf98e214b240468d987873164ff688`.
- `NIST-SSE-001` - transport `captured`; captured bytes `47984`; snapshot `sha256:64c8d41d9a78268d596a0a60aad18eead50fd375e8628486f7202788a207edaa`.

The comparison is structural and source-bound. It checks that a relevant primary handbook, standard, guidance record or open engineering project can represent the declared requirement, artifact, environment, operating boundary, version, test and lifecycle without rewriting the Fold law. It does not claim that a standard proves conformity, that one handbook exhausts Engineering, or that a custodian's reputation proves the result. New records can extend or challenge the surface only with source, version, configuration, method, failures and adverse rows retained.

Scientific meaning

For **engineering efficiency**, the consequence is practical and exact: Efficiency compares useful output with all registered input and loss inside an explicit boundary. An Engineering claim therefore has to expose the carrier `input-useful-output-loss-carrier`, the relation `accounted-output-to-input-relation`, the record `boundary-input-output-loss-time-and-uncertainty-held`, and the boundary `efficiency-boundary-declared` together. If any one is hidden, the result may remain a concept, prototype, simulation, demonstration or product choice, but it is not this admitted law.

This distinction protects users, maintainers and science. It prevents a successful output from erasing requirements, failed alternatives, hidden resources, inaccessible users, hazards, maintenance burdens or end-of-life costs. It prevents an application from selecting the upstream law and prevents engineering failure from becoming automatic scientific falsification. A replacement must still generate its forms, eliminate alternatives, survive controls, preserve custody and reproduce through the shared engine.

Machine certificate

Candidate count: `256`; survivor count: `1`; derivation seal: `sha256:e53a9e26a49a45d121b041d7d5dca7fd764e2e89536dce4b62844a9f4ec7049a`; independent certificate: `sha256:fa408247f9657a5b863130a1d5d0252e8130ec365b9ae26688e7fb7e48a9af91`; engine receipt: `sha256:1160782f2e7324856d0fe651e21519afd0368a759e827edd0e790b9b146c72b3`. The independent implementation recomputed the candidate product: `True`. All external rows were preserved: `True`.

16. Reliability

Claim: `SFT-ENG-RELIABILITY-001`

Family: `resources_efficiency_reliability`

Exact statement: Reliability is a demand-, duration- and population-bound record of successful and failed service.

Forward chain and exact carrier

The generated carrier is `system-demand-time-population-carrier`. Its required relation is `required-service-over-trials-relation`. The reconstruction must retain `units-demands-duration-failures-censoring-and-repair-held`. Its declared empirical boundary is `reliability-population-bounded`. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is `SFT-FOUNDATION-ONE-001`, `SFT-FOUNDATION-FOLD-001`, `SFT-FOUNDATION-FOLD-DYNAMICS-001`, `SFT-MATH-EXACT-ARITHMETIC-001`, `SFT-MATH-GRAPH-NETWORK-001`, `SFT-MATH-OPTIMIZATION-001`, `SFT-MATH-DYNAMICAL-SYSTEMS-001`, `SFT-MATH-LOGIC-PROOF-001`, `SFT-INFO-QUANTITY-001`, `SFT-INFO-CHANNEL-CAPACITY-001`, `SFT-COMP-FORM-STATE-TRANSITION-001`, `SFT-COMP-SEM-CORRECTNESS-001`, `SFT-COMP-SEM-COMPILATION-001`, `SFT-COMP-DIST-CAUSALITY-001`, `SFT-PHYS-MEAS-VALUE-RECORD-001`, `SFT-PHYS-MEAS-CALIBRATION-001`, `SFT-CHEM-MEAS-TRACEABILITY-001`, `SFT-MAT-SUST-LIFECYCLE-001`, `SFT-BIO-BIO-HANDOFF-001`, `SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001`, `SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001`, `SFT-EARTH-HAZARD-RISK-HANDOFF-001`, `SFT-ASTRO-ASTRO-HANDOFF-001`, `SFT-SOCIAL-ENGINEERING-HANDOFF-001`, `SFT-ENG-INTEGRATION-001`, `SFT-ENG-EFFICIENCY-001`. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external standard, application, performance target, product or prior implementation to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external standards, products, tests or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite components, versions, platforms, tests and lifecycle records; performance remains artifact, environment, user, method and operating-boundary limited.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - which requirement, artifact, component, user, environment or test bears the claim: `system-demand-time-population-carrier`.
- **boundary** - which operating envelope, version, platform, user and lifecycle stage bound the result: `reliability-population-bounded`.
- **relation** - which generated transformation, interface, comparison or handoff connects the carriers: `required-service-over-trials-relation`.
- **record** - which requirements, versions, alternatives, failures, uncertainty and rollback remain reconstructible: `units-demands-duration-failures-censoring-and-repair-held`.
- **evidence** - whether law, requirement, design, simulation, test, demonstration, performance and anomaly remain distinct: `science-design-test-and-decision-classes-explicit`.
- **provenance** - whether the result traces to the One and names every upstream receipt and design choice: `root-bound-forward-translation`.
- **generality** - whether finite extension preserves platforms, versions, failures and operating limits: `positive-finite-extension-preserves-platforms-failures-and-versions`.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: `no-extra-rule`.

Their ordered product gives the unique survivor:

```
system-demand-time-population-carrier__reliability-population-bounded__required-servi
ce-over-trials-relation__units-demands-duration-failures-censoring-and-repair-held__s
cience-design-test-and-decision-classes-explicit__root-bound-forward-translation__pos
itive-finite-extension-preserves-platforms-failures-and-versions__no-extra-rule
```

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses the requirement, artifact, user or environment. Erasing the boundary universalises a version, platform or test. Importing the relation lets a target or preferred application choose the result. Retaining only successful output removes requirements, alternatives, failures and rollback. Conflating evidence classes turns simulation, demonstration or performance into observation or law. Breaking provenance removes the trace to the One and upstream receipts. Erasing finite context makes one implementation universal. Adding a silent dependency or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite requirement or admitted law, versioned artifact, boundary, testable relation and complete record. The successor adds one lawful component, interface, platform, requirement, test, failure, lifecycle stage or anomaly while preserving all earlier identities, adverse rows and receipts.

Falsification and deliberately unfavourable controls

The registered falsification condition is: Reject if demonstrations omit failures, censored units or changed operating conditions.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: `source_bound_engineering_requirement_design_test_lifecycle_or_access_record/primary_standard_handbook_guidance_or_project_record_structural_correspondence`. Captured registered sources: 3. Unresolved transports retained: 0. The registered structural consequence corresponded: `True`. External evidence selected the survivor: `false`. Application or performance selected the law: `false`. Successful performance was relabelled scientific proof: `false`. Failed implementation was relabelled scientific falsification: `false`. Simulation or demonstration was relabelled observation: `false`.

- `EPA-LIFECYCLE-001` - transport `captured`; captured bytes `980`; snapshot `sha256:1d271583636a679563a046e767d0575a92d23eeafbf99f0da3fef2582e102c4e`.
- `NASA-SE-HANDBOOK-001` - transport `captured`; captured bytes `3773440`; snapshot `sha256:8eeb4887a4dc57a23049da7dd2ed556833cf98e214b240468d987873164ff688`.
- `NIST-SSE-001` - transport `captured`; captured bytes `47984`; snapshot `sha256:64c8d41d9a78268d596a0a60aad18eead50fd375e8628486f7202788a207edaa`.

The comparison is structural and source-bound. It checks that a relevant primary handbook, standard, guidance record or open engineering project can represent the declared requirement, artifact, environment, operating boundary, version, test and lifecycle without rewriting the Fold law. It does not claim that a standard proves conformity, that one handbook exhausts Engineering, or that a custodian's reputation proves the result. New records can extend or challenge the surface only with source, version, configuration, method, failures and adverse rows retained.

Scientific meaning

For **reliability**, the consequence is practical and exact: Reliability is a demand-, duration- and population-bound record of successful and failed service. An Engineering claim therefore has to expose the carrier `system-demand-time-population-carrier`, the relation `required-service-over-trials-relation`, the record `units-demands-duration-failures-censoring-and-repair-held`, and the boundary `reliability-population-bounded` together. If any one is hidden, the result may remain a concept, prototype, simulation, demonstration or product choice, but it is not this admitted law.

This distinction protects users, maintainers and science. It prevents a successful output from erasing requirements, failed alternatives, hidden resources, inaccessible users, hazards, maintenance burdens or end-of-life costs. It prevents an application

from selecting the upstream law and prevents engineering failure from becoming automatic scientific falsification. A replacement must still generate its forms, eliminate alternatives, survive controls, preserve custody and reproduce through the shared engine.

Machine certificate

Candidate count: `256`; survivor count: `1`; derivation seal: `sha256:51436d1eaa447173a6376c90077efa00b1568168760653a420c311885ceea999`; independent certificate: `sha256:9e71ccc1c969419b25482a652a9938a014891842be83bec619ea04ffd3f78d71`; engine receipt: `sha256:dff52b462bd4a1e798eccae2125213e8dd3341bd2aabd73935967952ad3ab060`. The independent implementation recomputed the candidate product: `True`. All external rows were preserved: `True`.

17. Availability

Claim: `SFT-ENG-AVAILABILITY-001`

Family: `resources_efficiency_reliability`

Exact statement: Availability retains serviceable and unavailable intervals, causes, maintenance and demand windows.

Forward chain and exact carrier

The generated carrier is `system-service-time-state-carrier`. Its required relation is `serviceable-time-relation`. The reconstruction must retain `uptime-downtime-cause-maintenance-demand-and-window-held`. Its declared empirical boundary is `availability-window-bounded`. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is `SFT-FOUNDATION-ONE-001`, `SFT-FOUNDATION-FOLD-001`, `SFT-FOUNDATION-FOLD-DYNAMICS-001`, `SFT-MATH-EXACT-ARITHMETIC-001`, `SFT-MATH-GRAPH-NETWORK-001`, `SFT-MATH-OPTIMIZATION-001`, `SFT-MATH-DYNAMICAL-SYSTEMS-001`, `SFT-MATH-LOGIC-PROOF-001`, `SFT-INFO-QUANTITY-001`, `SFT-INFO-CHANNEL-CAPACITY-001`, `SFT-COMP-FORM-STATE-TRANSITION-001`, `SFT-COMP-SEM-CORRECTNESS-001`, `SFT-COMP-SEM-COMPILATION-001`, `SFT-COMP-DIST-CAUSALITY-001`, `SFT-PHYS-MEAS-VALUE-RECORD-001`, `SFT-PHYS-MEAS-CALIBRATION-001`, `SFT-CHEM-MEAS-TRACEABILITY-001`, `SFT-MAT-SUST-LIFECYCLE-001`, `SFT-BIO-BIO-HANDOFF-001`, `SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001`, `SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001`, `SFT-EARTH-HAZARD-RISK-HANDOFF-001`, `SFT-ASTRO-ASTRO-HANDOFF-001`, `SFT-SOCIAL-ENGINEERING-HANDOFF-001`, `SFT-ENG-INTEGRATION-001`, `SFT-ENG-RELIABILITY-001`. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external standard, application, performance target, product or prior implementation to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external standards, products, tests or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite components, versions, platforms, tests and lifecycle records; performance remains artifact, environment, user, method and operating-boundary limited.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - which requirement, artifact, component, user, environment or test bears the claim: `system-service-time-state-carrier`.
- **boundary** - which operating envelope, version, platform, user and lifecycle stage bound the result: `availability-window-bounded`.
- **relation** - which generated transformation, interface, comparison or handoff connects the carriers: `serviceable-time-relation`.

- **record** - which requirements, versions, alternatives, failures, uncertainty and rollback remain reconstructible: `uptime-downtime-cause-maintenance-demand-and-window-held`.
- **evidence** - whether law, requirement, design, simulation, test, demonstration, performance and anomaly remain distinct: `science-design-test-and-decision-classes-explicit`.
- **provenance** - whether the result traces to the One and names every upstream receipt and design choice: `root-bound-forward-translation`.
- **generality** - whether finite extension preserves platforms, versions, failures and operating limits: `positive-finite-extension-preserves-platforms-failures-and-versions`.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: `no-extra-rule`.

Their ordered product gives the unique survivor:

```
system-service-time-state-carrier__availability-window-bounded__serviceable-time-relation__uptime-downtime-cause-maintenance-demand-and-window-held__science-design-test-and-decision-classes-explicit__root-bound-forward-translation__positive-finite-extension-preserves-platforms-failures-and-versions__no-extra-rule
```

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses the requirement, artifact, user or environment. Erasing the boundary universalises a version, platform or test. Importing the relation lets a target or preferred application choose the result. Retaining only successful output removes requirements, alternatives, failures and rollback. Conflating evidence classes turns simulation, demonstration or performance into observation or law. Breaking provenance removes the trace to the One and upstream receipts. Erasing finite context makes one implementation universal. Adding a silent dependency or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite requirement or admitted law, versioned artifact, boundary, testable relation and complete record. The successor adds one lawful component, interface, platform, requirement, test, failure, lifecycle stage or anomaly while preserving all earlier identities, adverse rows and receipts.

Falsification and deliberately unfavourable controls

The registered falsification condition is: Reject if scheduled observation or missing monitoring becomes uptime.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: `source_bound_engineering_requirement_design_test_lifecycle_or_access_record/primary_standard_handbook_guidance_or_project_record_structural_correspondence`. Captured registered sources: 3. Unresolved transports retained: 0. The registered structural consequence corresponded: `True`. External evidence selected the survivor: `false`. Application or performance selected the law: `false`. Successful performance was relabelled scientific proof: `false`. Failed implementation was relabelled scientific falsification: `false`. Simulation or demonstration was relabelled observation: `false`.

- `EPA-LIFECYCLE-001` - transport `captured`; captured bytes `980`; snapshot `sha256:1d271583636a679563a046e767d0575a92d23eeafbf99f0da3fef2582e102c4e`.
- `NASA-SE-HANDBOOK-001` - transport `captured`; captured bytes `3773440`; snapshot `sha256:8eeb4887a4dc57a23049da7dd2ed556833cf98e214b240468d987873164ff688`.

- `NIST-SSE-001` - transport `captured`; captured bytes `47984`; snapshot `sha256:64c8d41d9a78268d596a0a60aad18eead50fd375e8628486f7202788a207edaa`.

The comparison is structural and source-bound. It checks that a relevant primary handbook, standard, guidance record or open engineering project can represent the declared requirement, artifact, environment, operating boundary, version, test and lifecycle without rewriting the Fold law. It does not claim that a standard proves conformity, that one handbook exhausts Engineering, or that a custodian's reputation proves the result. New records can extend or challenge the surface only with source, version, configuration, method, failures and adverse rows retained.

Scientific meaning

For **availability**, the consequence is practical and exact: Availability retains serviceable and unavailable intervals, causes, maintenance and demand windows. An Engineering claim therefore has to expose the carrier `system-service-time-state-carrier`, the relation `serviceable-time-relation`, the record `uptime-downtime-cause-maintenance-demand-and-window-held`, and the boundary `availability-window-bounded` together. If any one is hidden, the result may remain a concept, prototype, simulation, demonstration or product choice, but it is not this admitted law.

This distinction protects users, maintainers and science. It prevents a successful output from erasing requirements, failed alternatives, hidden resources, inaccessible users, hazards, maintenance burdens or end-of-life costs. It prevents an application from selecting the upstream law and prevents engineering failure from becoming automatic scientific falsification. A replacement must still generate its forms, eliminate alternatives, survive controls, preserve custody and reproduce through the shared engine.

Machine certificate

Candidate count: `256`; survivor count: `1`; derivation seal: `sha256:d6949264c4d9cf38876a0391316e39f9f83b6a2826abec9f9bdd8df717e1c561`; independent certificate: `sha256:8d849b1918e5fe12c530a66bb071805cfe03a582cad47760e899a71d25042dfc`; engine receipt: `sha256:4c1848211e1cce6edb226b77981fb85e148de4025bdf865727f40cb1c51f2647`. The independent implementation recomputed the candidate product: `True`. All external rows were preserved: `True`.

18. Failure mode

Claim: `SFT-ENG-FAILURE-MODE-001`

Family: `resources_efficiency_reliability`

Exact statement: A failure mode is an observed or generated loss-of-function path retaining cause, detectability, effect and recovery.

Forward chain and exact carrier

The generated carrier is `component-condition-transition-consequence-carrier`. Its required relation is `condition-to-loss-of-function-relation`. The reconstruction must retain `cause-detection-state-effect-severity-and-recovery-held`. Its declared empirical boundary is `failure-definition-bounded`. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is `SFT-FOUNDATION-ONE-001`, `SFT-FOUNDATION-FOLD-001`, `SFT-FOUNDATION-FOLD-DYNAMICS-001`, `SFT-MATH-EXACT-ARITHMETIC-001`, `SFT-MATH-GRAPH-NETWORK-001`, `SFT-MATH-OPTIMIZATION-001`, `SFT-MATH-DYNAMICAL-SYSTEMS-001`, `SFT-MATH-LOGIC-PROOF-001`, `SFT-INFO-QUANTITY-001`, `SFT-INFO-CHANNEL-CAPACITY-001`, `SFT-COMP-FORM-STATE-TRANSITION-001`, `SFT-COMP-SEM-CORRECTNESS-001`, `SFT-COMP-SEM-COMPILATION-001`, `SFT-COMP-DIST-CAUSALITY-001`, `SFT-PHYS-MEAS-VALUE-RECORD-001`, `SFT-PHYS-MEAS-CALIBRATION-001`, `SFT-CHEM-MEAS-TRACEABILITY-001`, `SFT-MAT-SUST-LIFECYCLE-001`, `SFT-BIO-BIO-HANDOFF-001`, `SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001`, `SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001`, `SFT-EARTH-HAZARD-RISK-HANDOFF-001`, `SFT-ASTRO-ASTRO-HANDOFF-001`, `SFT-SOCIAL-ENGINEERING-HANDOFF-001`, `SFT-ENG-INTEGRATION-001`, `SFT-ENG-AVAILABILITY-001`. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external standard, application, performance target, product or prior implementation to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external standards, products, tests or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite components, versions, platforms, tests and lifecycle records; performance remains artifact, environment, user, method and operating-boundary limited.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - which requirement, artifact, component, user, environment or test bears the claim: `component-condition-transition-consequence-carrier`.
- **boundary** - which operating envelope, version, platform, user and lifecycle stage bound the result: `failure-definition-bounded`.
- **relation** - which generated transformation, interface, comparison or handoff connects the carriers: `condition-to-loss-of-function-relation`.
- **record** - which requirements, versions, alternatives, failures, uncertainty and rollback remain reconstructible: `cause-detection-state-effect-severity-and-recovery-held`.
- **evidence** - whether law, requirement, design, simulation, test, demonstration, performance and anomaly remain distinct: `science-design-test-and-decision-classes-explicit`.
- **provenance** - whether the result traces to the One and names every upstream receipt and design choice: `root-bound-forward-translation`.
- **generality** - whether finite extension preserves platforms, versions, failures and operating limits: `positive-finite-extension-preserves-platforms-failures-and-versions`.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: `no-extra-rule`.

Their ordered product gives the unique survivor:

```
component-condition-transition-consequence-carrier__failure-definition-bounded__condi
tion-to-loss-of-function-relation__cause-detection-state-effect-severity-and-recovery
-held__science-design-test-and-decision-classes-explicit__root-bound-forward-translat
ion__positive-finite-extension-preserves-platforms-failures-and-versions__no-extra-ru
le
```

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses the requirement, artifact, user or environment. Erasing the boundary universalises a version, platform or test. Importing the relation lets a target or preferred application choose the result. Retaining only successful output removes requirements, alternatives, failures and rollback. Conflating evidence classes turns simulation, demonstration or performance into observation or law. Breaking provenance removes the trace to the One and upstream receipts. Erasing finite context makes one implementation universal. Adding a silent dependency or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite requirement or admitted law, versioned artifact, boundary, testable relation and complete record. The successor adds one lawful component, interface, platform, requirement, test, failure, lifecycle stage or anomaly while preserving all earlier identities, adverse rows and receipts.

Falsification and deliberately unfavourable controls

The registered falsification condition is: Reject if absence of observed failure proves impossibility or all failures share one cause.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor

- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: `source_bound_engineering_requirement_design_test_lifecycle_or_access_record/primary_standard_handbook_guidance_or_project_record_structural_correspondence`. Captured registered sources: 3. Unresolved transports retained: 0. The registered structural consequence corresponded: `True`. External evidence selected the survivor: `false`. Application or performance selected the law: `false`. Successful performance was relabelled scientific proof: `false`. Failed implementation was relabelled scientific falsification: `false`. Simulation or demonstration was relabelled observation: `false`.

- `EPA-LIFECYCLE-001` - transport `captured`; captured bytes `980`; snapshot `sha256:1d271583636a679563a046e767d0575a92d23eeafbf99f0da3fef2582e102c4e`.
- `NASA-SE-HANDBOOK-001` - transport `captured`; captured bytes `3773440`; snapshot `sha256:8eeb4887a4dc57a23049da7dd2ed556833cf98e214b240468d987873164ff688`.
- `NIST-SSE-001` - transport `captured`; captured bytes `47984`; snapshot `sha256:64c8d41d9a78268d596a0a60aad18eead50fd375e8628486f7202788a207edaa`.

The comparison is structural and source-bound. It checks that a relevant primary handbook, standard, guidance record or open engineering project can represent the declared requirement, artifact, environment, operating boundary, version, test and lifecycle without rewriting the Fold law. It does not claim that a standard proves conformity, that one handbook exhausts Engineering, or that a custodian's reputation proves the result. New records can extend or challenge the surface only with source, version, configuration, method, failures and adverse rows retained.

Scientific meaning

For **failure mode**, the consequence is practical and exact: A failure mode is an observed or generated loss-of-function path retaining cause, detectability, effect and recovery. An Engineering claim therefore has to expose the carrier `component-condition-transition-consequence-carrier`, the relation `condition-to-loss-of-function-relation`, the record `cause-detection-state-effect-severity-and-recovery-held`, and the boundary `failure-definition-bounded` together. If any one is hidden, the result may remain a concept, prototype, simulation, demonstration or product choice, but it is not this admitted law.

This distinction protects users, maintainers and science. It prevents a successful output from erasing requirements, failed alternatives, hidden resources, inaccessible users, hazards, maintenance burdens or end-of-life costs. It prevents an application from selecting the upstream law and prevents engineering failure from becoming automatic scientific falsification. A replacement must still generate its forms, eliminate alternatives, survive controls, preserve custody and reproduce through the shared engine.

Machine certificate

Candidate count: `256`; survivor count: `1`; derivation seal: `sha256:f50b4a2d6994e8b2af609d70e499a04f26790fb292ee2f4423cf4feb6af8254b`; independent certificate: `sha256:ad61253b92b8638cd9175b8efa74ab52d6aa63f402abe4c0bc806af0bb22fad7`; engine receipt: `sha256:099c3b3650fe1fe4e8b8b05eaa25d413a880079796df52aea10404c61c92b12d`. The independent implementation recomputed the candidate product: `True`. All external rows were preserved: `True`.

19. Engineering measurement

Claim: `SFT-ENG-MEASUREMENT-001`

Family: `measurement_calibration_acceptance`

Exact statement: Engineering measurement retains measurand, unit, instrument, conditions, uncertainty and record custody.

Forward chain and exact carrier

The generated carrier is `measurand-instrument-method-record-carrier`. Its required relation is `property-to-record-relation`. The reconstruction must retain `definition-unit-instrument-condition-uncertainty-and-custody-held`. Its declared empirical boundary is `measurement-procedure-bounded`. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is `SFT-FOUNDATION-ONE-001`, `SFT-FOUNDATION-FOLD-001`, `SFT-FOUNDATION-FOLD-DYNAMICS-001`, `SFT-MATH-EXACT-ARITHMETIC-001`, `SFT-MATH-GRAPH-NETWORK-001`, `SFT-MATH-OPTIMIZATION-001`, `SFT-MATH-DYNAMICAL-SYSTEMS-001`, `SFT-MATH-LOGIC-PROOF-001`, `SFT-INFO-QUANTITY-001`, `SFT-INFO-CHANNEL-CAPACITY-001`, `SFT-COMP-FORM-STATE-TRANSITION-001`, `SFT-COMP-SEM-CORRECTNESS-001`, `SFT-COMP-SEM-COMPILATION-001`, `SFT-COMP-DIST-CAUSALITY-001`, `SFT-PHYS-MEAS-VALUE-RECORD-001`, `SFT-PHYS-MEAS-CALIBRATION-001`, `SFT-CHEM-MEAS-TRACEABILITY-001`, `SFT-MAT-SUST-LIFECYCLE-001`, `SFT-BIO-BIO-HANDOFF-001`, `SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001`, `SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001`, `SFT-EARTH-HAZARD-RISK-HANDOFF-001`, `SFT-ASTRO-ASTRO-HANDOFF-001`, `SFT-SOCIAL-ENGINEERING-HANDOFF-001`, `SFT-ENG-FAILURE-MODE-001`. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external standard, application, performance target, product or prior implementation to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external standards, products, tests or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite components, versions, platforms, tests and lifecycle records; performance remains artifact, environment, user, method and operating-boundary limited.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - which requirement, artifact, component, user, environment or test bears the claim: `measurand-instrument-method-record-carrier`.
- **boundary** - which operating envelope, version, platform, user and lifecycle stage bound the result: `measurement-procedure-bounded`.
- **relation** - which generated transformation, interface, comparison or handoff connects the carriers: `property-to-record-relation`.
- **record** - which requirements, versions, alternatives, failures, uncertainty and rollback remain reconstructible: `definition-unit-instrument-condition-uncertainty-and-custody-held`.
- **evidence** - whether law, requirement, design, simulation, test, demonstration, performance and anomaly remain distinct: `science-design-test-and-decision-classes-explicit`.
- **provenance** - whether the result traces to the One and names every upstream receipt and design choice: `root-bound-forward-translation`.
- **generality** - whether finite extension preserves platforms, versions, failures and operating limits: `positive-finite-extension-preserves-platforms-failures-and-versions`.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: `no-extra-rule`.

Their ordered product gives the unique survivor:

```
measurand-instrument-method-record-carrier__measurement-procedure-bounded__property-t
o-record-relation__definition-unit-instrument-condition-uncertainty-and-custody-held_
_science-design-test-and-decision-classes-explicit__root-bound-forward-translation__p
ositive-finite-extension-preserves-platforms-failures-and-versions__no-extra-rule
```

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses the requirement, artifact, user or environment. Erasing the boundary universalises a version, platform or test. Importing the relation lets a target or preferred application choose the result. Retaining only successful output removes requirements, alternatives, failures and rollback. Conflating evidence classes turns simulation, demonstration or performance into observation or law. Breaking provenance removes the trace to the One and upstream receipts. Erasing finite context makes one implementation universal. Adding a silent dependency or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite requirement or admitted law, versioned artifact, boundary, testable relation and complete record. The successor adds one lawful component, interface, platform, requirement, test, failure, lifecycle stage or anomaly while preserving all earlier identities, adverse rows and receipts.

Falsification and deliberately unfavourable controls

The registered falsification condition is: Reject if display output alone proves the measurand or uncertainty is erased.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: `source_bound_engineering_requirement_design_test_lifecycle_or_access_record/primary_standard_handbook_guidance_or_project_record_structural_correspondence`. Captured registered sources: 3. Unresolved transports retained: 0. The registered structural consequence corresponded: `True`. External evidence selected the survivor: `false`. Application or performance selected the law: `false`. Successful performance was relabelled scientific proof: `false`. Failed implementation was relabelled scientific falsification: `false`. Simulation or demonstration was relabelled observation: `false`.

- `NIST-TRACEABILITY-001` - transport `captured`; captured bytes `180987`; snapshot `sha256:4e79d3a99a9367842daf0f8910cff625de901d95b4b46c5fe2845d7512e70cd7`.
- `NASA-SE-APPENDIX-001` - transport `captured`; captured bytes `578114`; snapshot `sha256:c1db2db970b9fd14a7ef98aa0c23575326dedae46cc7cc56c468821ca89edd60`.
- `NASA-SE-HANDBOOK-001` - transport `captured`; captured bytes `3773440`; snapshot `sha256:8eeb4887a4dc57a23049da7dd2ed556833cf98e214b240468d987873164ff688`.

The comparison is structural and source-bound. It checks that a relevant primary handbook, standard, guidance record or open engineering project can represent the declared requirement, artifact, environment, operating boundary, version, test and lifecycle without rewriting the Fold law. It does not claim that a standard proves conformity, that one handbook exhausts Engineering, or that a custodian's reputation proves the result. New records can extend or challenge the surface only with source, version, configuration, method, failures and adverse rows retained.

Scientific meaning

For **engineering measurement**, the consequence is practical and exact: Engineering measurement retains measurand, unit, instrument, conditions, uncertainty and record custody. An Engineering claim therefore has to expose the carrier `measurand-instrument-method-record-carrier`, the relation `property-to-record-relation`, the record `definition-unit-instrument-condition-uncertainty-and-custody-held`, and the boundary `measurement-procedure-bounded` together. If any one is hidden, the result may remain a concept, prototype, simulation, demonstration or product choice, but it is not this admitted law.

This distinction protects users, maintainers and science. It prevents a successful output from erasing requirements, failed alternatives, hidden resources, inaccessible users, hazards, maintenance burdens or end-of-life costs. It prevents an application

from selecting the upstream law and prevents engineering failure from becoming automatic scientific falsification. A replacement must still generate its forms, eliminate alternatives, survive controls, preserve custody and reproduce through the shared engine.

Machine certificate

Candidate count: `256`; survivor count: `1`; derivation seal: `sha256:b94f749c54bf8ed1bba51dcd8d6f4f82be59404013216b52605a022d20252c8f`; independent certificate: `sha256:afbd8f029ce6086c69dada70e87818b55bbfb4ebef086363aa3571bd875a160f`; engine receipt: `sha256:1608c9591220707596f110129e8e2067a7af9be542c55bc33c5c220b95208558`. The independent implementation recomputed the candidate product: `True`. All external rows were preserved: `True`.

20. Calibration

Claim: `SFT-ENG-CALIBRATION-001`

Family: `measurement_calibration_acceptance`

Exact statement: Calibration is a dated comparison to a traceable reference across a declared range and condition.

Forward chain and exact carrier

The generated carrier is `instrument-reference-condition-result-carrier`. Its required relation is `reference-to-indication-comparison`. The reconstruction must retain `reference-traceability-range-condition-result-and-date-held`. Its declared empirical boundary is `calibration-range-bounded`. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is `SFT-FOUNDATION-ONE-001`, `SFT-FOUNDATION-FOLD-001`, `SFT-FOUNDATION-FOLD-DYNAMICS-001`, `SFT-MATH-EXACT-ARITHMETIC-001`, `SFT-MATH-GRAPH-NETWORK-001`, `SFT-MATH-OPTIMIZATION-001`, `SFT-MATH-DYNAMICAL-SYSTEMS-001`, `SFT-MATH-LOGIC-PROOF-001`, `SFT-INFO-QUANTITY-001`, `SFT-INFO-CHANNEL-CAPACITY-001`, `SFT-COMP-FORM-STATE-TRANSITION-001`, `SFT-COMP-SEM-CORRECTNESS-001`, `SFT-COMP-SEM-COMPILATION-001`, `SFT-COMP-DIST-CAUSALITY-001`, `SFT-PHYS-MEAS-VALUE-RECORD-001`, `SFT-PHYS-MEAS-CALIBRATION-001`, `SFT-CHEM-MEAS-TRACEABILITY-001`, `SFT-MAT-SUST-LIFECYCLE-001`, `SFT-BIO-BIO-HANDOFF-001`, `SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001`, `SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001`, `SFT-EARTH-HAZARD-RISK-HANDOFF-001`, `SFT-ASTRO-ASTRO-HANDOFF-001`, `SFT-SOCIAL-ENGINEERING-HANDOFF-001`, `SFT-ENG-FAILURE-MODE-001`, `SFT-ENG-MEASUREMENT-001`. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external standard, application, performance target, product or prior implementation to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external standards, products, tests or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite components, versions, platforms, tests and lifecycle records; performance remains artifact, environment, user, method and operating-boundary limited.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - which requirement, artifact, component, user, environment or test bears the claim: `instrument-reference-condition-result-carrier`.
- **boundary** - which operating envelope, version, platform, user and lifecycle stage bound the result: `calibration-range-bounded`.
- **relation** - which generated transformation, interface, comparison or handoff connects the carriers: `reference-to-indication-comparison`.

- **record** - which requirements, versions, alternatives, failures, uncertainty and rollback remain reconstructible: `reference-traceability-range-condition-result-and-date-held`.
- **evidence** - whether law, requirement, design, simulation, test, demonstration, performance and anomaly remain distinct: `science-design-test-and-decision-classes-explicit`.
- **provenance** - whether the result traces to the One and names every upstream receipt and design choice: `root-bound-forward-translation`.
- **generality** - whether finite extension preserves platforms, versions, failures and operating limits: `positive-finite-extension-preserves-platforms-failures-and-versions`.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: `no-extra-rule`.

Their ordered product gives the unique survivor:

```
instrument-reference-condition-result-carrier__calibration-range-bounded__reference-t
o-indication-comparison__reference-traceability-range-condition-result-and-date-held_
_science-design-test-and-decision-classes-explicit__root-bound-forward-translation__p
ositive-finite-extension-preserves-platforms-failures-and-versions__no-extra-rule
```

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses the requirement, artifact, user or environment. Erasing the boundary universalises a version, platform or test. Importing the relation lets a target or preferred application choose the result. Retaining only successful output removes requirements, alternatives, failures and rollback. Conflating evidence classes turns simulation, demonstration or performance into observation or law. Breaking provenance removes the trace to the One and upstream receipts. Erasing finite context makes one implementation universal. Adding a silent dependency or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite requirement or admitted law, versioned artifact, boundary, testable relation and complete record. The successor adds one lawful component, interface, platform, requirement, test, failure, lifecycle stage or anomaly while preserving all earlier identities, adverse rows and receipts.

Falsification and deliberately unfavourable controls

The registered falsification condition is: Reject if adjustment, self-test or manufacturer status is called calibration without comparison.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: `source_bound_engineering_requirement_design_test_lifecycle_or_access_record/primary_standard_handbook_guidance_or_project_record_structural_correspondence`. Captured registered sources: 3. Unresolved transports retained: 0. The registered structural consequence corresponded: `True`. External evidence selected the survivor: `false`. Application or performance selected the law: `false`. Successful performance was relabelled scientific proof: `false`. Failed implementation was relabelled scientific falsification: `false`. Simulation or demonstration was relabelled observation: `false`.

- `NIST-TRACEABILITY-001` - transport `captured`; captured bytes `180987`; snapshot `sha256:4e79d3a99a9367842daf0f8910cff625de901d95b4b46c5fe2845d7512e70cd7`.
- `NASA-SE-APPENDIX-001` - transport `captured`; captured bytes `578114`; snapshot `sha256:c1db2db970b9fd14a7ef98aa0c23575326dedae46cc7cc56c468821ca89edd60`.

- NASA-SE-HANDBOOK-001 - transport captured; captured bytes 3773440; snapshot sha256:8eeb4887a4dc57a23049da7dd2ed556833cf98e214b240468d987873164ff688.

The comparison is structural and source-bound. It checks that a relevant primary handbook, standard, guidance record or open engineering project can represent the declared requirement, artifact, environment, operating boundary, version, test and lifecycle without rewriting the Fold law. It does not claim that a standard proves conformity, that one handbook exhausts Engineering, or that a custodian's reputation proves the result. New records can extend or challenge the surface only with source, version, configuration, method, failures and adverse rows retained.

Scientific meaning

For **calibration**, the consequence is practical and exact: Calibration is a dated comparison to a traceable reference across a declared range and condition. An Engineering claim therefore has to expose the carrier instrument-reference-condition-result-carrier, the relation reference-to-indication-comparison, the record reference-traceability-range-condition-result-and-date-held, and the boundary calibration-range-bounded together. If any one is hidden, the result may remain a concept, prototype, simulation, demonstration or product choice, but it is not this admitted law.

This distinction protects users, maintainers and science. It prevents a successful output from erasing requirements, failed alternatives, hidden resources, inaccessible users, hazards, maintenance burdens or end-of-life costs. It prevents an application from selecting the upstream law and prevents engineering failure from becoming automatic scientific falsification. A replacement must still generate its forms, eliminate alternatives, survive controls, preserve custody and reproduce through the shared engine.

Machine certificate

Candidate count: 256; survivor count: 1; derivation seal: sha256:fbff96a9d04df72a02b7634abc16b8a6f64993acb980cff044b67001be8eedb3; independent certificate: sha256:39f53a80e0be1b3dd661fa5ce4156a699eb589e857a1f563cc7b530707165e54; engine receipt: sha256:deca4c00620d9e161f9255ca3d1b0baa74856c5e48964c32b1ac6d9be5fa0b9c. The independent implementation recomputed the candidate product: True. All external rows were preserved: True.

21. Tolerance

Claim: SFT-ENG-TOLERANCE-001

Family: measurement_calibration_acceptance

Exact statement: A tolerance is a purpose- and uncertainty-bound allowed region registered before acceptance data.

Forward chain and exact carrier

The generated carrier is target-allowed-region-measurement-carrier. Its required relation is measurement-to-acceptance-region-relation. The reconstruction must retain nominal-bounds-unit-uncertainty-rationale-and-version-held. Its declared empirical boundary is tolerance-purpose-bounded. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-OPTIMIZATION-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-CHANNEL-CAPACITY-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-COMP-SEM-CORRECTNESS-001, SFT-COMP-SEM-COMPILATION-001, SFT-COMP-DIST-CAUSALITY-001, SFT-PHYS-MEAS-VALUE-RECORD-001, SFT-PHYS-MEAS-CALIBRATION-001, SFT-CHEM-MEAS-TRACEABILITY-001, SFT-MAT-SUST-LIFECYCLE-001, SFT-BIO-BIO-HANDOFF-001, SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001, SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-ASTRO-HANDOFF-001, SFT-SOCIAL-ENGINEERING-HANDOFF-001, SFT-ENG-FAILURE-MODE-001, SFT-ENG-CALIBRATION-001. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external standard, application, performance target, product or prior implementation to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external standards, products, tests or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite components, versions, platforms, tests and lifecycle records; performance remains artifact, environment, user, method and operating-boundary limited.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - which requirement, artifact, component, user, environment or test bears the claim: `target-allowed-region-measurement-carrier`.
- **boundary** - which operating envelope, version, platform, user and lifecycle stage bound the result: `tolerance-purpose-bounded`.
- **relation** - which generated transformation, interface, comparison or handoff connects the carriers: `measurement-to-acceptance-region-relation`.
- **record** - which requirements, versions, alternatives, failures, uncertainty and rollback remain reconstructible: `nominal-bounds-unit-uncertainty-rationale-and-version-held`.
- **evidence** - whether law, requirement, design, simulation, test, demonstration, performance and anomaly remain distinct: `science-design-test-and-decision-classes-explicit`.
- **provenance** - whether the result traces to the One and names every upstream receipt and design choice: `root-bound-forward-translation`.
- **generality** - whether finite extension preserves platforms, versions, failures and operating limits: `positive-finite-extension-preserves-platforms-failures-and-versions`.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: `no-extra-rule`.

Their ordered product gives the unique survivor:

`target-allowed-region-measurement-carrier__tolerance-purpose-bounded__measurement-to-acceptance-region-relation__nominal-bounds-unit-uncertainty-rationale-and-version-held__science-design-test-and-decision-classes-explicit__root-bound-forward-translation__positive-finite-extension-preserves-platforms-failures-and-versions__no-extra-rule`

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses the requirement, artifact, user or environment. Erasing the boundary universalises a version, platform or test. Importing the relation lets a target or preferred application choose the result. Retaining only successful output removes requirements, alternatives, failures and rollback. Conflating evidence classes turns simulation, demonstration or performance into observation or law. Breaking provenance removes the trace to the One and upstream receipts. Erasing finite context makes one implementation universal. Adding a silent dependency or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite requirement or admitted law, versioned artifact, boundary, testable relation and complete record. The successor adds one lawful component, interface, platform, requirement, test, failure, lifecycle stage or anomaly while preserving all earlier identities, adverse rows and receipts.

Falsification and deliberately unfavourable controls

The registered falsification condition is: Reject if tolerance is widened after measurement or used as a fitted physical constant.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: `source_bound_engineering_requirement_design_test_lifecycle_or_access_record/primary_standard_handbook_guidance_or_project_record_structural_correspondence`. Captured registered sources: 3. Unresolved transports retained: 0. The registered structural consequence corresponded: `True`. External evidence selected the survivor: `false`. Application or performance selected the law: `false`. Successful performance was relabelled scientific proof: `false`. Failed implementation was relabelled scientific falsification: `false`. Simulation or demonstration was relabelled observation: `false`.

- `NIST-TRACEABILITY-001` - transport `captured`; captured bytes `180987`; snapshot `sha256:4e79d3a99a9367842daf0f8910cff625de901d95b4b46c5fe2845d7512e70cd7`.
- `NASA-SE-APPENDIX-001` - transport `captured`; captured bytes `578114`; snapshot `sha256:c1db2db970b9fd14a7ef98aa0c23575326dedae46cc7cc56c468821ca89edd60`.
- `NASA-SE-HANDBOOK-001` - transport `captured`; captured bytes `3773440`; snapshot `sha256:8eeb4887a4dc57a23049da7dd2ed556833cf98e214b240468d987873164ff688`.

The comparison is structural and source-bound. It checks that a relevant primary handbook, standard, guidance record or open engineering project can represent the declared requirement, artifact, environment, operating boundary, version, test and lifecycle without rewriting the Fold law. It does not claim that a standard proves conformity, that one handbook exhausts Engineering, or that a custodian's reputation proves the result. New records can extend or challenge the surface only with source, version, configuration, method, failures and adverse rows retained.

Scientific meaning

For **tolerance**, the consequence is practical and exact: A tolerance is a purpose- and uncertainty-bound allowed region registered before acceptance data. An Engineering claim therefore has to expose the carrier `target-allowed-region-measurement-carrier`, the relation `measurement-to-acceptance-region-relation`, the record `nominal-bounds-unit-uncertainty-rationale-and-version-held`, and the boundary `tolerance-purpose-bounded` together. If any one is hidden, the result may remain a concept, prototype, simulation, demonstration or product choice, but it is not this admitted law.

This distinction protects users, maintainers and science. It prevents a successful output from erasing requirements, failed alternatives, hidden resources, inaccessible users, hazards, maintenance burdens or end-of-life costs. It prevents an application from selecting the upstream law and prevents engineering failure from becoming automatic scientific falsification. A replacement must still generate its forms, eliminate alternatives, survive controls, preserve custody and reproduce through the shared engine.

Machine certificate

Candidate count: `256`; survivor count: `1`; derivation seal: `sha256:a6c1f4eb36d93b90146171e80e8a7306ab905ea55ac7ed29c0c38d1458093230`; independent certificate: `sha256:005a309e79e1e635f1850c2b3f450a7569f12366d7f3c52e737113a76d16be39`; engine receipt: `sha256:4ba2c2f415d392ae660698be08d0c89c2567d84b3f76e47be776a8cfe5fbfd2b`. The independent implementation recomputed the candidate product: `True`. All external rows were preserved: `True`.

22. Engineering test

Claim: `SFT-ENG-TEST-001`

Family: `measurement_calibration_acceptance`

Exact statement: A test retains article version, setup, procedure, raw observations, deviations and unfavorable outcomes.

Forward chain and exact carrier

The generated carrier is `article-condition-procedure-observation-carrier`. Its required relation is `stimulus-to-response-comparison`. The reconstruction must retain `article-version-setup-procedure-raw-data-failures-and-deviations-held`. Its declared empirical boundary is `test-protocol-bounded`. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is `SFT-FOUNDATION-ONE-001`, `SFT-FOUNDATION-FOLD-001`, `SFT-FOUNDATION-FOLD-DYNAMICS-001`, `SFT-MATH-EXACT-ARITHMETIC-001`, `SFT-MATH-GRAPH-NETWORK-001`, `SFT-MATH-OPTIMIZATION-001`, `SFT-MATH-DYNAMICAL-SYSTEMS-001`, `SFT-MATH-LOGIC-PROOF-001`, `SFT-INFO-QUANTITY-001`, `SFT-INFO-CHANNEL-CAPACITY-001`, `SFT-COMP-FORM-STATE-TRANSITION-001`, `SFT-COMP-SEM-CORRECTNESS-001`, `SFT-COMP-SEM-COMPILATION-001`, `SFT-COMP-DIST-CAUSALITY-001`, `SFT-PHYS-MEAS-VALUE-RECORD-001`, `SFT-PHYS-MEAS-CALIBRATION-001`, `SFT-CHEM-MEAS-TRACEABILITY-001`, `SFT-MAT-SUST-LIFECYCLE-001`, `SFT-BIO-BIO-HANDOFF-001`, `SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001`, `SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001`, `SFT-EARTH-HAZARD-RISK-HANDOFF-001`, `SFT-ASTRO-ASTRO-HANDOFF-001`, `SFT-SOCIAL-ENGINEERING-HANDOFF-001`, `SFT-ENG-FAILURE-MODE-001`, `SFT-ENG-TOLERANCE-001`. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external standard, application, performance target, product or prior implementation to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external standards, products, tests or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite components, versions, platforms, tests and lifecycle records; performance remains artifact, environment, user, method and operating-boundary limited.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - which requirement, artifact, component, user, environment or test bears the claim: `article-condition-procedure-observation-carrier`.
- **boundary** - which operating envelope, version, platform, user and lifecycle stage bound the result: `test-protocol-bounded`.
- **relation** - which generated transformation, interface, comparison or handoff connects the carriers: `stimulus-to-response-comparison`.
- **record** - which requirements, versions, alternatives, failures, uncertainty and rollback remain reconstructible: `article-version-setup-procedure-raw-data-failures-and-deviations-held`.
- **evidence** - whether law, requirement, design, simulation, test, demonstration, performance and anomaly remain distinct: `science-design-test-and-decision-classes-explicit`.
- **provenance** - whether the result traces to the One and names every upstream receipt and design choice: `root-bound-forward-translation`.
- **generality** - whether finite extension preserves platforms, versions, failures and operating limits: `positive-finite-extension-preserves-platforms-failures-and-versions`.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: `no-extra-rule`.

Their ordered product gives the unique survivor:

```
article-condition-procedure-observation-carrier__test-protocol-bounded__stimulus-to-r
esponse-comparison__article-version-setup-procedure-raw-data-failures-and-deviations-
held__science-design-test-and-decision-classes-explicit__root-bound-forward-translati
on__positive-finite-extension-preserves-platforms-failures-and-versions__no-extra-rul
e
```

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses the requirement, artifact, user or environment. Erasing the boundary universalises a version, platform or test. Importing the relation lets a target or preferred application choose the result. Retaining only successful output removes requirements, alternatives, failures and rollback. Conflating evidence classes turns simulation, demonstration or performance into observation or law. Breaking provenance removes the trace to the One and upstream receipts. Erasing finite context makes one implementation universal. Adding a silent dependency or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite requirement or admitted law, versioned artifact, boundary, testable relation and complete record. The successor adds one lawful component, interface, platform, requirement, test, failure, lifecycle stage or anomaly while preserving all earlier identities, adverse rows and receipts.

Falsification and deliberately unfavourable controls

The registered falsification condition is: Reject if a demonstration without a criterion or preserved raw record is called a passed test.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: `source_bound_engineering_requirement_design_test_lifecycle_or_access_record/primary_standard_handbook_guidance_or_project_record_structural_correspondence`. Captured registered sources: 3. Unresolved transports retained: 0. The registered structural consequence corresponded: `True`. External evidence selected the survivor: `false`. Application or performance selected the law: `false`. Successful performance was relabelled scientific proof: `false`. Failed implementation was relabelled scientific falsification: `false`. Simulation or demonstration was relabelled observation: `false`.

- `NIST-TRACEABILITY-001` - transport `captured`; captured bytes `180987`; snapshot `sha256:4e79d3a99a9367842daf0f8910cff625de901d95b4b46c5fe2845d7512e70cd7`.
- `NASA-SE-APPENDIX-001` - transport `captured`; captured bytes `578114`; snapshot `sha256:c1db2db970b9fd14a7ef98aa0c23575326dedae46cc7cc56c468821ca89edd60`.
- `NASA-SE-HANDBOOK-001` - transport `captured`; captured bytes `3773440`; snapshot `sha256:8eeb4887a4dc57a23049da7dd2ed556833cf98e214b240468d987873164ff688`.

The comparison is structural and source-bound. It checks that a relevant primary handbook, standard, guidance record or open engineering project can represent the declared requirement, artifact, environment, operating boundary, version, test and lifecycle without rewriting the Fold law. It does not claim that a standard proves conformity, that one handbook exhausts Engineering, or that a custodian's reputation proves the result. New records can extend or challenge the surface only with source, version, configuration, method, failures and adverse rows retained.

Scientific meaning

For **engineering test**, the consequence is practical and exact: A test retains article version, setup, procedure, raw observations, deviations and unfavourable outcomes. An Engineering claim therefore has to expose the carrier `article-condition-procedure-observation-carrier`, the relation `stimulus-to-response-comparison`, the record `article-version-setup-procedure-raw-data-failures-and-deviations-held`, and the boundary `test-protocol-bounded` together. If any one is hidden, the result may remain a concept, prototype, simulation, demonstration or product choice, but it is not this admitted law.

This distinction protects users, maintainers and science. It prevents a successful output from erasing requirements, failed alternatives, hidden resources, inaccessible users, hazards, maintenance burdens or end-of-life costs. It prevents an application from selecting the upstream law and prevents engineering failure from becoming automatic scientific falsification. A replacement must still generate its forms, eliminate alternatives, survive controls, preserve custody and reproduce through the shared engine.

Machine certificate

Candidate count: `256`; survivor count: `1`; derivation seal: `sha256:f581033639fb35ea0d1292393909adb6eef3ea11717948326d748bdbcae7f403`; independent certificate: `sha256:53c0121d1f36cabc740a9a5172c6d854e12dd8eee80bc1cd2526b88ad062ea8c`; engine receipt: `sha256:5957316c89ac4ea53b2ac9314e2dfbd7f1353a162dd94743f8a08da293e69276`. The independent implementation recomputed the candidate product: `True`. All external rows were preserved: `True`.

23. Acceptance test

Claim: `SFT-ENG-ACCEPTANCE-TEST-001`

Family: `measurement_calibration_acceptance`

Exact statement: An acceptance test compares a frozen requirement and threshold to a source-custodied result.

Forward chain and exact carrier

The generated carrier is `requirement-test-result-decision-carrier`. Its required relation is `registered-criterion-to-result-relation`. The reconstruction must retain `requirement-method-threshold-uncertainty-result-and-decision-held`. Its declared empirical boundary is `acceptance-test-bounded`. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is `SFT-FOUNDATION-ONE-001`, `SFT-FOUNDATION-FOLD-001`, `SFT-FOUNDATION-FOLD-DYNAMICS-001`, `SFT-MATH-EXACT-ARITHMETIC-001`, `SFT-MATH-GRAPH-NETWORK-001`, `SFT-MATH-OPTIMIZATION-001`, `SFT-MATH-DYNAMICAL-SYSTEMS-001`, `SFT-MATH-LOGIC-PROOF-001`, `SFT-INFO-QUANTITY-001`, `SFT-INFO-CHANNEL-CAPACITY-001`, `SFT-COMP-FORM-STATE-TRANSITION-001`, `SFT-COMP-SEM-CORRECTNESS-001`, `SFT-COMP-SEM-COMPILATION-001`, `SFT-COMP-DIST-CAUSALITY-001`, `SFT-PHYS-MEAS-VALUE-RECORD-001`, `SFT-PHYS-MEAS-CALIBRATION-001`, `SFT-CHEM-MEAS-TRACEABILITY-001`, `SFT-MAT-SUST-LIFECYCLE-001`, `SFT-BIO-BIO-HANDOFF-001`, `SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001`, `SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001`, `SFT-EARTH-HAZARD-RISK-HANDOFF-001`, `SFT-ASTRO-ASTRO-HANDOFF-001`, `SFT-SOCIAL-ENGINEERING-HANDOFF-001`, `SFT-ENG-FAILURE-MODE-001`, `SFT-ENG-TEST-001`. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external standard, application, performance target, product or prior implementation to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external standards, products, tests or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite components, versions, platforms, tests and lifecycle records; performance remains artifact, environment, user, method and operating-boundary limited.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - which requirement, artifact, component, user, environment or test bears the claim: `requirement-test-result-decision-carrier`.
- **boundary** - which operating envelope, version, platform, user and lifecycle stage bound the result: `acceptance-test-bounded`.
- **relation** - which generated transformation, interface, comparison or handoff connects the carriers: `registered-criterion-to-result-relation`.

- **record** - which requirements, versions, alternatives, failures, uncertainty and rollback remain reconstructible: `requirement-method-threshold-uncertainty-result-and-decision-held`.
- **evidence** - whether law, requirement, design, simulation, test, demonstration, performance and anomaly remain distinct: `science-design-test-and-decision-classes-explicit`.
- **provenance** - whether the result traces to the One and names every upstream receipt and design choice: `root-bound-forward-translation`.
- **generality** - whether finite extension preserves platforms, versions, failures and operating limits: `positive-finite-extension-preserves-platforms-failures-and-versions`.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: `no-extra-rule`.

Their ordered product gives the unique survivor:

```
requirement-test-result-decision-carrier__acceptance-test-bounded__registered-criteri
on-to-result-relation__requirement-method-threshold-uncertainty-result-and-decision-h
eld__science-design-test-and-decision-classes-explicit__root-bound-forward-translatio
n__positive-finite-extension-preserves-platforms-failures-and-versions__no-extra-rule
```

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses the requirement, artifact, user or environment. Erasing the boundary universalises a version, platform or test. Importing the relation lets a target or preferred application choose the result. Retaining only successful output removes requirements, alternatives, failures and rollback. Conflating evidence classes turns simulation, demonstration or performance into observation or law. Breaking provenance removes the trace to the One and upstream receipts. Erasing finite context makes one implementation universal. Adding a silent dependency or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite requirement or admitted law, versioned artifact, boundary, testable relation and complete record. The successor adds one lawful component, interface, platform, requirement, test, failure, lifecycle stage or anomaly while preserving all earlier identities, adverse rows and receipts.

Falsification and deliberately unfavourable controls

The registered falsification condition is: Reject if target identity, threshold or scoring rule changes after the result opens.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: `source_bound_engineering_requirement_design_test_lifecycle_or_access_record/primary_standard_handbook_guidance_or_project_record_structural_correspondence`. Captured registered sources: 3. Unresolved transports retained: 0. The registered structural consequence corresponded: `True`. External evidence selected the survivor: `false`. Application or performance selected the law: `false`. Successful performance was relabelled scientific proof: `false`. Failed implementation was relabelled scientific falsification: `false`. Simulation or demonstration was relabelled observation: `false`.

- `NIST-TRACEABILITY-001` - transport `captured`; captured bytes `180987`; snapshot `sha256:4e79d3a99a9367842daf0f8910cff625de901d95b4b46c5fe2845d7512e70cd7`.
- `NASA-SE-APPENDIX-001` - transport `captured`; captured bytes `578114`; snapshot `sha256:c1db2db970b9fd14a7ef98aa0c23575326dedae46cc7cc56c468821ca89edd60`.

- `NASA-SE-HANDBOOK-001` - transport `captured`; captured bytes `3773440`; snapshot `sha256:8eeb4887a4dc57a23049da7dd2ed556833cf98e214b240468d987873164ff688`.

The comparison is structural and source-bound. It checks that a relevant primary handbook, standard, guidance record or open engineering project can represent the declared requirement, artifact, environment, operating boundary, version, test and lifecycle without rewriting the Fold law. It does not claim that a standard proves conformity, that one handbook exhausts Engineering, or that a custodian's reputation proves the result. New records can extend or challenge the surface only with source, version, configuration, method, failures and adverse rows retained.

Scientific meaning

For **acceptance test**, the consequence is practical and exact: An acceptance test compares a frozen requirement and threshold to a source-custodied result. An Engineering claim therefore has to expose the carrier `requirement-test-result-decision-carrier`, the relation `registered-criterion-to-result-relation`, the record `requirement-method-threshold-uncertainty-result-and-decision-held`, and the boundary `acceptance-test-bounded` together. If any one is hidden, the result may remain a concept, prototype, simulation, demonstration or product choice, but it is not this admitted law.

This distinction protects users, maintainers and science. It prevents a successful output from erasing requirements, failed alternatives, hidden resources, inaccessible users, hazards, maintenance burdens or end-of-life costs. It prevents an application from selecting the upstream law and prevents engineering failure from becoming automatic scientific falsification. A replacement must still generate its forms, eliminate alternatives, survive controls, preserve custody and reproduce through the shared engine.

Machine certificate

Candidate count: `256`; survivor count: `1`; derivation seal: `sha256:5b04bc37723f96757b2704da43d80f3f65f6adce5269b630fd0a44b070ad22b0`; independent certificate: `sha256:7181e7e948295d14b29f8d2b43b7548d3a81b43aedac70c3d2c5b82cc131ad23`; engine receipt: `sha256:b12c54d4d4f0836beb387a1d91fb21dec030c3d945d33f5058ea8ace80eff987`. The independent implementation recomputed the candidate product: `True`. All external rows were preserved: `True`.

24. Measurement uncertainty

Claim: `SFT-ENG-MEASUREMENT-UNCERTAINTY-001`

Family: `measurement_calibration_acceptance`

Exact statement: Measurement uncertainty retains declared sources, correlations, method, coverage and revision.

Forward chain and exact carrier

The generated carrier is `result-source-variation-bound-carrier`. Its required relation is `sources-to-declared-result-region-relation`. The reconstruction must retain `components-method-correlation-coverage-boundary-and-version-held`. Its declared empirical boundary is `uncertainty-model-bounded`. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is `SFT-FOUNDATION-ONE-001`, `SFT-FOUNDATION-FOLD-001`, `SFT-FOUNDATION-FOLD-DYNAMICS-001`, `SFT-MATH-EXACT-ARITHMETIC-001`, `SFT-MATH-GRAPH-NETWORK-001`, `SFT-MATH-OPTIMIZATION-001`, `SFT-MATH-DYNAMICAL-SYSTEMS-001`, `SFT-MATH-LOGIC-PROOF-001`, `SFT-INFO-QUANTITY-001`, `SFT-INFO-CHANNEL-CAPACITY-001`, `SFT-COMP-FORM-STATE-TRANSITION-001`, `SFT-COMP-SEM-CORRECTNESS-001`, `SFT-COMP-SEM-COMPILATION-001`, `SFT-COMP-DIST-CAUSALITY-001`, `SFT-PHYS-MEAS-VALUE-RECORD-001`, `SFT-PHYS-MEAS-CALIBRATION-001`, `SFT-CHEM-MEAS-TRACEABILITY-001`, `SFT-MAT-SUST-LIFECYCLE-001`, `SFT-BIO-BIO-HANDOFF-001`, `SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001`, `SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001`, `SFT-EARTH-HAZARD-RISK-HANDOFF-001`, `SFT-ASTRO-ASTRO-HANDOFF-001`, `SFT-SOCIAL-ENGINEERING-HANDOFF-001`, `SFT-ENG-FAILURE-MODE-001`, `SFT-ENG-ACCEPTANCE-TEST-001`. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external standard, application, performance target, product or prior implementation to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external standards, products, tests or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite components, versions, platforms, tests and lifecycle records; performance remains artifact, environment, user, method and operating-boundary limited.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - which requirement, artifact, component, user, environment or test bears the claim: `result-source-variation-bound-carrier`.
- **boundary** - which operating envelope, version, platform, user and lifecycle stage bound the result: `uncertainty-model-bounded`.
- **relation** - which generated transformation, interface, comparison or handoff connects the carriers: `sources-to-declared-result-region-relation`.
- **record** - which requirements, versions, alternatives, failures, uncertainty and rollback remain reconstructible: `components-method-correlation-coverage-boundary-and-version-held`.
- **evidence** - whether law, requirement, design, simulation, test, demonstration, performance and anomaly remain distinct: `science-design-test-and-decision-classes-explicit`.
- **provenance** - whether the result traces to the One and names every upstream receipt and design choice: `root-bound-forward-translation`.
- **generality** - whether finite extension preserves platforms, versions, failures and operating limits: `positive-finite-extension-preserves-platforms-failures-and-versions`.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: `no-extra-rule`.

Their ordered product gives the unique survivor:

```
result-source-variation-bound-carrier__uncertainty-model-bounded__sources-to-declared
-result-region-relation__components-method-correlation-coverage-boundary-and-version-
held__science-design-test-and-decision-classes-explicit__root-bound-forward-translati
on__positive-finite-extension-preserves-platforms-failures-and-versions__no-extra-rul
e
```

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses the requirement, artifact, user or environment. Erasing the boundary universalises a version, platform or test. Importing the relation lets a target or preferred application choose the result. Retaining only successful output removes requirements, alternatives, failures and rollback. Conflating evidence classes turns simulation, demonstration or performance into observation or law. Breaking provenance removes the trace to the One and upstream receipts. Erasing finite context makes one implementation universal. Adding a silent dependency or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite requirement or admitted law, versioned artifact, boundary, testable relation and complete record. The successor adds one lawful component, interface, platform, requirement, test, failure, lifecycle stage or anomaly while preserving all earlier identities, adverse rows and receipts.

Falsification and deliberately unfavourable controls

The registered falsification condition is: Reject if uncertainty is omitted, tuned to force agreement or treated as proof error.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity

- tampered_artifact - passed True: reject a missing, duplicate or additional survivor
- boundary - passed True: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: source_bound_engineering_requirement_design_test_lifecycle_or_access_record/ primary_standard_handbook_guidance_or_project_record_structural_correspondence. Captured registered sources: 3. Unresolved transports retained: 0. The registered structural consequence corresponded: True. External evidence selected the survivor: false. Application or performance selected the law: false. Successful performance was relabelled scientific proof: false. Failed implementation was relabelled scientific falsification: false. Simulation or demonstration was relabelled observation: false.

- NIST-TRACEABILITY-001 - transport captured; captured bytes 180987; snapshot sha256:4e79d3a99a9367842daf0f8910cff625de901d95b4b46c5fe2845d7512e70cd7.
- NASA-SE-APPENDIX-001 - transport captured; captured bytes 578114; snapshot sha256:c1db2db970b9fd14a7ef98aa0c23575326dedae46cc7cc56c468821ca89edd60.
- NASA-SE-HANDBOOK-001 - transport captured; captured bytes 3773440; snapshot sha256:8eeb4887a4dc57a23049da7dd2ed556833cf98e214b240468d987873164ff688.

The comparison is structural and source-bound. It checks that a relevant primary handbook, standard, guidance record or open engineering project can represent the declared requirement, artifact, environment, operating boundary, version, test and lifecycle without rewriting the Fold law. It does not claim that a standard proves conformity, that one handbook exhausts Engineering, or that a custodian's reputation proves the result. New records can extend or challenge the surface only with source, version, configuration, method, failures and adverse rows retained.

Scientific meaning

For **measurement uncertainty**, the consequence is practical and exact: Measurement uncertainty retains declared sources, correlations, method, coverage and revision. An Engineering claim therefore has to expose the carrier result-source-variation-bound-carrier, the relation sources-to-declared-result-region-relation, the record components-method-correlation-coverage-boundary-and-version-held, and the boundary uncertainty-model-bounded together. If any one is hidden, the result may remain a concept, prototype, simulation, demonstration or product choice, but it is not this admitted law.

This distinction protects users, maintainers and science. It prevents a successful output from erasing requirements, failed alternatives, hidden resources, inaccessible users, hazards, maintenance burdens or end-of-life costs. It prevents an application from selecting the upstream law and prevents engineering failure from becoming automatic scientific falsification. A replacement must still generate its forms, eliminate alternatives, survive controls, preserve custody and reproduce through the shared engine.

Machine certificate

Candidate count: 256; survivor count: 1; derivation seal: sha256:977b6ebdfc15297e6e6879107628aa632aa2184cc1cf3059ac5ecf604a46caaa; independent certificate: sha256:ee04337e955247e1e4ca6df367c2413446943460c9846940a5c3d642e9275abb; engine receipt: sha256:45a61872d6bea9c23660d4b649ae931ce6c00a23c36e1150c4941aa749098863. The independent implementation recomputed the candidate product: True. All external rows were preserved: True.

25. Design alternative

Claim: SFT-ENG-ALTERNATIVE-001

Family: alternatives_tradeoffs_optimization

Exact statement: A design alternative must be generated from the frozen requirement grammar with every rejected option retained.

Forward chain and exact carrier

The generated carrier is `requirement-generated-design-carrier`. Its required relation is `design-to-evaluation-relation`. The reconstruction must retain `generator-assumptions-resources-results-failures-and-rejection-held`. Its declared empirical boundary is `alternative-grammar-bounded`. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is `SFT-FOUNDATION-ONE-001`, `SFT-FOUNDATION-FOLD-001`, `SFT-FOUNDATION-FOLD-DYNAMICS-001`, `SFT-MATH-EXACT-ARITHMETIC-001`, `SFT-MATH-GRAPH-NETWORK-001`, `SFT-MATH-OPTIMIZATION-001`, `SFT-MATH-DYNAMICAL-SYSTEMS-001`, `SFT-MATH-LOGIC-PROOF-001`, `SFT-INFO-QUANTITY-001`, `SFT-INFO-CHANNEL-CAPACITY-001`, `SFT-COMP-FORM-STATE-TRANSITION-001`, `SFT-COMP-SEM-CORRECTNESS-001`, `SFT-COMP-SEM-COMPILATION-001`, `SFT-COMP-DIST-CAUSALITY-001`, `SFT-PHYS-MEAS-VALUE-RECORD-001`, `SFT-PHYS-MEAS-CALIBRATION-001`, `SFT-CHEM-MEAS-TRACEABILITY-001`, `SFT-MAT-SUST-LIFECYCLE-001`, `SFT-BIO-BIO-HANDOFF-001`, `SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001`, `SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001`, `SFT-EARTH-HAZARD-RISK-HANDOFF-001`, `SFT-ASTRO-ASTRO-HANDOFF-001`, `SFT-SOCIAL-ENGINEERING-HANDOFF-001`, `SFT-ENG-MEASUREMENT-UNCERTAINTY-001`. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external standard, application, performance target, product or prior implementation to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external standards, products, tests or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite components, versions, platforms, tests and lifecycle records; performance remains artifact, environment, user, method and operating-boundary limited.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - which requirement, artifact, component, user, environment or test bears the claim: `requirement-generated-design-carrier`.
- **boundary** - which operating envelope, version, platform, user and lifecycle stage bound the result: `alternative-grammar-bounded`.
- **relation** - which generated transformation, interface, comparison or handoff connects the carriers: `design-to-evaluation-relation`.
- **record** - which requirements, versions, alternatives, failures, uncertainty and rollback remain reconstructible: `generator-assumptions-resources-results-failures-and-rejection-held`.
- **evidence** - whether law, requirement, design, simulation, test, demonstration, performance and anomaly remain distinct: `science-design-test-and-decision-classes-explicit`.
- **provenance** - whether the result traces to the One and names every upstream receipt and design choice: `root-bound-forward-translation`.
- **generality** - whether finite extension preserves platforms, versions, failures and operating limits: `positive-finite-extension-preserves-platforms-failures-and-versions`.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: `no-extra-rule`.

Their ordered product gives the unique survivor:

```
requirement-generated-design-carrier__alternative-grammar-bounded__design-to-evaluati
on-relation__generator-assumptions-resources-results-failures-and-rejection-held__sci
```

`ence-design-test-and-decision-classes-explicit__root-bound-forward-translation__positive-finite-extension-preserves-platforms-failures-and-versions__no-extra-rule`

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses the requirement, artifact, user or environment. Erasing the boundary universalises a version, platform or test. Importing the relation lets a target or preferred application choose the result. Retaining only successful output removes requirements, alternatives, failures and rollback. Conflating evidence classes turns simulation, demonstration or performance into observation or law. Breaking provenance removes the trace to the One and upstream receipts. Erasing finite context makes one implementation universal. Adding a silent dependency or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite requirement or admitted law, versioned artifact, boundary, testable relation and complete record. The successor adds one lawful component, interface, platform, requirement, test, failure, lifecycle stage or anomaly while preserving all earlier identities, adverse rows and receipts.

Falsification and deliberately unfavourable controls

The registered falsification condition is: Reject if only the preferred design is generated or alternatives are invented after outcome.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: `source_bound_engineering_requirement_design_test_lifecycle_or_access_record/primary_standard_handbook_guidance_or_project_record_structural_correspondence`. Captured registered sources: 3. Unresolved transports retained: 0. The registered structural consequence corresponded: `True`. External evidence selected the survivor: `false`. Application or performance selected the law: `false`. Successful performance was relabelled scientific proof: `false`. Failed implementation was relabelled scientific falsification: `false`. Simulation or demonstration was relabelled observation: `false`.

- `EPA-LIFECYCLE-001` - transport `captured`; captured bytes `980`; snapshot `sha256:1d271583636a679563a046e767d0575a92d23eeafbf99f0da3fef2582e102c4e`.
- `HSE-COMAH-001` - transport `captured`; captured bytes `52092`; snapshot `sha256:928bb33ae445b9eb3be9784ade179d832b3506a96bba0d3011460070de9053e9`.
- `NASA-SE-HANDBOOK-001` - transport `captured`; captured bytes `3773440`; snapshot `sha256:8eeb4887a4dc57a23049da7dd2ed556833cf98e214b240468d987873164ff688`.

The comparison is structural and source-bound. It checks that a relevant primary handbook, standard, guidance record or open engineering project can represent the declared requirement, artifact, environment, operating boundary, version, test and lifecycle without rewriting the Fold law. It does not claim that a standard proves conformity, that one handbook exhausts Engineering, or that a custodian's reputation proves the result. New records can extend or challenge the surface only with source, version, configuration, method, failures and adverse rows retained.

Scientific meaning

For **design alternative**, the consequence is practical and exact: A design alternative must be generated from the frozen requirement grammar with every rejected option retained. An Engineering claim therefore has to expose the carrier `requirement-generated-design-carrier`, the relation `design-to-evaluation-relation`, the record `generator-assumptions-resources-results-failures-and-rejection-held`, and the boundary `alternative-grammar-bounded` together. If any one is hidden, the result may remain a concept, prototype, simulation, demonstration or product choice, but it is not this admitted law.

This distinction protects users, maintainers and science. It prevents a successful output from erasing requirements, failed alternatives, hidden resources, inaccessible users, hazards, maintenance burdens or end-of-life costs. It prevents an application from selecting the upstream law and prevents engineering failure from becoming automatic scientific falsification. A replacement must still generate its forms, eliminate alternatives, survive controls, preserve custody and reproduce through the shared engine.

Machine certificate

Candidate count: `256`; survivor count: `1`; derivation seal: `sha256:ff0ab5d3fb2b73b28f7880a9e8628617303f50e53fecb8722a5e459784606ea4`; independent certificate: `sha256:ada90572dfae3d2d97889a6741e25421602f53559e9e48d26e270a33e8b0617e`; engine receipt: `sha256:cc6584f3217d9d2a73f7914dd439848c0307390734cbb3420abd421e209802d4`. The independent implementation recomputed the candidate product: `True`. All external rows were preserved: `True`.

26. Engineering trade-off

Claim: `SFT-ENG-TRADEOFF-001`

Family: `alternatives_tradeoffs_optimization`

Exact statement: A trade-off retains distinct criteria, evidence, conflicts, stakeholders and losses without collapsing them into a hidden scalar.

Forward chain and exact carrier

The generated carrier is `alternatives-criteria-context-carrier`. Its required relation is `criteria-wise-comparison-relation`. The reconstruction must retain `criteria-units-evidence-conflicts-stakeholders-and-losses-held`. Its declared empirical boundary is `tradeoff-context-bounded`. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is `SFT-FOUNDATION-ONE-001`, `SFT-FOUNDATION-FOLD-001`, `SFT-FOUNDATION-FOLD-DYNAMICS-001`, `SFT-MATH-EXACT-ARITHMETIC-001`, `SFT-MATH-GRAPH-NETWORK-001`, `SFT-MATH-OPTIMIZATION-001`, `SFT-MATH-DYNAMICAL-SYSTEMS-001`, `SFT-MATH-LOGIC-PROOF-001`, `SFT-INFO-QUANTITY-001`, `SFT-INFO-CHANNEL-CAPACITY-001`, `SFT-COMP-FORM-STATE-TRANSITION-001`, `SFT-COMP-SEM-CORRECTNESS-001`, `SFT-COMP-SEM-COMPILATION-001`, `SFT-COMP-DIST-CAUSALITY-001`, `SFT-PHYS-MEAS-VALUE-RECORD-001`, `SFT-PHYS-MEAS-CALIBRATION-001`, `SFT-CHEM-MEAS-TRACEABILITY-001`, `SFT-MAT-SUST-LIFECYCLE-001`, `SFT-BIO-BIO-HANDOFF-001`, `SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001`, `SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001`, `SFT-EARTH-HAZARD-RISK-HANDOFF-001`, `SFT-ASTRO-ASTRO-HANDOFF-001`, `SFT-SOCIAL-ENGINEERING-HANDOFF-001`, `SFT-ENG-MEASUREMENT-UNCERTAINTY-001`, `SFT-ENG-ALTERNATIVE-001`. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external standard, application, performance target, product or prior implementation to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external standards, products, tests or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite components, versions, platforms, tests and lifecycle records; performance remains artifact, environment, user, method and operating-boundary limited.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - which requirement, artifact, component, user, environment or test bears the claim: `alternatives-criteria-context-carrier`.
- **boundary** - which operating envelope, version, platform, user and lifecycle stage bound the result: `tradeoff-context-bounded`.
- **relation** - which generated transformation, interface, comparison or handoff connects the carriers: `criteria-wise-comparison-relation`.
- **record** - which requirements, versions, alternatives, failures, uncertainty and rollback remain reconstructible: `criteria-units-evidence-conflicts-stakeholders-and-losses-held`.
- **evidence** - whether law, requirement, design, simulation, test, demonstration, performance and anomaly remain distinct: `science-design-test-and-decision-classes-explicit`.
- **provenance** - whether the result traces to the One and names every upstream receipt and design choice: `root-bound-forward-translation`.
- **generality** - whether finite extension preserves platforms, versions, failures and operating limits: `positive-finite-extension-preserves-platforms-failures-and-versions`.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: `no-extra-rule`.

Their ordered product gives the unique survivor:

```
alternatives-criteria-context-carrier__tradeoff-context-bounded__criteria-wise-compar
ison-relation__criteria-units-evidence-conflicts-stakeholders-and-losses-held__scienc
e-design-test-and-decision-classes-explicit__root-bound-forward-translation__positive
-finite-extension-preserves-platforms-failures-and-versions__no-extra-rule
```

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses the requirement, artifact, user or environment. Erasing the boundary universalises a version, platform or test. Importing the relation lets a target or preferred application choose the result. Retaining only successful output removes requirements, alternatives, failures and rollback. Conflating evidence classes turns simulation, demonstration or performance into observation or law. Breaking provenance removes the trace to the One and upstream receipts. Erasing finite context makes one implementation universal. Adding a silent dependency or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite requirement or admitted law, versioned artifact, boundary, testable relation and complete record. The successor adds one lawful component, interface, platform, requirement, test, failure, lifecycle stage or anomaly while preserving all earlier identities, adverse rows and receipts.

Falsification and deliberately unfavourable controls

The registered falsification condition is: Reject if an unexplained weight converts value judgment into an exact law.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: `source_bound_engineering_requirement_design_test_lifecycle_or_access_record/primary_standard_handbook_guidance_or_project_record_structural_correspondence`. Captured registered sources: 3. Unresolved transports retained: 0. The registered structural consequence corresponded: `True`. External evidence selected the survivor: `false`. Application or performance selected the law: `false`. Successful performance was relabelled scientific proof: `false`. Failed implementation was relabelled scientific falsification: `false`. Simulation or demonstration was relabelled observation: `false`.

- `EPA-LIFECYCLE-001` - transport `captured`; captured bytes `980`; snapshot `sha256:1d271583636a679563a046e767d0575a92d23eeafbf99f0da3fef2582e102c4e`.
- `HSE-COMAH-001` - transport `captured`; captured bytes `52092`; snapshot `sha256:928bb33ae445b9eb3be9784ade179d832b3506a96bba0d3011460070de9053e9`.
- `NASA-SE-HANDBOOK-001` - transport `captured`; captured bytes `3773440`; snapshot `sha256:8eeb4887a4dc57a23049da7dd2ed556833cf98e214b240468d987873164ff688`.

The comparison is structural and source-bound. It checks that a relevant primary handbook, standard, guidance record or open engineering project can represent the declared requirement, artifact, environment, operating boundary, version, test and lifecycle without rewriting the Fold law. It does not claim that a standard proves conformity, that one handbook exhausts Engineering, or that a custodian's reputation proves the result. New records can extend or challenge the surface only with source, version, configuration, method, failures and adverse rows retained.

Scientific meaning

For **engineering trade-off**, the consequence is practical and exact: A trade-off retains distinct criteria, evidence, conflicts, stakeholders and losses without collapsing them into a hidden scalar. An Engineering claim therefore has to expose the carrier `alternatives-criteria-context-carrier`, the relation `criteria-wise-comparison-relation`, the record `criteria-units-evidence-conflicts-stakeholders-and-losses-held`, and the boundary `tradeoff-context-bounded` together. If any one is hidden, the result may remain a concept, prototype, simulation, demonstration or product choice, but it is not this admitted law.

This distinction protects users, maintainers and science. It prevents a successful output from erasing requirements, failed alternatives, hidden resources, inaccessible users, hazards, maintenance burdens or end-of-life costs. It prevents an application from selecting the upstream law and prevents engineering failure from becoming automatic scientific falsification. A replacement must still generate its forms, eliminate alternatives, survive controls, preserve custody and reproduce through the shared engine.

Machine certificate

Candidate count: `256`; survivor count: `1`; derivation seal: `sha256:b4b9dd010a074c399fd247ba04d611f6d3124fb42398b79ac92befc32be3dc15`; independent certificate: `sha256:110d1426847b94ce8df96b169ecea685536b5c564de31672e1740c51f3b36d72`; engine receipt: `sha256:dbb7ef3b75368e8e8eddb02c80dd7397f13edf88ffacc3f8dfadbb9746eca0ac`. The independent implementation recomputed the candidate product: `True`. All external rows were preserved: `True`.

27. Bounded engineering optimisation

Claim: `SFT-ENG-OPTIMIZATION-001`

Family: `alternatives_tradeoffs_optimization`

Exact statement: Engineering optimization enumerates a declared finite design set and preserves ties, failures and constraint violations.

Forward chain and exact carrier

The generated carrier is `finite-design-set-objective-constraint-carrier`. Its required relation is `enumeration-to-admissible-optimum-relation`. The reconstruction must retain `designs-objective-constraints-ties-failures-and-boundary-held`. Its declared empirical boundary is `finite-search-bounded`. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is `SFT-FOUNDATION-ONE-001`, `SFT-FOUNDATION-FOLD-001`, `SFT-FOUNDATION-FOLD-DYNAMICS-001`, `SFT-MATH-EXACT-ARITHMETIC-001`, `SFT-MATH-GRAPH-NETWORK-001`, `SFT-MATH-OPTIMIZATION-001`, `SFT-MATH-DYNAMICAL-SYSTEMS-001`, `SFT-MATH-LOGIC-PROOF-001`, `SFT-INFO-QUANTITY-001`, `SFT-INFO-CHANNEL-CAPACITY-001`, `SFT-COMP-FORM-STATE-TRANSITION-001`, `SFT-COMP-SEM-CORRECTNESS-001`, `SFT-COMP-SEM-COMPILATION-001`, `SFT-COMP-DIST-CAUSALITY-001`, `SFT-PHYS-MEAS-VALUE-RECORD-001`, `SFT-PHYS-MEAS-CALIBRATION-001`, `SFT-CHEM-MEAS-TRACEABILITY-001`, `SFT-MAT-SUST-LIFECYCLE-001`, `SFT-BIO-BIO-HANDOFF-001`, `SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001`, `SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001`, `SFT-EARTH-HAZARD-RISK-HANDOFF-001`, `SFT-ASTRO-ASTRO-HANDOFF-001`, `SFT-SOCIAL-ENGINEERING-HANDOFF-001`, `SFT-ENG-MEASUREMENT-UNCERTAINTY-001`, `SFT-ENG-TRADEOFF-001`. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external standard, application, performance target, product or prior implementation to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external standards, products, tests or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite components, versions, platforms, tests and lifecycle records; performance remains artifact, environment, user, method and operating-boundary limited.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - which requirement, artifact, component, user, environment or test bears the claim: `finite-design-set-objective-constraint-carrier`.
- **boundary** - which operating envelope, version, platform, user and lifecycle stage bound the result: `finite-search-bounded`.
- **relation** - which generated transformation, interface, comparison or handoff connects the carriers: `enumeration-to-admissible-optimum-relation`.
- **record** - which requirements, versions, alternatives, failures, uncertainty and rollback remain reconstructible: `designs-objective-constraints-ties-failures-and-boundary-held`.
- **evidence** - whether law, requirement, design, simulation, test, demonstration, performance and anomaly remain distinct: `science-design-test-and-decision-classes-explicit`.
- **provenance** - whether the result traces to the One and names every upstream receipt and design choice: `root-bound-forward-translation`.
- **generality** - whether finite extension preserves platforms, versions, failures and operating limits: `positive-finite-extension-preserves-platforms-failures-and-versions`.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: `no-extra-rule`.

Their ordered product gives the unique survivor:

```
finite-design-set-objective-constraint-carrier__finite-search-bounded__enumeration-to
-admissible-optimum-relation__designs-objective-constraints-ties-failures-and-boundar
y-held__science-design-test-and-decision-classes-explicit__root-bound-forward-transla
tion__positive-finite-extension-preserves-platforms-failures-and-versions__no-extra-r
ule
```

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses the requirement, artifact, user or environment. Erasing the boundary universalises a version, platform or test. Importing the relation lets a target or preferred application choose the result. Retaining only successful output removes requirements, alternatives, failures and rollback. Conflating evidence classes turns simulation, demonstration or performance into observation or law. Breaking provenance removes the trace to the One and upstream receipts. Erasing finite context makes one implementation universal. Adding a silent

dependency or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite requirement or admitted law, versioned artifact, boundary, testable relation and complete record. The successor adds one lawful component, interface, platform, requirement, test, failure, lifecycle stage or anomaly while preserving all earlier identities, adverse rows and receipts.

Falsification and deliberately unfavourable controls

The registered falsification condition is: Reject if an ungenerated design space, fitted weight or opaque oracle selects the optimum.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: `source_bound_engineering_requirement_design_test_lifecycle_or_access_record/primary_standard_handbook_guidance_or_project_record_structural_correspondence`. Captured registered sources: 3. Unresolved transports retained: 0. The registered structural consequence corresponded: `True`. External evidence selected the survivor: `false`. Application or performance selected the law: `false`. Successful performance was relabelled scientific proof: `false`. Failed implementation was relabelled scientific falsification: `false`. Simulation or demonstration was relabelled observation: `false`.

- `EPA-LIFECYCLE-001` - transport `captured`; captured bytes `980`; snapshot `sha256:1d271583636a679563a046e767d0575a92d23eeafbf99f0da3fef2582e102c4e`.
- `HSE-COMAH-001` - transport `captured`; captured bytes `52092`; snapshot `sha256:928bb33ae445b9eb3be9784ade179d832b3506a96bba0d3011460070de9053e9`.
- `NASA-SE-HANDBOOK-001` - transport `captured`; captured bytes `3773440`; snapshot `sha256:8eeb4887a4dc57a23049da7dd2ed556833cf98e214b240468d987873164ff688`.

The comparison is structural and source-bound. It checks that a relevant primary handbook, standard, guidance record or open engineering project can represent the declared requirement, artifact, environment, operating boundary, version, test and lifecycle without rewriting the Fold law. It does not claim that a standard proves conformity, that one handbook exhausts Engineering, or that a custodian's reputation proves the result. New records can extend or challenge the surface only with source, version, configuration, method, failures and adverse rows retained.

Scientific meaning

For **bounded engineering optimisation**, the consequence is practical and exact: Engineering optimisation enumerates a declared finite design set and preserves ties, failures and constraint violations. An Engineering claim therefore has to expose the carrier `finite-design-set-objective-constraint-carrier`, the relation `enumeration-to-admissible-optimum-relation`, the record `designs-objective-constraints-ties-failures-and-boundary-held`, and the boundary `finite-search-bounded` together. If any one is hidden, the result may remain a concept, prototype, simulation, demonstration or product choice, but it is not this admitted law.

This distinction protects users, maintainers and science. It prevents a successful output from erasing requirements, failed alternatives, hidden resources, inaccessible users, hazards, maintenance burdens or end-of-life costs. It prevents an application from selecting the upstream law and prevents engineering failure from becoming automatic scientific falsification. A replacement must still generate its forms, eliminate alternatives, survive controls, preserve custody and reproduce through the shared engine.

Machine certificate

Candidate count: 256; survivor count: 1; derivation seal: sha256:3c6c508b77506d97613c1630ba1967591703eb0381abfd3dd959acf9c78aa690; independent certificate: sha256:8a8d6c719722d6f121a597d9c8bceea5bb238ba929b76e6f0f53ad9fa3556d3b; engine receipt: sha256:acfdc9809f00fd4f3c22a9dda5e1d3a58d4a9cacb2f8e64aa92ec87bc02db2ae. The independent implementation recomputed the candidate product: True. All external rows were preserved: True.

28. Multi-criteria decision record

Claim: SFT-ENG-MULTI-CRITERIA-001

Family: alternatives_tradeoffs_optimization

Exact statement: Multi-criteria choice keeps measured evidence separate from declared values, priorities and dissent.

Forward chain and exact carrier

The generated carrier is alternatives-criteria-decision-maker-carrier. Its required relation is evidence-and-value-to-decision-relation. The reconstruction must retain evidence-values-priorities-conflicts-dissent-and-version-held. Its declared empirical boundary is decision-context-bounded. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-OPTIMIZATION-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-CHANNEL-CAPACITY-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-COMP-SEM-CORRECTNESS-001, SFT-COMP-SEM-COMPILATION-001, SFT-COMP-DIST-CAUSALITY-001, SFT-PHYS-MEAS-VALUE-RECORD-001, SFT-PHYS-MEAS-CALIBRATION-001, SFT-CHEM-MEAS-TRACEABILITY-001, SFT-MAT-SUST-LIFECYCLE-001, SFT-BIO-BIO-HANDOFF-001, SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001, SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-ASTRO-HANDOFF-001, SFT-SOCIAL-ENGINEERING-HANDOFF-001, SFT-ENG-MEASUREMENT-UNCERTAINTY-001, SFT-ENG-OPTIMIZATION-001. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external standard, application, performance target, product or prior implementation to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external standards, products, tests or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite components, versions, platforms, tests and lifecycle records; performance remains artifact, environment, user, method and operating-boundary limited.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - which requirement, artifact, component, user, environment or test bears the claim: alternatives-criteria-decision-maker-carrier.
- **boundary** - which operating envelope, version, platform, user and lifecycle stage bound the result: decision-context-bounded.
- **relation** - which generated transformation, interface, comparison or handoff connects the carriers: evidence-and-value-to-decision-relation.
- **record** - which requirements, versions, alternatives, failures, uncertainty and rollback remain reconstructible: evidence-values-priorities-conflicts-dissent-and-version-held.

- **evidence** - whether law, requirement, design, simulation, test, demonstration, performance and anomaly remain distinct: `science-design-test-and-decision-classes-explicit`.
- **provenance** - whether the result traces to the One and names every upstream receipt and design choice: `root-bound-forward-translation`.
- **generality** - whether finite extension preserves platforms, versions, failures and operating limits: `positive-finite-extension-preserves-platforms-failures-and-versions`.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: `no-extra-rule`.

Their ordered product gives the unique survivor:

```
alternatives-criteria-decision-maker-carrier__decision-context-bounded__evidence-and-value-to-decision-relation__evidence-values-priorities-conflicts-dissent-and-version-held__science-design-test-and-decision-classes-explicit__root-bound-forward-translation__positive-finite-extension-preserves-platforms-failures-and-versions__no-extra-rule
```

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses the requirement, artifact, user or environment. Erasing the boundary universalises a version, platform or test. Importing the relation lets a target or preferred application choose the result. Retaining only successful output removes requirements, alternatives, failures and rollback. Conflating evidence classes turns simulation, demonstration or performance into observation or law. Breaking provenance removes the trace to the One and upstream receipts. Erasing finite context makes one implementation universal. Adding a silent dependency or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite requirement or admitted law, versioned artifact, boundary, testable relation and complete record. The successor adds one lawful component, interface, platform, requirement, test, failure, lifecycle stage or anomaly while preserving all earlier identities, adverse rows and receipts.

Falsification and deliberately unfavourable controls

The registered falsification condition is: Reject if the decision is presented as a uniquely forced scientific truth.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: `source_bound_engineering_requirement_design_test_lifecycle_or_access_record/primary_standard_handbook_guidance_or_project_record_structural_correspondence`. Captured registered sources: 3. Unresolved transports retained: 0. The registered structural consequence corresponded: `True`. External evidence selected the survivor: `false`. Application or performance selected the law: `false`. Successful performance was relabelled scientific proof: `false`. Failed implementation was relabelled scientific falsification: `false`. Simulation or demonstration was relabelled observation: `false`.

- `EPA-LIFECYCLE-001` - transport `captured`; captured bytes `980`; snapshot `sha256:1d271583636a679563a046e767d0575a92d23eeafbf99f0da3fef2582e102c4e`.
- `HSE-COMAH-001` - transport `captured`; captured bytes `52092`; snapshot `sha256:928bb33ae445b9eb3be9784ade179d832b3506a96bba0d3011460070de9053e9`.
- `NASA-SE-HANDBOOK-001` - transport `captured`; captured bytes `3773440`; snapshot `sha256:8eeb4887a4dc57a23049da7dd2ed556833cf98e214b240468d987873164ff688`.

The comparison is structural and source-bound. It checks that a relevant primary handbook, standard, guidance record or open engineering project can represent the declared requirement, artifact, environment, operating boundary, version, test and lifecycle without rewriting the Fold law. It does not claim that a standard proves conformity, that one handbook exhausts Engineering, or that a custodian's reputation proves the result. New records can extend or challenge the surface only with source, version, configuration, method, failures and adverse rows retained.

Scientific meaning

For **multi-criteria decision record**, the consequence is practical and exact: Multi-criteria choice keeps measured evidence separate from declared values, priorities and dissent. An Engineering claim therefore has to expose the carrier `alternatives-criteria-decision-maker-carrier`, the relation `evidence-and-value-to-decision-relation`, the record `evidence-values-priorities-conflicts-dissent-and-version-held`, and the boundary `decision-context-bounded` together. If any one is hidden, the result may remain a concept, prototype, simulation, demonstration or product choice, but it is not this admitted law.

This distinction protects users, maintainers and science. It prevents a successful output from erasing requirements, failed alternatives, hidden resources, inaccessible users, hazards, maintenance burdens or end-of-life costs. It prevents an application from selecting the upstream law and prevents engineering failure from becoming automatic scientific falsification. A replacement must still generate its forms, eliminate alternatives, survive controls, preserve custody and reproduce through the shared engine.

Machine certificate

Candidate count: `256`; survivor count: `1`; derivation seal: `sha256:f913ff8b07db0158869ee32b158500541c28c072fc2aac03be08342f8d44c658`; independent certificate: `sha256:765312db97e1824688f6d1562a3e95714307bb18ac8e7a9d90c6bf2185a870c9`; engine receipt: `sha256:a050fbf706e359069160b952cb3a7e60442b6cb499251554190d3337ad71a55f`. The independent implementation recomputed the candidate product: `True`. All external rows were preserved: `True`.

29. Robust design

Claim: `SFT-ENG-ROBUST-DESIGN-001`

Family: `alternatives_tradeoffs_optimization`

Exact statement: A robust design retains performance across the preregistered condition family, including adverse corners.

Forward chain and exact carrier

The generated carrier is `design-condition-family-performance-carrier`. Its required relation is `condition-variation-to-performance-relation`. The reconstruction must retain `conditions-uncertainties-failures-margins-and-boundary-held`. Its declared empirical boundary is `registered-variation-bounded`. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is `SFT-FOUNDATION-ONE-001`, `SFT-FOUNDATION-FOLD-001`, `SFT-FOUNDATION-FOLD-DYNAMICS-001`, `SFT-MATH-EXACT-ARITHMETIC-001`, `SFT-MATH-GRAPH-NETWORK-001`, `SFT-MATH-OPTIMIZATION-001`, `SFT-MATH-DYNAMICAL-SYSTEMS-001`, `SFT-MATH-LOGIC-PROOF-001`, `SFT-INFO-QUANTITY-001`, `SFT-INFO-CHANNEL-CAPACITY-001`, `SFT-COMP-FORM-STATE-TRANSITION-001`, `SFT-COMP-SEM-CORRECTNESS-001`, `SFT-COMP-SEM-COMPILATION-001`, `SFT-COMP-DIST-CAUSALITY-001`, `SFT-PHYS-MEAS-VALUE-RECORD-001`, `SFT-PHYS-MEAS-CALIBRATION-001`, `SFT-CHEM-MEAS-TRACEABILITY-001`, `SFT-MAT-SUST-LIFECYCLE-001`, `SFT-BIO-BIO-HANDOFF-001`, `SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001`, `SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001`, `SFT-EARTH-HAZARD-RISK-HANDOFF-001`, `SFT-ASTRO-ASTRO-HANDOFF-001`, `SFT-SOCIAL-ENGINEERING-HANDOFF-001`, `SFT-ENG-MEASUREMENT-UNCERTAINTY-001`, `SFT-ENG-MULTI-CRITERIA-001`. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external standard, application, performance target, product or prior implementation to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external standards, products, tests or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite components, versions, platforms, tests and lifecycle records; performance remains artifact, environment, user, method and operating-boundary limited.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - which requirement, artifact, component, user, environment or test bears the claim: `design-condition-family-performance-carrier`.
- **boundary** - which operating envelope, version, platform, user and lifecycle stage bound the result: `registered-variation-bounded`.
- **relation** - which generated transformation, interface, comparison or handoff connects the carriers: `condition-variation-to-performance-relation`.
- **record** - which requirements, versions, alternatives, failures, uncertainty and rollback remain reconstructible: `conditions-uncertainties-failures-margins-and-boundary-held`.
- **evidence** - whether law, requirement, design, simulation, test, demonstration, performance and anomaly remain distinct: `science-design-test-and-decision-classes-explicit`.
- **provenance** - whether the result traces to the One and names every upstream receipt and design choice: `root-bound-forward-translation`.
- **generality** - whether finite extension preserves platforms, versions, failures and operating limits: `positive-finite-extension-preserves-platforms-failures-and-versions`.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: `no-extra-rule`.

Their ordered product gives the unique survivor:

```
design-condition-family-performance-carrier__registered-variation-bounded__condition-variation-to-performance-relation__conditions-uncertainties-failures-margins-and-boundary-held__science-design-test-and-decision-classes-explicit__root-bound-forward-translation__positive-finite-extension-preserves-platforms-failures-and-versions__no-extra-rule
```

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses the requirement, artifact, user or environment. Erasing the boundary universalises a version, platform or test. Importing the relation lets a target or preferred application choose the result. Retaining only successful output removes requirements, alternatives, failures and rollback. Conflating evidence classes turns simulation, demonstration or performance into observation or law. Breaking provenance removes the trace to the One and upstream receipts. Erasing finite context makes one implementation universal. Adding a silent dependency or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite requirement or admitted law, versioned artifact, boundary, testable relation and complete record. The successor adds one lawful component, interface, platform, requirement, test, failure, lifecycle stage or anomaly while preserving all earlier identities, adverse rows and receipts.

Falsification and deliberately unfavourable controls

The registered falsification condition is: Reject if favourable nominal operation proves robustness.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: `source_bound_engineering_requirement_design_test_lifecycle_or_access_record/primary_standard_handbook_guidance_or_project_record_structural_correspondence`. Captured registered sources: 3. Unresolved transports retained: 0. The registered structural consequence corresponded: `True`. External evidence selected the survivor: `false`. Application or performance selected the law: `false`. Successful performance was relabelled scientific proof: `false`. Failed implementation was relabelled scientific falsification: `false`. Simulation or demonstration was relabelled observation: `false`.

- `EPA-LIFECYCLE-001` - transport `captured`; captured bytes `980`; snapshot `sha256:1d271583636a679563a046e767d0575a92d23eeafbf99f0da3fef2582e102c4e`.
- `HSE-COMAH-001` - transport `captured`; captured bytes `52092`; snapshot `sha256:928bb33ae445b9eb3be9784ade179d832b3506a96bba0d3011460070de9053e9`.
- `NASA-SE-HANDBOOK-001` - transport `captured`; captured bytes `3773440`; snapshot `sha256:8eeb4887a4dc57a23049da7dd2ed556833cf98e214b240468d987873164ff688`.

The comparison is structural and source-bound. It checks that a relevant primary handbook, standard, guidance record or open engineering project can represent the declared requirement, artifact, environment, operating boundary, version, test and lifecycle without rewriting the Fold law. It does not claim that a standard proves conformity, that one handbook exhausts Engineering, or that a custodian's reputation proves the result. New records can extend or challenge the surface only with source, version, configuration, method, failures and adverse rows retained.

Scientific meaning

For **robust design**, the consequence is practical and exact: A robust design retains performance across the preregistered condition family, including adverse corners. An Engineering claim therefore has to expose the carrier `design-condition-family-performance-carrier`, the relation `condition-variation-to-performance-relation`, the record `conditions-uncertainties-failures-margins-and-boundary-held`, and the boundary `registered-variation-bounded` together. If any one is hidden, the result may remain a concept, prototype, simulation, demonstration or product choice, but it is not this admitted law.

This distinction protects users, maintainers and science. It prevents a successful output from erasing requirements, failed alternatives, hidden resources, inaccessible users, hazards, maintenance burdens or end-of-life costs. It prevents an application from selecting the upstream law and prevents engineering failure from becoming automatic scientific falsification. A replacement must still generate its forms, eliminate alternatives, survive controls, preserve custody and reproduce through the shared engine.

Machine certificate

Candidate count: `256`; survivor count: `1`; derivation seal: `sha256:eb20497a498de6741adb5cf51991b201c18a8a01a955be3acfaa53b10e97bfa0`; independent certificate: `sha256:7693990c06c7cffeac2ffd3875807673b468639288679bc409aac6dfda6bca18`; engine receipt: `sha256:0d0782c2fd5ddd62c75ceb0b924c32f028f03e8cbe8f120f60a02dff8a9a2016`. The independent implementation recomputed the candidate product: `True`. All external rows were preserved: `True`.

30. Design freeze and change control

Claim: `SFT-ENG-DESIGN-FREEZE-001`

Family: `alternatives_tradeoffs_optimization`

Exact statement: A design freeze is a versioned baseline whose later changes require explicit impact, retest and provenance.

Forward chain and exact carrier

The generated carrier is `design-version-baseline-change-carrier`. Its required relation is `proposal-to-approved-version-relation`. The reconstruction must retain `baseline-rationale-review-impact-tests-and-rollback-held`. Its declared empirical boundary is `configuration-boundary-declared`. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is `SFT-FOUNDATION-ONE-001`, `SFT-FOUNDATION-FOLD-001`, `SFT-FOUNDATION-FOLD-DYNAMICS-001`, `SFT-MATH-EXACT-ARITHMETIC-001`, `SFT-MATH-GRAPH-NETWORK-001`, `SFT-MATH-OPTIMIZATION-001`, `SFT-MATH-DYNAMICAL-SYSTEMS-001`, `SFT-MATH-LOGIC-PROOF-001`, `SFT-INFO-QUANTITY-001`, `SFT-INFO-CHANNEL-CAPACITY-001`, `SFT-COMP-FORM-STATE-TRANSITION-001`, `SFT-COMP-SEM-CORRECTNESS-001`, `SFT-COMP-SEM-COMPILATION-001`, `SFT-COMP-DIST-CAUSALITY-001`, `SFT-PHYS-MEAS-VALUE-RECORD-001`, `SFT-PHYS-MEAS-CALIBRATION-001`, `SFT-CHEM-MEAS-TRACEABILITY-001`, `SFT-MAT-SUST-LIFECYCLE-001`, `SFT-BIO-BIO-HANDOFF-001`, `SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001`, `SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001`, `SFT-EARTH-HAZARD-RISK-HANDOFF-001`, `SFT-ASTRO-ASTRO-HANDOFF-001`, `SFT-SOCIAL-ENGINEERING-HANDOFF-001`, `SFT-ENG-MEASUREMENT-UNCERTAINTY-001`, `SFT-ENG-ROBUST-DESIGN-001`. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external standard, application, performance target, product or prior implementation to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external standards, products, tests or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite components, versions, platforms, tests and lifecycle records; performance remains artifact, environment, user, method and operating-boundary limited.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - which requirement, artifact, component, user, environment or test bears the claim: `design-version-baseline-change-carrier`.
- **boundary** - which operating envelope, version, platform, user and lifecycle stage bound the result: `configuration-boundary-declared`.
- **relation** - which generated transformation, interface, comparison or handoff connects the carriers: `proposal-to-approved-version-relation`.
- **record** - which requirements, versions, alternatives, failures, uncertainty and rollback remain reconstructible: `baseline-rationale-review-impact-tests-and-rollback-held`.
- **evidence** - whether law, requirement, design, simulation, test, demonstration, performance and anomaly remain distinct: `science-design-test-and-decision-classes-explicit`.
- **provenance** - whether the result traces to the One and names every upstream receipt and design choice: `root-bound-forward-translation`.
- **generality** - whether finite extension preserves platforms, versions, failures and operating limits: `positive-finite-extension-preserves-platforms-failures-and-versions`.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: `no-extra-rule`.

Their ordered product gives the unique survivor:

```
design-version-baseline-change-carrier__configuration-boundary-declared__proposal-to-
approved-version-relation__baseline-rationale-review-impact-tests-and-rollback-held__
science-design-test-and-decision-classes-explicit__root-bound-forward-translation__po
sitive-finite-extension-preserves-platforms-failures-and-versions__no-extra-rule
```

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses the requirement, artifact, user or environment. Erasing the boundary universalises a version, platform or test. Importing the relation lets a target or preferred application choose the result. Retaining only successful output removes requirements, alternatives, failures and rollback. Conflating evidence classes turns simulation, demonstration or performance into observation or law. Breaking provenance removes the trace to the One and upstream receipts. Erasing finite context makes one implementation universal. Adding a silent dependency or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite requirement or admitted law, versioned artifact, boundary, testable relation and complete record. The successor adds one lawful component, interface, platform, requirement, test, failure, lifecycle stage or anomaly while preserving all earlier identities, adverse rows and receipts.

Falsification and deliberately unfavourable controls

The registered falsification condition is: Reject silent changes, retroactive baselines or unchanged receipts after source modification.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: `source_bound_engineering_requirement_design_test_lifecycle_or_access_record/primary_standard_handbook_guidance_or_project_record_structural_correspondence`. Captured registered sources: 3. Unresolved transports retained: 0. The registered structural consequence corresponded: `True`. External evidence selected the survivor: `false`. Application or performance selected the law: `false`. Successful performance was relabelled scientific proof: `false`. Failed implementation was relabelled scientific falsification: `false`. Simulation or demonstration was relabelled observation: `false`.

- `EPA-LIFECYCLE-001` - transport `captured`; captured bytes `980`; snapshot `sha256:1d271583636a679563a046e767d0575a92d23eeafbf99f0da3fef2582e102c4e`.
- `HSE-COMAH-001` - transport `captured`; captured bytes `52092`; snapshot `sha256:928bb33ae445b9eb3be9784ade179d832b3506a96bba0d3011460070de9053e9`.
- `NASA-SE-HANDBOOK-001` - transport `captured`; captured bytes `3773440`; snapshot `sha256:8eeb4887a4dc57a23049da7dd2ed556833cf98e214b240468d987873164ff688`.

The comparison is structural and source-bound. It checks that a relevant primary handbook, standard, guidance record or open engineering project can represent the declared requirement, artifact, environment, operating boundary, version, test and lifecycle without rewriting the Fold law. It does not claim that a standard proves conformity, that one handbook exhausts Engineering, or that a custodian's reputation proves the result. New records can extend or challenge the surface only with source, version, configuration, method, failures and adverse rows retained.

Scientific meaning

For **design freeze and change control**, the consequence is practical and exact: A design freeze is a versioned baseline whose later changes require explicit impact, retest and provenance. An Engineering claim therefore has to expose the carrier `design-version-baseline-change-carrier`, the relation `proposal-to-approved-version-relation`, the record `baseline-rationale-review-impact-tests-and-rollback-held`, and the boundary `configuration-boundary-declared` together. If any one is hidden, the result may remain a concept, prototype, simulation, demonstration or product choice, but it is not this admitted law.

This distinction protects users, maintainers and science. It prevents a successful output from erasing requirements, failed alternatives, hidden resources, inaccessible users, hazards, maintenance burdens or end-of-life costs. It prevents an application from selecting the upstream law and prevents engineering failure from becoming automatic scientific falsification. A replacement must still generate its forms, eliminate alternatives, survive controls, preserve custody and reproduce through the shared engine.

Machine certificate

Candidate count: `256`; survivor count: `1`; derivation seal: `sha256:03ba94f222dc4c5e9eaa33ed9cd2a4cd3d4f96dfeda6863fa4539892ceebab06`; independent certificate: `sha256:9a8b16096ee20fe445b18ea503c4fc9dc619146930bb9166730b148191362b75`; engine receipt: `sha256:8a48d8e7705c051d36098baa026a8d7aa3fee6522b9ac2f1a0399f577127006d`. The independent implementation recomputed the candidate product: `True`. All external rows were preserved: `True`.

31. Control action

Claim: `SFT-ENG-CONTROL-001`

Family: `control_feedback_stability`

Exact statement: Control maps an observed state to a bounded action with target, latency, limits and effects retained.

Forward chain and exact carrier

The generated carrier is `system-state-target-actuator-carrier`. Its required relation is `observed-state-to-bounded-action-relation`. The reconstruction must retain `target-state-measurement-action-limit-time-and-effect-held`. Its declared empirical boundary is `control-envelope-bounded`. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is `SFT-FOUNDATION-ONE-001`, `SFT-FOUNDATION-FOLD-001`, `SFT-FOUNDATION-FOLD-DYNAMICS-001`, `SFT-MATH-EXACT-ARITHMETIC-001`, `SFT-MATH-GRAPH-NETWORK-001`, `SFT-MATH-OPTIMIZATION-001`, `SFT-MATH-DYNAMICAL-SYSTEMS-001`, `SFT-MATH-LOGIC-PROOF-001`, `SFT-INFO-QUANTITY-001`, `SFT-INFO-CHANNEL-CAPACITY-001`, `SFT-COMP-FORM-STATE-TRANSITION-001`, `SFT-COMP-SEM-CORRECTNESS-001`, `SFT-COMP-SEM-COMPILATION-001`, `SFT-COMP-DIST-CAUSALITY-001`, `SFT-PHYS-MEAS-VALUE-RECORD-001`, `SFT-PHYS-MEAS-CALIBRATION-001`, `SFT-CHEM-MEAS-TRACEABILITY-001`, `SFT-MAT-SUST-LIFECYCLE-001`, `SFT-BIO-BIO-HANDOFF-001`, `SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001`, `SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001`, `SFT-EARTH-HAZARD-RISK-HANDOFF-001`, `SFT-ASTRO-ASTRO-HANDOFF-001`, `SFT-SOCIAL-ENGINEERING-HANDOFF-001`, `SFT-ENG-DESIGN-FREEZE-001`. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external standard, application, performance target, product or prior implementation to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external standards, products, tests or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite components, versions, platforms, tests and lifecycle records; performance remains artifact, environment, user, method and operating-boundary limited.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - which requirement, artifact, component, user, environment or test bears the claim: `system-state-target-actuator-carrier`.
- **boundary** - which operating envelope, version, platform, user and lifecycle stage bound the result: `control-envelope-bounded`.
- **relation** - which generated transformation, interface, comparison or handoff connects the carriers: `observed-state-to-bounded-action-relation`.

- **record** - which requirements, versions, alternatives, failures, uncertainty and rollback remain reconstructible: `target-state-measurement-action-limit-time-and-effect-held`.
- **evidence** - whether law, requirement, design, simulation, test, demonstration, performance and anomaly remain distinct: `science-design-test-and-decision-classes-explicit`.
- **provenance** - whether the result traces to the One and names every upstream receipt and design choice: `root-bound-forward-translation`.
- **generality** - whether finite extension preserves platforms, versions, failures and operating limits: `positive-finite-extension-preserves-platforms-failures-and-versions`.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: `no-extra-rule`.

Their ordered product gives the unique survivor:

```
system-state-target-actuator-carrier__control-envelope-bounded__observed-state-to-bou
nded-action-relation__target-state-measurement-action-limit-time-and-effect-held__sci
ence-design-test-and-decision-classes-explicit__root-bound-forward-translation__posit
ive-finite-extension-preserves-platforms-failures-and-versions__no-extra-rule
```

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses the requirement, artifact, user or environment. Erasing the boundary universalises a version, platform or test. Importing the relation lets a target or preferred application choose the result. Retaining only successful output removes requirements, alternatives, failures and rollback. Conflating evidence classes turns simulation, demonstration or performance into observation or law. Breaking provenance removes the trace to the One and upstream receipts. Erasing finite context makes one implementation universal. Adding a silent dependency or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite requirement or admitted law, versioned artifact, boundary, testable relation and complete record. The successor adds one lawful component, interface, platform, requirement, test, failure, lifecycle stage or anomaly while preserving all earlier identities, adverse rows and receipts.

Falsification and deliberately unfavourable controls

The registered falsification condition is: Reject if an unobserved state, hidden controller or unlimited action is assumed.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: `source_bound_engineering_requirement_design_test_lifecycle_or_access_record/primary_standard_handbook_guidance_or_project_record_structural_correspondence`. Captured registered sources: 3. Unresolved transports retained: 0. The registered structural consequence corresponded: `True`. External evidence selected the survivor: `false`. Application or performance selected the law: `false`. Successful performance was relabelled scientific proof: `false`. Failed implementation was relabelled scientific falsification: `false`. Simulation or demonstration was relabelled observation: `false`.

- `NASA-SE-HANDBOOK-001` - transport `captured`; captured bytes `3773440`; snapshot `sha256:8eeb4887a4dc57a23049da7dd2ed556833cf98e214b240468d987873164ff688`.
- `NIST-SSE-001` - transport `captured`; captured bytes `47984`; snapshot `sha256:64c8d41d9a78268d596a0a60aad18eead50fd375e8628486f7202788a207edaa`.

- NASA-SE-APPENDIX-001 - transport captured; captured bytes 578114; snapshot sha256:c1db2db970b9fd14a7ef98aa0c23575326dedae46cc7cc56c468821ca89edd60.

The comparison is structural and source-bound. It checks that a relevant primary handbook, standard, guidance record or open engineering project can represent the declared requirement, artifact, environment, operating boundary, version, test and lifecycle without rewriting the Fold law. It does not claim that a standard proves conformity, that one handbook exhausts Engineering, or that a custodian's reputation proves the result. New records can extend or challenge the surface only with source, version, configuration, method, failures and adverse rows retained.

Scientific meaning

For **control action**, the consequence is practical and exact: Control maps an observed state to a bounded action with target, latency, limits and effects retained. An Engineering claim therefore has to expose the carrier `system-state-target-actuator-carrier`, the relation `observed-state-to-bounded-action-relation`, the record `target-state-measurement-action-limit-time-and-effect-held`, and the boundary `control-envelope-bounded` together. If any one is hidden, the result may remain a concept, prototype, simulation, demonstration or product choice, but it is not this admitted law.

This distinction protects users, maintainers and science. It prevents a successful output from erasing requirements, failed alternatives, hidden resources, inaccessible users, hazards, maintenance burdens or end-of-life costs. It prevents an application from selecting the upstream law and prevents engineering failure from becoming automatic scientific falsification. A replacement must still generate its forms, eliminate alternatives, survive controls, preserve custody and reproduce through the shared engine.

Machine certificate

Candidate count: 256; survivor count: 1; derivation seal: sha256:c366ee27597a721b9790b0ae9c3b4df29d7ced59e52e01d8a99db893c934e61c; independent certificate: sha256:5886cac0ccdd1653fef339362887f981ae87d2df20dc92f084ebe77849564cf6; engine receipt: sha256:d4509325413d99106b345719a0e9e1274199be38329a4de49297b3b483c914e7. The independent implementation recomputed the candidate product: True. All external rows were preserved: True.

32. Feedback loop

Claim: SFT-ENG-FEEDBACK-001

Family: control_feedback_stability

Exact statement: A feedback loop retains plant, sensor, controller, actuator, delay, bounds, noise and failure paths.

Forward chain and exact carrier

The generated carrier is `plant-sensor-controller-actuator-carrier`. Its required relation is `output-record-to-next-action-relation`. The reconstruction must retain `components-delays-gains-bounds-noise-and-failures-held`. Its declared empirical boundary is `loop-configuration-bounded`. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-OPTIMIZATION-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-CHANNEL-CAPACITY-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-COMP-SEM-CORRECTNESS-001, SFT-COMP-SEM-COMPILATION-001, SFT-COMP-DIST-CAUSALITY-001, SFT-PHYS-MEAS-VALUE-RECORD-001, SFT-PHYS-MEAS-CALIBRATION-001, SFT-CHEM-MEAS-TRACEABILITY-001, SFT-MAT-SUST-LIFECYCLE-001, SFT-BIO-BIO-HANDOFF-001, SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001, SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-ASTRO-HANDOFF-001, SFT-SOCIAL-ENGINEERING-HANDOFF-001, SFT-ENG-DESIGN-FREEZE-001, SFT-ENG-CONTROL-001. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external standard, application, performance target, product or prior implementation to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external standards, products, tests or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite components, versions, platforms, tests and lifecycle records; performance remains artifact, environment, user, method and operating-boundary limited.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - which requirement, artifact, component, user, environment or test bears the claim: `plant-sensor-controller-actuator-carrier`.
- **boundary** - which operating envelope, version, platform, user and lifecycle stage bound the result: `loop-configuration-bounded`.
- **relation** - which generated transformation, interface, comparison or handoff connects the carriers: `output-record-to-next-action-relation`.
- **record** - which requirements, versions, alternatives, failures, uncertainty and rollback remain reconstructible: `components-delays-gains-bounds-noise-and-failures-held`.
- **evidence** - whether law, requirement, design, simulation, test, demonstration, performance and anomaly remain distinct: `science-design-test-and-decision-classes-explicit`.
- **provenance** - whether the result traces to the One and names every upstream receipt and design choice: `root-bound-forward-translation`.
- **generality** - whether finite extension preserves platforms, versions, failures and operating limits: `positive-finite-extension-preserves-platforms-failures-and-versions`.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: `no-extra-rule`.

Their ordered product gives the unique survivor:

```
plant-sensor-controller-actuator-carrier__loop-configuration-bounded__output-record-t
o-next-action-relation__components-delays-gains-bounds-noise-and-failures-held__scien
ce-design-test-and-decision-classes-explicit__root-bound-forward-translation__positiv
e-finite-extension-preserves-platforms-failures-and-versions__no-extra-rule
```

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses the requirement, artifact, user or environment. Erasing the boundary universalises a version, platform or test. Importing the relation lets a target or preferred application choose the result. Retaining only successful output removes requirements, alternatives, failures and rollback. Conflating evidence classes turns simulation, demonstration or performance into observation or law. Breaking provenance removes the trace to the One and upstream receipts. Erasing finite context makes one implementation universal. Adding a silent dependency or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite requirement or admitted law, versioned artifact, boundary, testable relation and complete record. The successor adds one lawful component, interface, platform, requirement, test, failure, lifecycle stage or anomaly while preserving all earlier identities, adverse rows and receipts.

Falsification and deliberately unfavourable controls

The registered falsification condition is: Reject if correlation or repeated adjustment proves closed-loop feedback.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: `source_bound_engineering_requirement_design_test_lifecycle_or_access_record/primary_standard_handbook_guidance_or_project_record_structural_correspondence`. Captured registered sources: 3. Unresolved transports retained: 0. The registered structural consequence corresponded: `True`. External evidence selected the survivor: `false`. Application or performance selected the law: `false`. Successful performance was relabelled scientific proof: `false`. Failed implementation was relabelled scientific falsification: `false`. Simulation or demonstration was relabelled observation: `false`.

- `NASA-SE-HANDBOOK-001` - transport `captured`; captured bytes `3773440`; snapshot `sha256:8eeb4887a4dc57a23049da7dd2ed556833cf98e214b240468d987873164ff688`.
- `NIST-SSE-001` - transport `captured`; captured bytes `47984`; snapshot `sha256:64c8d41d9a78268d596a0a60aad18eead50fd375e8628486f7202788a207edaa`.
- `NASA-SE-APPENDIX-001` - transport `captured`; captured bytes `578114`; snapshot `sha256:c1db2db970b9fd14a7ef98aa0c23575326dedae46cc7cc56c468821ca89edd60`.

The comparison is structural and source-bound. It checks that a relevant primary handbook, standard, guidance record or open engineering project can represent the declared requirement, artifact, environment, operating boundary, version, test and lifecycle without rewriting the Fold law. It does not claim that a standard proves conformity, that one handbook exhausts Engineering, or that a custodian's reputation proves the result. New records can extend or challenge the surface only with source, version, configuration, method, failures and adverse rows retained.

Scientific meaning

For **feedback loop**, the consequence is practical and exact: A feedback loop retains plant, sensor, controller, actuator, delay, bounds, noise and failure paths. An Engineering claim therefore has to expose the carrier `plant-sensor-controller-actuator-carrier`, the relation `output-record-to-next-action-relation`, the record `components-delays-gains-bounds-noise-and-failures-held`, and the boundary `loop-configuration-bounded` together. If any one is hidden, the result may remain a concept, prototype, simulation, demonstration or product choice, but it is not this admitted law.

This distinction protects users, maintainers and science. It prevents a successful output from erasing requirements, failed alternatives, hidden resources, inaccessible users, hazards, maintenance burdens or end-of-life costs. It prevents an application from selecting the upstream law and prevents engineering failure from becoming automatic scientific falsification. A replacement must still generate its forms, eliminate alternatives, survive controls, preserve custody and reproduce through the shared engine.

Machine certificate

Candidate count: `256`; survivor count: `1`; derivation seal: `sha256:88cb021cdb8ae721a122a4aca2e2aa12addfa3ed8e03a20d81879e5df09201d0`; independent certificate: `sha256:f56f573de080b61a76f81c6ec82e4b0de4f2c962b2722bfae0f4be6855c47052`; engine receipt: `sha256:7ea37de19f49afc9b337d3e7bb3fb9ae98b109adc65faa3bf222abdff96deae3`. The independent implementation recomputed the candidate product: `True`. All external rows were preserved: `True`.

33. Operational stability

Claim: `SFT-ENG-STABILITY-001`

Family: `control_feedback_stability`

Exact statement: Stability is a finite operating-region claim retaining perturbation, return bound, overshoot and failures.

Forward chain and exact carrier

The generated carrier is `system-perturbation-bound-return-carrier`. Its required relation is `perturbed-trajectory-to-operating-region-relation`. The reconstruction must retain `initial-state-disturbance-time-bound-overshoot-and-failure-held`. Its declared empirical boundary is `stability-region-bounded`. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is `SFT-FOUNDATION-ONE-001`, `SFT-FOUNDATION-FOLD-001`, `SFT-FOUNDATION-FOLD-DYNAMICS-001`, `SFT-MATH-EXACT-ARITHMETIC-001`, `SFT-MATH-GRAPH-NETWORK-001`, `SFT-MATH-OPTIMIZATION-001`, `SFT-MATH-DYNAMICAL-SYSTEMS-001`, `SFT-MATH-LOGIC-PROOF-001`, `SFT-INFO-QUANTITY-001`, `SFT-INFO-CHANNEL-CAPACITY-001`, `SFT-COMP-FORM-STATE-TRANSITION-001`, `SFT-COMP-SEM-CORRECTNESS-001`, `SFT-COMP-SEM-COMPILATION-001`, `SFT-COMP-DIST-CAUSALITY-001`, `SFT-PHYS-MEAS-VALUE-RECORD-001`, `SFT-PHYS-MEAS-CALIBRATION-001`, `SFT-CHEM-MEAS-TRACEABILITY-001`, `SFT-MAT-SUST-LIFECYCLE-001`, `SFT-BIO-BIO-HANDOFF-001`, `SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001`, `SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001`, `SFT-EARTH-HAZARD-RISK-HANDOFF-001`, `SFT-ASTRO-ASTRO-HANDOFF-001`, `SFT-SOCIAL-ENGINEERING-HANDOFF-001`, `SFT-ENG-DESIGN-FREEZE-001`, `SFT-ENG-FEEDBACK-001`. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external standard, application, performance target, product or prior implementation to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external standards, products, tests or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite components, versions, platforms, tests and lifecycle records; performance remains artifact, environment, user, method and operating-boundary limited.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - which requirement, artifact, component, user, environment or test bears the claim: `system-perturbation-bound-return-carrier`.
- **boundary** - which operating envelope, version, platform, user and lifecycle stage bound the result: `stability-region-bounded`.
- **relation** - which generated transformation, interface, comparison or handoff connects the carriers: `perturbed-trajectory-to-operating-region-relation`.
- **record** - which requirements, versions, alternatives, failures, uncertainty and rollback remain reconstructible: `initial-state-disturbance-time-bound-overshoot-and-failure-held`.
- **evidence** - whether law, requirement, design, simulation, test, demonstration, performance and anomaly remain distinct: `science-design-test-and-decision-classes-explicit`.
- **provenance** - whether the result traces to the One and names every upstream receipt and design choice: `root-bound-forward-translation`.
- **generality** - whether finite extension preserves platforms, versions, failures and operating limits: `positive-finite-extension-preserves-platforms-failures-and-versions`.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: `no-extra-rule`.

Their ordered product gives the unique survivor:

```
system-perturbation-bound-return-carrier__stability-region-bounded__perturbed-traject
ory-to-operating-region-relation__initial-state-disturbance-time-bound-overshoot-and-
failure-held__science-design-test-and-decision-classes-explicit__root-bound-forward-t
ranslation__positive-finite-extension-preserves-platforms-failures-and-versions__no-e
xtra-rule
```

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses the requirement, artifact, user or environment. Erasing the boundary universalises a version, platform or test. Importing the relation lets a target or preferred application choose the result. Retaining only successful output removes requirements, alternatives, failures and rollback. Conflating evidence classes turns simulation, demonstration or performance into observation or law. Breaking provenance removes the trace to the One and upstream receipts. Erasing finite context makes one implementation universal. Adding a silent dependency or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite requirement or admitted law, versioned artifact, boundary, testable relation and complete record. The successor adds one lawful component, interface, platform, requirement, test, failure, lifecycle stage or anomaly while preserving all earlier identities, adverse rows and receipts.

Falsification and deliberately unfavourable controls

The registered falsification condition is: Reject if one settled trace proves global or infinite-time stability.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: `source_bound_engineering_requirement_design_test_lifecycle_or_access_record/primary_standard_handbook_guidance_or_project_record_structural_correspondence`. Captured registered sources: 3. Unresolved transports retained: 0. The registered structural consequence corresponded: `True`. External evidence selected the survivor: `false`. Application or performance selected the law: `false`. Successful performance was relabelled scientific proof: `false`. Failed implementation was relabelled scientific falsification: `false`. Simulation or demonstration was relabelled observation: `false`.

- `NASA-SE-HANDBOOK-001` - transport `captured`; captured bytes `3773440`; snapshot `sha256:8eeb4887a4dc57a23049da7dd2ed556833cf98e214b240468d987873164ff688`.
- `NIST-SSE-001` - transport `captured`; captured bytes `47984`; snapshot `sha256:64c8d41d9a78268d596a0a60aad18eead50fd375e8628486f7202788a207edaa`.
- `NASA-SE-APPENDIX-001` - transport `captured`; captured bytes `578114`; snapshot `sha256:c1db2db970b9fd14a7ef98aa0c23575326dedae46cc7cc56c468821ca89edd60`.

The comparison is structural and source-bound. It checks that a relevant primary handbook, standard, guidance record or open engineering project can represent the declared requirement, artifact, environment, operating boundary, version, test and lifecycle without rewriting the Fold law. It does not claim that a standard proves conformity, that one handbook exhausts Engineering, or that a custodian's reputation proves the result. New records can extend or challenge the surface only with source, version, configuration, method, failures and adverse rows retained.

Scientific meaning

For **operational stability**, the consequence is practical and exact: Stability is a finite operating-region claim retaining perturbation, return bound, overshoot and failures. An Engineering claim therefore has to expose the carrier `system-perturbation-bound-return-carrier`, the relation `perturbed-trajectory-to-operating-region-relation`, the record `initial-state-disturbance-time-bound-overshoot-and-failure-held`, and the boundary `stability-region-bounded` together. If any one is hidden, the result may remain a concept, prototype, simulation, demonstration or product choice, but it is not this admitted law.

This distinction protects users, maintainers and science. It prevents a successful output from erasing requirements, failed alternatives, hidden resources, inaccessible users, hazards, maintenance burdens or end-of-life costs. It prevents an application from selecting the upstream law and prevents engineering failure from becoming automatic scientific falsification. A replacement must still generate its forms, eliminate alternatives, survive controls, preserve custody and reproduce through the shared engine.

Machine certificate

Candidate count: `256`; survivor count: `1`; derivation seal: `sha256:08f9cc443e6952a37f7cf4d11bee6f885b07c56e179eddb1eeede7e6557045a2`; independent certificate: `sha256:97fe2e6be74179c186ca446f57d1830dccd8290d4b169edbe231e6861b5f4439`; engine receipt: `sha256:e8358a5b9b7cfeb32c863a943ecdc1eb31836146432fc2d1fa18a4388b749aa7`. The independent implementation recomputed the candidate product: `True`. All external rows were preserved: `True`.

34. Engineering observability

Claim: `SFT-ENG-OBSERVABILITY-001`

Family: `control_feedback_stability`

Exact statement: Observability requires declared output histories to distinguish the registered internal state set.

Forward chain and exact carrier

The generated carrier is `internal-state-output-history-carrier`. Its required relation is `state-distinction-to-record-relation`. The reconstruction must retain `state-set-sensors-history-resolution-and-ambiguity-held`. Its declared empirical boundary is `observation-depth-bounded`. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is `SFT-FOUNDATION-ONE-001`, `SFT-FOUNDATION-FOLD-001`, `SFT-FOUNDATION-FOLD-DYNAMICS-001`, `SFT-MATH-EXACT-ARITHMETIC-001`, `SFT-MATH-GRAPH-NETWORK-001`, `SFT-MATH-OPTIMIZATION-001`, `SFT-MATH-DYNAMICAL-SYSTEMS-001`, `SFT-MATH-LOGIC-PROOF-001`, `SFT-INFO-QUANTITY-001`, `SFT-INFO-CHANNEL-CAPACITY-001`, `SFT-COMP-FORM-STATE-TRANSITION-001`, `SFT-COMP-SEM-CORRECTNESS-001`, `SFT-COMP-SEM-COMPILATION-001`, `SFT-COMP-DIST-CAUSALITY-001`, `SFT-PHYS-MEAS-VALUE-RECORD-001`, `SFT-PHYS-MEAS-CALIBRATION-001`, `SFT-CHEM-MEAS-TRACEABILITY-001`, `SFT-MAT-SUST-LIFECYCLE-001`, `SFT-BIO-BIO-HANDOFF-001`, `SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001`, `SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001`, `SFT-EARTH-HAZARD-RISK-HANDOFF-001`, `SFT-ASTRO-ASTRO-HANDOFF-001`, `SFT-SOCIAL-ENGINEERING-HANDOFF-001`, `SFT-ENG-DESIGN-FREEZE-001`, `SFT-ENG-STABILITY-001`. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external standard, application, performance target, product or prior implementation to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external standards, products, tests or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite components, versions, platforms, tests and lifecycle records; performance remains artifact, environment, user, method and operating-boundary limited.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - which requirement, artifact, component, user, environment or test bears the claim: `internal-state-output-history-carrier`.
- **boundary** - which operating envelope, version, platform, user and lifecycle stage bound the result: `observation-depth-bounded`.
- **relation** - which generated transformation, interface, comparison or handoff connects the carriers: `state-distinction-to-record-relation`.

- **record** - which requirements, versions, alternatives, failures, uncertainty and rollback remain reconstructible: `state-set-sensors-history-resolution-and-ambiguity-held`.
- **evidence** - whether law, requirement, design, simulation, test, demonstration, performance and anomaly remain distinct: `science-design-test-and-decision-classes-explicit`.
- **provenance** - whether the result traces to the One and names every upstream receipt and design choice: `root-bound-forward-translation`.
- **generality** - whether finite extension preserves platforms, versions, failures and operating limits: `positive-finite-extension-preserves-platforms-failures-and-versions`.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: `no-extra-rule`.

Their ordered product gives the unique survivor:

```
internal-state-output-history-carrier__observation-depth-bounded__state-distinction-t
o-record-relation__state-set-sensors-history-resolution-and-ambiguity-held__science-d
esign-test-and-decision-classes-explicit__root-bound-forward-translation__positive-fi
nite-extension-preserves-platforms-failures-and-versions__no-extra-rule
```

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses the requirement, artifact, user or environment. Erasing the boundary universalises a version, platform or test. Importing the relation lets a target or preferred application choose the result. Retaining only successful output removes requirements, alternatives, failures and rollback. Conflating evidence classes turns simulation, demonstration or performance into observation or law. Breaking provenance removes the trace to the One and upstream receipts. Erasing finite context makes one implementation universal. Adding a silent dependency or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite requirement or admitted law, versioned artifact, boundary, testable relation and complete record. The successor adds one lawful component, interface, platform, requirement, test, failure, lifecycle stage or anomaly while preserving all earlier identities, adverse rows and receipts.

Falsification and deliberately unfavourable controls

The registered falsification condition is: Reject if an unheld predecessor is claimed recoverable from a merged output.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: `source_bound_engineering_requirement_design_test_lifecycle_or_access_record/primary_standard_handbook_guidance_or_project_record_structural_correspondence`. Captured registered sources: 3. Unresolved transports retained: 0. The registered structural consequence corresponded: `True`. External evidence selected the survivor: `false`. Application or performance selected the law: `false`. Successful performance was relabelled scientific proof: `false`. Failed implementation was relabelled scientific falsification: `false`. Simulation or demonstration was relabelled observation: `false`.

- `NASA-SE-HANDBOOK-001` - transport `captured`; captured bytes `3773440`; snapshot `sha256:8eeb4887a4dc57a23049da7dd2ed556833cf98e214b240468d987873164ff688`.
- `NIST-SSE-001` - transport `captured`; captured bytes `47984`; snapshot `sha256:64c8d41d9a78268d596a0a60aad18eead50fd375e8628486f7202788a207edaa`.

- NASA-SE-APPENDIX-001 - transport captured; captured bytes 578114; snapshot sha256:c1db2db970b9fd14a7ef98aa0c23575326dedae46cc7cc56c468821ca89edd60.

The comparison is structural and source-bound. It checks that a relevant primary handbook, standard, guidance record or open engineering project can represent the declared requirement, artifact, environment, operating boundary, version, test and lifecycle without rewriting the Fold law. It does not claim that a standard proves conformity, that one handbook exhausts Engineering, or that a custodian's reputation proves the result. New records can extend or challenge the surface only with source, version, configuration, method, failures and adverse rows retained.

Scientific meaning

For **engineering observability**, the consequence is practical and exact: Observability requires declared output histories to distinguish the registered internal state set. An Engineering claim therefore has to expose the carrier internal-state-output-history-carrier, the relation state-distinction-to-record-relation, the record state-set-sensors-history-resolution-and-ambiguity-held, and the boundary observation-depth-bounded together. If any one is hidden, the result may remain a concept, prototype, simulation, demonstration or product choice, but it is not this admitted law.

This distinction protects users, maintainers and science. It prevents a successful output from erasing requirements, failed alternatives, hidden resources, inaccessible users, hazards, maintenance burdens or end-of-life costs. It prevents an application from selecting the upstream law and prevents engineering failure from becoming automatic scientific falsification. A replacement must still generate its forms, eliminate alternatives, survive controls, preserve custody and reproduce through the shared engine.

Machine certificate

Candidate count: 256; survivor count: 1; derivation seal: sha256:ca79a23f7da3207c4e805923b17f3d18f0f624dfef13136c935b4c06388ba7ad; independent certificate: sha256:5d82e85f9ff5e4ab290f49c38b8bcb0d7db9ee2d2fb76e0116b896596861aa3e; engine receipt: sha256:aac08906b9ea57a0e48ca7f1da0ea10f952914cfa6737f39d35a11e5cf552eba. The independent implementation recomputed the candidate product: True. All external rows were preserved: True.

35. Engineering controllability

Claim: SFT-ENG-CONTROLLABILITY-001

Family: control_feedback_stability

Exact statement: Controllability is reachability under declared actions, limits and duration with failed paths retained.

Forward chain and exact carrier

The generated carrier is initial-target-action-sequence-carrier. Its required relation is bounded-input-to-state-transition-relation. The reconstruction must retain states-actions-limits-duration-path-and-failures-held. Its declared empirical boundary is reachable-set-bounded. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-OPTIMIZATION-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-CHANNEL-CAPACITY-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-COMP-SEM-CORRECTNESS-001, SFT-COMP-SEM-COMPILATION-001, SFT-COMP-DIST-CAUSALITY-001, SFT-PHYS-MEAS-VALUE-RECORD-001, SFT-PHYS-MEAS-CALIBRATION-001, SFT-CHEM-MEAS-TRACEABILITY-001, SFT-MAT-SUST-LIFECYCLE-001, SFT-BIO-BIO-HANDOFF-001, SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001, SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-ASTRO-HANDOFF-001, SFT-SOCIAL-ENGINEERING-HANDOFF-001, SFT-ENG-DESIGN-FREEZE-001, SFT-ENG-OBSERVABILITY-001. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external standard, application, performance target, product or prior implementation to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external standards, products, tests or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite components, versions, platforms, tests and lifecycle records; performance remains artifact, environment, user, method and operating-boundary limited.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - which requirement, artifact, component, user, environment or test bears the claim: `initial-target-action-sequence-carrier`.
- **boundary** - which operating envelope, version, platform, user and lifecycle stage bound the result: `reachable-set-bounded`.
- **relation** - which generated transformation, interface, comparison or handoff connects the carriers: `bounded-input-to-state-transition-relation`.
- **record** - which requirements, versions, alternatives, failures, uncertainty and rollback remain reconstructible: `states-actions-limits-duration-path-and-failures-held`.
- **evidence** - whether law, requirement, design, simulation, test, demonstration, performance and anomaly remain distinct: `science-design-test-and-decision-classes-explicit`.
- **provenance** - whether the result traces to the One and names every upstream receipt and design choice: `root-bound-forward-translation`.
- **generality** - whether finite extension preserves platforms, versions, failures and operating limits: `positive-finite-extension-preserves-platforms-failures-and-versions`.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: `no-extra-rule`.

Their ordered product gives the unique survivor:

```
initial-target-action-sequence-carrier__reachable-set-bounded__bounded-input-to-state
-transition-relation__states-actions-limits-duration-path-and-failures-held__science-
design-test-and-decision-classes-explicit__root-bound-forward-translation__positive-f
inite-extension-preserves-platforms-failures-and-versions__no-extra-rule
```

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses the requirement, artifact, user or environment. Erasing the boundary universalises a version, platform or test. Importing the relation lets a target or preferred application choose the result. Retaining only successful output removes requirements, alternatives, failures and rollback. Conflating evidence classes turns simulation, demonstration or performance into observation or law. Breaking provenance removes the trace to the One and upstream receipts. Erasing finite context makes one implementation universal. Adding a silent dependency or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite requirement or admitted law, versioned artifact, boundary, testable relation and complete record. The successor adds one lawful component, interface, platform, requirement, test, failure, lifecycle stage or anomaly while preserving all earlier identities, adverse rows and receipts.

Falsification and deliberately unfavourable controls

The registered falsification condition is: Reject if an unreachable state is assumed or an unlimited actuator proves reachability.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: `source_bound_engineering_requirement_design_test_lifecycle_or_access_record/primary_standard_handbook_guidance_or_project_record_structural_correspondence`. Captured registered sources: 3. Unresolved transports retained: 0. The registered structural consequence corresponded: `True`. External evidence selected the survivor: `false`. Application or performance selected the law: `false`. Successful performance was relabelled scientific proof: `false`. Failed implementation was relabelled scientific falsification: `false`. Simulation or demonstration was relabelled observation: `false`.

- `NASA-SE-HANDBOOK-001` - transport `captured`; captured bytes `3773440`; snapshot `sha256:8eeb4887a4dc57a23049da7dd2ed556833cf98e214b240468d987873164ff688`.
- `NIST-SSE-001` - transport `captured`; captured bytes `47984`; snapshot `sha256:64c8d41d9a78268d596a0a60aad18eead50fd375e8628486f7202788a207edaa`.
- `NASA-SE-APPENDIX-001` - transport `captured`; captured bytes `578114`; snapshot `sha256:c1db2db970b9fd14a7ef98aa0c23575326dedae46cc7cc56c468821ca89edd60`.

The comparison is structural and source-bound. It checks that a relevant primary handbook, standard, guidance record or open engineering project can represent the declared requirement, artifact, environment, operating boundary, version, test and lifecycle without rewriting the Fold law. It does not claim that a standard proves conformity, that one handbook exhausts Engineering, or that a custodian's reputation proves the result. New records can extend or challenge the surface only with source, version, configuration, method, failures and adverse rows retained.

Scientific meaning

For **engineering controllability**, the consequence is practical and exact: Controllability is reachability under declared actions, limits and duration with failed paths retained. An Engineering claim therefore has to expose the carrier `initial-target-action-sequence-carrier`, the relation `bounded-input-to-state-transition-relation`, the record `states-actions-limits-duration-path-and-failures-held`, and the boundary `reachable-set-bounded` together. If any one is hidden, the result may remain a concept, prototype, simulation, demonstration or product choice, but it is not this admitted law.

This distinction protects users, maintainers and science. It prevents a successful output from erasing requirements, failed alternatives, hidden resources, inaccessible users, hazards, maintenance burdens or end-of-life costs. It prevents an application from selecting the upstream law and prevents engineering failure from becoming automatic scientific falsification. A replacement must still generate its forms, eliminate alternatives, survive controls, preserve custody and reproduce through the shared engine.

Machine certificate

Candidate count: `256`; survivor count: `1`; derivation seal: `sha256:fcede7a41b161cf74535677f72affe49c5c4d8fedea505e28b87164fd0960ac6`; independent certificate: `sha256:76e157ad7e729241fd15f61eaf7b6f2fa44fe289b96619c50ce1798785c803db`; engine receipt: `sha256:71a66eb5016f1a8d2e40ad11705b561255a2c1b9816025c64093e7eee9a5fea8`. The independent implementation recomputed the candidate product: `True`. All external rows were preserved: `True`.

36. Fault detection and response

Claim: `SFT-ENG-FAULT-RESPONSE-001`

Family: `control_feedback_stability`

Exact statement: Fault response retains detection, misses, latency, bounded action, residual state and recovery.

Forward chain and exact carrier

The generated carrier is `system-fault-sensor-response-carrier`. Its required relation is `fault-evidence-to-bounded-response-relation`. The reconstruction must retain `fault-detection-latency-action-state-recovery-and-misses-held`. Its declared empirical boundary is `fault-family-bounded`. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is `SFT-FOUNDATION-ONE-001`, `SFT-FOUNDATION-FOLD-001`, `SFT-FOUNDATION-FOLD-DYNAMICS-001`, `SFT-MATH-EXACT-ARITHMETIC-001`, `SFT-MATH-GRAPH-NETWORK-001`, `SFT-MATH-OPTIMIZATION-001`, `SFT-MATH-DYNAMICAL-SYSTEMS-001`, `SFT-MATH-LOGIC-PROOF-001`, `SFT-INFO-QUANTITY-001`, `SFT-INFO-CHANNEL-CAPACITY-001`, `SFT-COMP-FORM-STATE-TRANSITION-001`, `SFT-COMP-SEM-CORRECTNESS-001`, `SFT-COMP-SEM-COMPILATION-001`, `SFT-COMP-DIST-CAUSALITY-001`, `SFT-PHYS-MEAS-VALUE-RECORD-001`, `SFT-PHYS-MEAS-CALIBRATION-001`, `SFT-CHEM-MEAS-TRACEABILITY-001`, `SFT-MAT-SUST-LIFECYCLE-001`, `SFT-BIO-BIO-HANDOFF-001`, `SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001`, `SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001`, `SFT-EARTH-HAZARD-RISK-HANDOFF-001`, `SFT-ASTRO-ASTRO-HANDOFF-001`, `SFT-SOCIAL-ENGINEERING-HANDOFF-001`, `SFT-ENG-DESIGN-FREEZE-001`, `SFT-ENG-CONTROLLABILITY-001`. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external standard, application, performance target, product or prior implementation to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external standards, products, tests or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite components, versions, platforms, tests and lifecycle records; performance remains artifact, environment, user, method and operating-boundary limited.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - which requirement, artifact, component, user, environment or test bears the claim: `system-fault-sensor-response-carrier`.
- **boundary** - which operating envelope, version, platform, user and lifecycle stage bound the result: `fault-family-bounded`.
- **relation** - which generated transformation, interface, comparison or handoff connects the carriers: `fault-evidence-to-bounded-response-relation`.
- **record** - which requirements, versions, alternatives, failures, uncertainty and rollback remain reconstructible: `fault-detection-latency-action-state-recovery-and-misses-held`.
- **evidence** - whether law, requirement, design, simulation, test, demonstration, performance and anomaly remain distinct: `science-design-test-and-decision-classes-explicit`.
- **provenance** - whether the result traces to the One and names every upstream receipt and design choice: `root-bound-forward-translation`.
- **generality** - whether finite extension preserves platforms, versions, failures and operating limits: `positive-finite-extension-preserves-platforms-failures-and-versions`.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: `no-extra-rule`.

Their ordered product gives the unique survivor:

```
system-fault-sensor-response-carrier__fault-family-bounded__fault-evidence-to-bounded
-response-relation__fault-detection-latency-action-state-recovery-and-misses-held__sc
ience-design-test-and-decision-classes-explicit__root-bound-forward-translation__posi
tive-finite-extension-preserves-platforms-failures-and-versions__no-extra-rule
```

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses the requirement, artifact, user or environment. Erasing the boundary universalises a version, platform or test. Importing the relation lets a target or preferred application choose the result. Retaining only successful output removes requirements, alternatives, failures and rollback. Conflating evidence classes turns simulation, demonstration or performance into observation or law. Breaking provenance removes the trace to the One and upstream receipts. Erasing finite context makes one implementation universal. Adding a silent dependency or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite requirement or admitted law, versioned artifact, boundary, testable relation and complete record. The successor adds one lawful component, interface, platform, requirement, test, failure, lifecycle stage or anomaly while preserving all earlier identities, adverse rows and receipts.

Falsification and deliberately unfavourable controls

The registered falsification condition is: Reject if no alarm proves no fault or response success erases missed detections.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: `source_bound_engineering_requirement_design_test_lifecycle_or_access_record/primary_standard_handbook_guidance_or_project_record_structural_correspondence`. Captured registered sources: 3. Unresolved transports retained: 0. The registered structural consequence corresponded: `True`. External evidence selected the survivor: `false`. Application or performance selected the law: `false`. Successful performance was relabelled scientific proof: `false`. Failed implementation was relabelled scientific falsification: `false`. Simulation or demonstration was relabelled observation: `false`.

- `NASA-SE-HANDBOOK-001` - transport `captured`; captured bytes `3773440`; snapshot `sha256:8eeb4887a4dc57a23049da7dd2ed556833cf98e214b240468d987873164ff688`.
- `NIST-SSE-001` - transport `captured`; captured bytes `47984`; snapshot `sha256:64c8d41d9a78268d596a0a60aad18eead50fd375e8628486f7202788a207edaa`.
- `NASA-SE-APPENDIX-001` - transport `captured`; captured bytes `578114`; snapshot `sha256:c1db2db970b9fd14a7ef98aa0c23575326dedae46cc7cc56c468821ca89edd60`.

The comparison is structural and source-bound. It checks that a relevant primary handbook, standard, guidance record or open engineering project can represent the declared requirement, artifact, environment, operating boundary, version, test and lifecycle without rewriting the Fold law. It does not claim that a standard proves conformity, that one handbook exhausts Engineering, or that a custodian's reputation proves the result. New records can extend or challenge the surface only with source, version, configuration, method, failures and adverse rows retained.

Scientific meaning

For **fault detection and response**, the consequence is practical and exact: Fault response retains detection, misses, latency, bounded action, residual state and recovery. An Engineering claim therefore has to expose the carrier `system-fault-sensor-response-carrier`, the relation `fault-evidence-to-bounded-response-relation`, the record `fault-detection-latency-action-state-recovery-and-misses-held`, and the boundary `fault-family-bounded` together. If any one is hidden, the result may remain a concept, prototype, simulation, demonstration or product choice, but it is not this admitted law.

This distinction protects users, maintainers and science. It prevents a successful output from erasing requirements, failed alternatives, hidden resources, inaccessible users, hazards, maintenance burdens or end-of-life costs. It prevents an application from selecting the upstream law and prevents engineering failure from becoming automatic scientific falsification. A replacement must still generate its forms, eliminate alternatives, survive controls, preserve custody and reproduce through the shared engine.

Machine certificate

Candidate count: `256`; survivor count: `1`; derivation seal: `sha256:fde4b490c31d2daf29a92fec6454accd9fb7b642e7240f10570969a17cec1005`; independent certificate: `sha256:4329781e6cb8988480269e32c63089a43ef3fe869542e7ffeff68690d6f1d9f0`; engine receipt: `sha256:e56021c6ea58b1e74425c159fbaf64a2350d1f04f65de97fbf31adc41a39322b`. The independent implementation recomputed the candidate product: `True`. All external rows were preserved: `True`.

37. Engineering hazard

Claim: `SFT-ENG-HAZARD-001`

Family: `safety_hazard_resilience`

Exact statement: A hazard is a source-to-harm path retaining exposure, severity, likelihood and uncertainty.

Forward chain and exact carrier

The generated carrier is `source-condition-exposed-carrier`. Its required relation is `hazard-to-harm-path-relation`. The reconstruction must retain `source-path-exposure-severity-likelihood-and-uncertainty-held`. Its declared empirical boundary is `hazard-scenario-bounded`. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is `SFT-FOUNDATION-ONE-001`, `SFT-FOUNDATION-FOLD-001`, `SFT-FOUNDATION-FOLD-DYNAMICS-001`, `SFT-MATH-EXACT-ARITHMETIC-001`, `SFT-MATH-GRAPH-NETWORK-001`, `SFT-MATH-OPTIMIZATION-001`, `SFT-MATH-DYNAMICAL-SYSTEMS-001`, `SFT-MATH-LOGIC-PROOF-001`, `SFT-INFO-QUANTITY-001`, `SFT-INFO-CHANNEL-CAPACITY-001`, `SFT-COMP-FORM-STATE-TRANSITION-001`, `SFT-COMP-SEM-CORRECTNESS-001`, `SFT-COMP-SEM-COMPILATION-001`, `SFT-COMP-DIST-CAUSALITY-001`, `SFT-PHYS-MEAS-VALUE-RECORD-001`, `SFT-PHYS-MEAS-CALIBRATION-001`, `SFT-CHEM-MEAS-TRACEABILITY-001`, `SFT-MAT-SUST-LIFECYCLE-001`, `SFT-BIO-BIO-HANDOFF-001`, `SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001`, `SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001`, `SFT-EARTH-HAZARD-RISK-HANDOFF-001`, `SFT-ASTRO-ASTRO-HANDOFF-001`, `SFT-SOCIAL-ENGINEERING-HANDOFF-001`, `SFT-ENG-FAULT-RESPONSE-001`. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external standard, application, performance target, product or prior implementation to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external standards, products, tests or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite components, versions, platforms, tests and lifecycle records; performance remains artifact, environment, user, method and operating-boundary limited.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - which requirement, artifact, component, user, environment or test bears the claim: `source-condition-exposed-carrier`.
- **boundary** - which operating envelope, version, platform, user and lifecycle stage bound the result: `hazard-scenario-bounded`.
- **relation** - which generated transformation, interface, comparison or handoff connects the carriers: `hazard-to-harm-path-relation`.

- **record** - which requirements, versions, alternatives, failures, uncertainty and rollback remain reconstructible: `source-path-exposure-severity-likelihood-and-uncertainty-held`.
- **evidence** - whether law, requirement, design, simulation, test, demonstration, performance and anomaly remain distinct: `science-design-test-and-decision-classes-explicit`.
- **provenance** - whether the result traces to the One and names every upstream receipt and design choice: `root-bound-forward-translation`.
- **generality** - whether finite extension preserves platforms, versions, failures and operating limits: `positive-finite-extension-preserves-platforms-failures-and-versions`.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: `no-extra-rule`.

Their ordered product gives the unique survivor:

```
source-condition-exposed-carrier__hazard-scenario-bounded__hazard-to-harm-path-relati
on__source-path-exposure-severity-likelihood-and-uncertainty-held__science-design-tes
t-and-decision-classes-explicit__root-bound-forward-translation__positive-finite-exte
nsion-preserves-platforms-failures-and-versions__no-extra-rule
```

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses the requirement, artifact, user or environment. Erasing the boundary universalises a version, platform or test. Importing the relation lets a target or preferred application choose the result. Retaining only successful output removes requirements, alternatives, failures and rollback. Conflating evidence classes turns simulation, demonstration or performance into observation or law. Breaking provenance removes the trace to the One and upstream receipts. Erasing finite context makes one implementation universal. Adding a silent dependency or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite requirement or admitted law, versioned artifact, boundary, testable relation and complete record. The successor adds one lawful component, interface, platform, requirement, test, failure, lifecycle stage or anomaly while preserving all earlier identities, adverse rows and receipts.

Falsification and deliberately unfavourable controls

The registered falsification condition is: Reject if absence of prior harm proves safety or hazard is equated with realised harm.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: `source_bound_engineering_requirement_design_test_lifecycle_or_access_record/primary_standard_handbook_guidance_or_project_record_structural_correspondence`. Captured registered sources: 3. Unresolved transports retained: 0. The registered structural consequence corresponded: `True`. External evidence selected the survivor: `false`. Application or performance selected the law: `false`. Successful performance was relabelled scientific proof: `false`. Failed implementation was relabelled scientific falsification: `false`. Simulation or demonstration was relabelled observation: `false`.

- `HSE-COMAH-001` - transport `captured`; captured bytes `52092`; snapshot `sha256:928bb33ae445b9eb3be9784ade179d832b3506a96bba0d3011460070de9053e9`.
- `NIST-SSE-001` - transport `captured`; captured bytes `47984`; snapshot `sha256:64c8d41d9a78268d596a0a60aad18eead50fd375e8628486f7202788a207edaa`.

- `CISA-SECURE-BY-DESIGN-001` - transport `captured`; captured bytes `64467`; snapshot `sha256:b4ad9124bddafabd552dc29d861694336c8af6d64f135fa44ad3b1bf24fb4ee2`.

The comparison is structural and source-bound. It checks that a relevant primary handbook, standard, guidance record or open engineering project can represent the declared requirement, artifact, environment, operating boundary, version, test and lifecycle without rewriting the Fold law. It does not claim that a standard proves conformity, that one handbook exhausts Engineering, or that a custodian's reputation proves the result. New records can extend or challenge the surface only with source, version, configuration, method, failures and adverse rows retained.

Scientific meaning

For **engineering hazard**, the consequence is practical and exact: A hazard is a source-to-harm path retaining exposure, severity, likelihood and uncertainty. An Engineering claim therefore has to expose the carrier `source-condition-exposed-carrier`, the relation `hazard-to-harm-path-relation`, the record `source-path-exposure-severity-likelihood-and-uncertainty-held`, and the boundary `hazard-scenario-bounded` together. If any one is hidden, the result may remain a concept, prototype, simulation, demonstration or product choice, but it is not this admitted law.

This distinction protects users, maintainers and science. It prevents a successful output from erasing requirements, failed alternatives, hidden resources, inaccessible users, hazards, maintenance burdens or end-of-life costs. It prevents an application from selecting the upstream law and prevents engineering failure from becoming automatic scientific falsification. A replacement must still generate its forms, eliminate alternatives, survive controls, preserve custody and reproduce through the shared engine.

Machine certificate

Candidate count: `256`; survivor count: `1`; derivation seal: `sha256:44c27044a999c7f13539ed05214539f0860447788c7bd26f847f1b8d856403e0`; independent certificate: `sha256:3ef85436064299053c9dbb453f2adc02a1a32c75bb255cf734c04eb37a28dbc2`; engine receipt: `sha256:440b90261973a86e637ed8c25a2991f332aaad7eabbf3f16dc730162b4ab58af`. The independent implementation recomputed the candidate product: `True`. All external rows were preserved: `True`.

38. Engineering risk record

Claim: `SFT-ENG-RISK-001`

Family: `safety_hazard_resilience`

Exact statement: Risk retains scenario evidence, exposure, consequence, uncertainty and declared value judgments separately.

Forward chain and exact carrier

The generated carrier is `hazard-population-consequence-carrier`. Its required relation is `scenario-to-consequence-comparison`. The reconstruction must retain `scenarios-exposure-severity-frequency-uncertainty-and-values-held`. Its declared empirical boundary is `risk-model-bounded`. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is `SFT-FOUNDATION-ONE-001`, `SFT-FOUNDATION-FOLD-001`, `SFT-FOUNDATION-FOLD-DYNAMICS-001`, `SFT-MATH-EXACT-ARITHMETIC-001`, `SFT-MATH-GRAPH-NETWORK-001`, `SFT-MATH-OPTIMIZATION-001`, `SFT-MATH-DYNAMICAL-SYSTEMS-001`, `SFT-MATH-LOGIC-PROOF-001`, `SFT-INFO-QUANTITY-001`, `SFT-INFO-CHANNEL-CAPACITY-001`, `SFT-COMP-FORM-STATE-TRANSITION-001`, `SFT-COMP-SEM-CORRECTNESS-001`, `SFT-COMP-SEM-COMPILATION-001`, `SFT-COMP-DIST-CAUSALITY-001`, `SFT-PHYS-MEAS-VALUE-RECORD-001`, `SFT-PHYS-MEAS-CALIBRATION-001`, `SFT-CHEM-MEAS-TRACEABILITY-001`, `SFT-MAT-SUST-LIFECYCLE-001`, `SFT-BIO-BIO-HANDOFF-001`, `SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001`, `SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001`, `SFT-EARTH-HAZARD-RISK-HANDOFF-001`, `SFT-ASTRO-ASTRO-HANDOFF-001`, `SFT-SOCIAL-ENGINEERING-HANDOFF-001`, `SFT-ENG-FAULT-RESPONSE-001`, `SFT-ENG-HAZARD-001`. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external standard, application, performance target, product or prior implementation to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external standards, products, tests or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite components, versions, platforms, tests and lifecycle records; performance remains artifact, environment, user, method and operating-boundary limited.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - which requirement, artifact, component, user, environment or test bears the claim: `hazard-population-consequence-carrier`.
- **boundary** - which operating envelope, version, platform, user and lifecycle stage bound the result: `risk-model-bounded`.
- **relation** - which generated transformation, interface, comparison or handoff connects the carriers: `scenario-to-consequence-comparison`.
- **record** - which requirements, versions, alternatives, failures, uncertainty and rollback remain reconstructible: `scenarios-exposure-severity-frequency-uncertainty-and-values-held`.
- **evidence** - whether law, requirement, design, simulation, test, demonstration, performance and anomaly remain distinct: `science-design-test-and-decision-classes-explicit`.
- **provenance** - whether the result traces to the One and names every upstream receipt and design choice: `root-bound-forward-translation`.
- **generality** - whether finite extension preserves platforms, versions, failures and operating limits: `positive-finite-extension-preserves-platforms-failures-and-versions`.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: `no-extra-rule`.

Their ordered product gives the unique survivor:

```
hazard-population-consequence-carrier__risk-model-bounded__scenario-to-consequence-co
mparison__scenarios-exposure-severity-frequency-uncertainty-and-values-held__science-
design-test-and-decision-classes-explicit__root-bound-forward-translation__positive-f
inite-extension-preserves-platforms-failures-and-versions__no-extra-rule
```

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses the requirement, artifact, user or environment. Erasing the boundary universalises a version, platform or test. Importing the relation lets a target or preferred application choose the result. Retaining only successful output removes requirements, alternatives, failures and rollback. Conflating evidence classes turns simulation, demonstration or performance into observation or law. Breaking provenance removes the trace to the One and upstream receipts. Erasing finite context makes one implementation universal. Adding a silent dependency or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite requirement or admitted law, versioned artifact, boundary, testable relation and complete record. The successor adds one lawful component, interface, platform, requirement, test, failure, lifecycle stage or anomaly while preserving all earlier identities, adverse rows and receipts.

Falsification and deliberately unfavourable controls

The registered falsification condition is: Reject if one scalar hides population distribution or ethical choice.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: `source_bound_engineering_requirement_design_test_lifecycle_or_access_record/primary_standard_handbook_guidance_or_project_record_structural_correspondence`. Captured registered sources: 3. Unresolved transports retained: 0. The registered structural consequence corresponded: `True`. External evidence selected the survivor: `false`. Application or performance selected the law: `false`. Successful performance was relabelled scientific proof: `false`. Failed implementation was relabelled scientific falsification: `false`. Simulation or demonstration was relabelled observation: `false`.

- `HSE-COMAH-001` - transport `captured`; captured bytes `52092`; snapshot `sha256:928bb33ae445b9eb3be9784ade179d832b3506a96bba0d3011460070de9053e9`.
- `NIST-SSE-001` - transport `captured`; captured bytes `47984`; snapshot `sha256:64c8d41d9a78268d596a0a60aad18eead50fd375e8628486f7202788a207edaa`.
- `CISA-SECURE-BY-DESIGN-001` - transport `captured`; captured bytes `64467`; snapshot `sha256:b4ad9124bddafabd552dc29d861694336c8af6d64f135fa44ad3b1bf24fb4ee2`.

The comparison is structural and source-bound. It checks that a relevant primary handbook, standard, guidance record or open engineering project can represent the declared requirement, artifact, environment, operating boundary, version, test and lifecycle without rewriting the Fold law. It does not claim that a standard proves conformity, that one handbook exhausts Engineering, or that a custodian's reputation proves the result. New records can extend or challenge the surface only with source, version, configuration, method, failures and adverse rows retained.

Scientific meaning

For **engineering risk record**, the consequence is practical and exact: Risk retains scenario evidence, exposure, consequence, uncertainty and declared value judgments separately. An Engineering claim therefore has to expose the carrier `hazard-population-consequence-carrier`, the relation `scenario-to-consequence-comparison`, the record `scenarios-exposure-severity-frequency-uncertainty-and-values-held`, and the boundary `risk-model-bounded` together. If any one is hidden, the result may remain a concept, prototype, simulation, demonstration or product choice, but it is not this admitted law.

This distinction protects users, maintainers and science. It prevents a successful output from erasing requirements, failed alternatives, hidden resources, inaccessible users, hazards, maintenance burdens or end-of-life costs. It prevents an application from selecting the upstream law and prevents engineering failure from becoming automatic scientific falsification. A replacement must still generate its forms, eliminate alternatives, survive controls, preserve custody and reproduce through the shared engine.

Machine certificate

Candidate count: `256`; survivor count: `1`; derivation seal: `sha256:242d0f3536af67256eb8038d9258b7704fa5d219ba6a851820645d51d8c6dab2`; independent certificate: `sha256:766d24440d4bf7f257640cd1c234c288bb95bb983871cab5c7f7d1b12e111974`; engine receipt: `sha256:dc952d2c85cf201ff0cdef942030347e25f5842a6fdeec510e6a3a5bcd57ae21`. The independent implementation recomputed the candidate product: `True`. All external rows were preserved: `True`.

39. Safety case

Claim: `SFT-ENG-SAFETY-001`

Family: `safety_hazard_resilience`

Exact statement: A safety case is a scoped claim supported by traceable evidence, controls, gaps and review.

Forward chain and exact carrier

The generated carrier is `system-claim-argument-evidence-carrier`. Its required relation is `evidence-to-bounded-safety-claim-relation`. The reconstruction must retain `claim-scope-hazards-controls-evidence-gaps-and-review-held`. Its declared empirical boundary is `safety-case-bounded`. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is `SFT-FOUNDATION-ONE-001`, `SFT-FOUNDATION-FOLD-001`, `SFT-FOUNDATION-FOLD-DYNAMICS-001`, `SFT-MATH-EXACT-ARITHMETIC-001`, `SFT-MATH-GRAPH-NETWORK-001`, `SFT-MATH-OPTIMIZATION-001`, `SFT-MATH-DYNAMICAL-SYSTEMS-001`, `SFT-MATH-LOGIC-PROOF-001`, `SFT-INFO-QUANTITY-001`, `SFT-INFO-CHANNEL-CAPACITY-001`, `SFT-COMP-FORM-STATE-TRANSITION-001`, `SFT-COMP-SEM-CORRECTNESS-001`, `SFT-COMP-SEM-COMPILATION-001`, `SFT-COMP-DIST-CAUSALITY-001`, `SFT-PHYS-MEAS-VALUE-RECORD-001`, `SFT-PHYS-MEAS-CALIBRATION-001`, `SFT-CHEM-MEAS-TRACEABILITY-001`, `SFT-MAT-SUST-LIFECYCLE-001`, `SFT-BIO-BIO-HANDOFF-001`, `SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001`, `SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001`, `SFT-EARTH-HAZARD-RISK-HANDOFF-001`, `SFT-ASTRO-ASTRO-HANDOFF-001`, `SFT-SOCIAL-ENGINEERING-HANDOFF-001`, `SFT-ENG-FAULT-RESPONSE-001`, `SFT-ENG-RISK-001`. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external standard, application, performance target, product or prior implementation to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external standards, products, tests or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite components, versions, platforms, tests and lifecycle records; performance remains artifact, environment, user, method and operating-boundary limited.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - which requirement, artifact, component, user, environment or test bears the claim: `system-claim-argument-evidence-carrier`.
- **boundary** - which operating envelope, version, platform, user and lifecycle stage bound the result: `safety-case-bounded`.
- **relation** - which generated transformation, interface, comparison or handoff connects the carriers: `evidence-to-bounded-safety-claim-relation`.
- **record** - which requirements, versions, alternatives, failures, uncertainty and rollback remain reconstructible: `claim-scope-hazards-controls-evidence-gaps-and-review-held`.
- **evidence** - whether law, requirement, design, simulation, test, demonstration, performance and anomaly remain distinct: `science-design-test-and-decision-classes-explicit`.
- **provenance** - whether the result traces to the One and names every upstream receipt and design choice: `root-bound-forward-translation`.
- **generality** - whether finite extension preserves platforms, versions, failures and operating limits: `positive-finite-extension-preserves-platforms-failures-and-versions`.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: `no-extra-rule`.

Their ordered product gives the unique survivor:

```
system-claim-argument-evidence-carrier__safety-case-bounded__evidence-to-bounded-safe
ty-claim-relation__claim-scope-hazards-controls-evidence-gaps-and-review-held__scienc
e-design-test-and-decision-classes-explicit__root-bound-forward-translation__positive
-finite-extension-preserves-platforms-failures-and-versions__no-extra-rule
```

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses the requirement, artifact, user or environment. Erasing the boundary universalises a version, platform or test. Importing the relation lets a target or preferred application choose the result. Retaining only successful output removes requirements, alternatives, failures and rollback. Conflating evidence classes turns simulation, demonstration or performance into observation or law. Breaking provenance removes the trace to the One and upstream receipts. Erasing finite context makes one implementation universal. Adding a silent dependency or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite requirement or admitted law, versioned artifact, boundary, testable relation and complete record. The successor adds one lawful component, interface, platform, requirement, test, failure, lifecycle stage or anomaly while preserving all earlier identities, adverse rows and receipts.

Falsification and deliberately unfavourable controls

The registered falsification condition is: Reject if compliance, past success or authority alone proves safety.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: `source_bound_engineering_requirement_design_test_lifecycle_or_access_record/primary_standard_handbook_guidance_or_project_record_structural_correspondence`. Captured registered sources: 3. Unresolved transports retained: 0. The registered structural consequence corresponded: `True`. External evidence selected the survivor: `false`. Application or performance selected the law: `false`. Successful performance was relabelled scientific proof: `false`. Failed implementation was relabelled scientific falsification: `false`. Simulation or demonstration was relabelled observation: `false`.

- `HSE-COMAH-001` - transport `captured`; captured bytes `52092`; snapshot `sha256:928bb33ae445b9eb3be9784ade179d832b3506a96bba0d3011460070de9053e9`.
- `NIST-SSE-001` - transport `captured`; captured bytes `47984`; snapshot `sha256:64c8d41d9a78268d596a0a60aad18eead50fd375e8628486f7202788a207edaa`.
- `CISA-SECURE-BY-DESIGN-001` - transport `captured`; captured bytes `64467`; snapshot `sha256:b4ad9124bddafabd552dc29d861694336c8af6d64f135fa44ad3b1bf24fb4ee2`.

The comparison is structural and source-bound. It checks that a relevant primary handbook, standard, guidance record or open engineering project can represent the declared requirement, artifact, environment, operating boundary, version, test and lifecycle without rewriting the Fold law. It does not claim that a standard proves conformity, that one handbook exhausts Engineering, or that a custodian's reputation proves the result. New records can extend or challenge the surface only with source, version, configuration, method, failures and adverse rows retained.

Scientific meaning

For **safety case**, the consequence is practical and exact: A safety case is a scoped claim supported by traceable evidence, controls, gaps and review. An Engineering claim therefore has to expose the carrier `system-claim-argument-evidence-carrier`, the relation `evidence-to-bounded-safety-claim-relation`, the record `claim-scope-hazards-controls-evidence-gaps-and-review-held`, and the boundary `safety-case-bounded` together. If any one is hidden, the result may remain a concept, prototype, simulation, demonstration or product choice, but it is not this admitted law.

This distinction protects users, maintainers and science. It prevents a successful output from erasing requirements, failed alternatives, hidden resources, inaccessible users, hazards, maintenance burdens or end-of-life costs. It prevents an application from selecting the upstream law and prevents engineering failure from becoming automatic scientific falsification. A replacement must still generate its forms, eliminate alternatives, survive controls, preserve custody and reproduce through the shared engine.

Machine certificate

Candidate count: `256`; survivor count: `1`; derivation seal: `sha256:f9f283d4a866944957de16528a3baf2e202bb07f9c8cb62e25fabc08db2af6fa`; independent certificate: `sha256:193f7253e9e41063aefd79548056e92dc85ddfdf6bea955f40e92ce9722c3b69`; engine receipt: `sha256:68e77efa7e20c3522e74a3f7ca910bc7d0c6baeaf95b5f9f0e9a3d66f7c3fec2`. The independent implementation recomputed the candidate product: `True`. All external rows were preserved: `True`.

40. Fail-safe operation

Claim: `SFT-ENG-FAIL-SAFE-001`

Family: `safety_hazard_resilience`

Exact statement: Fail-safe operation requires every declared fault to reach a tested bounded safe state or preserve a visible halt.

Forward chain and exact carrier

The generated carrier is `system-fault-safe-state-carrier`. Its required relation is `detected-or-bounded-fault-to-safe-transition-relation`. The reconstruction must retain `faults-detection-actions-safe-state-residual-harm-and-test-held`. Its declared empirical boundary is `declared-fault-set-bounded`. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is `SFT-FOUNDATION-ONE-001`, `SFT-FOUNDATION-FOLD-001`, `SFT-FOUNDATION-FOLD-DYNAMICS-001`, `SFT-MATH-EXACT-ARITHMETIC-001`, `SFT-MATH-GRAPH-NETWORK-001`, `SFT-MATH-OPTIMIZATION-001`, `SFT-MATH-DYNAMICAL-SYSTEMS-001`, `SFT-MATH-LOGIC-PROOF-001`, `SFT-INFO-QUANTITY-001`, `SFT-INFO-CHANNEL-CAPACITY-001`, `SFT-COMP-FORM-STATE-TRANSITION-001`, `SFT-COMP-SEM-CORRECTNESS-001`, `SFT-COMP-SEM-COMPILATION-001`, `SFT-COMP-DIST-CAUSALITY-001`, `SFT-PHYS-MEAS-VALUE-RECORD-001`, `SFT-PHYS-MEAS-CALIBRATION-001`, `SFT-CHEM-MEAS-TRACEABILITY-001`, `SFT-MAT-SUST-LIFECYCLE-001`, `SFT-BIO-BIO-HANDOFF-001`, `SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001`, `SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001`, `SFT-EARTH-HAZARD-RISK-HANDOFF-001`, `SFT-ASTRO-ASTRO-HANDOFF-001`, `SFT-SOCIAL-ENGINEERING-HANDOFF-001`, `SFT-ENG-FAULT-RESPONSE-001`, `SFT-ENG-SAFETY-001`. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external standard, application, performance target, product or prior implementation to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external standards, products, tests or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite components, versions, platforms, tests and lifecycle records; performance remains artifact, environment, user, method and operating-boundary limited.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - which requirement, artifact, component, user, environment or test bears the claim: `system-fault-safe-state-carrier`.
- **boundary** - which operating envelope, version, platform, user and lifecycle stage bound the result: `declared-fault-set-bounded`.
- **relation** - which generated transformation, interface, comparison or handoff connects the carriers: `detected-or-bounded-fault-to-safe-transition-relation`.

- **record** - which requirements, versions, alternatives, failures, uncertainty and rollback remain reconstructible: `faults-detection-actions-safe-state-residual-harm-and-test-held`.
- **evidence** - whether law, requirement, design, simulation, test, demonstration, performance and anomaly remain distinct: `science-design-test-and-decision-classes-explicit`.
- **provenance** - whether the result traces to the One and names every upstream receipt and design choice: `root-bound-forward-translation`.
- **generality** - whether finite extension preserves platforms, versions, failures and operating limits: `positive-finite-extension-preserves-platforms-failures-and-versions`.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: `no-extra-rule`.

Their ordered product gives the unique survivor:

```
system-fault-safe-state-carrier__declared-fault-set-bounded__detected-or-bounded-faul
t-to-safe-transition-relation__faults-detection-actions-safe-state-residual-harm-and-
test-held__science-design-test-and-decision-classes-explicit__root-bound-forward-tran
slation__positive-finite-extension-preserves-platforms-failures-and-versions__no-extr
a-rule
```

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses the requirement, artifact, user or environment. Erasing the boundary universalises a version, platform or test. Importing the relation lets a target or preferred application choose the result. Retaining only successful output removes requirements, alternatives, failures and rollback. Conflating evidence classes turns simulation, demonstration or performance into observation or law. Breaking provenance removes the trace to the One and upstream receipts. Erasing finite context makes one implementation universal. Adding a silent dependency or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite requirement or admitted law, versioned artifact, boundary, testable relation and complete record. The successor adds one lawful component, interface, platform, requirement, test, failure, lifecycle stage or anomaly while preserving all earlier identities, adverse rows and receipts.

Falsification and deliberately unfavourable controls

The registered falsification condition is: Reject if an untested fault or silent continuation is called fail-safe.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: `source_bound_engineering_requirement_design_test_lifecycle_or_access_record/primary_standard_handbook_guidance_or_project_record_structural_correspondence`. Captured registered sources: 3. Unresolved transports retained: 0. The registered structural consequence corresponded: `True`. External evidence selected the survivor: `false`. Application or performance selected the law: `false`. Successful performance was relabelled scientific proof: `false`. Failed implementation was relabelled scientific falsification: `false`. Simulation or demonstration was relabelled observation: `false`.

- `HSE-COMAH-001` - transport `captured`; captured bytes `52092`; snapshot `sha256:928bb33ae445b9eb3be9784ade179d832b3506a96bba0d3011460070de9053e9`.
- `NIST-SSE-001` - transport `captured`; captured bytes `47984`; snapshot `sha256:64c8d41d9a78268d596a0a60aad18eead50fd375e8628486f7202788a207edaa`.

- `CISA-SECURE-BY-DESIGN-001` - transport `captured`; captured bytes `64467`; snapshot `sha256:b4ad9124bddafabd552dc29d861694336c8af6d64f135fa44ad3b1bf24fb4ee2`.

The comparison is structural and source-bound. It checks that a relevant primary handbook, standard, guidance record or open engineering project can represent the declared requirement, artifact, environment, operating boundary, version, test and lifecycle without rewriting the Fold law. It does not claim that a standard proves conformity, that one handbook exhausts Engineering, or that a custodian's reputation proves the result. New records can extend or challenge the surface only with source, version, configuration, method, failures and adverse rows retained.

Scientific meaning

For **fail-safe operation**, the consequence is practical and exact: Fail-safe operation requires every declared fault to reach a tested bounded safe state or preserve a visible halt. An Engineering claim therefore has to expose the carrier `system-fault-safe-state-carrier`, the relation `detected-or-bounded-fault-to-safe-transition-relation`, the record `faults-detection-actions-safe-state-residual-harm-and-test-held`, and the boundary `declared-fault-set-bounded` together. If any one is hidden, the result may remain a concept, prototype, simulation, demonstration or product choice, but it is not this admitted law.

This distinction protects users, maintainers and science. It prevents a successful output from erasing requirements, failed alternatives, hidden resources, inaccessible users, hazards, maintenance burdens or end-of-life costs. It prevents an application from selecting the upstream law and prevents engineering failure from becoming automatic scientific falsification. A replacement must still generate its forms, eliminate alternatives, survive controls, preserve custody and reproduce through the shared engine.

Machine certificate

Candidate count: `256`; survivor count: `1`; derivation seal: `sha256:8d1fbc8307f6763381a316672a4807e4d16576af49313826dc767aa97fb3a695`; independent certificate: `sha256:2ac1a52948e3bd9b426d421802c8f4c490f863a7a9c3acd32d38f34c5975a739`; engine receipt: `sha256:9134a2270e5cf3534e086e680450f7008be5f32965c812a138e227ced6323261`. The independent implementation recomputed the candidate product: `True`. All external rows were preserved: `True`.

41. Engineering resilience

Claim: `SFT-ENG-RESILIENCE-001`

Family: `safety_hazard_resilience`

Exact statement: Resilience retains service loss, adaptation resources, recovery time, residual degradation and failed recovery.

Forward chain and exact carrier

The generated carrier is `system-disruption-service-recovery-carrier`. Its required relation is `disruption-to-bounded-recovery-relation`. The reconstruction must retain `service-loss-adaptation-resources-time-residual-and-learning-held`. Its declared empirical boundary is `disruption-family-bounded`. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is `SFT-FOUNDATION-ONE-001`, `SFT-FOUNDATION-FOLD-001`, `SFT-FOUNDATION-FOLD-DYNAMICS-001`, `SFT-MATH-EXACT-ARITHMETIC-001`, `SFT-MATH-GRAPH-NETWORK-001`, `SFT-MATH-OPTIMIZATION-001`, `SFT-MATH-DYNAMICAL-SYSTEMS-001`, `SFT-MATH-LOGIC-PROOF-001`, `SFT-INFO-QUANTITY-001`, `SFT-INFO-CHANNEL-CAPACITY-001`, `SFT-COMP-FORM-STATE-TRANSITION-001`, `SFT-COMP-SEM-CORRECTNESS-001`, `SFT-COMP-SEM-COMPILATION-001`, `SFT-COMP-DIST-CAUSALITY-001`, `SFT-PHYS-MEAS-VALUE-RECORD-001`, `SFT-PHYS-MEAS-CALIBRATION-001`, `SFT-CHEM-MEAS-TRACEABILITY-001`, `SFT-MAT-SUST-LIFECYCLE-001`, `SFT-BIO-BIO-HANDOFF-001`, `SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001`, `SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001`, `SFT-EARTH-HAZARD-RISK-HANDOFF-001`, `SFT-ASTRO-ASTRO-HANDOFF-001`, `SFT-SOCIAL-ENGINEERING-HANDOFF-001`, `SFT-ENG-FAULT-RESPONSE-001`, `SFT-ENG-FAIL-SAFE-001`. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external standard, application, performance target, product or prior implementation to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external standards, products, tests or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite components, versions, platforms, tests and lifecycle records; performance remains artifact, environment, user, method and operating-boundary limited.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - which requirement, artifact, component, user, environment or test bears the claim: `system-disruption-service-recovery-carrier`.
- **boundary** - which operating envelope, version, platform, user and lifecycle stage bound the result: `disruption-family-bounded`.
- **relation** - which generated transformation, interface, comparison or handoff connects the carriers: `disruption-to-bounded-recovery-relation`.
- **record** - which requirements, versions, alternatives, failures, uncertainty and rollback remain reconstructible: `service-loss-adaptation-resources-time-residual-and-learning-held`.
- **evidence** - whether law, requirement, design, simulation, test, demonstration, performance and anomaly remain distinct: `science-design-test-and-decision-classes-explicit`.
- **provenance** - whether the result traces to the One and names every upstream receipt and design choice: `root-bound-forward-translation`.
- **generality** - whether finite extension preserves platforms, versions, failures and operating limits: `positive-finite-extension-preserves-platforms-failures-and-versions`.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: `no-extra-rule`.

Their ordered product gives the unique survivor:

```
system-disruption-service-recovery-carrier__disruption-family-bounded__disruption-to-
bounded-recovery-relation__service-loss-adaptation-resources-time-residual-and-learni
ng-held__science-design-test-and-decision-classes-explicit__root-bound-forward-transl
ation__positive-finite-extension-preserves-platforms-failures-and-versions__no-extra-
rule
```

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses the requirement, artifact, user or environment. Erasing the boundary universalises a version, platform or test. Importing the relation lets a target or preferred application choose the result. Retaining only successful output removes requirements, alternatives, failures and rollback. Conflating evidence classes turns simulation, demonstration or performance into observation or law. Breaking provenance removes the trace to the One and upstream receipts. Erasing finite context makes one implementation universal. Adding a silent dependency or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite requirement or admitted law, versioned artifact, boundary, testable relation and complete record. The successor adds one lawful component, interface, platform, requirement, test, failure, lifecycle stage or anomaly while preserving all earlier identities, adverse rows and receipts.

Falsification and deliberately unfavourable controls

The registered falsification condition is: Reject if survival alone proves full recovery or hidden external aid is omitted.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: `source_bound_engineering_requirement_design_test_lifecycle_or_access_record/primary_standard_handbook_guidance_or_project_record_structural_correspondence`. Captured registered sources: 3. Unresolved transports retained: 0. The registered structural consequence corresponded: `True`. External evidence selected the survivor: `false`. Application or performance selected the law: `false`. Successful performance was relabelled scientific proof: `false`. Failed implementation was relabelled scientific falsification: `false`. Simulation or demonstration was relabelled observation: `false`.

- `HSE-COMAH-001` - transport `captured`; captured bytes `52092`; snapshot `sha256:928bb33ae445b9eb3be9784ade179d832b3506a96bba0d3011460070de9053e9`.
- `NIST-SSE-001` - transport `captured`; captured bytes `47984`; snapshot `sha256:64c8d41d9a78268d596a0a60aad18eead50fd375e8628486f7202788a207edaa`.
- `CISA-SECURE-BY-DESIGN-001` - transport `captured`; captured bytes `64467`; snapshot `sha256:b4ad9124bddafabd552dc29d861694336c8af6d64f135fa44ad3b1bf24fb4ee2`.

The comparison is structural and source-bound. It checks that a relevant primary handbook, standard, guidance record or open engineering project can represent the declared requirement, artifact, environment, operating boundary, version, test and lifecycle without rewriting the Fold law. It does not claim that a standard proves conformity, that one handbook exhausts Engineering, or that a custodian's reputation proves the result. New records can extend or challenge the surface only with source, version, configuration, method, failures and adverse rows retained.

Scientific meaning

For **engineering resilience**, the consequence is practical and exact: Resilience retains service loss, adaptation resources, recovery time, residual degradation and failed recovery. An Engineering claim therefore has to expose the carrier `system-disruption-service-recovery-carrier`, the relation `disruption-to-bounded-recovery-relation`, the record `service-loss-adaptation-resources-time-residual-and-learning-held`, and the boundary `disruption-family-bounded` together. If any one is hidden, the result may remain a concept, prototype, simulation, demonstration or product choice, but it is not this admitted law.

This distinction protects users, maintainers and science. It prevents a successful output from erasing requirements, failed alternatives, hidden resources, inaccessible users, hazards, maintenance burdens or end-of-life costs. It prevents an application from selecting the upstream law and prevents engineering failure from becoming automatic scientific falsification. A replacement must still generate its forms, eliminate alternatives, survive controls, preserve custody and reproduce through the shared engine.

Machine certificate

Candidate count: `256`; survivor count: `1`; derivation seal: `sha256:dbcc2c7ba67bb2ea3cfb7a795a8166d6aa99525e064422aae35b1d2fd277b257`; independent certificate: `sha256:66efa13d42cf03d8b904690c0a42d4218cdf90fe400bbb3f0e5716324cc271c5`; engine receipt: `sha256:1765e8ffe9b823048f0ec4b8bebc88653b82e9d60e1a37886db1dcb1bd16ddba`. The independent implementation recomputed the candidate product: `True`. All external rows were preserved: `True`.

42. Defence in depth

Claim: `SFT-ENG-DEFENCE-IN-DEPTH-001`

Family: `safety_hazard_resilience`

Exact statement: Defence in depth requires separately testable barriers with dependencies, bypasses and common causes retained.

Forward chain and exact carrier

The generated carrier is `hazard-barrier-chain-carrier`. Its required relation is `independent-barrier-to-residual-path-relation`. The reconstruction must retain `barriers-dependencies-common-causes-bypasses-tests-and-failures-held`. Its declared empirical boundary is `threat-and-hazard-bounded`. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is `SFT-FOUNDATION-ONE-001`, `SFT-FOUNDATION-FOLD-001`, `SFT-FOUNDATION-FOLD-DYNAMICS-001`, `SFT-MATH-EXACT-ARITHMETIC-001`, `SFT-MATH-GRAPH-NETWORK-001`, `SFT-MATH-OPTIMIZATION-001`, `SFT-MATH-DYNAMICAL-SYSTEMS-001`, `SFT-MATH-LOGIC-PROOF-001`, `SFT-INFO-QUANTITY-001`, `SFT-INFO-CHANNEL-CAPACITY-001`, `SFT-COMP-FORM-STATE-TRANSITION-001`, `SFT-COMP-SEM-CORRECTNESS-001`, `SFT-COMP-SEM-COMPILATION-001`, `SFT-COMP-DIST-CAUSALITY-001`, `SFT-PHYS-MEAS-VALUE-RECORD-001`, `SFT-PHYS-MEAS-CALIBRATION-001`, `SFT-CHEM-MEAS-TRACEABILITY-001`, `SFT-MAT-SUST-LIFECYCLE-001`, `SFT-BIO-BIO-HANDOFF-001`, `SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001`, `SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001`, `SFT-EARTH-HAZARD-RISK-HANDOFF-001`, `SFT-ASTRO-ASTRO-HANDOFF-001`, `SFT-SOCIAL-ENGINEERING-HANDOFF-001`, `SFT-ENG-FAULT-RESPONSE-001`, `SFT-ENG-RESILIENCE-001`. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external standard, application, performance target, product or prior implementation to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external standards, products, tests or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite components, versions, platforms, tests and lifecycle records; performance remains artifact, environment, user, method and operating-boundary limited.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - which requirement, artifact, component, user, environment or test bears the claim: `hazard-barrier-chain-carrier`.
- **boundary** - which operating envelope, version, platform, user and lifecycle stage bound the result: `threat-and-hazard-bounded`.
- **relation** - which generated transformation, interface, comparison or handoff connects the carriers: `independent-barrier-to-residual-path-relation`.
- **record** - which requirements, versions, alternatives, failures, uncertainty and rollback remain reconstructible: `barriers-dependencies-common-causes-bypasses-tests-and-failures-held`.
- **evidence** - whether law, requirement, design, simulation, test, demonstration, performance and anomaly remain distinct: `science-design-test-and-decision-classes-explicit`.
- **provenance** - whether the result traces to the One and names every upstream receipt and design choice: `root-bound-forward-translation`.
- **generality** - whether finite extension preserves platforms, versions, failures and operating limits: `positive-finite-extension-preserves-platforms-failures-and-versions`.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: `no-extra-rule`.

Their ordered product gives the unique survivor:

```
hazard-barrier-chain-carrier__threat-and-hazard-bounded__independent-barrier-to-resid
ual-path-relation__barriers-dependencies-common-causes-bypasses-tests-and-failures-he
ld__science-design-test-and-decision-classes-explicit__root-bound-forward-translation
__positive-finite-extension-preserves-platforms-failures-and-versions__no-extra-rule
```

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses the requirement, artifact, user or environment. Erasing the boundary universalises a version, platform or test. Importing the relation lets a target or preferred application choose the result. Retaining only successful output removes requirements, alternatives, failures and rollback. Conflating evidence classes turns simulation, demonstration or performance into observation or law. Breaking provenance removes the trace to the One and upstream receipts. Erasing finite context makes one implementation universal. Adding a silent dependency or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite requirement or admitted law, versioned artifact, boundary, testable relation and complete record. The successor adds one lawful component, interface, platform, requirement, test, failure, lifecycle stage or anomaly while preserving all earlier identities, adverse rows and receipts.

Falsification and deliberately unfavourable controls

The registered falsification condition is: Reject if repeated copies with one common failure are called independent layers.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: `source_bound_engineering_requirement_design_test_lifecycle_or_access_record/primary_standard_handbook_guidance_or_project_record_structural_correspondence`. Captured registered sources: 3. Unresolved transports retained: 0. The registered structural consequence corresponded: `True`. External evidence selected the survivor: `false`. Application or performance selected the law: `false`. Successful performance was relabelled scientific proof: `false`. Failed implementation was relabelled scientific falsification: `false`. Simulation or demonstration was relabelled observation: `false`.

- `HSE-COMAH-001` - transport `captured`; captured bytes `52092`; snapshot `sha256:928bb33ae445b9eb3be9784ade179d832b3506a96bba0d3011460070de9053e9`.
- `NIST-SSE-001` - transport `captured`; captured bytes `47984`; snapshot `sha256:64c8d41d9a78268d596a0a60aad18eead50fd375e8628486f7202788a207edaa`.
- `CISA-SECURE-BY-DESIGN-001` - transport `captured`; captured bytes `64467`; snapshot `sha256:b4ad9124bddafabd552dc29d861694336c8af6d64f135fa44ad3b1bf24fb4ee2`.

The comparison is structural and source-bound. It checks that a relevant primary handbook, standard, guidance record or open engineering project can represent the declared requirement, artifact, environment, operating boundary, version, test and lifecycle without rewriting the Fold law. It does not claim that a standard proves conformity, that one handbook exhausts Engineering, or that a custodian's reputation proves the result. New records can extend or challenge the surface only with source, version, configuration, method, failures and adverse rows retained.

Scientific meaning

For **defence in depth**, the consequence is practical and exact: Defence in depth requires separately testable barriers with dependencies, bypasses and common causes retained. An Engineering claim therefore has to expose the carrier `hazard-barrier-chain-carrier`, the relation `independent-barrier-to-residual-path-relation`, the record `barriers-dependencies-common-causes-bypasses-tests-and-failures-held`, and the boundary `threat-and-hazard-bounded` together. If any one is hidden, the result may remain a concept, prototype, simulation, demonstration or product choice, but it is not this admitted law.

This distinction protects users, maintainers and science. It prevents a successful output from erasing requirements, failed alternatives, hidden resources, inaccessible users, hazards, maintenance burdens or end-of-life costs. It prevents an application

from selecting the upstream law and prevents engineering failure from becoming automatic scientific falsification. A replacement must still generate its forms, eliminate alternatives, survive controls, preserve custody and reproduce through the shared engine.

Machine certificate

Candidate count: `256`; survivor count: `1`; derivation seal: `sha256:1a45454ba4c9dad78ec2457658a99d82f15c8157264cd347e873d391aa645587`; independent certificate: `sha256:a5f76d598fa5bececfadbf9374463c5f1fe41e3bac8e8940a3c5439e8c6a003d`; engine receipt: `sha256:940904d1b6fa7f7bd79be5a92c7ee6f90ebad6d2a28812c1c102fe4a5f4cf259`. The independent implementation recomputed the candidate product: `True`. All external rows were preserved: `True`.

43. Engineering verification

Claim: `SFT-ENG-VERIFICATION-001`

Family: `verification_validation_traceability`

Exact statement: Verification tests whether a versioned artifact satisfies its frozen specification.

Forward chain and exact carrier

The generated carrier is `artifact-specification-evidence-carrier`. Its required relation is `artifact-to-specified-condition-comparison`. The reconstruction must retain `version-requirement-method-result-deviation-and-review-held`. Its declared empirical boundary is `verification-baseline-bounded`. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is `SFT-FOUNDATION-ONE-001`, `SFT-FOUNDATION-FOLD-001`, `SFT-FOUNDATION-FOLD-DYNAMICS-001`, `SFT-MATH-EXACT-ARITHMETIC-001`, `SFT-MATH-GRAPH-NETWORK-001`, `SFT-MATH-OPTIMIZATION-001`, `SFT-MATH-DYNAMICAL-SYSTEMS-001`, `SFT-MATH-LOGIC-PROOF-001`, `SFT-INFO-QUANTITY-001`, `SFT-INFO-CHANNEL-CAPACITY-001`, `SFT-COMP-FORM-STATE-TRANSITION-001`, `SFT-COMP-SEM-CORRECTNESS-001`, `SFT-COMP-SEM-COMPILATION-001`, `SFT-COMP-DIST-CAUSALITY-001`, `SFT-PHYS-MEAS-VALUE-RECORD-001`, `SFT-PHYS-MEAS-CALIBRATION-001`, `SFT-CHEM-MEAS-TRACEABILITY-001`, `SFT-MAT-SUST-LIFECYCLE-001`, `SFT-BIO-BIO-HANDOFF-001`, `SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001`, `SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001`, `SFT-EARTH-HAZARD-RISK-HANDOFF-001`, `SFT-ASTRO-ASTRO-HANDOFF-001`, `SFT-SOCIAL-ENGINEERING-HANDOFF-001`, `SFT-ENG-DEFENCE-IN-DEPTH-001`. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external standard, application, performance target, product or prior implementation to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external standards, products, tests or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite components, versions, platforms, tests and lifecycle records; performance remains artifact, environment, user, method and operating-boundary limited.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - which requirement, artifact, component, user, environment or test bears the claim: `artifact-specification-evidence-carrier`.
- **boundary** - which operating envelope, version, platform, user and lifecycle stage bound the result: `verification-baseline-bounded`.
- **relation** - which generated transformation, interface, comparison or handoff connects the carriers: `artifact-to-specified-condition-comparison`.

- **record** - which requirements, versions, alternatives, failures, uncertainty and rollback remain reconstructible: `version-requirement-method-result-deviation-and-review-held`.
- **evidence** - whether law, requirement, design, simulation, test, demonstration, performance and anomaly remain distinct: `science-design-test-and-decision-classes-explicit`.
- **provenance** - whether the result traces to the One and names every upstream receipt and design choice: `root-bound-forward-translation`.
- **generality** - whether finite extension preserves platforms, versions, failures and operating limits: `positive-finite-extension-preserves-platforms-failures-and-versions`.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: `no-extra-rule`.

Their ordered product gives the unique survivor:

```
artifact-specification-evidence-carrier__verification-baseline-bounded__artifact-to-s
pecified-condition-comparison__version-requirement-method-result-deviation-and-review
-held__science-design-test-and-decision-classes-explicit__root-bound-forward-translat
ion__positive-finite-extension-preserves-platforms-failures-and-versions__no-extra-ru
le
```

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses the requirement, artifact, user or environment. Erasing the boundary universalises a version, platform or test. Importing the relation lets a target or preferred application choose the result. Retaining only successful output removes requirements, alternatives, failures and rollback. Conflating evidence classes turns simulation, demonstration or performance into observation or law. Breaking provenance removes the trace to the One and upstream receipts. Erasing finite context makes one implementation universal. Adding a silent dependency or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite requirement or admitted law, versioned artifact, boundary, testable relation and complete record. The successor adds one lawful component, interface, platform, requirement, test, failure, lifecycle stage or anomaly while preserving all earlier identities, adverse rows and receipts.

Falsification and deliberately unfavourable controls

The registered falsification condition is: Reject if intended use, user approval or later performance substitutes for specification evidence.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: `source_bound_engineering_requirement_design_test_lifecycle_or_access_record/primary_standard_handbook_guidance_or_project_record_structural_correspondence`. Captured registered sources: 3. Unresolved transports retained: 0. The registered structural consequence corresponded: `True`. External evidence selected the survivor: `false`. Application or performance selected the law: `false`. Successful performance was relabelled scientific proof: `false`. Failed implementation was relabelled scientific falsification: `false`. Simulation or demonstration was relabelled observation: `false`.

- `NASA-SE-APPENDIX-001` - transport `captured`; captured bytes `578114`; snapshot `sha256:c1db2db970b9fd14a7ef98aa0c23575326dedae46cc7cc56c468821ca89edd60`.

- NIST-TRACEABILITY-001 - transport captured; captured bytes 180987; snapshot sha256:4e79d3a99a9367842daf0f8910cff625de901d95b4b46c5fe2845d7512e70cd7.
- REPRODUCIBLE-BUILDS-001 - transport captured; captured bytes 27924; snapshot sha256:857885a380c3f0b545ae0b69e99c3f061b60ea7e3e99e36d1edcbc3e44bada73.

The comparison is structural and source-bound. It checks that a relevant primary handbook, standard, guidance record or open engineering project can represent the declared requirement, artifact, environment, operating boundary, version, test and lifecycle without rewriting the Fold law. It does not claim that a standard proves conformity, that one handbook exhausts Engineering, or that a custodian's reputation proves the result. New records can extend or challenge the surface only with source, version, configuration, method, failures and adverse rows retained.

Scientific meaning

For **engineering verification**, the consequence is practical and exact: Verification tests whether a versioned artifact satisfies its frozen specification. An Engineering claim therefore has to expose the carrier `artifact-specification-evidence-carrier`, the relation `artifact-to-specified-condition-comparison`, the record `version-requirement-method-result-deviation-and-review-held`, and the boundary `verification-baseline-bounded` together. If any one is hidden, the result may remain a concept, prototype, simulation, demonstration or product choice, but it is not this admitted law.

This distinction protects users, maintainers and science. It prevents a successful output from erasing requirements, failed alternatives, hidden resources, inaccessible users, hazards, maintenance burdens or end-of-life costs. It prevents an application from selecting the upstream law and prevents engineering failure from becoming automatic scientific falsification. A replacement must still generate its forms, eliminate alternatives, survive controls, preserve custody and reproduce through the shared engine.

Machine certificate

Candidate count: `256`; survivor count: `1`; derivation seal: `sha256:3cf7a8e6fc77761b56586516002c6f7f96453e396f722aa66e14b00166b711ea`; independent certificate: `sha256:d5f632cfcdce5c732288398b75ff3b1f853fd97d79ea923aae6aabba67af25e2`; engine receipt: `sha256:393c68fee236e0f4ec90c0e2c998e690154236e8073ff2653e1d1e3bb72799ec`. The independent implementation recomputed the candidate product: `True`. All external rows were preserved: `True`.

44. Engineering validation

Claim: `SFT-ENG-VALIDATION-001`

Family: `verification_validation_traceability`

Exact statement: Validation tests whether the verified system serves its declared use in its declared context.

Forward chain and exact carrier

The generated carrier is `system-use-context-user-evidence-carrier`. Its required relation is `implemented-system-to-intended-use-comparison`. The reconstruction must retain `users-context-tasks-results-harms-limitations-and-review-held`. Its declared empirical boundary is `validation-use-bounded`. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is `SFT-FOUNDATION-ONE-001`, `SFT-FOUNDATION-FOLD-001`, `SFT-FOUNDATION-FOLD-DYNAMICS-001`, `SFT-MATH-EXACT-ARITHMETIC-001`, `SFT-MATH-GRAPH-NETWORK-001`, `SFT-MATH-OPTIMIZATION-001`, `SFT-MATH-DYNAMICAL-SYSTEMS-001`, `SFT-MATH-LOGIC-PROOF-001`, `SFT-INFO-QUANTITY-001`, `SFT-INFO-CHANNEL-CAPACITY-001`, `SFT-COMP-FORM-STATE-TRANSITION-001`, `SFT-COMP-SEM-CORRECTNESS-001`, `SFT-COMP-SEM-COMPILATION-001`, `SFT-COMP-DIST-CAUSALITY-001`, `SFT-PHYS-MEAS-VALUE-RECORD-001`, `SFT-PHYS-MEAS-CALIBRATION-001`, `SFT-CHEM-MEAS-TRACEABILITY-001`, `SFT-MAT-SUST-LIFECYCLE-001`, `SFT-BIO-BIO-HANDOFF-001`, `SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001`, `SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001`, `SFT-EARTH-HAZARD-RISK-HANDOFF-001`, `SFT-ASTRO-ASTRO-HANDOFF-001`, `SFT-SOCIAL-ENGINEERING-HANDOFF-001`, `SFT-ENG-DEFENCE-IN-DEPTH-001`, `SFT-ENG-VERIFICATION-001`. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external standard, application, performance target, product or prior implementation to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external standards, products, tests or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite components, versions, platforms, tests and lifecycle records; performance remains artifact, environment, user, method and operating-boundary limited.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - which requirement, artifact, component, user, environment or test bears the claim: `system-use-context-user-evidence-carrier`.
- **boundary** - which operating envelope, version, platform, user and lifecycle stage bound the result: `validation-use-bounded`.
- **relation** - which generated transformation, interface, comparison or handoff connects the carriers: `implemented-system-to-intended-use-comparison`.
- **record** - which requirements, versions, alternatives, failures, uncertainty and rollback remain reconstructible: `users-context-tasks-results-harms-limitations-and-review-held`.
- **evidence** - whether law, requirement, design, simulation, test, demonstration, performance and anomaly remain distinct: `science-design-test-and-decision-classes-explicit`.
- **provenance** - whether the result traces to the One and names every upstream receipt and design choice: `root-bound-forward-translation`.
- **generality** - whether finite extension preserves platforms, versions, failures and operating limits: `positive-finite-extension-preserves-platforms-failures-and-versions`.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: `no-extra-rule`.

Their ordered product gives the unique survivor:

```
system-use-context-user-evidence-carrier__validation-use-bounded__implemented-system-
to-intended-use-comparison__users-context-tasks-results-harms-limitations-and-review-
held__science-design-test-and-decision-classes-explicit__root-bound-forward-translati
on__positive-finite-extension-preserves-platforms-failures-and-versions__no-extra-rul
e
```

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses the requirement, artifact, user or environment. Erasing the boundary universalises a version, platform or test. Importing the relation lets a target or preferred application choose the result. Retaining only successful output removes requirements, alternatives, failures and rollback. Conflating evidence classes turns simulation, demonstration or performance into observation or law. Breaking provenance removes the trace to the One and upstream receipts. Erasing finite context makes one implementation universal. Adding a silent dependency or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite requirement or admitted law, versioned artifact, boundary, testable relation and complete record. The successor adds one lawful component, interface, platform, requirement, test, failure, lifecycle stage or anomaly while preserving all earlier identities, adverse rows and receipts.

Falsification and deliberately unfavourable controls

The registered falsification condition is: Reject if verification alone proves fitness for every user or deployment.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier

- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: `source_bound_engineering_requirement_design_test_lifecycle_or_access_record`/ `primary_standard_handbook_guidance_or_project_record_structural_correspondence`. Captured registered sources: 3. Unresolved transports retained: 0. The registered structural consequence corresponded: `True`. External evidence selected the survivor: `false`. Application or performance selected the law: `false`. Successful performance was relabelled scientific proof: `false`. Failed implementation was relabelled scientific falsification: `false`. Simulation or demonstration was relabelled observation: `false`.

- `NASA-SE-APPENDIX-001` - transport `captured`; captured bytes `578114`; snapshot `sha256:c1db2db970b9fd14a7ef98aa0c23575326dedae46cc7cc56c468821ca89edd60`.
- `NIST-TRACEABILITY-001` - transport `captured`; captured bytes `180987`; snapshot `sha256:4e79d3a99a9367842daf0f8910cff625de901d95b4b46c5fe2845d7512e70cd7`.
- `REPRODUCIBLE-BUILDS-001` - transport `captured`; captured bytes `27924`; snapshot `sha256:857885a380c3f0b545ae0b69e99c3f061b60ea7e3e99e36d1edcbc3e44bada73`.

The comparison is structural and source-bound. It checks that a relevant primary handbook, standard, guidance record or open engineering project can represent the declared requirement, artifact, environment, operating boundary, version, test and lifecycle without rewriting the Fold law. It does not claim that a standard proves conformity, that one handbook exhausts Engineering, or that a custodian's reputation proves the result. New records can extend or challenge the surface only with source, version, configuration, method, failures and adverse rows retained.

Scientific meaning

For **engineering validation**, the consequence is practical and exact: Validation tests whether the verified system serves its declared use in its declared context. An Engineering claim therefore has to expose the carrier `system-use-context-user-evidence-carrier`, the relation `implemented-system-to-intended-use-comparison`, the record `users-context-tasks-results-harms-limitations-and-review-held`, and the boundary `validation-use-bounded` together. If any one is hidden, the result may remain a concept, prototype, simulation, demonstration or product choice, but it is not this admitted law.

This distinction protects users, maintainers and science. It prevents a successful output from erasing requirements, failed alternatives, hidden resources, inaccessible users, hazards, maintenance burdens or end-of-life costs. It prevents an application from selecting the upstream law and prevents engineering failure from becoming automatic scientific falsification. A replacement must still generate its forms, eliminate alternatives, survive controls, preserve custody and reproduce through the shared engine.

Machine certificate

Candidate count: `256`; survivor count: `1`; derivation seal: `sha256:146ef345e2b71dc5eecd3a819b266e3d8340456ba89d366141d02c5816e36b04`; independent certificate: `sha256:5c14201360fb247ed31d48993c6c80eb4daeecfda61a8d24b64e3baddea0b822`; engine receipt: `sha256:a8c982119f3bb9f4887036daa9f5d02ccc6b15a438b7cbf6ee9cad7d12a68f22`. The independent implementation recomputed the candidate product: `True`. All external rows were preserved: `True`.

45. Bidirectional traceability

Claim: `SFT-ENG-TRACEABILITY-001`

Family: `verification_validation_traceability`

Exact statement: Traceability connects need, requirement, design, implementation, test and result in both directions.

Forward chain and exact carrier

The generated carrier is `need-requirement-design-test-result-carrier`. Its required relation is `bidirectional-evidence-chain-relation`. The reconstruction must retain `source-versions-links-changes-gaps-and-owners-held`. Its declared empirical boundary is `trace-baseline-bounded`. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is `SFT-FOUNDATION-ONE-001`, `SFT-FOUNDATION-FOLD-001`, `SFT-FOUNDATION-FOLD-DYNAMICS-001`, `SFT-MATH-EXACT-ARITHMETIC-001`, `SFT-MATH-GRAPH-NETWORK-001`, `SFT-MATH-OPTIMIZATION-001`, `SFT-MATH-DYNAMICAL-SYSTEMS-001`, `SFT-MATH-LOGIC-PROOF-001`, `SFT-INFO-QUANTITY-001`, `SFT-INFO-CHANNEL-CAPACITY-001`, `SFT-COMP-FORM-STATE-TRANSITION-001`, `SFT-COMP-SEM-CORRECTNESS-001`, `SFT-COMP-SEM-COMPILATION-001`, `SFT-COMP-DIST-CAUSALITY-001`, `SFT-PHYS-MEAS-VALUE-RECORD-001`, `SFT-PHYS-MEAS-CALIBRATION-001`, `SFT-CHEM-MEAS-TRACEABILITY-001`, `SFT-MAT-SUST-LIFECYCLE-001`, `SFT-BIO-BIO-HANDOFF-001`, `SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001`, `SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001`, `SFT-EARTH-HAZARD-RISK-HANDOFF-001`, `SFT-ASTRO-ASTRO-HANDOFF-001`, `SFT-SOCIAL-ENGINEERING-HANDOFF-001`, `SFT-ENG-DEFENCE-IN-DEPTH-001`, `SFT-ENG-VALIDATION-001`. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external standard, application, performance target, product or prior implementation to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external standards, products, tests or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite components, versions, platforms, tests and lifecycle records; performance remains artifact, environment, user, method and operating-boundary limited.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - which requirement, artifact, component, user, environment or test bears the claim: `need-requirement-design-test-result-carrier`.
- **boundary** - which operating envelope, version, platform, user and lifecycle stage bound the result: `trace-baseline-bounded`.
- **relation** - which generated transformation, interface, comparison or handoff connects the carriers: `bidirectional-evidence-chain-relation`.
- **record** - which requirements, versions, alternatives, failures, uncertainty and rollback remain reconstructible: `source-versions-links-changes-gaps-and-owners-held`.
- **evidence** - whether law, requirement, design, simulation, test, demonstration, performance and anomaly remain distinct: `science-design-test-and-decision-classes-explicit`.
- **provenance** - whether the result traces to the One and names every upstream receipt and design choice: `root-bound-forward-translation`.
- **generality** - whether finite extension preserves platforms, versions, failures and operating limits: `positive-finite-extension-preserves-platforms-failures-and-versions`.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: `no-extra-rule`.

Their ordered product gives the unique survivor:

```
need-requirement-design-test-result-carrier__trace-baseline-bounded__bidirectional-ev
idence-chain-relation__source-versions-links-changes-gaps-and-owners-held__science-de
```

`sign-test-and-decision-classes-explicit__root-bound-forward-translation__positive-finite-extension-preserves-platforms-failures-and-versions__no-extra-rule`

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses the requirement, artifact, user or environment. Erasing the boundary universalises a version, platform or test. Importing the relation lets a target or preferred application choose the result. Retaining only successful output removes requirements, alternatives, failures and rollback. Conflating evidence classes turns simulation, demonstration or performance into observation or law. Breaking provenance removes the trace to the One and upstream receipts. Erasing finite context makes one implementation universal. Adding a silent dependency or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite requirement or admitted law, versioned artifact, boundary, testable relation and complete record. The successor adds one lawful component, interface, platform, requirement, test, failure, lifecycle stage or anomaly while preserving all earlier identities, adverse rows and receipts.

Falsification and deliberately unfavourable controls

The registered falsification condition is: Reject if an orphan requirement, untested design element or unowned result is admitted.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: `source_bound_engineering_requirement_design_test_lifecycle_or_access_record/primary_standard_handbook_guidance_or_project_record_structural_correspondence`. Captured registered sources: 3. Unresolved transports retained: 0. The registered structural consequence corresponded: `True`. External evidence selected the survivor: `false`. Application or performance selected the law: `false`. Successful performance was relabelled scientific proof: `false`. Failed implementation was relabelled scientific falsification: `false`. Simulation or demonstration was relabelled observation: `false`.

- `NASA-SE-APPENDIX-001` - transport `captured`; captured bytes `578114`; snapshot `sha256:c1db2db970b9fd14a7ef98aa0c23575326dedae46cc7cc56c468821ca89edd60`.
- `NIST-TRACEABILITY-001` - transport `captured`; captured bytes `180987`; snapshot `sha256:4e79d3a99a9367842daf0f8910cff625de901d95b4b46c5fe2845d7512e70cd7`.
- `REPRODUCIBLE-BUILDS-001` - transport `captured`; captured bytes `27924`; snapshot `sha256:857885a380c3f0b545ae0b69e99c3f061b60ea7e3e99e36d1edcbc3e44bada73`.

The comparison is structural and source-bound. It checks that a relevant primary handbook, standard, guidance record or open engineering project can represent the declared requirement, artifact, environment, operating boundary, version, test and lifecycle without rewriting the Fold law. It does not claim that a standard proves conformity, that one handbook exhausts Engineering, or that a custodian's reputation proves the result. New records can extend or challenge the surface only with source, version, configuration, method, failures and adverse rows retained.

Scientific meaning

For **bidirectional traceability**, the consequence is practical and exact: Traceability connects need, requirement, design, implementation, test and result in both directions. An Engineering claim therefore has to expose the carrier `need-requirement-design-test-result-carrier`, the relation `bidirectional-evidence-chain-relation`, the record `source-versions-links-changes-gaps-and-owners-held`, and the boundary `trace-baseline-bounded` together. If any one is hidden, the result may remain a concept, prototype, simulation, demonstration or product choice, but it is not this admitted law.

This distinction protects users, maintainers and science. It prevents a successful output from erasing requirements, failed alternatives, hidden resources, inaccessible users, hazards, maintenance burdens or end-of-life costs. It prevents an application from selecting the upstream law and prevents engineering failure from becoming automatic scientific falsification. A replacement must still generate its forms, eliminate alternatives, survive controls, preserve custody and reproduce through the shared engine.

Machine certificate

Candidate count: `256`; survivor count: `1`; derivation seal: `sha256:894e7728a471c3243a508807d1fd92c5dee12a751d4312a7a8f483884b45dcd1`; independent certificate: `sha256:4a3bd1d480b9beead34e254f2828157516cb95c61c5050aa971087cc473e8a7b`; engine receipt: `sha256:3c212e3f3108cf8129b362ef95fbdd6a044c441058748f8be3d50fbee1da7d52`. The independent implementation recomputed the candidate product: `True`. All external rows were preserved: `True`.

46. Engineering reproducibility

Claim: `SFT-ENG-REPRODUCIBILITY-001`

Family: `verification_validation_traceability`

Exact statement: Reproducibility retains source, dependencies, environment, commands, outputs, failures and identities.

Forward chain and exact carrier

The generated carrier is `artifact-environment-procedure-result-carrier`. Its required relation is `declared-rebuild-to-comparable-result-relation`. The reconstruction must retain `source-dependencies-platform-commands-outputs-failures-and-hashes-held`. Its declared empirical boundary is `reproduction-environment-bounded`. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is `SFT-FOUNDATION-ONE-001`, `SFT-FOUNDATION-FOLD-001`, `SFT-FOUNDATION-FOLD-DYNAMICS-001`, `SFT-MATH-EXACT-ARITHMETIC-001`, `SFT-MATH-GRAPH-NETWORK-001`, `SFT-MATH-OPTIMIZATION-001`, `SFT-MATH-DYNAMICAL-SYSTEMS-001`, `SFT-MATH-LOGIC-PROOF-001`, `SFT-INFO-QUANTITY-001`, `SFT-INFO-CHANNEL-CAPACITY-001`, `SFT-COMP-FORM-STATE-TRANSITION-001`, `SFT-COMP-SEM-CORRECTNESS-001`, `SFT-COMP-SEM-COMPILATION-001`, `SFT-COMP-DIST-CAUSALITY-001`, `SFT-PHYS-MEAS-VALUE-RECORD-001`, `SFT-PHYS-MEAS-CALIBRATION-001`, `SFT-CHEM-MEAS-TRACEABILITY-001`, `SFT-MAT-SUST-LIFECYCLE-001`, `SFT-BIO-BIO-HANDOFF-001`, `SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001`, `SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001`, `SFT-EARTH-HAZARD-RISK-HANDOFF-001`, `SFT-ASTRO-ASTRO-HANDOFF-001`, `SFT-SOCIAL-ENGINEERING-HANDOFF-001`, `SFT-ENG-DEFENCE-IN-DEPTH-001`, `SFT-ENG-TRACEABILITY-001`. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external standard, application, performance target, product or prior implementation to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external standards, products, tests or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite components, versions, platforms, tests and lifecycle records; performance remains artifact, environment, user, method and operating-boundary limited.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - which requirement, artifact, component, user, environment or test bears the claim: `artifact-environment-procedure-result-carrier`.
- **boundary** - which operating envelope, version, platform, user and lifecycle stage bound the result: `reproduction-environment-bounded`.
- **relation** - which generated transformation, interface, comparison or handoff connects the carriers: `declared-rebuild-to-comparable-result-relation`.
- **record** - which requirements, versions, alternatives, failures, uncertainty and rollback remain reconstructible: `source-dependencies-platform-commands-outputs-failures-and-hashes-held`.
- **evidence** - whether law, requirement, design, simulation, test, demonstration, performance and anomaly remain distinct: `science-design-test-and-decision-classes-explicit`.
- **provenance** - whether the result traces to the One and names every upstream receipt and design choice: `root-bound-forward-translation`.
- **generality** - whether finite extension preserves platforms, versions, failures and operating limits: `positive-finite-extension-preserves-platforms-failures-and-versions`.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: `no-extra-rule`.

Their ordered product gives the unique survivor:

```
artifact-environment-procedure-result-carrier__reproduction-environment-bounded__decl
ared-rebuild-to-comparable-result-relation__source-dependencies-platform-commands-out
puts-failures-and-hashes-held__science-design-test-and-decision-classes-explicit__roo
t-bound-forward-translation__positive-finite-extension-preserves-platforms-failures-a
nd-versions__no-extra-rule
```

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses the requirement, artifact, user or environment. Erasing the boundary universalises a version, platform or test. Importing the relation lets a target or preferred application choose the result. Retaining only successful output removes requirements, alternatives, failures and rollback. Conflating evidence classes turns simulation, demonstration or performance into observation or law. Breaking provenance removes the trace to the One and upstream receipts. Erasing finite context makes one implementation universal. Adding a silent dependency or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite requirement or admitted law, versioned artifact, boundary, testable relation and complete record. The successor adds one lawful component, interface, platform, requirement, test, failure, lifecycle stage or anomaly while preserving all earlier identities, adverse rows and receipts.

Falsification and deliberately unfavourable controls

The registered falsification condition is: Reject if a private environment, missing dependency or unreconstructible manual step is required.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: `source_bound_engineering_requirement_design_test_lifecycle_or_access_record/primary_standard_handbook_guidance_or_project_record_structural_correspondence`. Captured registered sources: 3. Unresolved transports retained: 0. The registered structural consequence corresponded: `True`. External evidence selected the survivor: `false`. Application or performance selected the law: `false`. Successful performance was relabelled scientific proof: `false`. Failed implementation was relabelled scientific falsification: `false`. Simulation or demonstration was relabelled observation: `false`.

- `NASA-SE-APPENDIX-001` - transport `captured`; captured bytes `578114`; snapshot `sha256:c1db2db970b9fd14a7ef98aa0c23575326dedae46cc7cc56c468821ca89edd60`.
- `NIST-TRACEABILITY-001` - transport `captured`; captured bytes `180987`; snapshot `sha256:4e79d3a99a9367842daf0f8910cff625de901d95b4b46c5fe2845d7512e70cd7`.
- `REPRODUCIBLE-BUILDS-001` - transport `captured`; captured bytes `27924`; snapshot `sha256:857885a380c3f0b545ae0b69e99c3f061b60ea7e3e99e36d1edcbc3e44bada73`.

The comparison is structural and source-bound. It checks that a relevant primary handbook, standard, guidance record or open engineering project can represent the declared requirement, artifact, environment, operating boundary, version, test and lifecycle without rewriting the Fold law. It does not claim that a standard proves conformity, that one handbook exhausts Engineering, or that a custodian's reputation proves the result. New records can extend or challenge the surface only with source, version, configuration, method, failures and adverse rows retained.

Scientific meaning

For **engineering reproducibility**, the consequence is practical and exact: Reproducibility retains source, dependencies, environment, commands, outputs, failures and identities. An Engineering claim therefore has to expose the carrier `artifact-environment-procedure-result-carrier`, the relation `declared-rebuild-to-comparable-result-relation`, the record `source-dependencies-platform-commands-outputs-failures-and-hashes-held`, and the boundary `reproduction-environment-bounded` together. If any one is hidden, the result may remain a concept, prototype, simulation, demonstration or product choice, but it is not this admitted law.

This distinction protects users, maintainers and science. It prevents a successful output from erasing requirements, failed alternatives, hidden resources, inaccessible users, hazards, maintenance burdens or end-of-life costs. It prevents an application from selecting the upstream law and prevents engineering failure from becoming automatic scientific falsification. A replacement must still generate its forms, eliminate alternatives, survive controls, preserve custody and reproduce through the shared engine.

Machine certificate

Candidate count: `256`; survivor count: `1`; derivation seal: `sha256:0bbaa4cec59acc2966634a89cdf2cd017b1bf9e73d7cf1c9f1f92db4b59eea40`; independent certificate: `sha256:300c1d0fe6e0e23eb1c58debfcf99555e175980c29b699bc91a4528ce8088acc`; engine receipt: `sha256:ef072bf0eab7aa89aee2642c033dfc38adb51ca0ff69321dfbba4fa62dfcd276`. The independent implementation recomputed the candidate product: `True`. All external rows were preserved: `True`.

47. Implementation-distinct check

Claim: `SFT-ENG-INDEPENDENT-CHECK-001`

Family: `verification_validation_traceability`

Exact statement: An independent check reconstructs the result through a separately identified implementation path.

Forward chain and exact carrier

The generated carrier is `claim-first-implementation-second-implementation-carrier`. Its required relation is `independent-reconstruction-comparison`. The reconstruction must retain `specifications-codepaths-results-disagreements-and-provenance-held`. Its declared empirical boundary is `independence-criterion-bounded`. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is `SFT-FOUNDATION-ONE-001`, `SFT-FOUNDATION-FOLD-001`, `SFT-FOUNDATION-FOLD-DYNAMICS-001`, `SFT-MATH-EXACT-ARITHMETIC-001`, `SFT-MATH-GRAPH-NETWORK-001`, `SFT-MATH-OPTIMIZATION-001`, `SFT-MATH-DYNAMICAL-SYSTEMS-001`, `SFT-MATH-LOGIC-PROOF-001`, `SFT-INFO-QUANTITY-001`, `SFT-INFO-CHANNEL-CAPACITY-001`, `SFT-COMP-FORM-STATE-TRANSITION-001`, `SFT-COMP-SEM-CORRECTNESS-001`, `SFT-COMP-SEM-COMPILATION-001`, `SFT-COMP-DIST-CAUSALITY-001`, `SFT-PHYS-MEAS-VALUE-RECORD-001`, `SFT-PHYS-MEAS-CALIBRATION-001`, `SFT-CHEM-MEAS-TRACEABILITY-001`, `SFT-MAT-SUST-LIFECYCLE-001`, `SFT-BIO-BIO-HANDOFF-001`, `SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001`, `SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001`, `SFT-EARTH-HAZARD-RISK-HANDOFF-001`, `SFT-ASTRO-ASTRO-HANDOFF-001`, `SFT-SOCIAL-ENGINEERING-HANDOFF-001`, `SFT-ENG-DEFENCE-IN-DEPTH-001`, `SFT-ENG-REPRODUCIBILITY-001`. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external standard, application, performance target, product or prior implementation to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external standards, products, tests or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite components, versions, platforms, tests and lifecycle records; performance remains artifact, environment, user, method and operating-boundary limited.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - which requirement, artifact, component, user, environment or test bears the claim: `claim-first-implementation-second-implementation-carrier`.
- **boundary** - which operating envelope, version, platform, user and lifecycle stage bound the result: `independence-criterion-bounded`.
- **relation** - which generated transformation, interface, comparison or handoff connects the carriers: `independent-reconstruction-comparison`.
- **record** - which requirements, versions, alternatives, failures, uncertainty and rollback remain reconstructible: `specifications-codepaths-results-disagreements-and-provenance-held`.
- **evidence** - whether law, requirement, design, simulation, test, demonstration, performance and anomaly remain distinct: `science-design-test-and-decision-classes-explicit`.
- **provenance** - whether the result traces to the One and names every upstream receipt and design choice: `root-bound-forward-translation`.
- **generality** - whether finite extension preserves platforms, versions, failures and operating limits: `positive-finite-extension-preserves-platforms-failures-and-versions`.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: `no-extra-rule`.

Their ordered product gives the unique survivor:

```
claim-first-implementation-second-implementation-carrier__independence-criterion-boun
ded__independent-reconstruction-comparison__specifications-codepaths-results-disagree
ments-and-provenance-held__science-design-test-and-decision-classes-explicit__root-bo
und-forward-translation__positive-finite-extension-preserves-platforms-failures-and-v
ersions__no-extra-rule
```

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses the requirement, artifact, user or environment. Erasing the boundary universalises a version, platform or test. Importing the relation lets a target or preferred application choose the result. Retaining only successful output removes requirements, alternatives, failures and rollback. Conflating evidence classes turns simulation, demonstration or performance into observation or law. Breaking provenance removes the trace to the One and upstream receipts. Erasing finite context makes one implementation universal. Adding a silent

dependency or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite requirement or admitted law, versioned artifact, boundary, testable relation and complete record. The successor adds one lawful component, interface, platform, requirement, test, failure, lifecycle stage or anomaly while preserving all earlier identities, adverse rows and receipts.

Falsification and deliberately unfavourable controls

The registered falsification condition is: Reject if the checker imports the first implementation's survivor or output.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: `source_bound_engineering_requirement_design_test_lifecycle_or_access_record/primary_standard_handbook_guidance_or_project_record_structural_correspondence`. Captured registered sources: 3. Unresolved transports retained: 0. The registered structural consequence corresponded: `True`. External evidence selected the survivor: `false`. Application or performance selected the law: `false`. Successful performance was relabelled scientific proof: `false`. Failed implementation was relabelled scientific falsification: `false`. Simulation or demonstration was relabelled observation: `false`.

- `NASA-SE-APPENDIX-001` - transport `captured`; captured bytes `578114`; snapshot `sha256:c1db2db970b9fd14a7ef98aa0c23575326dedae46cc7cc56c468821ca89edd60`.
- `NIST-TRACEABILITY-001` - transport `captured`; captured bytes `180987`; snapshot `sha256:4e79d3a99a9367842daf0f8910cff625de901d95b4b46c5fe2845d7512e70cd7`.
- `REPRODUCIBLE-BUILDS-001` - transport `captured`; captured bytes `27924`; snapshot `sha256:857885a380c3f0b545ae0b69e99c3f061b60ea7e3e99e36d1edcbc3e44bada73`.

The comparison is structural and source-bound. It checks that a relevant primary handbook, standard, guidance record or open engineering project can represent the declared requirement, artifact, environment, operating boundary, version, test and lifecycle without rewriting the Fold law. It does not claim that a standard proves conformity, that one handbook exhausts Engineering, or that a custodian's reputation proves the result. New records can extend or challenge the surface only with source, version, configuration, method, failures and adverse rows retained.

Scientific meaning

For **implementation-distinct check**, the consequence is practical and exact: An independent check reconstructs the result through a separately identified implementation path. An Engineering claim therefore has to expose the carrier `claim-first-implementation-second-implementation-carrier`, the relation `independent-reconstruction-comparison`, the record `specifications-codepaths-results-disagreements-and-provenance-held`, and the boundary `independence-criterion-bounded` together. If any one is hidden, the result may remain a concept, prototype, simulation, demonstration or product choice, but it is not this admitted law.

This distinction protects users, maintainers and science. It prevents a successful output from erasing requirements, failed alternatives, hidden resources, inaccessible users, hazards, maintenance burdens or end-of-life costs. It prevents an application from selecting the upstream law and prevents engineering failure from becoming automatic scientific falsification. A replacement must still generate its forms, eliminate alternatives, survive controls, preserve custody and reproduce through the shared engine.

Machine certificate

Candidate count: `256`; survivor count: `1`; derivation seal: `sha256:0e6a9ee4932d30ab806b591a96944ae8e87bf980953b7dc6e6c6013199910f51`; independent certificate: `sha256:8b3142e9f00b43890f952520f2d4850d792fd37e64bb6a3cfa0fb3cdf86f4da2`; engine receipt: `sha256:927d1b0f871cb92e06ae180a9c1fbcf6b2fe0a024135fb487fbf41a9b082b364`. The independent implementation recomputed the candidate product: `True`. All external rows were preserved: `True`.

48. End-to-end verification

Claim: `SFT-ENG-E2E-001`

Family: `verification_validation_traceability`

Exact statement: End-to-end verification exercises the declared public entry through every stage to its final receipt.

Forward chain and exact carrier

The generated carrier is `source-input-complete-pipeline-output-carrier`. Its required relation is `declared-entry-to-final-result-relation`. The reconstruction must retain `versions-stages-intermediates-failures-duration-and-receipts-held`. Its declared empirical boundary is `pipeline-boundary-declared`. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is `SFT-FOUNDATION-ONE-001`, `SFT-FOUNDATION-FOLD-001`, `SFT-FOUNDATION-FOLD-DYNAMICS-001`, `SFT-MATH-EXACT-ARITHMETIC-001`, `SFT-MATH-GRAPH-NETWORK-001`, `SFT-MATH-OPTIMIZATION-001`, `SFT-MATH-DYNAMICAL-SYSTEMS-001`, `SFT-MATH-LOGIC-PROOF-001`, `SFT-INFO-QUANTITY-001`, `SFT-INFO-CHANNEL-CAPACITY-001`, `SFT-COMP-FORM-STATE-TRANSITION-001`, `SFT-COMP-SEM-CORRECTNESS-001`, `SFT-COMP-SEM-COMPILATION-001`, `SFT-COMP-DIST-CAUSALITY-001`, `SFT-PHYS-MEAS-VALUE-RECORD-001`, `SFT-PHYS-MEAS-CALIBRATION-001`, `SFT-CHEM-MEAS-TRACEABILITY-001`, `SFT-MAT-SUST-LIFECYCLE-001`, `SFT-BIO-BIO-HANDOFF-001`, `SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001`, `SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001`, `SFT-EARTH-HAZARD-RISK-HANDOFF-001`, `SFT-ASTRO-ASTRO-HANDOFF-001`, `SFT-SOCIAL-ENGINEERING-HANDOFF-001`, `SFT-ENG-DEFENCE-IN-DEPTH-001`, `SFT-ENG-INDEPENDENT-CHECK-001`. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external standard, application, performance target, product or prior implementation to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external standards, products, tests or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite components, versions, platforms, tests and lifecycle records; performance remains artifact, environment, user, method and operating-boundary limited.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - which requirement, artifact, component, user, environment or test bears the claim: `source-input-complete-pipeline-output-carrier`.
- **boundary** - which operating envelope, version, platform, user and lifecycle stage bound the result: `pipeline-boundary-declared`.
- **relation** - which generated transformation, interface, comparison or handoff connects the carriers: `declared-entry-to-final-result-relation`.
- **record** - which requirements, versions, alternatives, failures, uncertainty and rollback remain reconstructible: `versions-stages-intermediates-failures-duration-and-receipts-held`.

- **evidence** - whether law, requirement, design, simulation, test, demonstration, performance and anomaly remain distinct: `science-design-test-and-decision-classes-explicit`.
- **provenance** - whether the result traces to the One and names every upstream receipt and design choice: `root-bound-forward-translation`.
- **generality** - whether finite extension preserves platforms, versions, failures and operating limits: `positive-finite-extension-preserves-platforms-failures-and-versions`.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: `no-extra-rule`.

Their ordered product gives the unique survivor:

```
source-input-complete-pipeline-output-carrier__pipeline-boundary-declared__declared-e
ntry-to-final-result-relation__versions-stages-intermediates-failures-duration-and-re
ceipts-held__science-design-test-and-decision-classes-explicit__root-bound-forward-tr
anslation__positive-finite-extension-preserves-platforms-failures-and-versions__no-ex
tra-rule
```

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses the requirement, artifact, user or environment. Erasing the boundary universalises a version, platform or test. Importing the relation lets a target or preferred application choose the result. Retaining only successful output removes requirements, alternatives, failures and rollback. Conflating evidence classes turns simulation, demonstration or performance into observation or law. Breaking provenance removes the trace to the One and upstream receipts. Erasing finite context makes one implementation universal. Adding a silent dependency or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite requirement or admitted law, versioned artifact, boundary, testable relation and complete record. The successor adds one lawful component, interface, platform, requirement, test, failure, lifecycle stage or anomaly while preserving all earlier identities, adverse rows and receipts.

Falsification and deliberately unfavourable controls

The registered falsification condition is: Reject if mocks, placeholders, skipped stages or hidden manual corrections stand in for the pipeline.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: `source_bound_engineering_requirement_design_test_lifecycle_or_access_record` / `primary_standard_handbook_guidance_or_project_record_structural_correspondence`. Captured registered sources: 3. Unresolved transports retained: 0. The registered structural consequence corresponded: `True`. External evidence selected the survivor: `false`. Application or performance selected the law: `false`. Successful performance was relabelled scientific proof: `false`. Failed implementation was relabelled scientific falsification: `false`. Simulation or demonstration was relabelled observation: `false`.

- `NASA-SE-APPENDIX-001` - transport `captured`; captured bytes `578114`; snapshot `sha256:c1db2db970b9fd14a7ef98aa0c23575326dedae46cc7cc56c468821ca89edd60`.
- `NIST-TRACEABILITY-001` - transport `captured`; captured bytes `180987`; snapshot `sha256:4e79d3a99a9367842daf0f8910cff625de901d95b4b46c5fe2845d7512e70cd7`.

- REPRODUCIBLE-BUILDS-001 - transport captured; captured bytes 27924; snapshot sha256:857885a380c3f0b545ae0b69e99c3f061b60ea7e3e99e36d1edcbc3e44bada73.

The comparison is structural and source-bound. It checks that a relevant primary handbook, standard, guidance record or open engineering project can represent the declared requirement, artifact, environment, operating boundary, version, test and lifecycle without rewriting the Fold law. It does not claim that a standard proves conformity, that one handbook exhausts Engineering, or that a custodian's reputation proves the result. New records can extend or challenge the surface only with source, version, configuration, method, failures and adverse rows retained.

Scientific meaning

For **end-to-end verification**, the consequence is practical and exact: End-to-end verification exercises the declared public entry through every stage to its final receipt. An Engineering claim therefore has to expose the carrier `source-input-complete-pipeline-output-carrier`, the relation `declared-entry-to-final-result-relation`, the record `versions-stages-intermediates-failures-duration-and-receipts-held`, and the boundary `pipeline-boundary-declared` together. If any one is hidden, the result may remain a concept, prototype, simulation, demonstration or product choice, but it is not this admitted law.

This distinction protects users, maintainers and science. It prevents a successful output from erasing requirements, failed alternatives, hidden resources, inaccessible users, hazards, maintenance burdens or end-of-life costs. It prevents an application from selecting the upstream law and prevents engineering failure from becoming automatic scientific falsification. A replacement must still generate its forms, eliminate alternatives, survive controls, preserve custody and reproduce through the shared engine.

Machine certificate

Candidate count: `256`; survivor count: `1`; derivation seal: `sha256:54d4bc71108c987a3a0bc77e6b2414158de1243ad97227757e5b6271358ba31b`; independent certificate: `sha256:5643409a19a0e9b44cec82ff4854a757ebde298355c7dd05edf8052fd0923231`; engine receipt: `sha256:7a4acab46f4caa1d114918096648f62bb3a71347b35d2345e1a0353f1fa67b6e`. The independent implementation recomputed the candidate product: `True`. All external rows were preserved: `True`.

49. Cross-platform behaviour

Claim: `SFT-ENG-CROSS-PLATFORM-001`

Family: `cross_platform_accessibility`

Exact statement: Cross-platform evidence preserves macOS, Windows and Linux environments, results and platform-specific differences.

Forward chain and exact carrier

The generated carrier is `artifact-platform-environment-result-carrier`. Its required relation is `same-specification-across-platforms-relation`. The reconstruction must retain `platform-version-dependencies-commands-results-and-differences-held`. Its declared empirical boundary is `declared-platform-matrix`. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is `SFT-FOUNDATION-ONE-001`, `SFT-FOUNDATION-FOLD-001`, `SFT-FOUNDATION-FOLD-DYNAMICS-001`, `SFT-MATH-EXACT-ARITHMETIC-001`, `SFT-MATH-GRAPH-NETWORK-001`, `SFT-MATH-OPTIMIZATION-001`, `SFT-MATH-DYNAMICAL-SYSTEMS-001`, `SFT-MATH-LOGIC-PROOF-001`, `SFT-INFO-QUANTITY-001`, `SFT-INFO-CHANNEL-CAPACITY-001`, `SFT-COMP-FORM-STATE-TRANSITION-001`, `SFT-COMP-SEM-CORRECTNESS-001`, `SFT-COMP-SEM-COMPILATION-001`, `SFT-COMP-DIST-CAUSALITY-001`, `SFT-PHYS-MEAS-VALUE-RECORD-001`, `SFT-PHYS-MEAS-CALIBRATION-001`, `SFT-CHEM-MEAS-TRACEABILITY-001`, `SFT-MAT-SUST-LIFECYCLE-001`, `SFT-BIO-BIO-HANDOFF-001`, `SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001`, `SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001`, `SFT-EARTH-HAZARD-RISK-HANDOFF-001`, `SFT-ASTRO-ASTRO-HANDOFF-001`, `SFT-SOCIAL-ENGINEERING-HANDOFF-001`, `SFT-ENG-E2E-001`. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external standard, application, performance target, product or prior implementation to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external standards, products, tests or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite components, versions, platforms, tests and lifecycle records; performance remains artifact, environment, user, method and operating-boundary limited.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - which requirement, artifact, component, user, environment or test bears the claim: `artifact-platform-environment-result-carrier`.
- **boundary** - which operating envelope, version, platform, user and lifecycle stage bound the result: `declared-platform-matrix`.
- **relation** - which generated transformation, interface, comparison or handoff connects the carriers: `same-specification-across-platforms-relation`.
- **record** - which requirements, versions, alternatives, failures, uncertainty and rollback remain reconstructible: `platform-version-dependencies-commands-results-and-differences-held`.
- **evidence** - whether law, requirement, design, simulation, test, demonstration, performance and anomaly remain distinct: `science-design-test-and-decision-classes-explicit`.
- **provenance** - whether the result traces to the One and names every upstream receipt and design choice: `root-bound-forward-translation`.
- **generality** - whether finite extension preserves platforms, versions, failures and operating limits: `positive-finite-extension-preserves-platforms-failures-and-versions`.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: `no-extra-rule`.

Their ordered product gives the unique survivor:

```
artifact-platform-environment-result-carrier__declared-platform-matrix__same-specific
ation-across-platforms-relation__platform-version-dependencies-commands-results-and-d
ifferences-held__science-design-test-and-decision-classes-explicit__root-bound-forwar
d-translation__positive-finite-extension-preserves-platforms-failures-and-versions__n
o-extra-rule
```

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses the requirement, artifact, user or environment. Erasing the boundary universalises a version, platform or test. Importing the relation lets a target or preferred application choose the result. Retaining only successful output removes requirements, alternatives, failures and rollback. Conflating evidence classes turns simulation, demonstration or performance into observation or law. Breaking provenance removes the trace to the One and upstream receipts. Erasing finite context makes one implementation universal. Adding a silent dependency or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite requirement or admitted law, versioned artifact, boundary, testable relation and complete record. The successor adds one lawful component, interface, platform, requirement, test, failure, lifecycle stage or anomaly while preserving all earlier identities, adverse rows and receipts.

Falsification and deliberately unfavourable controls

The registered falsification condition is: Reject if success on one operating system proves portability.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity

- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: `source_bound_engineering_requirement_design_test_lifecycle_or_access_record/primary_standard_handbook_guidance_or_project_record_structural_correspondence`. Captured registered sources: 3. Unresolved transports retained: 0. The registered structural consequence corresponded: `True`. External evidence selected the survivor: `false`. Application or performance selected the law: `false`. Successful performance was relabelled scientific proof: `false`. Failed implementation was relabelled scientific falsification: `false`. Simulation or demonstration was relabelled observation: `false`.

- `W3C-WCAG22-001` - transport `captured`; captured bytes `512457`; snapshot `sha256:6e3c5fe397257cae509a2fb4752b73062cf8cbeb92c2cec618989b17e4cf7057`.
- `REPRODUCIBLE-BUILDS-001` - transport `captured`; captured bytes `27924`; snapshot `sha256:857885a380c3f0b545ae0b69e99c3f061b60ea7e3e99e36d1edcbc3e44bada73`.
- `PYTHON-PLATFORM-001` - transport `captured`; captured bytes `63923`; snapshot `sha256:6873ff23f5e2e6985050e60024a477265ed779867e32542e133879aae0e01611`.

The comparison is structural and source-bound. It checks that a relevant primary handbook, standard, guidance record or open engineering project can represent the declared requirement, artifact, environment, operating boundary, version, test and lifecycle without rewriting the Fold law. It does not claim that a standard proves conformity, that one handbook exhausts Engineering, or that a custodian's reputation proves the result. New records can extend or challenge the surface only with source, version, configuration, method, failures and adverse rows retained.

Scientific meaning

For **cross-platform behaviour**, the consequence is practical and exact: Cross-platform evidence preserves macOS, Windows and Linux environments, results and platform-specific differences. An Engineering claim therefore has to expose the carrier `artifact-platform-environment-result-carrier`, the relation `same-specification-across-platforms-relation`, the record `platform-version-dependencies-commands-results-and-differences-held`, and the boundary `declared-platform-matrix` together. If any one is hidden, the result may remain a concept, prototype, simulation, demonstration or product choice, but it is not this admitted law.

This distinction protects users, maintainers and science. It prevents a successful output from erasing requirements, failed alternatives, hidden resources, inaccessible users, hazards, maintenance burdens or end-of-life costs. It prevents an application from selecting the upstream law and prevents engineering failure from becoming automatic scientific falsification. A replacement must still generate its forms, eliminate alternatives, survive controls, preserve custody and reproduce through the shared engine.

Machine certificate

Candidate count: `256`; survivor count: `1`; derivation seal: `sha256:446d0e018481a446b77fba2cb20f3d387e20a2db8e409f784569a0fda9299383`; independent certificate: `sha256:fc1442b2e3a409638323217ce507ef73bdecf8c70582f277b1498587006a16c1`; engine receipt: `sha256:a009ad70a8663a428d971e9cfd37765344ed0cff220fb991a83801e955b66572`. The independent implementation recomputed the candidate product: `True`. All external rows were preserved: `True`.

50. Portable data and record format

Claim: `SFT-ENG-PORTABLE-DATA-001`

Family: `cross_platform_accessibility`

Exact statement: A portable record has an open schema, explicit encoding, version, units, errors and migration path.

Forward chain and exact carrier

The generated carrier is `record-format-reader-platform-carrier`. Its required relation is `serialized-record-to-equivalent-reading-relation`. The reconstruction must retain `schema-encoding-version-units-errors-and-migrations-held`. Its declared empirical boundary is `format-version-bounded`. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is `SFT-FOUNDATION-ONE-001`, `SFT-FOUNDATION-FOLD-001`, `SFT-FOUNDATION-FOLD-DYNAMICS-001`, `SFT-MATH-EXACT-ARITHMETIC-001`, `SFT-MATH-GRAPH-NETWORK-001`, `SFT-MATH-OPTIMIZATION-001`, `SFT-MATH-DYNAMICAL-SYSTEMS-001`, `SFT-MATH-LOGIC-PROOF-001`, `SFT-INFO-QUANTITY-001`, `SFT-INFO-CHANNEL-CAPACITY-001`, `SFT-COMP-FORM-STATE-TRANSITION-001`, `SFT-COMP-SEM-CORRECTNESS-001`, `SFT-COMP-SEM-COMPILATION-001`, `SFT-COMP-DIST-CAUSALITY-001`, `SFT-PHYS-MEAS-VALUE-RECORD-001`, `SFT-PHYS-MEAS-CALIBRATION-001`, `SFT-CHEM-MEAS-TRACEABILITY-001`, `SFT-MAT-SUST-LIFECYCLE-001`, `SFT-BIO-BIO-HANDOFF-001`, `SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001`, `SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001`, `SFT-EARTH-HAZARD-RISK-HANDOFF-001`, `SFT-ASTRO-ASTRO-HANDOFF-001`, `SFT-SOCIAL-ENGINEERING-HANDOFF-001`, `SFT-ENG-E2E-001`, `SFT-ENG-CROSS-PLATFORM-001`. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external standard, application, performance target, product or prior implementation to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external standards, products, tests or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite components, versions, platforms, tests and lifecycle records; performance remains artifact, environment, user, method and operating-boundary limited.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - which requirement, artifact, component, user, environment or test bears the claim: `record-format-reader-platform-carrier`.
- **boundary** - which operating envelope, version, platform, user and lifecycle stage bound the result: `format-version-bounded`.
- **relation** - which generated transformation, interface, comparison or handoff connects the carriers: `serialized-record-to-equivalent-reading-relation`.
- **record** - which requirements, versions, alternatives, failures, uncertainty and rollback remain reconstructible: `schema-encoding-version-units-errors-and-migrations-held`.
- **evidence** - whether law, requirement, design, simulation, test, demonstration, performance and anomaly remain distinct: `science-design-test-and-decision-classes-explicit`.
- **provenance** - whether the result traces to the One and names every upstream receipt and design choice: `root-bound-forward-translation`.
- **generality** - whether finite extension preserves platforms, versions, failures and operating limits: `positive-finite-extension-preserves-platforms-failures-and-versions`.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: `no-extra-rule`.

Their ordered product gives the unique survivor:

```
record-format-reader-platform-carrier__format-version-bounded__serialized-record-to-e
quivalent-reading-relation__schema-encoding-version-units-errors-and-migrations-held_
_science-design-test-and-decision-classes-explicit__root-bound-forward-translation__p
ositive-finite-extension-preserves-platforms-failures-and-versions__no-extra-rule
```

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses the requirement, artifact, user or environment. Erasing the boundary universalises a version, platform or test. Importing the relation lets a target or preferred application choose the result. Retaining only successful output removes requirements, alternatives, failures and rollback. Conflating evidence classes turns simulation, demonstration or performance into observation or law. Breaking provenance removes the trace to the One and upstream receipts. Erasing finite context makes one implementation universal. Adding a silent dependency or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite requirement or admitted law, versioned artifact, boundary, testable relation and complete record. The successor adds one lawful component, interface, platform, requirement, test, failure, lifecycle stage or anomaly while preserving all earlier identities, adverse rows and receipts.

Falsification and deliberately unfavourable controls

The registered falsification condition is: Reject if one proprietary application or unstated locale is required to interpret it.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: `source_bound_engineering_requirement_design_test_lifecycle_or_access_record/primary_standard_handbook_guidance_or_project_record_structural_correspondence`. Captured registered sources: 3. Unresolved transports retained: 0. The registered structural consequence corresponded: `True`. External evidence selected the survivor: `false`. Application or performance selected the law: `false`. Successful performance was relabelled scientific proof: `false`. Failed implementation was relabelled scientific falsification: `false`. Simulation or demonstration was relabelled observation: `false`.

- `W3C-WCAG22-001` - transport `captured`; captured bytes `512457`; snapshot `sha256:6e3c5fe397257cae509a2fb4752b73062cf8cbeb92c2cec618989b17e4cf7057`.
- `REPRODUCIBLE-BUILDS-001` - transport `captured`; captured bytes `27924`; snapshot `sha256:857885a380c3f0b545ae0b69e99c3f061b60ea7e3e99e36d1edcbc3e44bada73`.
- `PYTHON-PLATFORM-001` - transport `captured`; captured bytes `63923`; snapshot `sha256:6873ff23f5e2e6985050e60024a477265ed779867e32542e133879aae0e01611`.

The comparison is structural and source-bound. It checks that a relevant primary handbook, standard, guidance record or open engineering project can represent the declared requirement, artifact, environment, operating boundary, version, test and lifecycle without rewriting the Fold law. It does not claim that a standard proves conformity, that one handbook exhausts Engineering, or that a custodian's reputation proves the result. New records can extend or challenge the surface only with source, version, configuration, method, failures and adverse rows retained.

Scientific meaning

For **portable data and record format**, the consequence is practical and exact: A portable record has an open schema, explicit encoding, version, units, errors and migration path. An Engineering claim therefore has to expose the carrier `record-format-reader-platform-carrier`, the relation `serialized-record-to-equivalent-reading-relation`, the record `schema-encoding-version-units-errors-and-migrations-held`, and the boundary `format-version-bounded` together. If any one is hidden, the result may remain a concept, prototype, simulation, demonstration or product choice, but it is not this admitted law.

This distinction protects users, maintainers and science. It prevents a successful output from erasing requirements, failed alternatives, hidden resources, inaccessible users, hazards, maintenance burdens or end-of-life costs. It prevents an application from selecting the upstream law and prevents engineering failure from becoming automatic scientific falsification. A replacement must still generate its forms, eliminate alternatives, survive controls, preserve custody and reproduce through the shared engine.

Machine certificate

Candidate count: `256`; survivor count: `1`; derivation seal: `sha256:d1ca87f9037ae94cf344d13bf922c8904bbcb8f891e6b191beb40b1794693be2`; independent certificate: `sha256:6fa009edc910c4b7ccddce2f560aec809fa3989ef99fe902b5c2342a30d7d336`; engine receipt: `sha256:a34ca2385523846818968bfb4f0faf86b00b8ca6205291cd0ea54fa82bfe2d5d`. The independent implementation recomputed the candidate product: `True`. All external rows were preserved: `True`.

51. Accessible human interface

Claim: `SFT-ENG-ACCESSIBLE-INTERFACE-001`

Family: `cross_platform_accessibility`

Exact statement: An accessible interface retains user diversity, input and output modes, errors, guidance and direct testing.

Forward chain and exact carrier

The generated carrier is `user-task-interface-context-carrier`. Its required relation is `action-to-understandable-result-relation`. The reconstruction must retain `users-input-modes-output-modes-errors-guidance-and-testing-held`. Its declared empirical boundary is `user-and-task-bounded`. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is `SFT-FOUNDATION-ONE-001`, `SFT-FOUNDATION-FOLD-001`, `SFT-FOUNDATION-FOLD-DYNAMICS-001`, `SFT-MATH-EXACT-ARITHMETIC-001`, `SFT-MATH-GRAPH-NETWORK-001`, `SFT-MATH-OPTIMIZATION-001`, `SFT-MATH-DYNAMICAL-SYSTEMS-001`, `SFT-MATH-LOGIC-PROOF-001`, `SFT-INFO-QUANTITY-001`, `SFT-INFO-CHANNEL-CAPACITY-001`, `SFT-COMP-FORM-STATE-TRANSITION-001`, `SFT-COMP-SEM-CORRECTNESS-001`, `SFT-COMP-SEM-COMPILATION-001`, `SFT-COMP-DIST-CAUSALITY-001`, `SFT-PHYS-MEAS-VALUE-RECORD-001`, `SFT-PHYS-MEAS-CALIBRATION-001`, `SFT-CHEM-MEAS-TRACEABILITY-001`, `SFT-MAT-SUST-LIFECYCLE-001`, `SFT-BIO-BIO-HANDOFF-001`, `SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001`, `SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001`, `SFT-EARTH-HAZARD-RISK-HANDOFF-001`, `SFT-ASTRO-ASTRO-HANDOFF-001`, `SFT-SOCIAL-ENGINEERING-HANDOFF-001`, `SFT-ENG-E2E-001`, `SFT-ENG-PORTABLE-DATA-001`. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external standard, application, performance target, product or prior implementation to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external standards, products, tests or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite components, versions, platforms, tests and lifecycle records; performance remains artifact, environment, user, method and operating-boundary limited.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - which requirement, artifact, component, user, environment or test bears the claim: `user-task-interface-context-carrier`.
- **boundary** - which operating envelope, version, platform, user and lifecycle stage bound the result: `user-and-task-bounded`.
- **relation** - which generated transformation, interface, comparison or handoff connects the carriers: `action-to-understandable-result-relation`.

- **record** - which requirements, versions, alternatives, failures, uncertainty and rollback remain reconstructible: `users-input-modes-output-modes-errors-guidance-and-testing-held`.
- **evidence** - whether law, requirement, design, simulation, test, demonstration, performance and anomaly remain distinct: `science-design-test-and-decision-classes-explicit`.
- **provenance** - whether the result traces to the One and names every upstream receipt and design choice: `root-bound-forward-translation`.
- **generality** - whether finite extension preserves platforms, versions, failures and operating limits: `positive-finite-extension-preserves-platforms-failures-and-versions`.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: `no-extra-rule`.

Their ordered product gives the unique survivor:

```
user-task-interface-context-carrier__user-and-task-bounded__action-to-understandable-
result-relation__users-input-modes-output-modes-errors-guidance-and-testing-held__sci
ence-design-test-and-decision-classes-explicit__root-bound-forward-translation__posit
ive-finite-extension-preserves-platforms-failures-and-versions__no-extra-rule
```

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses the requirement, artifact, user or environment. Erasing the boundary universalises a version, platform or test. Importing the relation lets a target or preferred application choose the result. Retaining only successful output removes requirements, alternatives, failures and rollback. Conflating evidence classes turns simulation, demonstration or performance into observation or law. Breaking provenance removes the trace to the One and upstream receipts. Erasing finite context makes one implementation universal. Adding a silent dependency or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite requirement or admitted law, versioned artifact, boundary, testable relation and complete record. The successor adds one lawful component, interface, platform, requirement, test, failure, lifecycle stage or anomaly while preserving all earlier identities, adverse rows and receipts.

Falsification and deliberately unfavourable controls

The registered falsification condition is: Reject if developer usability or visual presentation alone proves accessibility.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: `source_bound_engineering_requirement_design_test_lifecycle_or_access_record/primary_standard_handbook_guidance_or_project_record_structural_correspondence`. Captured registered sources: 3. Unresolved transports retained: 0. The registered structural consequence corresponded: `True`. External evidence selected the survivor: `false`. Application or performance selected the law: `false`. Successful performance was relabelled scientific proof: `false`. Failed implementation was relabelled scientific falsification: `false`. Simulation or demonstration was relabelled observation: `false`.

- `W3C-WCAG22-001` - transport `captured`; captured bytes `512457`; snapshot `sha256:6e3c5fe397257cae509a2fb4752b73062cf8cbeb92c2cec618989b17e4cf7057`.
- `REPRODUCIBLE-BUILDS-001` - transport `captured`; captured bytes `27924`; snapshot `sha256:857885a380c3f0b545ae0b69e99c3f061b60ea7e3e99e36d1edcbc3e44bada73`.

- PYTHON-PLATFORM-001 - transport captured; captured bytes 63923; snapshot sha256:6873ff23f5e2e6985050e60024a477265ed779867e32542e133879aae0e01611.

The comparison is structural and source-bound. It checks that a relevant primary handbook, standard, guidance record or open engineering project can represent the declared requirement, artifact, environment, operating boundary, version, test and lifecycle without rewriting the Fold law. It does not claim that a standard proves conformity, that one handbook exhausts Engineering, or that a custodian's reputation proves the result. New records can extend or challenge the surface only with source, version, configuration, method, failures and adverse rows retained.

Scientific meaning

For **accessible human interface**, the consequence is practical and exact: An accessible interface retains user diversity, input and output modes, errors, guidance and direct testing. An Engineering claim therefore has to expose the carrier user-task-interface-context-carrier, the relation action-to-understandable-result-relation, the record users-input-modes-output-modes-errors-guidance-and-testing-held, and the boundary user-and-task-bounded together. If any one is hidden, the result may remain a concept, prototype, simulation, demonstration or product choice, but it is not this admitted law.

This distinction protects users, maintainers and science. It prevents a successful output from erasing requirements, failed alternatives, hidden resources, inaccessible users, hazards, maintenance burdens or end-of-life costs. It prevents an application from selecting the upstream law and prevents engineering failure from becoming automatic scientific falsification. A replacement must still generate its forms, eliminate alternatives, survive controls, preserve custody and reproduce through the shared engine.

Machine certificate

Candidate count: 256; survivor count: 1; derivation seal: sha256:90ab79cc923b1158e392b655e727308e98f5bb5f6618140203f37323d1494b38; independent certificate: sha256:8410ab5a8b244a9e33bc7249736e717f5099b4334fc86bc9ebcfcc294c5bddbb; engine receipt: sha256:239e2b3219d5815d69bdd260db4406dee4d0494204d6156600f9c15a348e0ddc. The independent implementation recomputed the candidate product: True. All external rows were preserved: True.

52. Lightweight deployment

Claim: SFT-ENG-LIGHTWEIGHT-DEPLOYMENT-001

Family: cross_platform_accessibility

Exact statement: Lightweight deployment declares and minimizes runtime dependencies, permissions, downloads and infrastructure.

Forward chain and exact carrier

The generated carrier is artifact-minimal-runtime-install-carrier. Its required relation is declared-package-to-working-system-relation. The reconstruction must retain runtime-dependencies-size-permissions-network-and-failures-held. Its declared empirical boundary is installation-environment-bounded. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-OPTIMIZATION-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-CHANNEL-CAPACITY-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-COMP-SEM-CORRECTNESS-001, SFT-COMP-SEM-COMPILATION-001, SFT-COMP-DIST-CAUSALITY-001, SFT-PHYS-MEAS-VALUE-RECORD-001, SFT-PHYS-MEAS-CALIBRATION-001, SFT-CHEM-MEAS-TRACEABILITY-001, SFT-MAT-SUST-LIFECYCLE-001, SFT-BIO-BIO-HANDOFF-001, SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001, SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-ASTRO-HANDOFF-001, SFT-SOCIAL-ENGINEERING-HANDOFF-001, SFT-ENG-E2E-001, SFT-ENG-ACCESSIBLE-INTERFACE-001. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external standard, application, performance target, product or prior implementation to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external standards, products, tests or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite components, versions, platforms, tests and lifecycle records; performance remains artifact, environment, user, method and operating-boundary limited.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - which requirement, artifact, component, user, environment or test bears the claim: `artifact-minimal-runtime-install-carrier`.
- **boundary** - which operating envelope, version, platform, user and lifecycle stage bound the result: `installation-environment-bounded`.
- **relation** - which generated transformation, interface, comparison or handoff connects the carriers: `declared-package-to-working-system-relation`.
- **record** - which requirements, versions, alternatives, failures, uncertainty and rollback remain reconstructible: `runtime-dependencies-size-permissions-network-and-failures-held`.
- **evidence** - whether law, requirement, design, simulation, test, demonstration, performance and anomaly remain distinct: `science-design-test-and-decision-classes-explicit`.
- **provenance** - whether the result traces to the One and names every upstream receipt and design choice: `root-bound-forward-translation`.
- **generality** - whether finite extension preserves platforms, versions, failures and operating limits: `positive-finite-extension-preserves-platforms-failures-and-versions`.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: `no-extra-rule`.

Their ordered product gives the unique survivor:

`artifact-minimal-runtime-install-carrier__installation-environment-bounded__declared-package-to-working-system-relation__runtime-dependencies-size-permissions-network-and-failures-held__science-design-test-and-decision-classes-explicit__root-bound-forward-translation__positive-finite-extension-preserves-platforms-failures-and-versions__no-extra-rule`

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses the requirement, artifact, user or environment. Erasing the boundary universalises a version, platform or test. Importing the relation lets a target or preferred application choose the result. Retaining only successful output removes requirements, alternatives, failures and rollback. Conflating evidence classes turns simulation, demonstration or performance into observation or law. Breaking provenance removes the trace to the One and upstream receipts. Erasing finite context makes one implementation universal. Adding a silent dependency or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite requirement or admitted law, versioned artifact, boundary, testable relation and complete record. The successor adds one lawful component, interface, platform, requirement, test, failure, lifecycle stage or anomaly while preserving all earlier identities, adverse rows and receipts.

Falsification and deliberately unfavourable controls

The registered falsification condition is: Reject if Docker, a cloud service, administrator access or a hidden heavyweight dependency is silently required.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor

- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: `source_bound_engineering_requirement_design_test_lifecycle_or_access_record/primary_standard_handbook_guidance_or_project_record_structural_correspondence`. Captured registered sources: 3. Unresolved transports retained: 0. The registered structural consequence corresponded: `True`. External evidence selected the survivor: `false`. Application or performance selected the law: `false`. Successful performance was relabelled scientific proof: `false`. Failed implementation was relabelled scientific falsification: `false`. Simulation or demonstration was relabelled observation: `false`.

- `W3C-WCAG22-001` - transport `captured`; captured bytes `512457`; snapshot `sha256:6e3c5fe397257cae509a2fb4752b73062cf8cbeb92c2cec618989b17e4cf7057`.
- `REPRODUCIBLE-BUILDS-001` - transport `captured`; captured bytes `27924`; snapshot `sha256:857885a380c3f0b545ae0b69e99c3f061b60ea7e3e99e36d1edcbc3e44bada73`.
- `PYTHON-PLATFORM-001` - transport `captured`; captured bytes `63923`; snapshot `sha256:6873ff23f5e2e6985050e60024a477265ed779867e32542e133879aae0e01611`.

The comparison is structural and source-bound. It checks that a relevant primary handbook, standard, guidance record or open engineering project can represent the declared requirement, artifact, environment, operating boundary, version, test and lifecycle without rewriting the Fold law. It does not claim that a standard proves conformity, that one handbook exhausts Engineering, or that a custodian's reputation proves the result. New records can extend or challenge the surface only with source, version, configuration, method, failures and adverse rows retained.

Scientific meaning

For **lightweight deployment**, the consequence is practical and exact: Lightweight deployment declares and minimizes runtime dependencies, permissions, downloads and infrastructure. An Engineering claim therefore has to expose the carrier `artifact-minimal-runtime-install-carrier`, the relation `declared-package-to-working-system-relation`, the record `runtime-dependencies-size-permissions-network-and-failures-held`, and the boundary `installation-environment-bounded` together. If any one is hidden, the result may remain a concept, prototype, simulation, demonstration or product choice, but it is not this admitted law.

This distinction protects users, maintainers and science. It prevents a successful output from erasing requirements, failed alternatives, hidden resources, inaccessible users, hazards, maintenance burdens or end-of-life costs. It prevents an application from selecting the upstream law and prevents engineering failure from becoming automatic scientific falsification. A replacement must still generate its forms, eliminate alternatives, survive controls, preserve custody and reproduce through the shared engine.

Machine certificate

Candidate count: `256`; survivor count: `1`; derivation seal: `sha256:bd9711e7754d472a4fbf72f098f9894e040352f0cc577f1176ee864c7d40878d`; independent certificate: `sha256:4fa8103bb65e258b60b537f47ba3d9d57f19deda8395d74af2a1beb5bb596bf0`; engine receipt: `sha256:abeb57586a62a61b447c72cac396427666a79fcf0cafaba3f0d9c3b29e72569f`. The independent implementation recomputed the candidate product: `True`. All external rows were preserved: `True`.

53. One-command public verification

Claim: `SFT-ENG-ONE-COMMAND-001`

Family: `cross_platform_accessibility`

Exact statement: A single documented command reaches the complete verification route and reports progress, duration, failures and receipt.

Forward chain and exact carrier

The generated carrier is `repository-public-command-result-carrier`. Its required relation is `single-entry-to-complete-verification-relation`. The reconstruction must retain `command-platform-stages-progress-duration-failures-and-receipt-held`. Its declared empirical boundary is `repository-version-bounded`. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is `SFT-FOUNDATION-ONE-001`, `SFT-FOUNDATION-FOLD-001`, `SFT-FOUNDATION-FOLD-DYNAMICS-001`, `SFT-MATH-EXACT-ARITHMETIC-001`, `SFT-MATH-GRAPH-NETWORK-001`, `SFT-MATH-OPTIMIZATION-001`, `SFT-MATH-DYNAMICAL-SYSTEMS-001`, `SFT-MATH-LOGIC-PROOF-001`, `SFT-INFO-QUANTITY-001`, `SFT-INFO-CHANNEL-CAPACITY-001`, `SFT-COMP-FORM-STATE-TRANSITION-001`, `SFT-COMP-SEM-CORRECTNESS-001`, `SFT-COMP-SEM-COMPILATION-001`, `SFT-COMP-DIST-CAUSALITY-001`, `SFT-PHYS-MEAS-VALUE-RECORD-001`, `SFT-PHYS-MEAS-CALIBRATION-001`, `SFT-CHEM-MEAS-TRACEABILITY-001`, `SFT-MAT-SUST-LIFECYCLE-001`, `SFT-BIO-BIO-HANDOFF-001`, `SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001`, `SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001`, `SFT-EARTH-HAZARD-RISK-HANDOFF-001`, `SFT-ASTRO-ASTRO-HANDOFF-001`, `SFT-SOCIAL-ENGINEERING-HANDOFF-001`, `SFT-ENG-E2E-001`, `SFT-ENG-LIGHTWEIGHT-DEPLOYMENT-001`. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external standard, application, performance target, product or prior implementation to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external standards, products, tests or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite components, versions, platforms, tests and lifecycle records; performance remains artifact, environment, user, method and operating-boundary limited.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - which requirement, artifact, component, user, environment or test bears the claim: `repository-public-command-result-carrier`.
- **boundary** - which operating envelope, version, platform, user and lifecycle stage bound the result: `repository-version-bounded`.
- **relation** - which generated transformation, interface, comparison or handoff connects the carriers: `single-entry-to-complete-verification-relation`.
- **record** - which requirements, versions, alternatives, failures, uncertainty and rollback remain reconstructible: `command-platform-stages-progress-duration-failures-and-receipt-held`.
- **evidence** - whether law, requirement, design, simulation, test, demonstration, performance and anomaly remain distinct: `science-design-test-and-decision-classes-explicit`.
- **provenance** - whether the result traces to the One and names every upstream receipt and design choice: `root-bound-forward-translation`.
- **generality** - whether finite extension preserves platforms, versions, failures and operating limits: `positive-finite-extension-preserves-platforms-failures-and-versions`.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: `no-extra-rule`.

Their ordered product gives the unique survivor:

```
repository-public-command-result-carrier__repository-version-bounded__single-entry-to
-complete-verification-relation__command-platform-stages-progress-duration-failures-a
nd-receipt-held__science-design-test-and-decision-classes-explicit__root-bound-forwar
d-translation__positive-finite-extension-preserves-platforms-failures-and-versions__n
o-extra-rule
```

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses the requirement, artifact, user or environment. Erasing the boundary universalises a version, platform or test. Importing the relation lets a target or preferred application choose the result. Retaining only successful output removes requirements, alternatives, failures and rollback. Conflating evidence classes turns simulation, demonstration or performance into observation or law. Breaking provenance removes the trace to the One and upstream receipts. Erasing finite context makes one implementation universal. Adding a silent dependency or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite requirement or admitted law, versioned artifact, boundary, testable relation and complete record. The successor adds one lawful component, interface, platform, requirement, test, failure, lifecycle stage or anomaly while preserving all earlier identities, adverse rows and receipts.

Falsification and deliberately unfavourable controls

The registered falsification condition is: Reject if the command skips claims, mutates authority or needs undisclosed setup.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: `source_bound_engineering_requirement_design_test_lifecycle_or_access_record/primary_standard_handbook_guidance_or_project_record_structural_correspondence`. Captured registered sources: 3. Unresolved transports retained: 0. The registered structural consequence corresponded: `True`. External evidence selected the survivor: `false`. Application or performance selected the law: `false`. Successful performance was relabelled scientific proof: `false`. Failed implementation was relabelled scientific falsification: `false`. Simulation or demonstration was relabelled observation: `false`.

- `W3C-WCAG22-001` - transport `captured`; captured bytes `512457`; snapshot `sha256:6e3c5fe397257cae509a2fb4752b73062cf8cbeb92c2cec618989b17e4cf7057`.
- `REPRODUCIBLE-BUILDS-001` - transport `captured`; captured bytes `27924`; snapshot `sha256:857885a380c3f0b545ae0b69e99c3f061b60ea7e3e99e36d1edcbc3e44bada73`.
- `PYTHON-PLATFORM-001` - transport `captured`; captured bytes `63923`; snapshot `sha256:6873ff23f5e2e6985050e60024a477265ed779867e32542e133879aae0e01611`.

The comparison is structural and source-bound. It checks that a relevant primary handbook, standard, guidance record or open engineering project can represent the declared requirement, artifact, environment, operating boundary, version, test and lifecycle without rewriting the Fold law. It does not claim that a standard proves conformity, that one handbook exhausts Engineering, or that a custodian's reputation proves the result. New records can extend or challenge the surface only with source, version, configuration, method, failures and adverse rows retained.

Scientific meaning

For **one-command public verification**, the consequence is practical and exact: A single documented command reaches the complete verification route and reports progress, duration, failures and receipt. An Engineering claim therefore has to expose the carrier `repository-public-command-result-carrier`, the relation `single-entry-to-complete-verification-relation`, the record `command-platform-stages-progress-duration-failures-and-receipt-held`, and the boundary `repository-version-bounded` together. If any one is hidden, the result may remain a concept, prototype, simulation, demonstration or product choice, but it is not this admitted law.

This distinction protects users, maintainers and science. It prevents a successful output from erasing requirements, failed alternatives, hidden resources, inaccessible users, hazards, maintenance burdens or end-of-life costs. It prevents an application from selecting the upstream law and prevents engineering failure from becoming automatic scientific falsification. A replacement must still generate its forms, eliminate alternatives, survive controls, preserve custody and reproduce through the shared engine.

Machine certificate

Candidate count: `256`; survivor count: `1`; derivation seal: `sha256:0cd6d8b1210b76b2ef3394a8bd6909fee70ddd42e424f9a2b9489feac8d5e991`; independent certificate: `sha256:982c366c05494e670b1e0a95fec753b00b68f7dde07ca63d05a014325bc87caa`; engine receipt: `sha256:94b593700a97b650125e459e4e75a3600542aed875d7d75d986183080e19d4c8`. The independent implementation recomputed the candidate product: `True`. All external rows were preserved: `True`.

54. Learnable public operation

Claim: `SFT-ENG-LEARNABLE-USE-001`

Family: `cross_platform_accessibility`

Exact statement: Learnable operation retains audience, prerequisites, progressive guidance, errors, examples and user tests.

Forward chain and exact carrier

The generated carrier is `new-user-task-guidance-feedback-carrier`. Its required relation is `instruction-and-feedback-to-correct-use-relation`. The reconstruction must retain `audience-prerequisites-steps-errors-examples-and-tests-held`. Its declared empirical boundary is `declared-audience-bounded`. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is `SFT-FOUNDATION-ONE-001`, `SFT-FOUNDATION-FOLD-001`, `SFT-FOUNDATION-FOLD-DYNAMICS-001`, `SFT-MATH-EXACT-ARITHMETIC-001`, `SFT-MATH-GRAPH-NETWORK-001`, `SFT-MATH-OPTIMIZATION-001`, `SFT-MATH-DYNAMICAL-SYSTEMS-001`, `SFT-MATH-LOGIC-PROOF-001`, `SFT-INFO-QUANTITY-001`, `SFT-INFO-CHANNEL-CAPACITY-001`, `SFT-COMP-FORM-STATE-TRANSITION-001`, `SFT-COMP-SEM-CORRECTNESS-001`, `SFT-COMP-SEM-COMPILATION-001`, `SFT-COMP-DIST-CAUSALITY-001`, `SFT-PHYS-MEAS-VALUE-RECORD-001`, `SFT-PHYS-MEAS-CALIBRATION-001`, `SFT-CHEM-MEAS-TRACEABILITY-001`, `SFT-MAT-SUST-LIFECYCLE-001`, `SFT-BIO-BIO-HANDOFF-001`, `SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001`, `SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001`, `SFT-EARTH-HAZARD-RISK-HANDOFF-001`, `SFT-ASTRO-ASTRO-HANDOFF-001`, `SFT-SOCIAL-ENGINEERING-HANDOFF-001`, `SFT-ENG-E2E-001`, `SFT-ENG-ONE-COMMAND-001`. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external standard, application, performance target, product or prior implementation to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external standards, products, tests or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite components, versions, platforms, tests and lifecycle records; performance remains artifact, environment, user, method and operating-boundary limited.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - which requirement, artifact, component, user, environment or test bears the claim: `new-user-task-guidance-feedback-carrier`.
- **boundary** - which operating envelope, version, platform, user and lifecycle stage bound the result: `declared-audience-bounded`.
- **relation** - which generated transformation, interface, comparison or handoff connects the carriers: `instruction-and-feedback-to-correct-use-relation`.

- **record** - which requirements, versions, alternatives, failures, uncertainty and rollback remain reconstructible: `audience-prerequisites-steps-errors-examples-and-tests-held`.
- **evidence** - whether law, requirement, design, simulation, test, demonstration, performance and anomaly remain distinct: `science-design-test-and-decision-classes-explicit`.
- **provenance** - whether the result traces to the One and names every upstream receipt and design choice: `root-bound-forward-translation`.
- **generality** - whether finite extension preserves platforms, versions, failures and operating limits: `positive-finite-extension-preserves-platforms-failures-and-versions`.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: `no-extra-rule`.

Their ordered product gives the unique survivor:

```
new-user-task-guidance-feedback-carrier__declared-audience-bounded__instruction-and-f
eedback-to-correct-use-relation__audience-prerequisites-steps-errors-examples-and-tes
ts-held__science-design-test-and-decision-classes-explicit__root-bound-forward-transl
ation__positive-finite-extension-preserves-platforms-failures-and-versions__no-extra-
rule
```

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses the requirement, artifact, user or environment. Erasing the boundary universalises a version, platform or test. Importing the relation lets a target or preferred application choose the result. Retaining only successful output removes requirements, alternatives, failures and rollback. Conflating evidence classes turns simulation, demonstration or performance into observation or law. Breaking provenance removes the trace to the One and upstream receipts. Erasing finite context makes one implementation universal. Adding a silent dependency or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite requirement or admitted law, versioned artifact, boundary, testable relation and complete record. The successor adds one lawful component, interface, platform, requirement, test, failure, lifecycle stage or anomaly while preserving all earlier identities, adverse rows and receipts.

Falsification and deliberately unfavourable controls

The registered falsification condition is: Reject if expert familiarity is treated as evidence that any person can use the system.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: `source_bound_engineering_requirement_design_test_lifecycle_or_access_record/primary_standard_handbook_guidance_or_project_record_structural_correspondence`. Captured registered sources: 3. Unresolved transports retained: 0. The registered structural consequence corresponded: `True`. External evidence selected the survivor: `false`. Application or performance selected the law: `false`. Successful performance was relabelled scientific proof: `false`. Failed implementation was relabelled scientific falsification: `false`. Simulation or demonstration was relabelled observation: `false`.

- `W3C-WCAG22-001` - transport `captured`; captured bytes `512457`; snapshot `sha256:6e3c5fe397257cae509a2fb4752b73062cf8cbeb92c2cec618989b17e4cf7057`.
- `REPRODUCIBLE-BUILDS-001` - transport `captured`; captured bytes `27924`; snapshot `sha256:857885a380c3f0b545ae0b69e99c3f061b60ea7e3e99e36d1edcbc3e44bada73`.

- `PYTHON-PLATFORM-001` - transport `captured`; captured bytes `63923`; snapshot `sha256:6873ff23f5e2e6985050e60024a477265ed779867e32542e133879aae0e01611`.

The comparison is structural and source-bound. It checks that a relevant primary handbook, standard, guidance record or open engineering project can represent the declared requirement, artifact, environment, operating boundary, version, test and lifecycle without rewriting the Fold law. It does not claim that a standard proves conformity, that one handbook exhausts Engineering, or that a custodian's reputation proves the result. New records can extend or challenge the surface only with source, version, configuration, method, failures and adverse rows retained.

Scientific meaning

For **learnable public operation**, the consequence is practical and exact: Learnable operation retains audience, prerequisites, progressive guidance, errors, examples and user tests. An Engineering claim therefore has to expose the carrier `new-user-task-guidance-feedback-carrier`, the relation `instruction-and-feedback-to-correct-use-relation`, the record `audience-prerequisites-steps-errors-examples-and-tests-held`, and the boundary `declared-audience-bounded` together. If any one is hidden, the result may remain a concept, prototype, simulation, demonstration or product choice, but it is not this admitted law.

This distinction protects users, maintainers and science. It prevents a successful output from erasing requirements, failed alternatives, hidden resources, inaccessible users, hazards, maintenance burdens or end-of-life costs. It prevents an application from selecting the upstream law and prevents engineering failure from becoming automatic scientific falsification. A replacement must still generate its forms, eliminate alternatives, survive controls, preserve custody and reproduce through the shared engine.

Machine certificate

Candidate count: `256`; survivor count: `1`; derivation seal: `sha256:8e38599fab255df33a0fbcaa554571a8032fcb98dacea7deeaaf99ba34f8de3f`; independent certificate: `sha256:8e01c7290626d086a5b9f1f89886cca54eb73e0e1e2cfddaff5ab71e02d00e5a`; engine receipt: `sha256:1e8205c6d619a0b70399e6a4c68a28a5df3997933a6ace803aa5e1f648f2a2d8`. The independent implementation recomputed the candidate product: `True`. All external rows were preserved: `True`.

55. Configuration identity

Claim: `SFT-ENG-CONFIGURATION-001`

Family: `lifecycle_maintenance_sustainability`

Exact statement: Configuration identity binds source, dependencies, settings, platform and hashes to a dated behavior.

Forward chain and exact carrier

The generated carrier is `artifact-version-dependency-environment-carrier`. Its required relation is `configuration-to-reproduced-behaviour-relation`. The reconstruction must retain `source-dependencies-settings-platform-hashes-and-date-held`. Its declared empirical boundary is `configuration-baseline-bounded`. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is `SFT-FOUNDATION-ONE-001`, `SFT-FOUNDATION-FOLD-001`, `SFT-FOUNDATION-FOLD-DYNAMICS-001`, `SFT-MATH-EXACT-ARITHMETIC-001`, `SFT-MATH-GRAPH-NETWORK-001`, `SFT-MATH-OPTIMIZATION-001`, `SFT-MATH-DYNAMICAL-SYSTEMS-001`, `SFT-MATH-LOGIC-PROOF-001`, `SFT-INFO-QUANTITY-001`, `SFT-INFO-CHANNEL-CAPACITY-001`, `SFT-COMP-FORM-STATE-TRANSITION-001`, `SFT-COMP-SEM-CORRECTNESS-001`, `SFT-COMP-SEM-COMPILATION-001`, `SFT-COMP-DIST-CAUSALITY-001`, `SFT-PHYS-MEAS-VALUE-RECORD-001`, `SFT-PHYS-MEAS-CALIBRATION-001`, `SFT-CHEM-MEAS-TRACEABILITY-001`, `SFT-MAT-SUST-LIFECYCLE-001`, `SFT-BIO-BIO-HANDOFF-001`, `SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001`, `SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001`, `SFT-EARTH-HAZARD-RISK-HANDOFF-001`, `SFT-ASTRO-ASTRO-HANDOFF-001`, `SFT-SOCIAL-ENGINEERING-HANDOFF-001`, `SFT-ENG-LEARNABLE-USE-001`. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external standard, application, performance target, product or prior implementation to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external standards, products, tests or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite components, versions, platforms, tests and lifecycle records; performance remains artifact, environment, user, method and operating-boundary limited.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - which requirement, artifact, component, user, environment or test bears the claim: `artifact-version-dependency-environment-carrier`.
- **boundary** - which operating envelope, version, platform, user and lifecycle stage bound the result: `configuration-baseline-bounded`.
- **relation** - which generated transformation, interface, comparison or handoff connects the carriers: `configuration-to-reproduced-behaviour-relation`.
- **record** - which requirements, versions, alternatives, failures, uncertainty and rollback remain reconstructible: `source-dependencies-settings-platform-hashes-and-date-held`.
- **evidence** - whether law, requirement, design, simulation, test, demonstration, performance and anomaly remain distinct: `science-design-test-and-decision-classes-explicit`.
- **provenance** - whether the result traces to the One and names every upstream receipt and design choice: `root-bound-forward-translation`.
- **generality** - whether finite extension preserves platforms, versions, failures and operating limits: `positive-finite-extension-preserves-platforms-failures-and-versions`.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: `no-extra-rule`.

Their ordered product gives the unique survivor:

```
artifact-version-dependency-environment-carrier__configuration-baseline-bounded__conf
iguration-to-reproduced-behaviour-relation__source-dependencies-settings-platform-has
hes-and-date-held__science-design-test-and-decision-classes-explicit__root-bound-forw
ard-translation__positive-finite-extension-preserves-platforms-failures-and-versions_
_no-extra-rule
```

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses the requirement, artifact, user or environment. Erasing the boundary universalises a version, platform or test. Importing the relation lets a target or preferred application choose the result. Retaining only successful output removes requirements, alternatives, failures and rollback. Conflating evidence classes turns simulation, demonstration or performance into observation or law. Breaking provenance removes the trace to the One and upstream receipts. Erasing finite context makes one implementation universal. Adding a silent dependency or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite requirement or admitted law, versioned artifact, boundary, testable relation and complete record. The successor adds one lawful component, interface, platform, requirement, test, failure, lifecycle stage or anomaly while preserving all earlier identities, adverse rows and receipts.

Falsification and deliberately unfavourable controls

The registered falsification condition is: Reject if mutable latest versions or hidden settings preserve the same identity.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: `source_bound_engineering_requirement_design_test_lifecycle_or_access_record/primary_standard_handbook_guidance_or_project_record_structural_correspondence`. Captured registered sources: 3. Unresolved transports retained: 0. The registered structural consequence corresponded: `True`. External evidence selected the survivor: `false`. Application or performance selected the law: `false`. Successful performance was relabelled scientific proof: `false`. Failed implementation was relabelled scientific falsification: `false`. Simulation or demonstration was relabelled observation: `false`.

- `EPA-LIFECYCLE-001` - transport `captured`; captured bytes `980`; snapshot `sha256:1d271583636a679563a046e767d0575a92d23eeafbf99f0da3fef2582e102c4e`.
- `NIST-SSDF-001` - transport `captured`; captured bytes `48313`; snapshot `sha256:d5a8c59b4e3dca64e07dfc15f83141517039120bedfd0aca8d46a5cba2b9fcb7`.
- `NIST-SSE-001` - transport `captured`; captured bytes `47984`; snapshot `sha256:64c8d41d9a78268d596a0a60aad18eead50fd375e8628486f7202788a207edaa`.

The comparison is structural and source-bound. It checks that a relevant primary handbook, standard, guidance record or open engineering project can represent the declared requirement, artifact, environment, operating boundary, version, test and lifecycle without rewriting the Fold law. It does not claim that a standard proves conformity, that one handbook exhausts Engineering, or that a custodian's reputation proves the result. New records can extend or challenge the surface only with source, version, configuration, method, failures and adverse rows retained.

Scientific meaning

For **configuration identity**, the consequence is practical and exact: Configuration identity binds source, dependencies, settings, platform and hashes to a dated behaviour. An Engineering claim therefore has to expose the carrier `artifact-version-dependency-environment-carrier`, the relation `configuration-to-reproduced-behaviour-relation`, the record `source-dependencies-settings-platform-hashes-and-date-held`, and the boundary `configuration-baseline-bounded` together. If any one is hidden, the result may remain a concept, prototype, simulation, demonstration or product choice, but it is not this admitted law.

This distinction protects users, maintainers and science. It prevents a successful output from erasing requirements, failed alternatives, hidden resources, inaccessible users, hazards, maintenance burdens or end-of-life costs. It prevents an application from selecting the upstream law and prevents engineering failure from becoming automatic scientific falsification. A replacement must still generate its forms, eliminate alternatives, survive controls, preserve custody and reproduce through the shared engine.

Machine certificate

Candidate count: `256`; survivor count: `1`; derivation seal: `sha256:2b19de499bdeb7b4351fe3214fc7b424fac18a3740e73d5b0ba8cad0a435aa19`; independent certificate: `sha256:f50139eda6869c7968a6001e30156d45011530d01444fb462813ecf0ecba9665`; engine receipt: `sha256:b447c208278ac2fef2764deafc27e128f167bedbb7e17211b5e6b528791847de`. The independent implementation recomputed the candidate product: `True`. All external rows were preserved: `True`.

56. Maintenance action

Claim: `SFT-ENG-MAINTENANCE-001`

Family: `lifecycle_maintenance_sustainability`

Exact statement: Maintenance retains diagnosis, parts, action, test, downtime and resulting configuration.

Forward chain and exact carrier

The generated carrier is `system-condition-action-result-carrier`. Its required relation is `observed-condition-to-restorative-action-relation`. The reconstruction must retain `condition-diagnosis-parts-action-tests-downtime-and-result-held`. Its declared empirical boundary is `maintenance-event-bounded`. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is `SFT-FOUNDATION-ONE-001`, `SFT-FOUNDATION-FOLD-001`, `SFT-FOUNDATION-FOLD-DYNAMICS-001`, `SFT-MATH-EXACT-ARITHMETIC-001`, `SFT-MATH-GRAPH-NETWORK-001`, `SFT-MATH-OPTIMIZATION-001`, `SFT-MATH-DYNAMICAL-SYSTEMS-001`, `SFT-MATH-LOGIC-PROOF-001`, `SFT-INFO-QUANTITY-001`, `SFT-INFO-CHANNEL-CAPACITY-001`, `SFT-COMP-FORM-STATE-TRANSITION-001`, `SFT-COMP-SEM-CORRECTNESS-001`, `SFT-COMP-SEM-COMPILATION-001`, `SFT-COMP-DIST-CAUSALITY-001`, `SFT-PHYS-MEAS-VALUE-RECORD-001`, `SFT-PHYS-MEAS-CALIBRATION-001`, `SFT-CHEM-MEAS-TRACEABILITY-001`, `SFT-MAT-SUST-LIFECYCLE-001`, `SFT-BIO-BIO-HANDOFF-001`, `SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001`, `SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001`, `SFT-EARTH-HAZARD-RISK-HANDOFF-001`, `SFT-ASTRO-ASTRO-HANDOFF-001`, `SFT-SOCIAL-ENGINEERING-HANDOFF-001`, `SFT-ENG-LEARNABLE-USE-001`, `SFT-ENG-CONFIGURATION-001`. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external standard, application, performance target, product or prior implementation to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external standards, products, tests or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite components, versions, platforms, tests and lifecycle records; performance remains artifact, environment, user, method and operating-boundary limited.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - which requirement, artifact, component, user, environment or test bears the claim: `system-condition-action-result-carrier`.
- **boundary** - which operating envelope, version, platform, user and lifecycle stage bound the result: `maintenance-event-bounded`.
- **relation** - which generated transformation, interface, comparison or handoff connects the carriers: `observed-condition-to-restorative-action-relation`.
- **record** - which requirements, versions, alternatives, failures, uncertainty and rollback remain reconstructible: `condition-diagnosis-parts-action-tests-downtime-and-result-held`.
- **evidence** - whether law, requirement, design, simulation, test, demonstration, performance and anomaly remain distinct: `science-design-test-and-decision-classes-explicit`.
- **provenance** - whether the result traces to the One and names every upstream receipt and design choice: `root-bound-forward-translation`.
- **generality** - whether finite extension preserves platforms, versions, failures and operating limits: `positive-finite-extension-preserves-platforms-failures-and-versions`.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: `no-extra-rule`.

Their ordered product gives the unique survivor:

```
system-condition-action-result-carrier__maintenance-event-bounded__observed-condition
-to-restorative-action-relation__condition-diagnosis-parts-action-tests-downtime-and-
result-held__science-design-test-and-decision-classes-explicit__root-bound-forward-tr
```

`anslation__positive-finite-extension-preserves-platforms-failures-and-versions__no-extra-rule`

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses the requirement, artifact, user or environment. Erasing the boundary universalises a version, platform or test. Importing the relation lets a target or preferred application choose the result. Retaining only successful output removes requirements, alternatives, failures and rollback. Conflating evidence classes turns simulation, demonstration or performance into observation or law. Breaking provenance removes the trace to the One and upstream receipts. Erasing finite context makes one implementation universal. Adding a silent dependency or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite requirement or admitted law, versioned artifact, boundary, testable relation and complete record. The successor adds one lawful component, interface, platform, requirement, test, failure, lifecycle stage or anomaly while preserving all earlier identities, adverse rows and receipts.

Falsification and deliberately unfavourable controls

The registered falsification condition is: Reject if maintenance history or induced failures are erased.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: `source_bound_engineering_requirement_design_test_lifecycle_or_access_record/primary_standard_handbook_guidance_or_project_record_structural_correspondence`. Captured registered sources: 3. Unresolved transports retained: 0. The registered structural consequence corresponded: `True`. External evidence selected the survivor: `false`. Application or performance selected the law: `false`. Successful performance was relabelled scientific proof: `false`. Failed implementation was relabelled scientific falsification: `false`. Simulation or demonstration was relabelled observation: `false`.

- `EPA-LIFECYCLE-001` - transport `captured`; captured bytes `980`; snapshot `sha256:1d271583636a679563a046e767d0575a92d23eeafbf99f0da3fef2582e102c4e`.
- `NIST-SSDF-001` - transport `captured`; captured bytes `48313`; snapshot `sha256:d5a8c59b4e3dca64e07dfc15f83141517039120bedfd0aca8d46a5cba2b9fcb7`.
- `NIST-SSE-001` - transport `captured`; captured bytes `47984`; snapshot `sha256:64c8d41d9a78268d596a0a60aad18eead50fd375e8628486f7202788a207edaa`.

The comparison is structural and source-bound. It checks that a relevant primary handbook, standard, guidance record or open engineering project can represent the declared requirement, artifact, environment, operating boundary, version, test and lifecycle without rewriting the Fold law. It does not claim that a standard proves conformity, that one handbook exhausts Engineering, or that a custodian's reputation proves the result. New records can extend or challenge the surface only with source, version, configuration, method, failures and adverse rows retained.

Scientific meaning

For **maintenance action**, the consequence is practical and exact: Maintenance retains diagnosis, parts, action, test, downtime and resulting configuration. An Engineering claim therefore has to expose the carrier `system-condition-action-result-carrier`, the relation `observed-condition-to-restorative-action-relation`, the record `condition-diagnosis-parts-action-tests-downtime-and-result-held`, and the boundary `maintenance-event-bounded` together. If any one is hidden, the result may remain a concept, prototype, simulation, demonstration or product choice, but it is not this admitted law.

This distinction protects users, maintainers and science. It prevents a successful output from erasing requirements, failed alternatives, hidden resources, inaccessible users, hazards, maintenance burdens or end-of-life costs. It prevents an application from selecting the upstream law and prevents engineering failure from becoming automatic scientific falsification. A replacement must still generate its forms, eliminate alternatives, survive controls, preserve custody and reproduce through the shared engine.

Machine certificate

Candidate count: `256`; survivor count: `1`; derivation seal: `sha256:420c6d9c76c2283cf0921f3794dcb0cd2165356ff1aabf7f4097f23a90e80b6b`; independent certificate: `sha256:36ead34650b33ee4614e670b4394ab206934c96493d5054480eb4411124aae92`; engine receipt: `sha256:c2373257572192e77d76ba6c52eff6123c52e8a90abaad8a4e17489c9b86db65`. The independent implementation recomputed the candidate product: `True`. All external rows were preserved: `True`.

57. Change impact

Claim: `SFT-ENG-CHANGE-IMPACT-001`

Family: `lifecycle_maintenance_sustainability`

Exact statement: Change impact traces a proposed modification through affected requirements, interfaces, hazards, tests and releases.

Forward chain and exact carrier

The generated carrier is `baseline-change-dependency-carrier`. Its required relation is `changed-element-to-affected-evidence-relation`. The reconstruction must retain `proposal-links-risks-tests-results-and-rollback-held`. Its declared empirical boundary is `dependency-graph-bounded`. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is `SFT-FOUNDATION-ONE-001`, `SFT-FOUNDATION-FOLD-001`, `SFT-FOUNDATION-FOLD-DYNAMICS-001`, `SFT-MATH-EXACT-ARITHMETIC-001`, `SFT-MATH-GRAPH-NETWORK-001`, `SFT-MATH-OPTIMIZATION-001`, `SFT-MATH-DYNAMICAL-SYSTEMS-001`, `SFT-MATH-LOGIC-PROOF-001`, `SFT-INFO-QUANTITY-001`, `SFT-INFO-CHANNEL-CAPACITY-001`, `SFT-COMP-FORM-STATE-TRANSITION-001`, `SFT-COMP-SEM-CORRECTNESS-001`, `SFT-COMP-SEM-COMPILATION-001`, `SFT-COMP-DIST-CAUSALITY-001`, `SFT-PHYS-MEAS-VALUE-RECORD-001`, `SFT-PHYS-MEAS-CALIBRATION-001`, `SFT-CHEM-MEAS-TRACEABILITY-001`, `SFT-MAT-SUST-LIFECYCLE-001`, `SFT-BIO-BIO-HANDOFF-001`, `SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001`, `SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001`, `SFT-EARTH-HAZARD-RISK-HANDOFF-001`, `SFT-ASTRO-ASTRO-HANDOFF-001`, `SFT-SOCIAL-ENGINEERING-HANDOFF-001`, `SFT-ENG-LEARNABLE-USE-001`, `SFT-ENG-MAINTENANCE-001`. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external standard, application, performance target, product or prior implementation to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external standards, products, tests or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite components, versions, platforms, tests and lifecycle records; performance remains artifact, environment, user, method and operating-boundary limited.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - which requirement, artifact, component, user, environment or test bears the claim: `baseline-change-dependency-carrier`.
- **boundary** - which operating envelope, version, platform, user and lifecycle stage bound the result: `dependency-graph-bounded`.
- **relation** - which generated transformation, interface, comparison or handoff connects the carriers: `changed-element-to-affected-evidence-relation`.
- **record** - which requirements, versions, alternatives, failures, uncertainty and rollback remain reconstructible: `proposal-links-risks-tests-results-and-rollback-held`.
- **evidence** - whether law, requirement, design, simulation, test, demonstration, performance and anomaly remain distinct: `science-design-test-and-decision-classes-explicit`.
- **provenance** - whether the result traces to the One and names every upstream receipt and design choice: `root-bound-forward-translation`.
- **generality** - whether finite extension preserves platforms, versions, failures and operating limits: `positive-finite-extension-preserves-platforms-failures-and-versions`.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: `no-extra-rule`.

Their ordered product gives the unique survivor:

```
baseline-change-dependency-carrier__dependency-graph-bounded__changed-element-to-affe
cted-evidence-relation__proposal-links-risks-tests-results-and-rollback-held__science
-design-test-and-decision-classes-explicit__root-bound-forward-translation__positive-
finite-extension-preserves-platforms-failures-and-versions__no-extra-rule
```

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses the requirement, artifact, user or environment. Erasing the boundary universalises a version, platform or test. Importing the relation lets a target or preferred application choose the result. Retaining only successful output removes requirements, alternatives, failures and rollback. Conflating evidence classes turns simulation, demonstration or performance into observation or law. Breaking provenance removes the trace to the One and upstream receipts. Erasing finite context makes one implementation universal. Adding a silent dependency or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite requirement or admitted law, versioned artifact, boundary, testable relation and complete record. The successor adds one lawful component, interface, platform, requirement, test, failure, lifecycle stage or anomaly while preserving all earlier identities, adverse rows and receipts.

Falsification and deliberately unfavourable controls

The registered falsification condition is: Reject if a changed dependency is declared irrelevant without trace evidence.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: source_bound_engineering_requirement_design_test_lifecycle_or_access_record/primary_standard_handbook_guidance_or_project_record_structural_correspondence. Captured registered sources: 3. Unresolved transports retained: 0. The registered structural consequence corresponded: True. External evidence selected the survivor: false. Application or performance selected the law: false. Successful performance was relabelled scientific proof: false. Failed implementation was relabelled scientific falsification: false. Simulation or demonstration was relabelled observation: false.

- EPA-LIFECYCLE-001 - transport captured; captured bytes 980; snapshot sha256:1d271583636a679563a046e767d0575a92d23eeafbf99f0da3fef2582e102c4e.
- NIST-SSDF-001 - transport captured; captured bytes 48313; snapshot sha256:d5a8c59b4e3dca64e07dfc15f83141517039120bedfd0aca8d46a5cba2b9fcb7.
- NIST-SSE-001 - transport captured; captured bytes 47984; snapshot sha256:64c8d41d9a78268d596a0a60aad18eead50fd375e8628486f7202788a207edaa.

The comparison is structural and source-bound. It checks that a relevant primary handbook, standard, guidance record or open engineering project can represent the declared requirement, artifact, environment, operating boundary, version, test and lifecycle without rewriting the Fold law. It does not claim that a standard proves conformity, that one handbook exhausts Engineering, or that a custodian's reputation proves the result. New records can extend or challenge the surface only with source, version, configuration, method, failures and adverse rows retained.

Scientific meaning

For **change impact**, the consequence is practical and exact: Change impact traces a proposed modification through affected requirements, interfaces, hazards, tests and releases. An Engineering claim therefore has to expose the carrier baseline-change-dependency-carrier, the relation changed-element-to-affected-evidence-relation, the record proposal-links-risks-tests-results-and-rollback-held, and the boundary dependency-graph-bounded together. If any one is hidden, the result may remain a concept, prototype, simulation, demonstration or product choice, but it is not this admitted law.

This distinction protects users, maintainers and science. It prevents a successful output from erasing requirements, failed alternatives, hidden resources, inaccessible users, hazards, maintenance burdens or end-of-life costs. It prevents an application from selecting the upstream law and prevents engineering failure from becoming automatic scientific falsification. A replacement must still generate its forms, eliminate alternatives, survive controls, preserve custody and reproduce through the shared engine.

Machine certificate

Candidate count: 256; survivor count: 1; derivation seal: sha256:e7b4b036ddd8cc533af1bac0e5f65563fd4008bcbcfcfe2dab113826d14695b7; independent certificate: sha256:517bcbb6233df287acc4e591f35963bb0aaa3a6c7d1c0b366e3853eed28cb9a7; engine receipt: sha256:30f84346b9018ccfa62f832883de26beb5bb7c39e9ebae32e5dc02db3daf9814. The independent implementation recomputed the candidate product: True. All external rows were preserved: True.

58. Decommissioning

Claim: SFT-ENG-DECOMMISSION-001

Family: lifecycle_maintenance_sustainability

Exact statement: Decommissioning retains authority, users, data, materials, hazards, recovery and archive custody.

Forward chain and exact carrier

The generated carrier is system-service-end-assets-record-carrier. Its required relation is active-to-retired-state-relation. The reconstruction must retain authority-users-data-materials-hazards-recovery-and-custody-held. Its declared empirical boundary is retirement-scope-bounded. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is `SFT-FOUNDATION-ONE-001`, `SFT-FOUNDATION-FOLD-001`, `SFT-FOUNDATION-FOLD-DYNAMICS-001`, `SFT-MATH-EXACT-ARITHMETIC-001`, `SFT-MATH-GRAPH-NETWORK-001`, `SFT-MATH-OPTIMIZATION-001`, `SFT-MATH-DYNAMICAL-SYSTEMS-001`, `SFT-MATH-LOGIC-PROOF-001`, `SFT-INFO-QUANTITY-001`, `SFT-INFO-CHANNEL-CAPACITY-001`, `SFT-COMP-FORM-STATE-TRANSITION-001`, `SFT-COMP-SEM-CORRECTNESS-001`, `SFT-COMP-SEM-COMPILATION-001`, `SFT-COMP-DIST-CAUSALITY-001`, `SFT-PHYS-MEAS-VALUE-RECORD-001`, `SFT-PHYS-MEAS-CALIBRATION-001`, `SFT-CHEM-MEAS-TRACEABILITY-001`, `SFT-MAT-SUST-LIFECYCLE-001`, `SFT-BIO-BIO-HANDOFF-001`, `SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001`, `SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001`, `SFT-EARTH-HAZARD-RISK-HANDOFF-001`, `SFT-ASTRO-ASTRO-HANDOFF-001`, `SFT-SOCIAL-ENGINEERING-HANDOFF-001`, `SFT-ENG-LEARNABLE-USE-001`, `SFT-ENG-CHANGE-IMPACT-001`. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external standard, application, performance target, product or prior implementation to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external standards, products, tests or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite components, versions, platforms, tests and lifecycle records; performance remains artifact, environment, user, method and operating-boundary limited.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - which requirement, artifact, component, user, environment or test bears the claim: `system-service-end-assets-record-carrier`.
- **boundary** - which operating envelope, version, platform, user and lifecycle stage bound the result: `retirement-scope-bounded`.
- **relation** - which generated transformation, interface, comparison or handoff connects the carriers: `active-to-retired-state-relation`.
- **record** - which requirements, versions, alternatives, failures, uncertainty and rollback remain reconstructible: `authority-users-data-materials-hazards-recovery-and-custody-held`.
- **evidence** - whether law, requirement, design, simulation, test, demonstration, performance and anomaly remain distinct: `science-design-test-and-decision-classes-explicit`.
- **provenance** - whether the result traces to the One and names every upstream receipt and design choice: `root-bound-forward-translation`.
- **generality** - whether finite extension preserves platforms, versions, failures and operating limits: `positive-finite-extension-preserves-platforms-failures-and-versions`.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: `no-extra-rule`.

Their ordered product gives the unique survivor:

```
system-service-end-assets-record-carrier__retirement-scope-bounded__active-to-retired
-state-relation__authority-users-data-materials-hazards-recovery-and-custody-held__sc
ience-design-test-and-decision-classes-explicit__root-bound-forward-translation__posi
tive-finite-extension-preserves-platforms-failures-and-versions__no-extra-rule
```

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses the requirement, artifact, user or environment. Erasing the boundary universalises a version, platform or test. Importing the relation lets a target or preferred application choose the result. Retaining only successful output removes requirements, alternatives, failures and rollback. Conflating evidence classes turns simulation, demonstration or performance into observation or law. Breaking provenance removes the trace to the One and upstream receipts. Erasing finite context makes one implementation universal. Adding a silent

dependency or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite requirement or admitted law, versioned artifact, boundary, testable relation and complete record. The successor adds one lawful component, interface, platform, requirement, test, failure, lifecycle stage or anomaly while preserving all earlier identities, adverse rows and receipts.

Falsification and deliberately unfavourable controls

The registered falsification condition is: Reject if disabling a service proves data, hazard or dependency removal.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: `source_bound_engineering_requirement_design_test_lifecycle_or_access_record/primary_standard_handbook_guidance_or_project_record_structural_correspondence`. Captured registered sources: 3. Unresolved transports retained: 0. The registered structural consequence corresponded: `True`. External evidence selected the survivor: `false`. Application or performance selected the law: `false`. Successful performance was relabelled scientific proof: `false`. Failed implementation was relabelled scientific falsification: `false`. Simulation or demonstration was relabelled observation: `false`.

- `EPA-LIFECYCLE-001` - transport `captured`; captured bytes `980`; snapshot `sha256:1d271583636a679563a046e767d0575a92d23eeafbf99f0da3fef2582e102c4e`.
- `NIST-SSDF-001` - transport `captured`; captured bytes `48313`; snapshot `sha256:d5a8c59b4e3dca64e07dfc15f83141517039120bedfd0aca8d46a5cba2b9fcb7`.
- `NIST-SSE-001` - transport `captured`; captured bytes `47984`; snapshot `sha256:64c8d41d9a78268d596a0a60aad18eead50fd375e8628486f7202788a207edaa`.

The comparison is structural and source-bound. It checks that a relevant primary handbook, standard, guidance record or open engineering project can represent the declared requirement, artifact, environment, operating boundary, version, test and lifecycle without rewriting the Fold law. It does not claim that a standard proves conformity, that one handbook exhausts Engineering, or that a custodian's reputation proves the result. New records can extend or challenge the surface only with source, version, configuration, method, failures and adverse rows retained.

Scientific meaning

For **decommissioning**, the consequence is practical and exact: Decommissioning retains authority, users, data, materials, hazards, recovery and archive custody. An Engineering claim therefore has to expose the carrier `system-service-end-assets-record-carrier`, the relation `active-to-retired-state-relation`, the record `authority-users-data-materials-hazards-recovery-and-custody-held`, and the boundary `retirement-scope-bounded` together. If any one is hidden, the result may remain a concept, prototype, simulation, demonstration or product choice, but it is not this admitted law.

This distinction protects users, maintainers and science. It prevents a successful output from erasing requirements, failed alternatives, hidden resources, inaccessible users, hazards, maintenance burdens or end-of-life costs. It prevents an application from selecting the upstream law and prevents engineering failure from becoming automatic scientific falsification. A replacement must still generate its forms, eliminate alternatives, survive controls, preserve custody and reproduce through the shared engine.

Machine certificate

Candidate count: `256`; survivor count: `1`; derivation seal: `sha256:fc014d9c5e3ae8192c3851020bab19c9b619b3dcd60f26dc10c74284cc1d601b`; independent certificate: `sha256:c3bca37d2dc576533bad23863d1e44b38f22fa87bdd2aac2351df9c8c9eef8ca`; engine receipt: `sha256:45be484f60e18ab440a2f956881937e921b3d52181cac0a7d16d49f8d82d4ccf`. The independent implementation recomputed the candidate product: `True`. All external rows were preserved: `True`.

59. Full lifecycle record

Claim: `SFT-ENG-LIFECYCLE-001`

Family: `lifecycle_maintenance_sustainability`

Exact statement: A lifecycle record connects need, design, build, use, maintenance and retirement with resources and impacts retained.

Forward chain and exact carrier

The generated carrier is `need-design-build-use-retire-carrier`. Its required relation is `versioned-lifecycle-transition-relation`. The reconstruction must retain `resources-decisions-configurations-failures-impacts-and-ownership-held`. Its declared empirical boundary is `lifecycle-boundary-declared`. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is `SFT-FOUNDATION-ONE-001`, `SFT-FOUNDATION-FOLD-001`, `SFT-FOUNDATION-FOLD-DYNAMICS-001`, `SFT-MATH-EXACT-ARITHMETIC-001`, `SFT-MATH-GRAPH-NETWORK-001`, `SFT-MATH-OPTIMIZATION-001`, `SFT-MATH-DYNAMICAL-SYSTEMS-001`, `SFT-MATH-LOGIC-PROOF-001`, `SFT-INFO-QUANTITY-001`, `SFT-INFO-CHANNEL-CAPACITY-001`, `SFT-COMP-FORM-STATE-TRANSITION-001`, `SFT-COMP-SEM-CORRECTNESS-001`, `SFT-COMP-SEM-COMPILATION-001`, `SFT-COMP-DIST-CAUSALITY-001`, `SFT-PHYS-MEAS-VALUE-RECORD-001`, `SFT-PHYS-MEAS-CALIBRATION-001`, `SFT-CHEM-MEAS-TRACEABILITY-001`, `SFT-MAT-SUST-LIFECYCLE-001`, `SFT-BIO-BIO-HANDOFF-001`, `SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001`, `SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001`, `SFT-EARTH-HAZARD-RISK-HANDOFF-001`, `SFT-ASTRO-ASTRO-HANDOFF-001`, `SFT-SOCIAL-ENGINEERING-HANDOFF-001`, `SFT-ENG-LEARNABLE-USE-001`, `SFT-ENG-DECOMMISSION-001`. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external standard, application, performance target, product or prior implementation to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external standards, products, tests or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite components, versions, platforms, tests and lifecycle records; performance remains artifact, environment, user, method and operating-boundary limited.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - which requirement, artifact, component, user, environment or test bears the claim: `need-design-build-use-retire-carrier`.
- **boundary** - which operating envelope, version, platform, user and lifecycle stage bound the result: `lifecycle-boundary-declared`.
- **relation** - which generated transformation, interface, comparison or handoff connects the carriers: `versioned-lifecycle-transition-relation`.
- **record** - which requirements, versions, alternatives, failures, uncertainty and rollback remain reconstructible: `resources-decisions-configurations-failures-impacts-and-ownership-held`.

- **evidence** - whether law, requirement, design, simulation, test, demonstration, performance and anomaly remain distinct: `science-design-test-and-decision-classes-explicit`.
- **provenance** - whether the result traces to the One and names every upstream receipt and design choice: `root-bound-forward-translation`.
- **generality** - whether finite extension preserves platforms, versions, failures and operating limits: `positive-finite-extension-preserves-platforms-failures-and-versions`.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: `no-extra-rule`.

Their ordered product gives the unique survivor:

```
need-design-build-use-retire-carrier__lifecycle-boundary-declared__versioned-lifecycl
e-transition-relation__resources-decisions-configurations-failures-impacts-and-owners
hip-held__science-design-test-and-decision-classes-explicit__root-bound-forward-trans
lation__positive-finite-extension-preserves-platforms-failures-and-versions__no-extra
-rule
```

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses the requirement, artifact, user or environment. Erasing the boundary universalises a version, platform or test. Importing the relation lets a target or preferred application choose the result. Retaining only successful output removes requirements, alternatives, failures and rollback. Conflating evidence classes turns simulation, demonstration or performance into observation or law. Breaking provenance removes the trace to the One and upstream receipts. Erasing finite context makes one implementation universal. Adding a silent dependency or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite requirement or admitted law, versioned artifact, boundary, testable relation and complete record. The successor adds one lawful component, interface, platform, requirement, test, failure, lifecycle stage or anomaly while preserving all earlier identities, adverse rows and receipts.

Falsification and deliberately unfavourable controls

The registered falsification condition is: Reject if manufacturing, operation, maintenance or end-of-life is omitted from a whole-life claim.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: `source_bound_engineering_requirement_design_test_lifecycle_or_access_record/primary_standard_handbook_guidance_or_project_record_structural_correspondence`. Captured registered sources: 3. Unresolved transports retained: 0. The registered structural consequence corresponded: `True`. External evidence selected the survivor: `false`. Application or performance selected the law: `false`. Successful performance was relabelled scientific proof: `false`. Failed implementation was relabelled scientific falsification: `false`. Simulation or demonstration was relabelled observation: `false`.

- `EPA-LIFECYCLE-001` - transport `captured`; captured bytes `980`; snapshot `sha256:1d271583636a679563a046e767d0575a92d23eeafbf99f0da3fef2582e102c4e`.
- `NIST-SSDF-001` - transport `captured`; captured bytes `48313`; snapshot `sha256:d5a8c59b4e3dca64e07dfc15f83141517039120bedfd0aca8d46a5cba2b9fcb7`.

- `NIST-SSE-001` - transport `captured`; captured bytes `47984`; snapshot `sha256:64c8d41d9a78268d596a0a60aad18eead50fd375e8628486f7202788a207edaa`.

The comparison is structural and source-bound. It checks that a relevant primary handbook, standard, guidance record or open engineering project can represent the declared requirement, artifact, environment, operating boundary, version, test and lifecycle without rewriting the Fold law. It does not claim that a standard proves conformity, that one handbook exhausts Engineering, or that a custodian's reputation proves the result. New records can extend or challenge the surface only with source, version, configuration, method, failures and adverse rows retained.

Scientific meaning

For **full lifecycle record**, the consequence is practical and exact: A lifecycle record connects need, design, build, use, maintenance and retirement with resources and impacts retained. An Engineering claim therefore has to expose the carrier `need-design-build-use-retire-carrier`, the relation `versioned-lifecycle-transition-relation`, the record `resources-decisions-configurations-failures-impacts-and-ownership-held`, and the boundary `lifecycle-boundary-declared` together. If any one is hidden, the result may remain a concept, prototype, simulation, demonstration or product choice, but it is not this admitted law.

This distinction protects users, maintainers and science. It prevents a successful output from erasing requirements, failed alternatives, hidden resources, inaccessible users, hazards, maintenance burdens or end-of-life costs. It prevents an application from selecting the upstream law and prevents engineering failure from becoming automatic scientific falsification. A replacement must still generate its forms, eliminate alternatives, survive controls, preserve custody and reproduce through the shared engine.

Machine certificate

Candidate count: `256`; survivor count: `1`; derivation seal: `sha256:9cd4eee70ae90d67802221f886d34a77c959b86b9c5414a0aa492d11a7b9ff6d`; independent certificate: `sha256:0ad4986898c3ffec2ff01555d1c80ae4f725ed1b16114a255a1813651f766bee`; engine receipt: `sha256:6eb58d7716a9da0c2a5b865c272db63eb090b31b56dd73890f9f5792604ea985`. The independent implementation recomputed the candidate product: `True`. All external rows were preserved: `True`.

60. Bounded sustainability claim

Claim: `SFT-ENG-SUSTAINABILITY-001`

Family: `lifecycle_maintenance_sustainability`

Exact statement: A sustainability claim retains lifecycle boundary, resources, emissions, labour, distribution, uncertainty and trade-offs.

Forward chain and exact carrier

The generated carrier is `system-lifecycle-resources-affected-carrier`. Its required relation is `lifecycle-input-output-impact-relation`. The reconstruction must retain `boundary-resources-emissions-labour-distribution-uncertainty-and-tradeoffs-held`. Its declared empirical boundary is `declared-lifecycle-boundary`. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is `SFT-FOUNDATION-ONE-001`, `SFT-FOUNDATION-FOLD-001`, `SFT-FOUNDATION-FOLD-DYNAMICS-001`, `SFT-MATH-EXACT-ARITHMETIC-001`, `SFT-MATH-GRAPH-NETWORK-001`, `SFT-MATH-OPTIMIZATION-001`, `SFT-MATH-DYNAMICAL-SYSTEMS-001`, `SFT-MATH-LOGIC-PROOF-001`, `SFT-INFO-QUANTITY-001`, `SFT-INFO-CHANNEL-CAPACITY-001`, `SFT-COMP-FORM-STATE-TRANSITION-001`, `SFT-COMP-SEM-CORRECTNESS-001`, `SFT-COMP-SEM-COMPILATION-001`, `SFT-COMP-DIST-CAUSALITY-001`, `SFT-PHYS-MEAS-VALUE-RECORD-001`, `SFT-PHYS-MEAS-CALIBRATION-001`, `SFT-CHEM-MEAS-TRACEABILITY-001`, `SFT-MAT-SUST-LIFECYCLE-001`, `SFT-BIO-BIO-HANDOFF-001`, `SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001`, `SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001`, `SFT-EARTH-HAZARD-RISK-HANDOFF-001`, `SFT-ASTRO-ASTRO-HANDOFF-001`, `SFT-SOCIAL-ENGINEERING-HANDOFF-001`, `SFT-ENG-LEARNABLE-USE-001`, `SFT-ENG-LIFECYCLE-001`. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external standard, application, performance target, product or prior implementation to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external standards, products, tests or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite components, versions, platforms, tests and lifecycle records; performance remains artifact, environment, user, method and operating-boundary limited.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - which requirement, artifact, component, user, environment or test bears the claim: `system-lifecycle-resources-affected-carrier`.
- **boundary** - which operating envelope, version, platform, user and lifecycle stage bound the result: `declared-lifecycle-boundary`.
- **relation** - which generated transformation, interface, comparison or handoff connects the carriers: `lifecycle-input-output-impact-relation`.
- **record** - which requirements, versions, alternatives, failures, uncertainty and rollback remain reconstructible: `boundary-resources-emissions-labour-distribution-uncertainty-and-tradeoffs-held`.
- **evidence** - whether law, requirement, design, simulation, test, demonstration, performance and anomaly remain distinct: `science-design-test-and-decision-classes-explicit`.
- **provenance** - whether the result traces to the One and names every upstream receipt and design choice: `root-bound-forward-translation`.
- **generality** - whether finite extension preserves platforms, versions, failures and operating limits: `positive-finite-extension-preserves-platforms-failures-and-versions`.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: `no-extra-rule`.

Their ordered product gives the unique survivor:

```
system-lifecycle-resources-affected-carrier__declared-lifecycle-boundary__lifecycle-input-output-impact-relation__boundary-resources-emissions-labour-distribution-uncertainty-and-tradeoffs-held__science-design-test-and-decision-classes-explicit__root-bound-forward-translation__positive-finite-extension-preserves-platforms-failures-and-versions__no-extra-rule
```

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses the requirement, artifact, user or environment. Erasing the boundary universalises a version, platform or test. Importing the relation lets a target or preferred application choose the result. Retaining only successful output removes requirements, alternatives, failures and rollback. Conflating evidence classes turns simulation, demonstration or performance into observation or law. Breaking provenance removes the trace to the One and upstream receipts. Erasing finite context makes one implementation universal. Adding a silent dependency or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite requirement or admitted law, versioned artifact, boundary, testable relation and complete record. The successor adds one lawful component, interface, platform, requirement, test, failure, lifecycle stage or anomaly while preserving all earlier identities, adverse rows and receipts.

Falsification and deliberately unfavourable controls

The registered falsification condition is: Reject if one favourable indicator, offset or displaced harm proves sustainability.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: `source_bound_engineering_requirement_design_test_lifecycle_or_access_record/primary_standard_handbook_guidance_or_project_record_structural_correspondence`. Captured registered sources: 3. Unresolved transports retained: 0. The registered structural consequence corresponded: `True`. External evidence selected the survivor: `false`. Application or performance selected the law: `false`. Successful performance was relabelled scientific proof: `false`. Failed implementation was relabelled scientific falsification: `false`. Simulation or demonstration was relabelled observation: `false`.

- `EPA-LIFECYCLE-001` - transport `captured`; captured bytes `980`; snapshot `sha256:1d271583636a679563a046e767d0575a92d23eeafbf99f0da3fef2582e102c4e`.
- `NIST-SSDF-001` - transport `captured`; captured bytes `48313`; snapshot `sha256:d5a8c59b4e3dca64e07dfc15f83141517039120bedfd0aca8d46a5cba2b9fcb7`.
- `NIST-SSE-001` - transport `captured`; captured bytes `47984`; snapshot `sha256:64c8d41d9a78268d596a0a60aad18eead50fd375e8628486f7202788a207edaa`.

The comparison is structural and source-bound. It checks that a relevant primary handbook, standard, guidance record or open engineering project can represent the declared requirement, artifact, environment, operating boundary, version, test and lifecycle without rewriting the Fold law. It does not claim that a standard proves conformity, that one handbook exhausts Engineering, or that a custodian's reputation proves the result. New records can extend or challenge the surface only with source, version, configuration, method, failures and adverse rows retained.

Scientific meaning

For **bounded sustainability claim**, the consequence is practical and exact: A sustainability claim retains lifecycle boundary, resources, emissions, labour, distribution, uncertainty and trade-offs. An Engineering claim therefore has to expose the carrier `system-lifecycle-resources-affected-carrier`, the relation `lifecycle-input-output-impact-relation`, the record `boundary-resources-emissions-labour-distribution-uncertainty-and-tradeoffs-held`, and the boundary `declared-lifecycle-boundary` together. If any one is hidden, the result may remain a concept, prototype, simulation, demonstration or product choice, but it is not this admitted law.

This distinction protects users, maintainers and science. It prevents a successful output from erasing requirements, failed alternatives, hidden resources, inaccessible users, hazards, maintenance burdens or end-of-life costs. It prevents an application from selecting the upstream law and prevents engineering failure from becoming automatic scientific falsification. A replacement must still generate its forms, eliminate alternatives, survive controls, preserve custody and reproduce through the shared engine.

Machine certificate

Candidate count: `256`; survivor count: `1`; derivation seal: `sha256:2ceea811d443e3d5e4f91856b376f7bce8d2bddf9a63e7d359de550132a30363`; independent certificate: `sha256:fa637b0536c5dc38a007c6fd229cff32da3631f1d385f3d70756ad40cccfb7b6`; engine receipt: `sha256:7a3510262c0aa4807573001ccfa38c1cf5e12f56a62977aefc85abce1cf09c42`. The independent implementation recomputed the candidate product: `True`. All external rows were preserved: `True`.

61. Scientific-law translation

Claim: `SFT-ENG-LAW-TRANSLATION-001`

Family: `science_design_evidence_boundary`

Exact statement: Engineering translation consumes an admitted law and records every added design choice, assumption, boundary and test.

Forward chain and exact carrier

The generated carrier is `admitted-law-requirement-design-test-carrier`. Its required relation is `law-to-bounded-implementation-relation`. The reconstruction must retain `upstream-receipt-assumptions-choices-boundary-results-and-failures-held`. Its declared empirical boundary is `translation-purpose-bounded`. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is `SFT-FOUNDATION-ONE-001`, `SFT-FOUNDATION-FOLD-001`, `SFT-FOUNDATION-FOLD-DYNAMICS-001`, `SFT-MATH-EXACT-ARITHMETIC-001`, `SFT-MATH-GRAPH-NETWORK-001`, `SFT-MATH-OPTIMIZATION-001`, `SFT-MATH-DYNAMICAL-SYSTEMS-001`, `SFT-MATH-LOGIC-PROOF-001`, `SFT-INFO-QUANTITY-001`, `SFT-INFO-CHANNEL-CAPACITY-001`, `SFT-COMP-FORM-STATE-TRANSITION-001`, `SFT-COMP-SEM-CORRECTNESS-001`, `SFT-COMP-SEM-COMPILATION-001`, `SFT-COMP-DIST-CAUSALITY-001`, `SFT-PHYS-MEAS-VALUE-RECORD-001`, `SFT-PHYS-MEAS-CALIBRATION-001`, `SFT-CHEM-MEAS-TRACEABILITY-001`, `SFT-MAT-SUST-LIFECYCLE-001`, `SFT-BIO-BIO-HANDOFF-001`, `SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001`, `SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001`, `SFT-EARTH-HAZARD-RISK-HANDOFF-001`, `SFT-ASTRO-ASTRO-HANDOFF-001`, `SFT-SOCIAL-ENGINEERING-HANDOFF-001`, `SFT-ENG-SUSTAINABILITY-001`. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external standard, application, performance target, product or prior implementation to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external standards, products, tests or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite components, versions, platforms, tests and lifecycle records; performance remains artifact, environment, user, method and operating-boundary limited.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - which requirement, artifact, component, user, environment or test bears the claim: `admitted-law-requirement-design-test-carrier`.
- **boundary** - which operating envelope, version, platform, user and lifecycle stage bound the result: `translation-purpose-bounded`.
- **relation** - which generated transformation, interface, comparison or handoff connects the carriers: `law-to-bounded-implementation-relation`.
- **record** - which requirements, versions, alternatives, failures, uncertainty and rollback remain reconstructible: `upstream-receipt-assumptions-choices-boundary-results-and-failures-held`.
- **evidence** - whether law, requirement, design, simulation, test, demonstration, performance and anomaly remain distinct: `science-design-test-and-decision-classes-explicit`.
- **provenance** - whether the result traces to the One and names every upstream receipt and design choice: `root-bound-forward-translation`.
- **generality** - whether finite extension preserves platforms, versions, failures and operating limits: `positive-finite-extension-preserves-platforms-failures-and-versions`.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: `no-extra-rule`.

Their ordered product gives the unique survivor:

```
admitted-law-requirement-design-test-carrier__translation-purpose-bounded__law-to-bou
nded-implementation-relation__upstream-receipt-assumptions-choices-boundary-results-a
nd-failures-held__science-design-test-and-decision-classes-explicit__root-bound-forwa
rd-translation__positive-finite-extension-preserves-platforms-failures-and-versions__
no-extra-rule
```

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses the requirement, artifact, user or environment. Erasing the boundary universalises a version, platform or test. Importing the relation lets a target or preferred application choose the result. Retaining only successful output removes requirements, alternatives, failures and rollback. Conflating evidence classes turns simulation, demonstration or performance into observation or law. Breaking provenance removes the trace to the One and upstream receipts. Erasing finite context makes one implementation universal. Adding a silent dependency or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite requirement or admitted law, versioned artifact, boundary, testable relation and complete record. The successor adds one lawful component, interface, platform, requirement, test, failure, lifecycle stage or anomaly while preserving all earlier identities, adverse rows and receipts.

Falsification and deliberately unfavourable controls

The registered falsification condition is: Reject if an application target selects or rewrites the upstream law.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: `source_bound_engineering_requirement_design_test_lifecycle_or_access_record/primary_standard_handbook_guidance_or_project_record_structural_correspondence`. Captured registered sources: 2. Unresolved transports retained: 1. The registered structural consequence corresponded: `True`. External evidence selected the survivor: `false`. Application or performance selected the law: `false`. Successful performance was relabelled scientific proof: `false`. Failed implementation was relabelled scientific falsification: `false`. Simulation or demonstration was relabelled observation: `false`.

- `NASA-SE-APPENDIX-001` - transport `captured`; captured bytes `578114`; snapshot `sha256:c1db2db970b9fd14a7ef98aa0c23575326dedae46cc7cc56c468821ca89edd60`.
- `FDA-DESIGN-CONTROLS-001` - transport `failed_preserved`; captured bytes `0`; snapshot `unresolved and preserved`.
- `CISA-SECURE-BY-DESIGN-001` - transport `captured`; captured bytes `64467`; snapshot `sha256:b4ad9124bddafabd552dc29d861694336c8af6d64f135fa44ad3b1bf24fb4ee2`.

The comparison is structural and source-bound. It checks that a relevant primary handbook, standard, guidance record or open engineering project can represent the declared requirement, artifact, environment, operating boundary, version, test and lifecycle without rewriting the Fold law. It does not claim that a standard proves conformity, that one handbook exhausts Engineering, or that a custodian's reputation proves the result. New records can extend or challenge the surface only with source, version, configuration, method, failures and adverse rows retained.

Scientific meaning

For **scientific-law translation**, the consequence is practical and exact: Engineering translation consumes an admitted law and records every added design choice, assumption, boundary and test. An Engineering claim therefore has to expose the carrier `admitted-law-requirement-design-test-carrier`, the relation `law-to-bounded-implementation-relation`, the record `upstream-receipt-assumptions-choices-boundary-results-and-failures-held`, and the boundary `translation-purpose-bounded` together. If any one is hidden, the result may remain a concept, prototype, simulation, demonstration or product choice, but it is not this admitted law.

This distinction protects users, maintainers and science. It prevents a successful output from erasing requirements, failed alternatives, hidden resources, inaccessible users, hazards, maintenance burdens or end-of-life costs. It prevents an application from selecting the upstream law and prevents engineering failure from becoming automatic scientific falsification. A replacement must still generate its forms, eliminate alternatives, survive controls, preserve custody and reproduce through the shared engine.

Machine certificate

Candidate count: `256`; survivor count: `1`; derivation seal: `sha256:37a39f4c03f4aab1aef95326886e2061a2ffaed85e76c3bc06c2e3cb4a18ce1a`; independent certificate: `sha256:1a83087a379c66d400cca03bc67524cc6daf9e21e920ed179176effccf133d5e`; engine receipt: `sha256:4d67b3f68c70fbf95c1a9f852e93782710b7821d02bc1e309cb7503c394faa1c`. The independent implementation recomputed the candidate product: `True`. All external rows were preserved: `True`.

62. Design performance

Claim: `SFT-ENG-PERFORMANCE-001`

Family: `science_design_evidence_boundary`

Exact statement: Performance is a version- and condition-bound implementation result, not a fundamental scientific law.

Forward chain and exact carrier

The generated carrier is `implementation-condition-metric-result-carrier`. Its required relation is `test-condition-to-performance-relation`. The reconstruction must retain `artifact-version-metric-boundary-results-failures-and-uncertainty-held`. Its declared empirical boundary is `performance-test-bounded`. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is `SFT-FOUNDATION-ONE-001`, `SFT-FOUNDATION-FOLD-001`, `SFT-FOUNDATION-FOLD-DYNAMICS-001`, `SFT-MATH-EXACT-ARITHMETIC-001`, `SFT-MATH-GRAPH-NETWORK-001`, `SFT-MATH-OPTIMIZATION-001`, `SFT-MATH-DYNAMICAL-SYSTEMS-001`, `SFT-MATH-LOGIC-PROOF-001`, `SFT-INFO-QUANTITY-001`, `SFT-INFO-CHANNEL-CAPACITY-001`, `SFT-COMP-FORM-STATE-TRANSITION-001`, `SFT-COMP-SEM-CORRECTNESS-001`, `SFT-COMP-SEM-COMPILATION-001`, `SFT-COMP-DIST-CAUSALITY-001`, `SFT-PHYS-MEAS-VALUE-RECORD-001`, `SFT-PHYS-MEAS-CALIBRATION-001`, `SFT-CHEM-MEAS-TRACEABILITY-001`, `SFT-MAT-SUST-LIFECYCLE-001`, `SFT-BIO-BIO-HANDOFF-001`, `SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001`, `SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001`, `SFT-EARTH-HAZARD-RISK-HANDOFF-001`, `SFT-ASTRO-ASTRO-HANDOFF-001`, `SFT-SOCIAL-ENGINEERING-HANDOFF-001`, `SFT-ENG-SUSTAINABILITY-001`, `SFT-ENG-LAW-TRANSLATION-001`. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external standard, application, performance target, product or prior implementation to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external standards, products, tests or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite components, versions, platforms, tests and lifecycle records; performance remains artifact, environment, user, method and operating-boundary limited.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - which requirement, artifact, component, user, environment or test bears the claim: `implementation-condition-metric-result-carrier`.
- **boundary** - which operating envelope, version, platform, user and lifecycle stage bound the result: `performance-test-bounded`.

- **relation** - which generated transformation, interface, comparison or handoff connects the carriers: `test-condition-to-performance-relation`.
- **record** - which requirements, versions, alternatives, failures, uncertainty and rollback remain reconstructible: `artifact-version-metric-boundary-results-failures-and-uncertainty-held`.
- **evidence** - whether law, requirement, design, simulation, test, demonstration, performance and anomaly remain distinct: `science-design-test-and-decision-classes-explicit`.
- **provenance** - whether the result traces to the One and names every upstream receipt and design choice: `root-bound-forward-translation`.
- **generality** - whether finite extension preserves platforms, versions, failures and operating limits: `positive-finite-extension-preserves-platforms-failures-and-versions`.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: `no-extra-rule`.

Their ordered product gives the unique survivor:

```
implementation-condition-metric-result-carrier__performance-test-bounded__test-condit
ion-to-performance-relation__artifact-version-metric-boundary-results-failures-and-un
certainty-held__science-design-test-and-decision-classes-explicit__root-bound-forward
-translation__positive-finite-extension-preserves-platforms-failures-and-versions__no
-extra-rule
```

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses the requirement, artifact, user or environment. Erasing the boundary universalises a version, platform or test. Importing the relation lets a target or preferred application choose the result. Retaining only successful output removes requirements, alternatives, failures and rollback. Conflating evidence classes turns simulation, demonstration or performance into observation or law. Breaking provenance removes the trace to the One and upstream receipts. Erasing finite context makes one implementation universal. Adding a silent dependency or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite requirement or admitted law, versioned artifact, boundary, testable relation and complete record. The successor adds one lawful component, interface, platform, requirement, test, failure, lifecycle stage or anomaly while preserving all earlier identities, adverse rows and receipts.

Falsification and deliberately unfavourable controls

The registered falsification condition is: Reject if successful performance is presented as retroactive proof of the law.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: `source_bound_engineering_requirement_design_test_lifecycle_or_access_record/primary_standard_handbook_guidance_or_project_record_structural_correspondence`. Captured registered sources: 2. Unresolved transports retained: 1. The registered structural consequence corresponded: `True`. External evidence selected the survivor: `false`. Application or performance selected the law: `false`. Successful performance was relabelled scientific proof: `false`. Failed implementation was relabelled scientific falsification: `false`. Simulation or demonstration was relabelled observation: `false`.

- `NASA-SE-APPENDIX-001` - transport `captured`; captured bytes `578114`; snapshot `sha256:c1db2db970b9fd14a7ef98aa0c23575326dedae46cc7cc56c468821ca89edd60`.

- FDA-DESIGN-CONTROLS-001 - transport failed_preserved; captured bytes 0; snapshot unresolved and preserved.
- CISA-SECURE-BY-DESIGN-001 - transport captured; captured bytes 64467; snapshot sha256:b4ad9124bddafabd552dc29d861694336c8af6d64f135fa44ad3b1bf24fb4ee2.

The comparison is structural and source-bound. It checks that a relevant primary handbook, standard, guidance record or open engineering project can represent the declared requirement, artifact, environment, operating boundary, version, test and lifecycle without rewriting the Fold law. It does not claim that a standard proves conformity, that one handbook exhausts Engineering, or that a custodian's reputation proves the result. New records can extend or challenge the surface only with source, version, configuration, method, failures and adverse rows retained.

Scientific meaning

For **design performance**, the consequence is practical and exact: Performance is a version- and condition-bound implementation result, not a fundamental scientific law. An Engineering claim therefore has to expose the carrier implementation-condition-metric-result-carrier, the relation test-condition-to-performance-relation, the record artifact-version-metric-boundary-results-failures-and-uncertainty-held, and the boundary performance-test-bounded together. If any one is hidden, the result may remain a concept, prototype, simulation, demonstration or product choice, but it is not this admitted law.

This distinction protects users, maintainers and science. It prevents a successful output from erasing requirements, failed alternatives, hidden resources, inaccessible users, hazards, maintenance burdens or end-of-life costs. It prevents an application from selecting the upstream law and prevents engineering failure from becoming automatic scientific falsification. A replacement must still generate its forms, eliminate alternatives, survive controls, preserve custody and reproduce through the shared engine.

Machine certificate

Candidate count: 256; survivor count: 1; derivation seal: sha256:e9f9272494e8eb0f1ee8d520349a58ae920a30d753370e36302e54352b8d0aa8; independent certificate: sha256:c37dbb54d750836a8ee9d1f9ced74a87e995e59b4deb6a84eda96ed76cd4d832; engine receipt: sha256:ed77a06d3f29985acf91f2b2f67a5d396fa0f25a56b1d2f1409b6a16de48e05a. The independent implementation recomputed the candidate product: True. All external rows were preserved: True.

63. Implementation failure boundary

Claim: SFT-ENG-IMPLEMENTATION-FAILURE-001

Family: science_design_evidence_boundary

Exact statement: A failed implementation can falsify its operating claim while leaving the upstream law unchanged unless a valid anomaly bridge is established.

Forward chain and exact carrier

The generated carrier is design-condition-failure-record-carrier. Its required relation is implementation-path-to-failed-result-relation. The reconstruction must retain requirements-choices-build-state-test-cause-and-uncertainty-held. Its declared empirical boundary is implementation-version-bounded. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-OPTIMIZATION-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-CHANNEL-CAPACITY-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-COMP-SEM-CORRECTNESS-001, SFT-COMP-SEM-COMPILATION-001, SFT-COMP-DIST-CAUSALITY-001, SFT-PHYS-MEAS-VALUE-RECORD-001, SFT-PHYS-MEAS-CALIBRATION-001, SFT-CHEM-MEAS-TRACEABILITY-001, SFT-MAT-SUST-LIFECYCLE-001, SFT-BIO-BIO-HANDOFF-001, SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001, SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-ASTRO-HANDOFF-001,

`SFT-SOCIAL-ENGINEERING-HANDOFF-001`, `SFT-ENG-SUSTAINABILITY-001`, `SFT-ENG-PERFORMANCE-001`. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external standard, application, performance target, product or prior implementation to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external standards, products, tests or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite components, versions, platforms, tests and lifecycle records; performance remains artifact, environment, user, method and operating-boundary limited.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - which requirement, artifact, component, user, environment or test bears the claim: `design-condition-failure-record-carrier`.
- **boundary** - which operating envelope, version, platform, user and lifecycle stage bound the result: `implementation-version-bounded`.
- **relation** - which generated transformation, interface, comparison or handoff connects the carriers: `implementation-path-to-failed-result-relation`.
- **record** - which requirements, versions, alternatives, failures, uncertainty and rollback remain reconstructible: `requirements-choices-build-state-test-cause-and-uncertainty-held`.
- **evidence** - whether law, requirement, design, simulation, test, demonstration, performance and anomaly remain distinct: `science-design-test-and-decision-classes-explicit`.
- **provenance** - whether the result traces to the One and names every upstream receipt and design choice: `root-bound-forward-translation`.
- **generality** - whether finite extension preserves platforms, versions, failures and operating limits: `positive-finite-extension-preserves-platforms-failures-and-versions`.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: `no-extra-rule`.

Their ordered product gives the unique survivor:

```
design-condition-failure-record-carrier__implementation-version-bounded__implementati
on-path-to-failed-result-relation__requirements-choices-build-state-test-cause-and-un
certainty-held__science-design-test-and-decision-classes-explicit__root-bound-forward
-translation__positive-finite-extension-preserves-platforms-failures-and-versions__no
-extra-rule
```

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses the requirement, artifact, user or environment. Erasing the boundary universalises a version, platform or test. Importing the relation lets a target or preferred application choose the result. Retaining only successful output removes requirements, alternatives, failures and rollback. Conflating evidence classes turns simulation, demonstration or performance into observation or law. Breaking provenance removes the trace to the One and upstream receipts. Erasing finite context makes one implementation universal. Adding a silent dependency or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite requirement or admitted law, versioned artifact, boundary, testable relation and complete record. The successor adds one lawful component, interface, platform, requirement, test, failure, lifecycle stage or anomaly while preserving all earlier identities, adverse rows and receipts.

Falsification and deliberately unfavourable controls

The registered falsification condition is: Reject if engineering failure is automatically relabelled scientific falsification.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier

- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: `source_bound_engineering_requirement_design_test_lifecycle_or_access_record` / `primary_standard_handbook_guidance_or_project_record_structural_correspondence`. Captured registered sources: 2. Unresolved transports retained: 1. The registered structural consequence corresponded: `True`. External evidence selected the survivor: `false`. Application or performance selected the law: `false`. Successful performance was relabelled scientific proof: `false`. Failed implementation was relabelled scientific falsification: `false`. Simulation or demonstration was relabelled observation: `false`.

- `NASA-SE-APPENDIX-001` - transport `captured`; captured bytes `578114`; snapshot `sha256:c1db2db970b9fd14a7ef98aa0c23575326dedae46cc7cc56c468821ca89edd60`.
- `FDA-DESIGN-CONTROLS-001` - transport `failed_preserved`; captured bytes `0`; snapshot `unresolved and preserved`.
- `CISA-SECURE-BY-DESIGN-001` - transport `captured`; captured bytes `64467`; snapshot `sha256:b4ad9124bddafabd552dc29d861694336c8af6d64f135fa44ad3b1bf24fb4ee2`.

The comparison is structural and source-bound. It checks that a relevant primary handbook, standard, guidance record or open engineering project can represent the declared requirement, artifact, environment, operating boundary, version, test and lifecycle without rewriting the Fold law. It does not claim that a standard proves conformity, that one handbook exhausts Engineering, or that a custodian's reputation proves the result. New records can extend or challenge the surface only with source, version, configuration, method, failures and adverse rows retained.

Scientific meaning

For **implementation failure boundary**, the consequence is practical and exact: A failed implementation can falsify its operating claim while leaving the upstream law unchanged unless a valid anomaly bridge is established. An Engineering claim therefore has to expose the carrier `design-condition-failure-record-carrier`, the relation `implementation-path-to-failed-result-relation`, the record `requirements-choices-build-state-test-cause-and-uncertainty-held`, and the boundary `implementation-version-bounded` together. If any one is hidden, the result may remain a concept, prototype, simulation, demonstration or product choice, but it is not this admitted law.

This distinction protects users, maintainers and science. It prevents a successful output from erasing requirements, failed alternatives, hidden resources, inaccessible users, hazards, maintenance burdens or end-of-life costs. It prevents an application from selecting the upstream law and prevents engineering failure from becoming automatic scientific falsification. A replacement must still generate its forms, eliminate alternatives, survive controls, preserve custody and reproduce through the shared engine.

Machine certificate

Candidate count: `256`; survivor count: `1`; derivation seal: `sha256:8fdcee779bf23e1a1fdec6f1ad11588ee2e724bf76cf1cbae51948f448b258fb`; independent certificate: `sha256:cd7f688943f87a79ee44da8a7410bfc83303f0b6b9112aff5dd3ff931a540b0c`; engine receipt: `sha256:d190bd5feedd40c3b9abae516143bec91fb27ccbfa4bf6e6ef8e902845c3c417`. The independent implementation recomputed the candidate product: `True`. All external rows were preserved: `True`.

64. Prototype evidence

Claim: `SFT-ENG-PROTOTYPE-001`

Family: `science_design_evidence_boundary`

Exact statement: Prototype evidence retains omissions, fidelity, setup, failures and the bounded question it answers.

Forward chain and exact carrier

The generated carrier is `prototype-version-purpose-test-carrier`. Its required relation is `partial-implementation-to-bounded-learning-relation`. The reconstruction must retain `omissions-fidelity-setup-results-failures-and-next-decision-held`. Its declared empirical boundary is `prototype-purpose-bounded`. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is `SFT-FOUNDATION-ONE-001`, `SFT-FOUNDATION-FOLD-001`, `SFT-FOUNDATION-FOLD-DYNAMICS-001`, `SFT-MATH-EXACT-ARITHMETIC-001`, `SFT-MATH-GRAPH-NETWORK-001`, `SFT-MATH-OPTIMIZATION-001`, `SFT-MATH-DYNAMICAL-SYSTEMS-001`, `SFT-MATH-LOGIC-PROOF-001`, `SFT-INFO-QUANTITY-001`, `SFT-INFO-CHANNEL-CAPACITY-001`, `SFT-COMP-FORM-STATE-TRANSITION-001`, `SFT-COMP-SEM-CORRECTNESS-001`, `SFT-COMP-SEM-COMPILATION-001`, `SFT-COMP-DIST-CAUSALITY-001`, `SFT-PHYS-MEAS-VALUE-RECORD-001`, `SFT-PHYS-MEAS-CALIBRATION-001`, `SFT-CHEM-MEAS-TRACEABILITY-001`, `SFT-MAT-SUST-LIFECYCLE-001`, `SFT-BIO-BIO-HANDOFF-001`, `SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001`, `SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001`, `SFT-EARTH-HAZARD-RISK-HANDOFF-001`, `SFT-ASTRO-ASTRO-HANDOFF-001`, `SFT-SOCIAL-ENGINEERING-HANDOFF-001`, `SFT-ENG-SUSTAINABILITY-001`, `SFT-ENG-IMPLEMENTATION-FAILURE-001`. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external standard, application, performance target, product or prior implementation to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external standards, products, tests or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite components, versions, platforms, tests and lifecycle records; performance remains artifact, environment, user, method and operating-boundary limited.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - which requirement, artifact, component, user, environment or test bears the claim: `prototype-version-purpose-test-carrier`.
- **boundary** - which operating envelope, version, platform, user and lifecycle stage bound the result: `prototype-purpose-bounded`.
- **relation** - which generated transformation, interface, comparison or handoff connects the carriers: `partial-implementation-to-bounded-learning-relation`.
- **record** - which requirements, versions, alternatives, failures, uncertainty and rollback remain reconstructible: `omissions-fidelity-setup-results-failures-and-next-decision-held`.
- **evidence** - whether law, requirement, design, simulation, test, demonstration, performance and anomaly remain distinct: `science-design-test-and-decision-classes-explicit`.
- **provenance** - whether the result traces to the One and names every upstream receipt and design choice: `root-bound-forward-translation`.
- **generality** - whether finite extension preserves platforms, versions, failures and operating limits: `positive-finite-extension-preserves-platforms-failures-and-versions`.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: `no-extra-rule`.

Their ordered product gives the unique survivor:

```
prototype-version-purpose-test-carrier__prototype-purpose-bounded__partial-implementa
tion-to-bounded-learning-relation__omissions-fidelity-setup-results-failures-and-next
-decision-held__science-design-test-and-decision-classes-explicit__root-bound-forward
```

`-translation__positive-finite-extension-preserves-platforms-failures-and-versions__no-extra-rule`

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses the requirement, artifact, user or environment. Erasing the boundary universalises a version, platform or test. Importing the relation lets a target or preferred application choose the result. Retaining only successful output removes requirements, alternatives, failures and rollback. Conflating evidence classes turns simulation, demonstration or performance into observation or law. Breaking provenance removes the trace to the One and upstream receipts. Erasing finite context makes one implementation universal. Adding a silent dependency or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite requirement or admitted law, versioned artifact, boundary, testable relation and complete record. The successor adds one lawful component, interface, platform, requirement, test, failure, lifecycle stage or anomaly while preserving all earlier identities, adverse rows and receipts.

Falsification and deliberately unfavourable controls

The registered falsification condition is: Reject if a prototype demonstration is called production validation or universal proof.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: `source_bound_engineering_requirement_design_test_lifecycle_or_access_record/primary_standard_handbook_guidance_or_project_record_structural_correspondence`. Captured registered sources: 2. Unresolved transports retained: 1. The registered structural consequence corresponded: `True`. External evidence selected the survivor: `false`. Application or performance selected the law: `false`. Successful performance was relabelled scientific proof: `false`. Failed implementation was relabelled scientific falsification: `false`. Simulation or demonstration was relabelled observation: `false`.

- `NASA-SE-APPENDIX-001` - transport `captured`; captured bytes `578114`; snapshot `sha256:c1db2db970b9fd14a7ef98aa0c23575326dedae46cc7cc56c468821ca89edd60`.
- `FDA-DESIGN-CONTROLS-001` - transport `failed_preserved`; captured bytes `0`; snapshot `unresolved and preserved`.
- `CISA-SECURE-BY-DESIGN-001` - transport `captured`; captured bytes `64467`; snapshot `sha256:b4ad9124bddafabd552dc29d861694336c8af6d64f135fa44ad3b1bf24fb4ee2`.

The comparison is structural and source-bound. It checks that a relevant primary handbook, standard, guidance record or open engineering project can represent the declared requirement, artifact, environment, operating boundary, version, test and lifecycle without rewriting the Fold law. It does not claim that a standard proves conformity, that one handbook exhausts Engineering, or that a custodian's reputation proves the result. New records can extend or challenge the surface only with source, version, configuration, method, failures and adverse rows retained.

Scientific meaning

For **prototype evidence**, the consequence is practical and exact: Prototype evidence retains omissions, fidelity, setup, failures and the bounded question it answers. An Engineering claim therefore has to expose the carrier `prototype-version-purpose-test-carrier`, the relation `partial-implementation-to-bounded-learning-relation`, the record `omissions-fidelity-setup-results-failures-and-next-decision-held`, and the boundary `prototype-purpose-bounded` together. If any one is hidden, the result may remain a concept, prototype, simulation, demonstration or product choice, but it is not this admitted law.

This distinction protects users, maintainers and science. It prevents a successful output from erasing requirements, failed alternatives, hidden resources, inaccessible users, hazards, maintenance burdens or end-of-life costs. It prevents an application from selecting the upstream law and prevents engineering failure from becoming automatic scientific falsification. A replacement must still generate its forms, eliminate alternatives, survive controls, preserve custody and reproduce through the shared engine.

Machine certificate

Candidate count: `256`; survivor count: `1`; derivation seal: `sha256:4dac3cd050c7b77766ec57ec773e4808896f13a06c6f4e1610d6c5d7647a9186`; independent certificate: `sha256:9efd7e13c4dc18cdfcb64dd12088d91583a776997c2f153bd0e50a0d2b6a494e`; engine receipt: `sha256:8f19503903dc28e39c772205296e92fb67a25ae0f4d3fd2de378e24d8e69bf5f`. The independent implementation recomputed the candidate product: `True`. All external rows were preserved: `True`.

65. Engineering simulation boundary

Claim: `SFT-ENG-SIMULATION-001`

Family: `science_design_evidence_boundary`

Exact statement: A simulation retains its model, discretization, inputs, assumptions, error and comparison; it is not an observation.

Forward chain and exact carrier

The generated carrier is `model-input-assumption-output-carrier`. Its required relation is `declared-model-to-computed-result-relation`. The reconstruction must retain `equations-discretization-inputs-assumptions-errors-and-comparison-held`. Its declared empirical boundary is `model-and-resolution-bounded`. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is `SFT-FOUNDATION-ONE-001`, `SFT-FOUNDATION-FOLD-001`, `SFT-FOUNDATION-FOLD-DYNAMICS-001`, `SFT-MATH-EXACT-ARITHMETIC-001`, `SFT-MATH-GRAPH-NETWORK-001`, `SFT-MATH-OPTIMIZATION-001`, `SFT-MATH-DYNAMICAL-SYSTEMS-001`, `SFT-MATH-LOGIC-PROOF-001`, `SFT-INFO-QUANTITY-001`, `SFT-INFO-CHANNEL-CAPACITY-001`, `SFT-COMP-FORM-STATE-TRANSITION-001`, `SFT-COMP-SEM-CORRECTNESS-001`, `SFT-COMP-SEM-COMPILATION-001`, `SFT-COMP-DIST-CAUSALITY-001`, `SFT-PHYS-MEAS-VALUE-RECORD-001`, `SFT-PHYS-MEAS-CALIBRATION-001`, `SFT-CHEM-MEAS-TRACEABILITY-001`, `SFT-MAT-SUST-LIFECYCLE-001`, `SFT-BIO-BIO-HANDOFF-001`, `SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001`, `SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001`, `SFT-EARTH-HAZARD-RISK-HANDOFF-001`, `SFT-ASTRO-ASTRO-HANDOFF-001`, `SFT-SOCIAL-ENGINEERING-HANDOFF-001`, `SFT-ENG-SUSTAINABILITY-001`, `SFT-ENG-PROTOTYPE-001`. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external standard, application, performance target, product or prior implementation to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external standards, products, tests or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite components, versions, platforms, tests and lifecycle records; performance remains artifact, environment, user, method and operating-boundary limited.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - which requirement, artifact, component, user, environment or test bears the claim: `model-input-assumption-output-carrier`.
- **boundary** - which operating envelope, version, platform, user and lifecycle stage bound the result: `model-and-resolution-bounded`.
- **relation** - which generated transformation, interface, comparison or handoff connects the carriers: `declared-model-to-computed-result-relation`.
- **record** - which requirements, versions, alternatives, failures, uncertainty and rollback remain reconstructible: `equations-discretization-inputs-assumptions-errors-and-comparison-held`.
- **evidence** - whether law, requirement, design, simulation, test, demonstration, performance and anomaly remain distinct: `science-design-test-and-decision-classes-explicit`.
- **provenance** - whether the result traces to the One and names every upstream receipt and design choice: `root-bound-forward-translation`.
- **generality** - whether finite extension preserves platforms, versions, failures and operating limits: `positive-finite-extension-preserves-platforms-failures-and-versions`.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: `no-extra-rule`.

Their ordered product gives the unique survivor:

```
model-input-assumption-output-carrier__model-and-resolution-bounded__declared-model-t
o-computed-result-relation__equations-discretization-inputs-assumptions-errors-and-co
mparison-held__science-design-test-and-decision-classes-explicit__root-bound-forward-
translation__positive-finite-extension-preserves-platforms-failures-and-versions__no-
extra-rule
```

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses the requirement, artifact, user or environment. Erasing the boundary universalises a version, platform or test. Importing the relation lets a target or preferred application choose the result. Retaining only successful output removes requirements, alternatives, failures and rollback. Conflating evidence classes turns simulation, demonstration or performance into observation or law. Breaking provenance removes the trace to the One and upstream receipts. Erasing finite context makes one implementation universal. Adding a silent dependency or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite requirement or admitted law, versioned artifact, boundary, testable relation and complete record. The successor adds one lawful component, interface, platform, requirement, test, failure, lifecycle stage or anomaly while preserving all earlier identities, adverse rows and receipts.

Falsification and deliberately unfavourable controls

The registered falsification condition is: Reject if simulated output is relabelled measured performance.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: `source_bound_engineering_requirement_design_test_lifecycle_or_access_record/primary_standard_handbook_guidance_or_project_record_structural_correspondence`. Captured registered sources: 2. Unresolved transports retained: 1. The registered structural consequence corresponded: `True`. External evidence selected the survivor: `false`. Application or performance selected the law: `false`. Successful performance was relabelled scientific proof: `false`. Failed implementation was relabelled scientific falsification: `false`. Simulation or demonstration was relabelled observation: `false`.

- `NASA-SE-APPENDIX-001` - transport `captured`; captured bytes `578114`; snapshot `sha256:c1db2db970b9fd14a7ef98aa0c23575326dedae46cc7cc56c468821ca89edd60`.
- `FDA-DESIGN-CONTROLS-001` - transport `failed_preserved`; captured bytes `0`; snapshot `unresolved and preserved`.
- `CISA-SECURE-BY-DESIGN-001` - transport `captured`; captured bytes `64467`; snapshot `sha256:b4ad9124bddafabd552dc29d861694336c8af6d64f135fa44ad3b1bf24fb4ee2`.

The comparison is structural and source-bound. It checks that a relevant primary handbook, standard, guidance record or open engineering project can represent the declared requirement, artifact, environment, operating boundary, version, test and lifecycle without rewriting the Fold law. It does not claim that a standard proves conformity, that one handbook exhausts Engineering, or that a custodian's reputation proves the result. New records can extend or challenge the surface only with source, version, configuration, method, failures and adverse rows retained.

Scientific meaning

For **engineering simulation boundary**, the consequence is practical and exact: A simulation retains its model, discretization, inputs, assumptions, error and comparison; it is not an observation. An Engineering claim therefore has to expose the carrier `model-input-assumption-output-carrier`, the relation `declared-model-to-computed-result-relation`, the record `equations-discretization-inputs-assumptions-errors-and-comparison-held`, and the boundary `model-and-resolution-bounded` together. If any one is hidden, the result may remain a concept, prototype, simulation, demonstration or product choice, but it is not this admitted law.

This distinction protects users, maintainers and science. It prevents a successful output from erasing requirements, failed alternatives, hidden resources, inaccessible users, hazards, maintenance burdens or end-of-life costs. It prevents an application from selecting the upstream law and prevents engineering failure from becoming automatic scientific falsification. A replacement must still generate its forms, eliminate alternatives, survive controls, preserve custody and reproduce through the shared engine.

Machine certificate

Candidate count: `256`; survivor count: `1`; derivation seal: `sha256:cb5367823b7944ede9909a643368058f49c65dad6267dd8e03a8ce4ed20c0835`; independent certificate: `sha256:76327bde061cdc0ed176d127b6985deca215e9695a489c84824b51cac0b200ad`; engine receipt: `sha256:c88333739def6fda69b8d90fc4337eaab45bc453ac7b5fe67f24d9dc8fe4e0a0`. The independent implementation recomputed the candidate product: `True`. All external rows were preserved: `True`.

66. Engineering demonstration

Claim: `SFT-ENG-DEMONSTRATION-001`

Family: `science_design_evidence_boundary`

Exact statement: A demonstration shows bounded behavior in a retained scenario and remains distinct from verification, validation and science admission.

Forward chain and exact carrier

The generated carrier is `artifact-scenario-observer-result-carrier`. Its required relation is `declared-scenario-to-visible-behaviour-relation`. The reconstruction must retain `version-setup-scenario-observations-failures-and-limits-held`. Its declared empirical boundary is `demonstration-scenario-bounded`. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is `SFT-FOUNDATION-ONE-001`, `SFT-FOUNDATION-FOLD-001`, `SFT-FOUNDATION-FOLD-DYNAMICS-001`, `SFT-MATH-EXACT-ARITHMETIC-001`, `SFT-MATH-GRAPH-NETWORK-001`, `SFT-MATH-OPTIMIZATION-001`, `SFT-MATH-DYNAMICAL-SYSTEMS-001`, `SFT-MATH-LOGIC-PROOF-001`, `SFT-INFO-QUANTITY-001`, `SFT-INFO-CHANNEL-CAPACITY-001`, `SFT-COMP-FORM-STATE-TRANSITION-001`, `SFT-COMP-SEM-CORRECTNESS-001`, `SFT-COMP-SEM-COMPILATION-001`, `SFT-COMP-DIST-CAUSALITY-001`, `SFT-PHYS-MEAS-VALUE-RECORD-001`, `SFT-PHYS-MEAS-CALIBRATION-001`, `SFT-CHEM-MEAS-TRACEABILITY-001`, `SFT-MAT-SUST-LIFECYCLE-001`, `SFT-BIO-BIO-HANDOFF-001`, `SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001`, `SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001`, `SFT-EARTH-HAZARD-RISK-HANDOFF-001`, `SFT-ASTRO-ASTRO-HANDOFF-001`, `SFT-SOCIAL-ENGINEERING-HANDOFF-001`, `SFT-ENG-SUSTAINABILITY-001`, `SFT-ENG-SIMULATION-001`. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external standard, application, performance target, product or prior implementation to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external standards, products, tests or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite components, versions, platforms, tests and lifecycle records; performance remains artifact, environment, user, method and operating-boundary limited.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - which requirement, artifact, component, user, environment or test bears the claim: `artifact-scenario-observer-result-carrier`.
- **boundary** - which operating envelope, version, platform, user and lifecycle stage bound the result: `demonstration-scenario-bounded`.
- **relation** - which generated transformation, interface, comparison or handoff connects the carriers: `declared-scenario-to-visible-behaviour-relation`.
- **record** - which requirements, versions, alternatives, failures, uncertainty and rollback remain reconstructible: `version-setup-scenario-observations-failures-and-limits-held`.
- **evidence** - whether law, requirement, design, simulation, test, demonstration, performance and anomaly remain distinct: `science-design-test-and-decision-classes-explicit`.
- **provenance** - whether the result traces to the One and names every upstream receipt and design choice: `root-bound-forward-translation`.
- **generality** - whether finite extension preserves platforms, versions, failures and operating limits: `positive-finite-extension-preserves-platforms-failures-and-versions`.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: `no-extra-rule`.

Their ordered product gives the unique survivor:

```
artifact-scenario-observer-result-carrier__demonstration-scenario-bounded__declared-s
cenario-to-visible-behaviour-relation__version-setup-scenario-observations-failures-a
nd-limits-held__science-design-test-and-decision-classes-explicit__root-bound-forward
-translation__positive-finite-extension-preserves-platforms-failures-and-versions__no
-extra-rule
```

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses the requirement, artifact, user or environment. Erasing the boundary universalises a version, platform or test. Importing the relation lets a target or preferred application choose the result. Retaining only successful output removes requirements, alternatives, failures and rollback. Conflating evidence classes turns simulation, demonstration or performance into observation or law. Breaking provenance removes the trace to the One and upstream receipts. Erasing finite context makes one implementation universal. Adding a silent dependency or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite requirement or admitted law, versioned artifact, boundary, testable relation and complete record. The successor adds one lawful component, interface, platform, requirement, test, failure, lifecycle stage or anomaly while preserving all earlier identities, adverse rows and receipts.

Falsification and deliberately unfavourable controls

The registered falsification condition is: Reject if a compelling display replaces complete testing or external evidence.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: `source_bound_engineering_requirement_design_test_lifecycle_or_access_record/primary_standard_handbook_guidance_or_project_record_structural_correspondence`. Captured registered sources: 2. Unresolved transports retained: 1. The registered structural consequence corresponded: `True`. External evidence selected the survivor: `false`. Application or performance selected the law: `false`. Successful performance was relabelled scientific proof: `false`. Failed implementation was relabelled scientific falsification: `false`. Simulation or demonstration was relabelled observation: `false`.

- `NASA-SE-APPENDIX-001` - transport `captured`; captured bytes `578114`; snapshot `sha256:c1db2db970b9fd14a7ef98aa0c23575326dedae46cc7cc56c468821ca89edd60`.
- `FDA-DESIGN-CONTROLS-001` - transport `failed_preserved`; captured bytes `0`; snapshot `unresolved and preserved`.
- `CISA-SECURE-BY-DESIGN-001` - transport `captured`; captured bytes `64467`; snapshot `sha256:b4ad9124bddafabd552dc29d861694336c8af6d64f135fa44ad3b1bf24fb4ee2`.

The comparison is structural and source-bound. It checks that a relevant primary handbook, standard, guidance record or open engineering project can represent the declared requirement, artifact, environment, operating boundary, version, test and lifecycle without rewriting the Fold law. It does not claim that a standard proves conformity, that one handbook exhausts Engineering, or that a custodian's reputation proves the result. New records can extend or challenge the surface only with source, version, configuration, method, failures and adverse rows retained.

Scientific meaning

For **engineering demonstration**, the consequence is practical and exact: A demonstration shows bounded behaviour in a retained scenario and remains distinct from verification, validation and science admission. An Engineering claim therefore has to expose the carrier `artifact-scenario-observer-result-carrier`, the relation `declared-scenario-to-visible-behaviour-relation`, the record `version-setup-scenario-observations-failures-and-limits-held`, and the boundary `demonstration-scenario-bounded` together. If any one is hidden, the result may remain a concept, prototype, simulation, demonstration or product choice, but it is not this admitted law.

This distinction protects users, maintainers and science. It prevents a successful output from erasing requirements, failed alternatives, hidden resources, inaccessible users, hazards, maintenance burdens or end-of-life costs. It prevents an application from selecting the upstream law and prevents engineering failure from becoming automatic scientific falsification. A replacement must still generate its forms, eliminate alternatives, survive controls, preserve custody and reproduce through the shared engine.

Machine certificate

Candidate count: `256`; survivor count: `1`; derivation seal: `sha256:3026156e959a7cffbd2ad11fc1f6b3ac26c7a851d6935047b9a7b34d9239fdbd`; independent certificate: `sha256:6dd09dc77df741f099dc61f1f37bf91c1f42fb97d67181fe5f937f6c606e9cf7`; engine receipt: `sha256:dec0c9f70709d6b4c1a799f9d27e8624d4a67af314594331ef174990aadb1f47`. The independent implementation recomputed the candidate product: `True`. All external rows were preserved: `True`.

67. Engineering anomaly

Claim: `SFT-ENG-ANOMALY-001`

Family: `ownership_anomaly_handoffs`

Exact statement: An anomaly is a source-custodied deviation from a preregistered engineering expectation with controls and repeats retained.

Forward chain and exact carrier

The generated carrier is `artifact-expected-observed-context-carrier`. Its required relation is `registered-expectation-to-deviation-relation`. The reconstruction must retain `versions-setup-raw-data-controls-repeats-and-uncertainty-held`. Its declared empirical boundary is `anomaly-protocol-bounded`. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is `SFT-FOUNDATION-ONE-001`, `SFT-FOUNDATION-FOLD-001`, `SFT-FOUNDATION-FOLD-DYNAMICS-001`, `SFT-MATH-EXACT-ARITHMETIC-001`, `SFT-MATH-GRAPH-NETWORK-001`, `SFT-MATH-OPTIMIZATION-001`, `SFT-MATH-DYNAMICAL-SYSTEMS-001`, `SFT-MATH-LOGIC-PROOF-001`, `SFT-INFO-QUANTITY-001`, `SFT-INFO-CHANNEL-CAPACITY-001`, `SFT-COMP-FORM-STATE-TRANSITION-001`, `SFT-COMP-SEM-CORRECTNESS-001`, `SFT-COMP-SEM-COMPILATION-001`, `SFT-COMP-DIST-CAUSALITY-001`, `SFT-PHYS-MEAS-VALUE-RECORD-001`, `SFT-PHYS-MEAS-CALIBRATION-001`, `SFT-CHEM-MEAS-TRACEABILITY-001`, `SFT-MAT-SUST-LIFECYCLE-001`, `SFT-BIO-BIO-HANDOFF-001`, `SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001`, `SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001`, `SFT-EARTH-HAZARD-RISK-HANDOFF-001`, `SFT-ASTRO-ASTRO-HANDOFF-001`, `SFT-SOCIAL-ENGINEERING-HANDOFF-001`, `SFT-ENG-DEMONSTRATION-001`. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external standard, application, performance target, product or prior implementation to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external standards, products, tests or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite components, versions, platforms, tests and lifecycle records; performance remains artifact, environment, user, method and operating-boundary limited.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - which requirement, artifact, component, user, environment or test bears the claim: `artifact-expected-observed-context-carrier`.
- **boundary** - which operating envelope, version, platform, user and lifecycle stage bound the result: `anomaly-protocol-bounded`.

- **relation** - which generated transformation, interface, comparison or handoff connects the carriers: `registered-expectation-to-deviation-relation`.
- **record** - which requirements, versions, alternatives, failures, uncertainty and rollback remain reconstructible: `versions-setup-raw-data-controls-repeats-and-uncertainty-held`.
- **evidence** - whether law, requirement, design, simulation, test, demonstration, performance and anomaly remain distinct: `science-design-test-and-decision-classes-explicit`.
- **provenance** - whether the result traces to the One and names every upstream receipt and design choice: `root-bound-forward-translation`.
- **generality** - whether finite extension preserves platforms, versions, failures and operating limits: `positive-finite-extension-preserves-platforms-failures-and-versions`.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: `no-extra-rule`.

Their ordered product gives the unique survivor:

```
artifact-expected-observed-context-carrier__anomaly-protocol-bounded__registered-expe
ctation-to-deviation-relation__versions-setup-raw-data-controls-repeats-and-uncertain
ty-held__science-design-test-and-decision-classes-explicit__root-bound-forward-transl
ation__positive-finite-extension-preserves-platforms-failures-and-versions__no-extra-
rule
```

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses the requirement, artifact, user or environment. Erasing the boundary universalises a version, platform or test. Importing the relation lets a target or preferred application choose the result. Retaining only successful output removes requirements, alternatives, failures and rollback. Conflating evidence classes turns simulation, demonstration or performance into observation or law. Breaking provenance removes the trace to the One and upstream receipts. Erasing finite context makes one implementation universal. Adding a silent dependency or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite requirement or admitted law, versioned artifact, boundary, testable relation and complete record. The successor adds one lawful component, interface, platform, requirement, test, failure, lifecycle stage or anomaly while preserving all earlier identities, adverse rows and receipts.

Falsification and deliberately unfavourable controls

The registered falsification condition is: Reject if surprise, noise, undocumented configuration or post-hoc expectation is called an anomaly.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: `source_bound_engineering_requirement_design_test_lifecycle_or_access_record/ primary_standard_handbook_guidance_or_project_record_structural_correspondence`. Captured registered sources: 2. Unresolved transports retained: 1. The registered structural consequence corresponded: `True`. External evidence selected the survivor: `false`. Application or performance selected the law: `false`. Successful performance was relabelled scientific proof: `false`. Failed implementation was relabelled scientific falsification: `false`. Simulation or demonstration was relabelled observation: `false`.

- FDA-DESIGN-CONTROLS-001 - transport failed_preserved; captured bytes 0; snapshot unresolved and preserved.
- CISA-SECURE-BY-DESIGN-001 - transport captured; captured bytes 64467; snapshot sha256:b4ad9124bddafabd552dc29d861694336c8af6d64f135fa44ad3b1bf24fb4ee2.
- NASA-SE-HANDBOOK-001 - transport captured; captured bytes 3773440; snapshot sha256:8eeb4887a4dc57a23049da7dd2ed556833cf98e214b240468d987873164ff688.

The comparison is structural and source-bound. It checks that a relevant primary handbook, standard, guidance record or open engineering project can represent the declared requirement, artifact, environment, operating boundary, version, test and lifecycle without rewriting the Fold law. It does not claim that a standard proves conformity, that one handbook exhausts Engineering, or that a custodian's reputation proves the result. New records can extend or challenge the surface only with source, version, configuration, method, failures and adverse rows retained.

Scientific meaning

For **engineering anomaly**, the consequence is practical and exact: An anomaly is a source-custodied deviation from a preregistered engineering expectation with controls and repeats retained. An Engineering claim therefore has to expose the carrier artifact-expected-observed-context-carrier, the relation registered-expectation-to-deviation-relation, the record versions-setup-raw-data-controls-repeats-and-uncertainty-held, and the boundary anomaly-protocol-bounded together. If any one is hidden, the result may remain a concept, prototype, simulation, demonstration or product choice, but it is not this admitted law.

This distinction protects users, maintainers and science. It prevents a successful output from erasing requirements, failed alternatives, hidden resources, inaccessible users, hazards, maintenance burdens or end-of-life costs. It prevents an application from selecting the upstream law and prevents engineering failure from becoming automatic scientific falsification. A replacement must still generate its forms, eliminate alternatives, survive controls, preserve custody and reproduce through the shared engine.

Machine certificate

Candidate count: 256; survivor count: 1; derivation seal: sha256:f86b71f2c1b9ddcf3ee04e04484797e4747d23178f2b2c4f550585525e8ec1a7; independent certificate: sha256:164b14d416c422271548ce908066456d955703d6f983f282a0268876f63c1b5f; engine receipt: sha256:1f0ad93282eef64f4d4fdd8e69f88c5fc50a8c614880548c2fc5e38bec522a54. The independent implementation recomputed the candidate product: True. All external rows were preserved: True.

68. Return of anomaly to science

Claim: SFT-ENG-SCIENCE-RETURN-001

Family: ownership_anomaly_handoffs

Exact statement: A validated anomaly returns as a question to its categorical science owner; it does not enter the model through Engineering.

Forward chain and exact carrier

The generated carrier is engineering-anomaly-owner-question-carrier. Its required relation is validated-deviation-to-categorical-handoff-relation. The reconstruction must retain upstream-receipts-implementation-audit-controls-data-and-open-question-held. Its declared empirical boundary is disciplinary-ownership-bounded. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-OPTIMIZATION-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-CHANNEL-CAPACITY-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-COMP-SEM-CORRECTNESS-001, SFT-COMP-SEM-COMPILATION-001, SFT-COMP-DIST-CAUSALITY-001, SFT-PHYS-MEAS-VALUE-RECORD-001, SFT-PHYS-MEAS-CALIBRATION-001,

SFT-CHEM-MEAS-TRACEABILITY-001, SFT-MAT-SUST-LIFECYCLE-001, SFT-BIO-BIO-HANDOFF-001, SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001, SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-ASTRO-HANDOFF-001, SFT-SOCIAL-ENGINEERING-HANDOFF-001, SFT-ENG-DEMONSTRATION-001, SFT-ENG-ANOMALY-001. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external standard, application, performance target, product or prior implementation to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external standards, products, tests or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite components, versions, platforms, tests and lifecycle records; performance remains artifact, environment, user, method and operating-boundary limited.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - which requirement, artifact, component, user, environment or test bears the claim: `engineering-anomaly-owner-question-carrier`.
- **boundary** - which operating envelope, version, platform, user and lifecycle stage bound the result: `disciplinary-ownership-bounded`.
- **relation** - which generated transformation, interface, comparison or handoff connects the carriers: `validated-deviation-to-categorical-handoff-relation`.
- **record** - which requirements, versions, alternatives, failures, uncertainty and rollback remain reconstructible: `upstream-receipts-implementation-audit-controls-data-and-open-question-held`.
- **evidence** - whether law, requirement, design, simulation, test, demonstration, performance and anomaly remain distinct: `science-design-test-and-decision-classes-explicit`.
- **provenance** - whether the result traces to the One and names every upstream receipt and design choice: `root-bound-forward-translation`.
- **generality** - whether finite extension preserves platforms, versions, failures and operating limits: `positive-finite-extension-preserves-platforms-failures-and-versions`.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: `no-extra-rule`.

Their ordered product gives the unique survivor:

```
engineering-anomaly-owner-question-carrier__disciplinary-ownership-bounded__validated
-deviation-to-categorical-handoff-relation__upstream-receipts-implementation-audit-co
ntrols-data-and-open-question-held__science-design-test-and-decision-classes-explicit
__root-bound-forward-translation__positive-finite-extension-preserves-platforms-failu
res-and-versions__no-extra-rule
```

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses the requirement, artifact, user or environment. Erasing the boundary universalises a version, platform or test. Importing the relation lets a target or preferred application choose the result. Retaining only successful output removes requirements, alternatives, failures and rollback. Conflating evidence classes turns simulation, demonstration or performance into observation or law. Breaking provenance removes the trace to the One and upstream receipts. Erasing finite context makes one implementation universal. Adding a silent dependency or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite requirement or admitted law, versioned artifact, boundary, testable relation and complete record. The successor adds one lawful component, interface, platform, requirement, test, failure, lifecycle stage or anomaly while preserving all earlier identities, adverse rows and receipts.

Falsification and deliberately unfavourable controls

The registered falsification condition is: Reject if the implementation directly edits a scientific law or bypasses branch admission.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: `source_bound_engineering_requirement_design_test_lifecycle_or_access_record/primary_standard_handbook_guidance_or_project_record_structural_correspondence`. Captured registered sources: 2. Unresolved transports retained: 1. The registered structural consequence corresponded: `True`. External evidence selected the survivor: `false`. Application or performance selected the law: `false`. Successful performance was relabelled scientific proof: `false`. Failed implementation was relabelled scientific falsification: `false`. Simulation or demonstration was relabelled observation: `false`.

- `FDA-DESIGN-CONTROLS-001` - transport `failed_preserved`; captured bytes `0`; snapshot `unresolved and preserved`.
- `CISA-SECURE-BY-DESIGN-001` - transport `captured`; captured bytes `64467`; snapshot `sha256:b4ad9124bddafabd552dc29d861694336c8af6d64f135fa44ad3b1bf24fb4ee2`.
- `NASA-SE-HANDBOOK-001` - transport `captured`; captured bytes `3773440`; snapshot `sha256:8eeb4887a4dc57a23049da7dd2ed556833cf98e214b240468d987873164ff688`.

The comparison is structural and source-bound. It checks that a relevant primary handbook, standard, guidance record or open engineering project can represent the declared requirement, artifact, environment, operating boundary, version, test and lifecycle without rewriting the Fold law. It does not claim that a standard proves conformity, that one handbook exhausts Engineering, or that a custodian's reputation proves the result. New records can extend or challenge the surface only with source, version, configuration, method, failures and adverse rows retained.

Scientific meaning

For **return of anomaly to science**, the consequence is practical and exact: A validated anomaly returns as a question to its categorical science owner; it does not enter the model through Engineering. An Engineering claim therefore has to expose the carrier `engineering-anomaly-owner-question-carrier`, the relation `validated-deviation-to-categorical-handoff-relation`, the record `upstream-receipts-implementation-audit-controls-data-and-open-question-held`, and the boundary `disciplinary-ownership-bounded` together. If any one is hidden, the result may remain a concept, prototype, simulation, demonstration or product choice, but it is not this admitted law.

This distinction protects users, maintainers and science. It prevents a successful output from erasing requirements, failed alternatives, hidden resources, inaccessible users, hazards, maintenance burdens or end-of-life costs. It prevents an application from selecting the upstream law and prevents engineering failure from becoming automatic scientific falsification. A replacement must still generate its forms, eliminate alternatives, survive controls, preserve custody and reproduce through the shared engine.

Machine certificate

Candidate count: `256`; survivor count: `1`; derivation seal: `sha256:26a8175caa1e805d2cdda1c29f3078fdae68f3c24934b107885d741492bf53aa`; independent certificate: `sha256:3aa3f8559ae06ad53a5877617243ab45fbf68f222de9b35d44853f94f0a413bf`; engine receipt: `sha256:34e0866bd57b15b2fb248cfc397b9ab24eaf242b57014034536f2d222be3a00a`. The independent implementation recomputed the candidate product: `True`. All external rows were preserved: `True`.

69. Software and computation handoff

Claim: `SFT-ENG-SOFTWARE-HANDOFF-001`

Family: `ownership_anomaly_handoffs`

Exact statement: Computation owns formal semantics; Engineering owns the versioned executable translation and deployment evidence.

Forward chain and exact carrier

The generated carrier is `formal-process-implementation-platform-carrier`. Its required relation is `semantics-to-executable-artifact-relation`. The reconstruction must retain `specification-source-build-platform-tests-and-resource-record-held`. Its declared empirical boundary is `software-implementation-bounded`. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is `SFT-FOUNDATION-ONE-001`, `SFT-FOUNDATION-FOLD-001`, `SFT-FOUNDATION-FOLD-DYNAMICS-001`, `SFT-MATH-EXACT-ARITHMETIC-001`, `SFT-MATH-GRAPH-NETWORK-001`, `SFT-MATH-OPTIMIZATION-001`, `SFT-MATH-DYNAMICAL-SYSTEMS-001`, `SFT-MATH-LOGIC-PROOF-001`, `SFT-INFO-QUANTITY-001`, `SFT-INFO-CHANNEL-CAPACITY-001`, `SFT-COMP-FORM-STATE-TRANSITION-001`, `SFT-COMP-SEM-CORRECTNESS-001`, `SFT-COMP-SEM-COMPILATION-001`, `SFT-COMP-DIST-CAUSALITY-001`, `SFT-PHYS-MEAS-VALUE-RECORD-001`, `SFT-PHYS-MEAS-CALIBRATION-001`, `SFT-CHEM-MEAS-TRACEABILITY-001`, `SFT-MAT-SUST-LIFECYCLE-001`, `SFT-BIO-BIO-HANDOFF-001`, `SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001`, `SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001`, `SFT-EARTH-HAZARD-RISK-HANDOFF-001`, `SFT-ASTRO-ASTRO-HANDOFF-001`, `SFT-SOCIAL-ENGINEERING-HANDOFF-001`, `SFT-ENG-DEMONSTRATION-001`, `SFT-ENG-SCIENCE-RETURN-001`. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external standard, application, performance target, product or prior implementation to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external standards, products, tests or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite components, versions, platforms, tests and lifecycle records; performance remains artifact, environment, user, method and operating-boundary limited.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - which requirement, artifact, component, user, environment or test bears the claim: `formal-process-implementation-platform-carrier`.
- **boundary** - which operating envelope, version, platform, user and lifecycle stage bound the result: `software-implementation-bounded`.
- **relation** - which generated transformation, interface, comparison or handoff connects the carriers: `semantics-to-executable-artifact-relation`.
- **record** - which requirements, versions, alternatives, failures, uncertainty and rollback remain reconstructible: `specification-source-build-platform-tests-and-resource-record-held`.
- **evidence** - whether law, requirement, design, simulation, test, demonstration, performance and anomaly remain distinct: `science-design-test-and-decision-classes-explicit`.
- **provenance** - whether the result traces to the One and names every upstream receipt and design choice: `root-bound-forward-translation`.
- **generality** - whether finite extension preserves platforms, versions, failures and operating limits: `positive-finite-extension-preserves-platforms-failures-and-versions`.

- **extension** - whether any fitted weight, exception or opaque oracle has been added: `no-extra-rule`.

Their ordered product gives the unique survivor:

`formal-process-implementation-platform-carrier__software-implementation-bounded__semantics-to-executable-artifact-relation__specification-source-build-platform-tests-and-resource-record-held__science-design-test-and-decision-classes-explicit__root-bound-forward-translation__positive-finite-extension-preserves-platforms-failures-and-versions__no-extra-rule`

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses the requirement, artifact, user or environment. Erasing the boundary universalises a version, platform or test. Importing the relation lets a target or preferred application choose the result. Retaining only successful output removes requirements, alternatives, failures and rollback. Conflating evidence classes turns simulation, demonstration or performance into observation or law. Breaking provenance removes the trace to the One and upstream receipts. Erasing finite context makes one implementation universal. Adding a silent dependency or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite requirement or admitted law, versioned artifact, boundary, testable relation and complete record. The successor adds one lawful component, interface, platform, requirement, test, failure, lifecycle stage or anomaly while preserving all earlier identities, adverse rows and receipts.

Falsification and deliberately unfavourable controls

The registered falsification condition is: Reject if platform behaviour rewrites computation law or formal correctness proves usable deployment.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: `source_bound_engineering_requirement_design_test_lifecycle_or_access_record/primary_standard_handbook_guidance_or_project_record_structural_correspondence`. Captured registered sources: 2. Unresolved transports retained: 1. The registered structural consequence corresponded: `True`. External evidence selected the survivor: `false`. Application or performance selected the law: `false`. Successful performance was relabelled scientific proof: `false`. Failed implementation was relabelled scientific falsification: `false`. Simulation or demonstration was relabelled observation: `false`.

- `FDA-DESIGN-CONTROLS-001` - transport `failed_preserved`; captured bytes `0`; snapshot `unresolved and preserved`.
- `CISA-SECURE-BY-DESIGN-001` - transport `captured`; captured bytes `64467`; snapshot `sha256:b4ad9124bddafabd552dc29d861694336c8af6d64f135fa44ad3b1bf24fb4ee2`.
- `NASA-SE-HANDBOOK-001` - transport `captured`; captured bytes `3773440`; snapshot `sha256:8eeb4887a4dc57a23049da7dd2ed556833cf98e214b240468d987873164ff688`.

The comparison is structural and source-bound. It checks that a relevant primary handbook, standard, guidance record or open engineering project can represent the declared requirement, artifact, environment, operating boundary, version, test and lifecycle without rewriting the Fold law. It does not claim that a standard proves conformity, that one handbook exhausts Engineering, or that a custodian's reputation proves the result. New records can extend or challenge the surface only with source, version, configuration, method, failures and adverse rows retained.

Scientific meaning

For **software and computation handoff**, the consequence is practical and exact: Computation owns formal semantics; Engineering owns the versioned executable translation and deployment evidence. An Engineering claim therefore has to expose the carrier `formal-process-implementation-platform-carrier`, the relation `semantics-to-executable-artifact-relation`, the record `specification-source-build-platform-tests-and-resource-record-held`, and the boundary `software-implementation-bounded` together. If any one is hidden, the result may remain a concept, prototype, simulation, demonstration or product choice, but it is not this admitted law.

This distinction protects users, maintainers and science. It prevents a successful output from erasing requirements, failed alternatives, hidden resources, inaccessible users, hazards, maintenance burdens or end-of-life costs. It prevents an application from selecting the upstream law and prevents engineering failure from becoming automatic scientific falsification. A replacement must still generate its forms, eliminate alternatives, survive controls, preserve custody and reproduce through the shared engine.

Machine certificate

Candidate count: `256`; survivor count: `1`; derivation seal: `sha256:807396dd909afd3dd8e2ebdfac0ee18219ed813895a841eb746bd345f3cc67f3`; independent certificate: `sha256:a05d5da5ba51c6523ebb4ef88d9afedfe0cca537fdc179695e63482a9cb2a7c5`; engine receipt: `sha256:45f670866891a5c3228a55768feae7ef15cf44776eaae1e92a7f4a1fee5b4e7a`. The independent implementation recomputed the candidate product: `True`. All external rows were preserved: `True`.

70. Physical sciences handoff

Claim: `SFT-ENG-PHYSICAL-HANDOFF-001`

Family: `ownership_anomaly_handoffs`

Exact statement: Physics, Chemistry, Materials, Earth and Astronomy own their laws; Engineering owns bounded artifacts using them.

Forward chain and exact carrier

The generated carrier is `physical-law-component-environment-carrier`. Its required relation is `law-to-physical-implementation-relation`. The reconstruction must retain `law-receipts-materials-geometry-boundary-measurements-and-failures-held`. Its declared empirical boundary is `physical-design-bounded`. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is `SFT-FOUNDATION-ONE-001`, `SFT-FOUNDATION-FOLD-001`, `SFT-FOUNDATION-FOLD-DYNAMICS-001`, `SFT-MATH-EXACT-ARITHMETIC-001`, `SFT-MATH-GRAPH-NETWORK-001`, `SFT-MATH-OPTIMIZATION-001`, `SFT-MATH-DYNAMICAL-SYSTEMS-001`, `SFT-MATH-LOGIC-PROOF-001`, `SFT-INFO-QUANTITY-001`, `SFT-INFO-CHANNEL-CAPACITY-001`, `SFT-COMP-FORM-STATE-TRANSITION-001`, `SFT-COMP-SEM-CORRECTNESS-001`, `SFT-COMP-SEM-COMPILATION-001`, `SFT-COMP-DIST-CAUSALITY-001`, `SFT-PHYS-MEAS-VALUE-RECORD-001`, `SFT-PHYS-MEAS-CALIBRATION-001`, `SFT-CHEM-MEAS-TRACEABILITY-001`, `SFT-MAT-SUST-LIFECYCLE-001`, `SFT-BIO-BIO-HANDOFF-001`, `SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001`, `SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001`, `SFT-EARTH-HAZARD-RISK-HANDOFF-001`, `SFT-ASTRO-ASTRO-HANDOFF-001`, `SFT-SOCIAL-ENGINEERING-HANDOFF-001`, `SFT-ENG-DEMONSTRATION-001`, `SFT-ENG-SOFTWARE-HANDOFF-001`. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external standard, application, performance target, product or prior implementation to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external standards, products, tests or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite components, versions, platforms, tests and lifecycle records; performance remains artifact, environment, user, method and operating-boundary limited.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - which requirement, artifact, component, user, environment or test bears the claim: `physical-law-component-environment-carrier`.
- **boundary** - which operating envelope, version, platform, user and lifecycle stage bound the result: `physical-design-bounded`.
- **relation** - which generated transformation, interface, comparison or handoff connects the carriers: `law-to-physical-implementation-relation`.
- **record** - which requirements, versions, alternatives, failures, uncertainty and rollback remain reconstructible: `law-receipts-materials-geometry-boundary-measurements-and-failures-held`.
- **evidence** - whether law, requirement, design, simulation, test, demonstration, performance and anomaly remain distinct: `science-design-test-and-decision-classes-explicit`.
- **provenance** - whether the result traces to the One and names every upstream receipt and design choice: `root-bound-forward-translation`.
- **generality** - whether finite extension preserves platforms, versions, failures and operating limits: `positive-finite-extension-preserves-platforms-failures-and-versions`.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: `no-extra-rule`.

Their ordered product gives the unique survivor:

`physical-law-component-environment-carrier__physical-design-bounded__law-to-physical-implementation-relation__law-receipts-materials-geometry-boundary-measurements-and-failures-held__science-design-test-and-decision-classes-explicit__root-bound-forward-translation__positive-finite-extension-preserves-platforms-failures-and-versions__no-extra-rule`

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses the requirement, artifact, user or environment. Erasing the boundary universalises a version, platform or test. Importing the relation lets a target or preferred application choose the result. Retaining only successful output removes requirements, alternatives, failures and rollback. Conflating evidence classes turns simulation, demonstration or performance into observation or law. Breaking provenance removes the trace to the One and upstream receipts. Erasing finite context makes one implementation universal. Adding a silent dependency or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite requirement or admitted law, versioned artifact, boundary, testable relation and complete record. The successor adds one lawful component, interface, platform, requirement, test, failure, lifecycle stage or anomaly while preserving all earlier identities, adverse rows and receipts.

Falsification and deliberately unfavourable controls

The registered falsification condition is: Reject if material or environmental behaviour is inferred from product success without scientific admission.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: `source_bound_engineering_requirement_design_test_lifecycle_or_access_record/primary_standard_handbook_guidance_or_project_record_structural_correspondence`. Captured registered sources: 2. Unresolved transports retained: 1. The registered structural consequence corresponded: `True`. External evidence selected the survivor: `false`. Application or performance selected the law: `false`. Successful performance was relabelled scientific proof: `false`. Failed implementation was relabelled scientific falsification: `false`. Simulation or demonstration was relabelled observation: `false`.

- `FDA-DESIGN-CONTROLS-001` - transport `failed_preserved`; captured bytes `0`; snapshot `unresolved and preserved`.
- `CISA-SECURE-BY-DESIGN-001` - transport `captured`; captured bytes `64467`; snapshot `sha256:b4ad9124bddafabd552dc29d861694336c8af6d64f135fa44ad3b1bf24fb4ee2`.
- `NASA-SE-HANDBOOK-001` - transport `captured`; captured bytes `3773440`; snapshot `sha256:8eeb4887a4dc57a23049da7dd2ed556833cf98e214b240468d987873164ff688`.

The comparison is structural and source-bound. It checks that a relevant primary handbook, standard, guidance record or open engineering project can represent the declared requirement, artifact, environment, operating boundary, version, test and lifecycle without rewriting the Fold law. It does not claim that a standard proves conformity, that one handbook exhausts Engineering, or that a custodian's reputation proves the result. New records can extend or challenge the surface only with source, version, configuration, method, failures and adverse rows retained.

Scientific meaning

For **physical sciences handoff**, the consequence is practical and exact: Physics, Chemistry, Materials, Earth and Astronomy own their laws; Engineering owns bounded artifacts using them. An Engineering claim therefore has to expose the carrier `physical-law-component-environment-carrier`, the relation `law-to-physical-implementation-relation`, the record `law-receipts-materials-geometry-boundary-measurements-and-failures-held`, and the boundary `physical-design-bounded` together. If any one is hidden, the result may remain a concept, prototype, simulation, demonstration or product choice, but it is not this admitted law.

This distinction protects users, maintainers and science. It prevents a successful output from erasing requirements, failed alternatives, hidden resources, inaccessible users, hazards, maintenance burdens or end-of-life costs. It prevents an application from selecting the upstream law and prevents engineering failure from becoming automatic scientific falsification. A replacement must still generate its forms, eliminate alternatives, survive controls, preserve custody and reproduce through the shared engine.

Machine certificate

Candidate count: `256`; survivor count: `1`; derivation seal: `sha256:ef0083d8543c7e6e4ffdde1f83b6670a371cfd76a2dc88ccff77ca741dbf7f02`; independent certificate: `sha256:d71d8daaa8776459d8ed3590d16e76694a2b7347b85d00a908d47485bcef6894`; engine receipt: `sha256:3c5e2bcfc0353b573de31a1f50e5b60e1983a348e9712b6ada307d889c5bbcf8`. The independent implementation recomputed the candidate product: `True`. All external rows were preserved: `True`.

71. Biological, medical and social handoff

Claim: `SFT-ENG-BIO-MED-SOCIAL-HANDOFF-001`

Family: `ownership_anomaly_handoffs`

Exact statement: Biology, Medicine, Consciousness and Social Science retain their evidence while Engineering translates a bounded intervention.

Forward chain and exact carrier

The generated carrier is `organism-user-population-device-carrier`. Its required relation is `science-and-context-to-intervention-relation`. The reconstruction must retain `biological-clinical-social-receipts-users-ethics-harms-and-access-held`. Its declared empirical boundary is `human-use-bounded`. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is `SFT-FOUNDATION-ONE-001`, `SFT-FOUNDATION-FOLD-001`, `SFT-FOUNDATION-FOLD-DYNAMICS-001`, `SFT-MATH-EXACT-ARITHMETIC-001`, `SFT-MATH-GRAPH-NETWORK-001`, `SFT-MATH-OPTIMIZATION-001`, `SFT-MATH-DYNAMICAL-SYSTEMS-001`, `SFT-MATH-LOGIC-PROOF-001`, `SFT-INFO-QUANTITY-001`, `SFT-INFO-CHANNEL-CAPACITY-001`, `SFT-COMP-FORM-STATE-TRANSITION-001`, `SFT-COMP-SEM-CORRECTNESS-001`, `SFT-COMP-SEM-COMPILATION-001`, `SFT-COMP-DIST-CAUSALITY-001`, `SFT-PHYS-MEAS-VALUE-RECORD-001`, `SFT-PHYS-MEAS-CALIBRATION-001`, `SFT-CHEM-MEAS-TRACEABILITY-001`, `SFT-MAT-SUST-LIFECYCLE-001`, `SFT-BIO-BIO-HANDOFF-001`, `SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001`, `SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001`, `SFT-EARTH-HAZARD-RISK-HANDOFF-001`, `SFT-ASTRO-ASTRO-HANDOFF-001`, `SFT-SOCIAL-ENGINEERING-HANDOFF-001`, `SFT-ENG-DEMONSTRATION-001`, `SFT-ENG-PHYSICAL-HANDOFF-001`. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external standard, application, performance target, product or prior implementation to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external standards, products, tests or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite components, versions, platforms, tests and lifecycle records; performance remains artifact, environment, user, method and operating-boundary limited.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - which requirement, artifact, component, user, environment or test bears the claim: `organism-user-population-device-carrier`.
- **boundary** - which operating envelope, version, platform, user and lifecycle stage bound the result: `human-use-bounded`.
- **relation** - which generated transformation, interface, comparison or handoff connects the carriers: `science-and-context-to-intervention-relation`.
- **record** - which requirements, versions, alternatives, failures, uncertainty and rollback remain reconstructible: `biological-clinical-social-receipts-users-ethics-harms-and-access-held`.
- **evidence** - whether law, requirement, design, simulation, test, demonstration, performance and anomaly remain distinct: `science-design-test-and-decision-classes-explicit`.
- **provenance** - whether the result traces to the One and names every upstream receipt and design choice: `root-bound-forward-translation`.
- **generality** - whether finite extension preserves platforms, versions, failures and operating limits: `positive-finite-extension-preserves-platforms-failures-and-versions`.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: `no-extra-rule`.

Their ordered product gives the unique survivor:

```
organism-user-population-device-carrier__human-use-bounded__science-and-context-to-in
tervention-relation__biological-clinical-social-receipts-users-ethics-harms-and-acces
s-held__science-design-test-and-decision-classes-explicit__root-bound-forward-transla
```

`tion__positive-finite-extension-preserves-platforms-failures-and-versions__no-extra-rule`

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses the requirement, artifact, user or environment. Erasing the boundary universalises a version, platform or test. Importing the relation lets a target or preferred application choose the result. Retaining only successful output removes requirements, alternatives, failures and rollback. Conflating evidence classes turns simulation, demonstration or performance into observation or law. Breaking provenance removes the trace to the One and upstream receipts. Erasing finite context makes one implementation universal. Adding a silent dependency or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite requirement or admitted law, versioned artifact, boundary, testable relation and complete record. The successor adds one lawful component, interface, platform, requirement, test, failure, lifecycle stage or anomaly while preserving all earlier identities, adverse rows and receipts.

Falsification and deliberately unfavourable controls

The registered falsification condition is: Reject if device performance proves diagnosis, subjective experience, equal access or ethical legitimacy.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: `source_bound_engineering_requirement_design_test_lifecycle_or_access_record/primary_standard_handbook_guidance_or_project_record_structural_correspondence`. Captured registered sources: 2. Unresolved transports retained: 1. The registered structural consequence corresponded: `True`. External evidence selected the survivor: `false`. Application or performance selected the law: `false`. Successful performance was relabelled scientific proof: `false`. Failed implementation was relabelled scientific falsification: `false`. Simulation or demonstration was relabelled observation: `false`.

- `FDA-DESIGN-CONTROLS-001` - transport `failed_preserved`; captured bytes `0`; snapshot `unresolved and preserved`.
- `CISA-SECURE-BY-DESIGN-001` - transport `captured`; captured bytes `64467`; snapshot `sha256:b4ad9124bddafabd552dc29d861694336c8af6d64f135fa44ad3b1bf24fb4ee2`.
- `NASA-SE-HANDBOOK-001` - transport `captured`; captured bytes `3773440`; snapshot `sha256:8eeb4887a4dc57a23049da7dd2ed556833cf98e214b240468d987873164ff688`.

The comparison is structural and source-bound. It checks that a relevant primary handbook, standard, guidance record or open engineering project can represent the declared requirement, artifact, environment, operating boundary, version, test and lifecycle without rewriting the Fold law. It does not claim that a standard proves conformity, that one handbook exhausts Engineering, or that a custodian's reputation proves the result. New records can extend or challenge the surface only with source, version, configuration, method, failures and adverse rows retained.

Scientific meaning

For **biological, medical and social handoff**, the consequence is practical and exact: Biology, Medicine, Consciousness and Social Science retain their evidence while Engineering translates a bounded intervention. An Engineering claim therefore has to expose the carrier `organism-user-population-device-carrier`, the relation `science-and-context-to-intervention-relation`, the record `biological-clinical-social-receipts-users-ethics-harms-and-access-held`, and the boundary `human-use-bounded` together. If any one is hidden, the result may remain a concept, prototype, simulation, demonstration or product choice, but it is not this admitted law.

This distinction protects users, maintainers and science. It prevents a successful output from erasing requirements, failed alternatives, hidden resources, inaccessible users, hazards, maintenance burdens or end-of-life costs. It prevents an application from selecting the upstream law and prevents engineering failure from becoming automatic scientific falsification. A replacement must still generate its forms, eliminate alternatives, survive controls, preserve custody and reproduce through the shared engine.

Machine certificate

Candidate count: `256`; survivor count: `1`; derivation seal: `sha256:19643457f66096c1fadc2723505b9d55eb837d3365aaaed304f8711470edcdb4`; independent certificate: `sha256:6bce824dd0bce46d6a41396731df34109f30ad1c1b6ddb74faf660d9a8f6f294`; engine receipt: `sha256:3feb705c20067692e2e8d4166858d32ca429a9144902b0fb43bedbce3418cacf`. The independent implementation recomputed the candidate product: `True`. All external rows were preserved: `True`.

72. SFT-derived self-hosted V4 handoff

Claim: `SFT-ENG-V4-HANDOFF-001`

Family: `ownership_anomaly_handoffs`

Exact statement: A future V4 self-hosted rebuild must reproduce V3 semantics, receipts and constraints through an independently auditable bootstrap.

Forward chain and exact carrier

The generated carrier is `v3-corpus-language-compiler-runtime-carrier`. Its required relation is `admitted-semantics-to-self-hosted-rebuild-relation`. The reconstruction must retain `all-upstream-receipts-specification-bootstrap-tests-divergences-and-provenance-held`. Its declared empirical boundary is `future-rebuild-authority-bounded`. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is `SFT-FOUNDATION-ONE-001`, `SFT-FOUNDATION-FOLD-001`, `SFT-FOUNDATION-FOLD-DYNAMICS-001`, `SFT-MATH-EXACT-ARITHMETIC-001`, `SFT-MATH-GRAPH-NETWORK-001`, `SFT-MATH-OPTIMIZATION-001`, `SFT-MATH-DYNAMICAL-SYSTEMS-001`, `SFT-MATH-LOGIC-PROOF-001`, `SFT-INFO-QUANTITY-001`, `SFT-INFO-CHANNEL-CAPACITY-001`, `SFT-COMP-FORM-STATE-TRANSITION-001`, `SFT-COMP-SEM-CORRECTNESS-001`, `SFT-COMP-SEM-COMPILATION-001`, `SFT-COMP-DIST-CAUSALITY-001`, `SFT-PHYS-MEAS-VALUE-RECORD-001`, `SFT-PHYS-MEAS-CALIBRATION-001`, `SFT-CHEM-MEAS-TRACEABILITY-001`, `SFT-MAT-SUST-LIFECYCLE-001`, `SFT-BIO-BIO-HANDOFF-001`, `SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001`, `SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001`, `SFT-EARTH-HAZARD-RISK-HANDOFF-001`, `SFT-ASTRO-ASTRO-HANDOFF-001`, `SFT-SOCIAL-ENGINEERING-HANDOFF-001`, `SFT-ENG-DEMONSTRATION-001`, `SFT-ENG-BIO-MED-SOCIAL-HANDOFF-001`. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external standard, application, performance target, product or prior implementation to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external standards, products, tests or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite components, versions, platforms, tests and lifecycle records; performance remains artifact, environment, user, method and operating-boundary limited.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - which requirement, artifact, component, user, environment or test bears the claim: `v3-corpus-language-compiler-runtime-carrier`.
- **boundary** - which operating envelope, version, platform, user and lifecycle stage bound the result: `future-rebuild-authority-bounded`.
- **relation** - which generated transformation, interface, comparison or handoff connects the carriers: `admitted-semantics-to-self-hosted-rebuild-relation`.
- **record** - which requirements, versions, alternatives, failures, uncertainty and rollback remain reconstructible: `all-upstream-receipts-specification-bootstrap-tests-divergences-and-provenance-held`.
- **evidence** - whether law, requirement, design, simulation, test, demonstration, performance and anomaly remain distinct: `science-design-test-and-decision-classes-explicit`.
- **provenance** - whether the result traces to the One and names every upstream receipt and design choice: `root-bound-forward-translation`.
- **generality** - whether finite extension preserves platforms, versions, failures and operating limits: `positive-finite-extension-preserves-platforms-failures-and-versions`.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: `no-extra-rule`.

Their ordered product gives the unique survivor:

```
v3-corpus-language-compiler-runtime-carrier__future-rebuild-authority-bounded__admitt
ed-semantics-to-self-hosted-rebuild-relation__all-upstream-receipts-specification-boo
tstrap-tests-divergences-and-provenance-held__science-design-test-and-decision-classe
s-explicit__root-bound-forward-translation__positive-finite-extension-preserves-platf
orms-failures-and-versions__no-extra-rule
```

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses the requirement, artifact, user or environment. Erasing the boundary universalises a version, platform or test. Importing the relation lets a target or preferred application choose the result. Retaining only successful output removes requirements, alternatives, failures and rollback. Conflating evidence classes turns simulation, demonstration or performance into observation or law. Breaking provenance removes the trace to the One and upstream receipts. Erasing finite context makes one implementation universal. Adding a silent dependency or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite requirement or admitted law, versioned artifact, boundary, testable relation and complete record. The successor adds one lawful component, interface, platform, requirement, test, failure, lifecycle stage or anomaly while preserving all earlier identities, adverse rows and receipts.

Falsification and deliberately unfavourable controls

The registered falsification condition is: Reject if a self-hosted implementation becomes authority merely by reproducing itself or changes the protected V3 record.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: `source_bound_engineering_requirement_design_test_lifecycle_or_access_record/primary_standard_handbook_guidance_or_project_record_structural_correspondence`. Captured registered sources: 2. Unresolved transports retained: 1. The registered structural consequence corresponded: `True`. External evidence selected the survivor: `false`. Application or performance selected the law: `false`. Successful performance was relabelled scientific proof: `false`. Failed implementation was relabelled scientific falsification: `false`. Simulation or demonstration was relabelled observation: `false`.

- `FDA-DESIGN-CONTROLS-001` - transport `failed_preserved`; captured bytes `0`; snapshot `unresolved and preserved`.
- `CISA-SECURE-BY-DESIGN-001` - transport `captured`; captured bytes `64467`; snapshot `sha256:b4ad9124bddafabd552dc29d861694336c8af6d64f135fa44ad3b1bf24fb4ee2`.
- `NASA-SE-HANDBOOK-001` - transport `captured`; captured bytes `3773440`; snapshot `sha256:8eeb4887a4dc57a23049da7dd2ed556833cf98e214b240468d987873164ff688`.

The comparison is structural and source-bound. It checks that a relevant primary handbook, standard, guidance record or open engineering project can represent the declared requirement, artifact, environment, operating boundary, version, test and lifecycle without rewriting the Fold law. It does not claim that a standard proves conformity, that one handbook exhausts Engineering, or that a custodian's reputation proves the result. New records can extend or challenge the surface only with source, version, configuration, method, failures and adverse rows retained.

Scientific meaning

For **sft-derived self-hosted v4 handoff**, the consequence is practical and exact: A future V4 self-hosted rebuild must reproduce V3 semantics, receipts and constraints through an independently auditable bootstrap. An Engineering claim therefore has to expose the carrier `v3-corpus-language-compiler-runtime-carrier`, the relation `admitted-semantics-to-self-hosted-rebuild-relation`, the record `all-upstream-receipts-specification-bootstrap-tests-divergences-and-provenance-held`, and the boundary `future-rebuild-authority-bounded` together. If any one is hidden, the result may remain a concept, prototype, simulation, demonstration or product choice, but it is not this admitted law.

This distinction protects users, maintainers and science. It prevents a successful output from erasing requirements, failed alternatives, hidden resources, inaccessible users, hazards, maintenance burdens or end-of-life costs. It prevents an application from selecting the upstream law and prevents engineering failure from becoming automatic scientific falsification. A replacement must still generate its forms, eliminate alternatives, survive controls, preserve custody and reproduce through the shared engine.

Machine certificate

Candidate count: `256`; survivor count: `1`; derivation seal: `sha256:adc9b2ebc17ca1bcc9233c9a1d6b8f9e565c29ee96ea6bbefd18d70c6343165e`; independent certificate: `sha256:50aa60a21e8d18dd48b22f8a37f5c3b29518aeac5c48861eee94b85faa0584b5`; engine receipt: `sha256:aedcf633fb62d3dc4fd8a18c5fbfed692746a0d2f57fd8b41ed2d87244544ded`. The independent implementation recomputed the candidate product: `True`. All external rows were preserved: `True`.

Historical reconciliation

The inherited V1/V2 census contains 763 entries. Every entry was classified atomically by present branch ownership. Seven Engineering-owned questions-reproduction, end-to-end audit, system architecture, accessible artifact, law translation, application validation and bounded control-were registered as questions and mapped to present claims. Their former answers and implementations were not imported as premises. Fold Protein, Fold Chess, Fold Go and Unison AI remain later application handoffs rather than foundation selectors.

What this foundation does and does not claim

It claims that the presently known foundational forms required to translate admitted science into bounded artifacts have been generated, uniquely closed, independently reconstructed and source-bound. It does not claim that every engineering domain or future technology is finished, that every standard is accessible, or that a successful implementation proves its upstream law. It does not turn simulations into observations, demonstrations into verification, verification into validation, performance into science admission, or applications into law selectors.

The branch is complete to Maria Smith's current standard and evidence, and open to lawful extension. A new translation may add a component, interface, platform, user, hazard, test, lifecycle record or anomaly bridge. It must preserve existing adverse, absent and failed records and pass the same engine. No branch is permanently locked against valid additions.

Full-field continuation roadmap

- systems requirements mechanical manufacturing and industrial engineering.
- electrical electronic photonic hardware software network control and telecommunications engineering.
- civil structural transport infrastructure aerospace space marine nuclear geological mining and subsurface engineering.
- chemical process agricultural food biological materials energy environmental resource and biomedical engineering.
- robotics autonomy human-machine microtechnology nanotechnology metrology laboratories safety security maintainability resilience and full lifecycle evidence.
- fresh Fold Protein Fold Chess Fold Go and Unison AI translations only after prerequisite branches are ready.
- implementation-distinct cross-platform reference artifacts for every scientific branch.
- future V4 reconstruction in SFT-derived self-hosted language compiler runtime and proof principles.

Every later extension will be decomposed into nonduplicate obligations, sealed before outcomes, admitted claim by claim and incorporated through a versioned paper update. Applications, public policy and engineered platforms may test the laws but may not select them.

Reproduction and present validation state

During branch construction the repository used proportionate checks: both immutable seals around each admission, one 256-form census and independent reconstruction per claim, and the six Engineering-focused tests after integration. After all fifteen foundations were integrated, the untouched repository command `python3 -m sft verify-all` passed on 28 July 2026: 1,319/1,319 derivations replayed, 969/969 unit and end-to-end tests passed, 1,264/1,264 core executable lines covered, 1,011 empirical claims audited, 114/114 formal Physics claims reaching measurement and 5/5 live exact NIST/CODATA checks. The measured runtime was 4,753.740 seconds.

This branch passed 72 sequential admissions, 72 independent validators, 18,432 candidate decisions, its integration audit and six focused tests. Neither the engine nor verification authority was modified. The paper and evidence release are published open access at DOI 10.5281/zenodo.21640816.

Rights, participation and scientific admission

Anyone may read, reproduce, criticize and extend the work. No credential is required to speak. The Ernos Labs designation requires the shared empirical constitution: visible root derivation, exact grammar, complete enumeration, no target selection, no hidden fit, preserved unfavourable evidence, independent reconstruction, unchanged protected authority, retained authorship and open licensing. Criticism is public participation; admission is a separately evidenced result.

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Discord: https://discord.gg/ucwGryVxGr

GitHub: https://github.com/MettaMazza

Code license: Apache-2.0

Machine identities

- Inventory: `sha256:8015e13a40e432153333967fc6c46576cc53db35099bb9648748fe6c6a2ccc50`
- Prior audit: `sha256:afe78a50e116317350fe94bcf5b2fab7591c74781c3429058601b19b1ae1dfc1`
- Integration audit: `sha256:3e61762ee704c25ce9a784d6ded5bc5416cce789c5837854f1129a68381f74af`
- Source registry: `sha256:5550d8a107d79ce68c88899c58db9c895b07765a6c07e2f11f882b6bd483cc6b`
- External target census: `sha256:ef96db6229b8b2e91cfcfb8a2a4ff44c456112af85da2827e766c33faf51d73b`
- Engine: `sha256:4f4cdd7986808e6a6102d650c85e6093d6425e49f14a5f05d70fa05e6031d46a`
- Verification authority: `sha256:bf810a190b504f0f874a778a52e23251904b17b40a7364135e74b34e8ba0c3b8`

Complete current-census successor supplement

The authoritative current scope of this paper is 80 live model-admitted claims (`engineering_translation` 80). Counts retained inside the preceding-version narrative describe the dated freeze at which those passages were written; this successor table and the machine census control the current total.

The preceding manuscript did not contain 8 later-admitted claim section(s). They are included below in dependency order so the successor covers the complete current branch surface.

Tesla resonant-transfer engineering protocol

Claim ID: `SFT-ENG-TESLA-RESONANT-TRANSFER-PROTOCOL-002`

Exact statement: A Tesla-family transfer test is lawful only as a sealed, calibrated connected-path protocol with phase, load, complete power ledger, adverse controls and visible halt retained.

Dependency route: `SFT-ENG-REQUIREMENT-001` -> `SFT-ENG-MEASUREMENT-001` -> `SFT-ENG-CALIBRATION-001` -> `SFT-ENG-ACCEPTANCE-TEST-001` -> `SFT-ENG-SAFETY-001` -> `SFT-ENG-TRACEABILITY-001` -> `SFT-ENG-REPRODUCIBILITY-001` -> `SFT-ENG-DEMONSTRATION-001` -> `SFT-PHYS-TESLA-RESONANT-TRANSFER-081` -> `SFT-PHYS-VALIDATION-TESLA-RESONANCE-FAMILY-082`

Candidate generator: Generate the complete eight-axis engineering protocol product for SFT-ENG-TESLA-RESONANT-TRANSFER-PROTOCOL-002, then reconstruct every protocol field and control independently.

Grammar boundary: Every bounded resonant-transfer apparatus and result class represented by the sealed upstream Tesla family.

Complete census: 256 candidates; 1 unique survivor; 4 controls

Survivor: `sealed-upstream-receipts__phase-bound-connected-path-complete-ledger-protocol__complete-common-and-domain-record__complete-declared-control-family__favourable-adverse-absent-unresolved__visible-halt-and-bounded-safe-state__protocol-seal-before-outcome__no-law-rewrite`

Exact result: The protocol requires all common records plus source phase, resonator mode, receiver load and stored, delivered, returned and loss carriers; it includes drive-absent, off-resonance, disconnected, reversed-phase, dummy-load and independent-ledger controls and makes no success claim before execution.

Closure: `depth_independent`; minimality `True`; named-shape uniqueness `True`

External status: `independently_replicated`

Engine receipt: `sha256:d5cc4adc3231b51af5e107ea7de0cb32d343245955a6ce9d2390d8af532163be` at `receipts/engine/model_admitted/SFT-ENG-TESLA-RESONANT-TRANSFER-PROTOCOL-002-d5cc4adc3231b51a.json`

Control	Passed	Expected	Observed
false_premise	True	reject the form missing the first structural preservation	reject the form missing the first structural preservation
tampered_source	True	reject a changed source identity	reject a changed source identity
tampered_artifact	True	reject a missing, duplicate or additional survivor	reject a missing, duplicate or additional survivor
boundary	True	reject forbidden values, target feedback and extra laws	reject forbidden values, target feedback and extra laws

Driven vacuum/inertial-response engineering protocol

Claim ID: `SFT-ENG-VACUUM-INERTIA-RESPONSE-PROTOCOL-002`

Exact statement: A driven vacuum/inertial test is lawful only when drive, paired response channels, all ordinary confounds, complete restoration and source accounting remain independently measurable and preregistered.

Dependency route: `SFT-ENG-TESLA-RESONANT-TRANSFER-PROTOCOL-002` -> `SFT-ENG-REQUIREMENT-001` -> `SFT-ENG-MEASUREMENT-001` -> `SFT-ENG-CALIBRATION-001` -> `SFT-ENG-ACCEPTANCE-TEST-001` -> `SFT-ENG-SAFETY-001` -> `SFT-ENG-TRACEABILITY-001` -> `SFT-ENG-REPRODUCIBILITY-001` -> `SFT-ENG-DEMONSTRATION-001` -> `SFT-PHYS-VACUUM-LOCAL-RESONANT-DRIVE-083` -> `SFT-PHYS-VACUUM-INERTIA-COVARIATION-084` -> `SFT-PHYS-VACUUM-INERTIA-COMPLETE-LEDGER-086` -> `SFT-PHYS-VALIDATION-VACUUM-INERTIA-DRIVE-FAMILY-087`

Candidate generator: Generate the complete eight-axis engineering protocol product for SFT-ENG-VACUUM-INERTIA-RESPONSE-PROTOCOL-002, then reconstruct every protocol field and control independently.

Grammar boundary: Every bounded source-driven apparatus state, confound, reversal and result class represented by the sealed vacuum/inertia family.

Complete census: 256 candidates; 1 unique survivor; 4 controls

Survivor: `sealed-upstream-receipts__paired-drive-response-restoration-protocol__complete-common-and-domain-record__complete-declared-control-family__favourable-adverse-absent-unresolved__visible-halt-and-bounded-safe-state__protocol-seal-before-outcome__no-law-rewrite`

Exact result: The protocol separates resonant drive, vacuum proxy, inertial response, restoration and ordinary thermal, electromagnetic, mechanical and gravitational confounds; no driven-inertia observation is asserted.

Closure: `depth_independent`; minimality `True`; named-shape uniqueness `True`

External status: `independently_replicated`

Engine receipt: `sha256:afe48fe26787fd9d8ac066b5568ce0cb8799828c690818b8bf4445f591aba1e4` at `receipts/engine/model_admitted/SFT-ENG-VACUUM-INERTIA-RESPONSE-PROTOCOL-002-afe48fe26787fd9d.json`

Control	Passed	Expected	Observed
false_premise	True	reject the form missing the first structural preservation	reject the form missing the first structural preservation
tampered_source	True	reject a changed source identity	reject a changed source identity
tampered_artifact	True	reject a missing, duplicate or additional survivor	reject a missing, duplicate or additional survivor
boundary	True	reject forbidden values, target feedback and extra laws	reject forbidden values, target feedback and extra laws

Vacuum-beat transfer and restoration protocol

Claim ID: `SFT-ENG-VACUUM-BEAT-RESTORATION-PROTOCOL-002`

Exact statement: A vacuum-beat test must distinguish the exact outward one-sixth transfer from complete returned-cycle restoration and must halt any net-gain claim whose source, pump, loss or final-state ledger remains open.

Dependency route: `SFT-ENG-VACUUM-INERTIA-RESPONSE-PROTOCOL-002` -> `SFT-ENG-REQUIREMENT-001` -> `SFT-ENG-MEASUREMENT-001` -> `SFT-ENG-CALIBRATION-001` -> `SFT-ENG-ACCEPTANCE-TEST-001` -> `SFT-ENG-SAFETY-001` -> `SFT-ENG-TRACEABILITY-001` -> `SFT-ENG-REPRODUCIBILITY-001` -> `SFT-ENG-DEMONSTRATION-001` -> `SFT-PHYS-VACUUM-ASYMMETRIC-BEAT-EXTRACTION-003` -> `SFT-PHYS-VACUUM-COMPLETE-CYCLE-LEDGER-003` -> `SFT-PHYS-VALIDATION-VACUUM-EXTRACTION-003`

Candidate generator: Generate the complete eight-axis engineering protocol product for SFT-ENG-VACUUM-BEAT-RESTORATION-PROTOCOL-002, then reconstruct every protocol field and control independently.

Grammar boundary: The exact Fold beat-transfer grammar and every apparatus source, pump, boundary, restoration and result class needed to test it.

Complete census: 256 candidates; 1 unique survivor; 4 controls

Survivor: `sealed-upstream-receipts__one-sixth-outward-and-one-sixth-return-ledger-protocol__complete-common-and-domain-record__complete-declared-control-family__favourable-adverse-absent-unresolved__visible-halt-and-bounded-safe-state__protocol-seal-before-outcome__no-law-rewrite`

Exact result: The protocol retains the half-One initial carrier, one-third residual, one-sixth outward work transfer, one-sixth restoration transfer and identical final state, with independent power ledgers and explicit no-unrecorded-gain acceptance boundary.

Closure: `depth_independent`; minimality `True`; named-shape uniqueness `True`

External status: `independently_replicated`

Engine receipt: `sha256:b9889a36c4ed3f9192d7b832dc7d831dd6d0fced428632192721d1d433223769` at `receipts/engine/model_admitted/SFT-ENG-VACUUM-BEAT-RESTORATION-PROTOCOL-002-b9889a36c4ed3f91.json`

Control	Passed	Expected	Observed
false_premise	True	reject the form missing the first structural preservation	reject the form missing the first structural preservation
tampered_source	True	reject a changed source identity	reject a changed source identity
tampered_artifact	True	reject a missing, duplicate or additional survivor	reject a missing, duplicate or additional survivor
boundary	True	reject forbidden values, target feedback and extra laws	reject forbidden values, target feedback and extra laws

Sector-five/seven blind detection protocol

Claim ID: `SFT-ENG-SECTOR-FIVE-SEVEN-DETECTION-PROTOCOL-002`

Exact statement: A sector-five/seven search seals the complete p-fibre signatures before unblinding and retains backgrounds, systematics, nulls, outside-list events and independent reconstructions without relabelling a standing prediction as discovery.

Dependency route: `SFT-ENG-REQUIREMENT-001` -> `SFT-ENG-MEASUREMENT-001` -> `SFT-ENG-CALIBRATION-001` -> `SFT-ENG-ACCEPTANCE-TEST-001` -> `SFT-ENG-SAFETY-001` -> `SFT-ENG-TRACEABILITY-001` -> `SFT-ENG-REPRODUCIBILITY-001` -> `SFT-ENG-DEMONSTRATION-001` -> `SFT-PHYS-FORCE-PRIME-SECTOR-LADDER-002` -> `SFT-PHYS-FORCE-COMPLETE-SECTOR-INVENTORY-003` -> `SFT-PHYS-NO-EXTRA-SECTOR-PARTICLE-BOUNDARY-093` -> `SFT-PHYS-VALIDATION-NEW-SECTOR-COMPLETE-FAMILY-095`

Candidate generator: Generate the complete eight-axis engineering protocol product for SFT-ENG-SECTOR-FIVE-SEVEN-DETECTION-PROTOCOL-002, then reconstruct every protocol field and control independently.

Grammar boundary: The complete sector-five/seven signature table, detector/background alternatives, blind analysis states and outside-list falsification boundary.

Complete census: 256 candidates; 1 unique survivor; 4 controls

Survivor: `sealed-upstream-receipts__sealed-p-five-p-seven-blind-signature-protocol__complete-common-and-domain-record__complete-declared-control-family__favourable-adverse-absent-unresolved__visible-halt-and-bounded-safe-state__protocol-seal-before-outcome__no-law-rewrite`

Exact result: The protocol preregisters p=5 with 5 charge labels, 24 mediators and coupling 4/5, and p=7 with 7 labels, 48 mediators and coupling 6/7; those are standing signatures, not observed particles.

Closure: `depth_independent`; minimality `True`; named-shape uniqueness `True`

External status: `independently_replicated`

Engine receipt: `sha256:c0c460e08c6164471534ee3ceb0ed2d549c70d6887d293f978c5ed140683a095` at `receipts/engine/model_admitted/SFT-ENG-SECTOR-FIVE-SEVEN-DETECTION-PROTOCOL-002-c0c460e08c616447.json`

Control	Passed	Expected	Observed
false_premise	True	reject the form missing the first structural preservation	reject the form missing the first structural preservation
tampered_source	True	reject a changed source identity	reject a changed source identity
tampered_artifact	True	reject a missing, duplicate or additional survivor	reject a missing, duplicate or additional survivor
boundary	True	reject forbidden values, target feedback and extra laws	reject forbidden values, target feedback and extra laws

Smithium synthesis and joint-identification protocol

Claim ID: `SFT-ENG-SMITHIUM-SYNTHESIS-IDENTIFICATION-PROTOCOL-002`

Exact statement: A Smithium test retains exact 126/184/310 conservation, apparatus and yield records, genetically linked decay, mass, ion, oxidation and spectroscopy channels, while preserving the distinction between SFT evidence and official discovery criteria.

Dependency route: `SFT-ENG-REQUIREMENT-001` -> `SFT-ENG-MEASUREMENT-001` -> `SFT-ENG-CALIBRATION-001` -> `SFT-ENG-ACCEPTANCE-TEST-001` -> `SFT-ENG-SAFETY-001` -> `SFT-ENG-TRACEABILITY-001` -> `SFT-ENG-REPRODUCIBILITY-001` -> `SFT-ENG-DEMONSTRATION-001` -> `SFT-CHEM-SMITHIUM-SYNTHESIS-CONSERVATION-001` -> `SFT-CHEM-SMITHIUM-JOINT-DETECTION-001` -> `SFT-CHEM-VALIDATION-SMITHIUM-COMPLETE-FAMILY-001`

Candidate generator: Generate the complete eight-axis engineering protocol product for SFT-ENG-SMITHIUM-SYNTHESIS-IDENTIFICATION-PROTOCOL-002, then reconstruct every protocol field and control independently.

Grammar boundary: Every conservation-valid Smithium attempt, complete joint-evidence class, control, ambiguity and result class inside the sealed Smithium family.

Complete census: 256 candidates; 1 unique survivor; 4 controls

Survivor: `sealed-upstream-receipts__Smithium-126-complete-synthesis-identification-protocol__complete-common-and-domain-record__complete-declared-control-family__favourable-adverse-absent-unresolved__visible-halt-and-bounded-safe-state__protocol-seal-before-outcome__no-law-rewrite`

Exact result: The protocol binds every synthesis attempt to Z=126, N=184 and A=310 conservation and requires joint mass, nuclear, decay, ion/oxidation and spectroscopic records; it does not claim Smithium has been produced.

Closure: `depth_independent`; minimality `True`; named-shape uniqueness `True`

External status: `independently_replicated`

Engine receipt: `sha256:b56e36103ed992219a22c4d9bedfaa034aedb02e98f8217d082f40ebcfafc7e9` at `receipts/engine/model_admitted/SFT-ENG-SMITHIUM-SYNTHESIS-IDENTIFICATION-PROTOCOL-002-b56e36103ed99221.json`

Control	Passed	Expected	Observed
false_premise	True	reject the form missing the first structural preservation	reject the form missing the first structural preservation
tampered_source	True	reject a changed source identity	reject a changed source identity
tampered_artifact	True	reject a missing, duplicate or additional survivor	reject a missing, duplicate or additional survivor
boundary	True	reject forbidden values, target feedback and extra laws	reject forbidden values, target feedback and extra laws

Complete prior-return engineering protocol family

Claim ID: `SFT-ENG-NOVEL-TRANSLATIONS-COMPLETE-FAMILY-002`

Exact statement: The five mandatory prior-return translations form one traceable engineering family whose complete records, controls, result classes, safe halts and non-selection boundary are independently reconstructible without asserting unperformed experiments.

Dependency route: `SFT-ENG-TESLA-RESONANT-TRANSFER-PROTOCOL-002` -> `SFT-ENG-VACUUM-INERTIA-RESPONSE-PROTOCOL-002` -> `SFT-ENG-VACUUM-BEAT-RESTORATION-PROTOCOL-002` -> `SFT-ENG-SECTOR-FIVE-SEVEN-DETECTION-PROTOCOL-002` -> `SFT-ENG-SMITHIUM-SYNTHESIS-IDENTIFICATION-PROTOCOL-002` -> `SFT-ENG-E2E-001` -> `SFT-ENG-INDEPENDENT-CHECK-001` -> `SFT-ENG-PORTABLE-DATA-001`

Candidate generator: Generate the complete eight-axis engineering protocol product for SFT-ENG-NOVEL-TRANSLATIONS-COMPLETE-FAMILY-002, then reconstruct every protocol field and control independently.

Grammar boundary: Exactly the five mandatory prior-corpus Engineering translations in the frozen 2026-07-28 return census; physical executions remain extension-open.

Complete census: 256 candidates; 1 unique survivor; 4 controls

Survivor: `sealed-upstream-receipts__five-protocol-traceable-complete-family__complete-common-and-domain-record__complete-declared-control-family__favourable-adverse-absent-unresolved__visible-halt-and-bounded-safe-state__protocol-seal-before-outcome__no-law-rewrite`

Exact result: Exactly five protocols close the registered Engineering novel-return boundary. Every one carries sealed upstream identities, complete records, controls, all four result classes, safety halts, independent reconstruction and an explicit unperformed-outcome status.

Closure: `depth_independent`; minimality `True`; named-shape uniqueness `True`

External status: `independently_replicated`

Engine receipt: `sha256:b29fcbcb46dbf68ed79bedc13a7aa2fb391ba487875c8fb21b973d5093fc1ea5` at `receipts/engine/model_admitted/SFT-ENG-NOVEL-TRANSLATIONS-COMPLETE-FAMILY-002-b29fcbcb46dbf68e.json`

Control	Passed	Expected	Observed
false_premise	True	reject the form missing the first structural preservation	reject the form missing the first structural preservation
tampered_source	True	reject a changed source identity	reject a changed source identity
tampered_artifact	True	reject a missing, duplicate or additional survivor	reject a missing, duplicate or additional survivor

Control	Passed	Expected	Observed
boundary	True	reject forbidden values, target feedback and extra laws	reject forbidden values, target feedback and extra laws

Open consciousness, placebo and cross-binding protocol

Claim ID: `SFT-ENG-CONSCIOUSNESS-PLACEBO-CROSS-BINDING-PROTOCOL-002`

Exact statement: A lawful human-participant test keeps interior report, objective physiology, modality binding, expectation, blinding, available-state limits, consent, privacy and adverse events distinct while preserving every result class.

Dependency route: `SFT-ENG-NOVEL-TRANSLATIONS-COMPLETE-FAMILY-002` -> `SFT-ENG-REQUIREMENT-001` -> `SFT-ENG-MEASUREMENT-001` -> `SFT-ENG-ACCEPTANCE-TEST-001` -> `SFT-ENG-SAFETY-001` -> `SFT-ENG-TRACEABILITY-001` -> `SFT-ENG-REPRODUCIBILITY-001` -> `SFT-MED-INFORMED-CONSENT-001` -> `SFT-MED-CLINICAL-PRIVACY-001` -> `SFT-CONSC-CROSS-MODAL-QUALIA-001` -> `SFT-CONSC-SYNAESTHESIA-DIRECTIONAL-LOCK-002` -> `SFT-CONSC-VALIDATION-NONORDINARY-COMPLETE-FAMILY-002` -> `SFT-MED-PLACEBO-AVAILABLE-STATE-BOUNDARY-002` -> `SFT-MED-PLACEBO-OBJECTIVE-REPORT-SEPARATION-002` -> `SFT-MED-VALIDATION-PLACEBO-NOCEBO-COMPLETE-FAMILY-002`

Candidate generator: Generate the complete eight-axis append-only engineering protocol product for SFT-ENG-CONSCIOUSNESS-PLACEBO-CROSS-BINDING-PROTOCOL-002 and independently reconstruct its no-omission witness.

Grammar boundary: Every open, reproducible cross-binding and placebo/nocebo test represented by the named Consciousness and Medicine receipts, with human-participant ethics retained.

Complete census: 256 candidates; 1 unique survivor; 4 controls

Survivor: `sealed-upstream-receipts__ethics-bound-interior-objective-cross-binding-protocol__complete-common-and-domain-record__complete-declared-control-family__favourable-adverse-absent-unresolved__visible-halt-and-bounded-safe-state__protocol-seal-before-outcome__no-law-rewrite`

Exact result: The protocol preregisters separate qualitative report, physiological and behavioural channels; directional and expectation controls; consent, withdrawal, privacy and adverse-event halts; and makes no experimental success claim before a lawful participant study.

Closure: `depth_independent`; minimality `True`; named-shape uniqueness `True`

External status: `independently_replicated`

Engine receipt: `sha256:52856443290a7ff3f0a058bf50aa8533462b7acfb743be21afa85edd484edade` at `receipts/engine/model_admitted/SFT-ENG-CONSCIOUSNESS-PLACEBO-CROSS-BINDING-PROTOCOL-002-52856443290a7ff3.json`

Control	Passed	Expected	Observed
false_premise	True	reject the form missing the first structural preservation	reject the form missing the first structural preservation
tampered_source	True	reject a changed source identity	reject a changed source identity
tampered_artifact	True	reject a missing, duplicate or additional survivor	reject a missing, duplicate or additional survivor
boundary	True	reject forbidden values, target feedback and extra laws	reject forbidden values, target feedback and extra laws

Engineering novel-translations no-omission addendum

Claim ID: `SFT-ENG-NOVEL-TRANSLATIONS-NO-OMISSION-ADDENDUM-002`

Exact statement: The five previously sealed physical and chemical protocols plus the separately sealed consciousness/placebo/cross-binding protocol exhaust the six Engineering obligations in the dated prior-return audit without changing any prior receipt.

Dependency route: `SFT-ENG-CONSCIOUSNESS-PLACEBO-CROSS-BINDING-PROTOCOL-002` -> `SFT-ENG-NOVEL-TRANSLATIONS-COMPLETE-FAMILY-002` -> `SFT-ENG-TRACEABILITY-001` -> `SFT-ENG-INDEPENDENT-CHECK-001`

Candidate generator: Generate the complete eight-axis append-only engineering protocol product for SFT-ENG-NOVEL-TRANSLATIONS-NO-OMISSION-ADDENDUM-002 and independently reconstruct its no-omission witness.

Grammar boundary: Exactly the six numbered Engineering translations in the V1/V2 novel-return audit dated 2026-07-28.

Complete census: 256 candidates; 1 unique survivor; 4 controls

Survivor: `sealed-upstream-receipts__six-obligation-append-only-no-omission-assembly__complete-common-and-domain-record__complete-declared-control-family__favourable-adverse-absent-unresolved__visible-halt-and-bounded-safe-state__protocol-seal-before-outcome__no-law-rewrite`

Exact result: Exactly six inherited protocol obligations are now represented: Tesla transfer, vacuum/inertia response, vacuum-beat restoration, sector-five/seven detection, Smithium synthesis/identification, and open consciousness/placebo/cross-binding tests. No protocol asserts an unperformed outcome.

Closure: `depth_independent`; minimality `True`; named-shape uniqueness `True`

External status: `independently_replicated`

Engine receipt: `sha256:ad3233db90a4586713e027eefb3d4a23a54aa652c88e89111091f487b88cd96a` at `receipts/engine/model_admitted/SFT-ENG-NOVEL-TRANSLATIONS-NO-OMISSION-ADDENDUM-002-ad3233db90a45867.json`

Control	Passed	Expected	Observed
false_premise	True	reject the form missing the first structural preservation	reject the form missing the first structural preservation
tampered_source	True	reject a changed source identity	reject a changed source identity
tampered_artifact	True	reject a missing, duplicate or additional survivor	reject a missing, duplicate or additional survivor
boundary	True	reject forbidden values, target feedback and extra laws	reject forbidden values, target feedback and extra laws

Preceding-version abstract preserved for chronology

The following abstract is reproduced from version 1.0.0. Its counts and status language describe that dated version and do not override the current abstract or census above.

Engineering fails scientifically when a desired output selects its own requirements, when a prototype is called validation, when a hidden service supplies uncounted resources, when a simulation is relabelled observation, or when successful performance is treated as proof of a fundamental law. This paper reconstructs Engineering Translation from the One through every admitted upstream branch. Twelve families and seventy-two laws govern requirements, components, resources, measurement, alternatives, control, safety, verification, accessibility, lifecycle, evidence classes and anomaly handoff. The proof layer adds no axiom, free parameter, fit, conventional numerical zero, negative proof scalar, irrational or imaginary proof number, completed infinity or ungenerated continuum. Every claim is a 256-form census across eight exact axes; all 18,432 forms were sealed before external engineering sources were registered. Independent reconstructions and source-bound comparisons preserve adverse, missing and failed records. The result is a public, cross-platform, extension-open constitution for turning science into working artifacts without confusing translation with discovery.

Registered source surface

Human-readable scholarly references remain in the inherited paper. Machine source IDs, custodians, releases, locators, source captures, access outcomes, hashes and transport status remain in the current claim packages and external-source registries. Source prestige is not evidential authority; each source is used only in its registered role. The Lean source-binding gate checks custody and identity, not empirical truth by reputation.

Limitations and open frontier

The current result is formally complete only to the named dated grammar, census and artifact graph. Native Lean theoremhood is presently established for the operational root and generic gate implications; the other scientific claims are artifact checked rather than individually restated as Lean propositions. Empirical validation remains claim specific, and adverse, unavailable and unresolved evidence remains open where the current records say so. New discoveries, falsifications, stronger comparisons and native claim-level formalisations may enter only through a later version without rewriting this one.

Data and code availability

The successor Markdown is `publications/successors/engineering_translation/FROM_ONE_LAW_TO_A_WORKING_WORLD_PAPER_001_V1_1.md` and its scientific predecessor is `publications/current/engineering_translation/FROM_ONE_LAW_TO_A_WORKING_WORLD.md`. The Lean sources and report are under `generated/lean4_validation/`; the report is `generated/lean4_validation/reports/whole_model_validation.json`. Current claims are indexed by `census/claims.json`; complete candidates, decisions, controls, certificates, observations and receipts remain under `claims/<claim-id>/` and `receipts/`. The paper's evidence map and draft deposit metadata sit beside the successor source. No remote action was performed.

References and source registry

1. Smith, Maria. The preceding version of this paper, preserved at the repository path identified above, 2026.
2. Smith, Maria. *There Is No Nothing*. Ernos Labs Methods Paper 00, versioned publication series, 2026.
3. de Moura, Leonardo, and Sebastian Ullrich. "The Lean 4 Theorem Prover and Programming Language." *Automated Deduction - CADE 28*, 2021. DOI: 10.1007/978-3-030-79876-5_37.
4. Lean FRO. *The Lean Language Reference*. lean-lang.org/doc/reference/latest.
5. Ernos Labs. Current claim packages, source registries and Lean verification report, repository snapshot bound by the machine identities below.

Successor machine-identity appendix

- Preceding source: `publications/current/engineering_translation/FROM_ONE_LAW_TO_A_WORKING_WORLD.md`
- Preceding source SHA-256: `sha256:e232febfafdd820c6ac4557b42bf9b9fcf0b894c316c272e87e19b34a6bc44d0`
- Lean report: `generated/lean4_validation/reports/whole_model_validation.json`
- Lean report SHA-256: `sha256:6b6a5a9a9a0b74687a6f97fb3b4fcc7455968e35c82395290837cda60d7501cf`
- Ordered census SHA-256: `sha256:f3f540f77a9b677bf7ddb9f5c6ea1ea41f08e57bb07163e25f385fd4d1dcde71`
- Execution manifest SHA-256: `sha256:517fd288f4484f43fdc0343b17fcabd06ea9e12c977a4512d4a891dd038fce41`
- Protected engine seal: `sha256:4f4cdd7986808e6a6102d650c85e6093d6425e49f14a5f05d70fa05e6031d46a`
- Verification-authority seal: `sha256:bf810a190b504f0f874a778a52e23251904b17b40a7364135e74b34e8ba0c3b8`
- Publication authorised: `false`
- Remote actions performed: `false`

Lean 4 independent verification update

Verified result

The pinned Lean toolchain `leanprover/lean4:v4.32.0` compiled the verification project and the executable verifier returned `PASS`. Its immutable report identity for this successor is `sha256:6b6a5a9a9a0b74687a6f97fb3b4fcc7455968e35c82395290837cda60d7501cf`.

Verified surface	Result
Ordered census claims	2,777
Proof-bearing accepted claim gates	2,777
Source-bound claims	2,777
Candidate records	898,902
Decision records	898,902
Passed controls	11,108
Registered branches	17
Issues	0

This paper's registered branch surface contributes **80** of the accepted claims.

Native theorem proof and artifact verification are different layers

`SFTValidation/Root.lean` formalises the exact registered two-member operational grammar: `unpresentedAbsence` and `presentedOccurrence`. The decision function rejects the former and accepts the latter. Lean proves existence by the accepted constructor and uniqueness by exhaustive case analysis over the two constructors. `#print axioms` reports no imported or user-declared axioms for the exported root theorems. This is a kernel-checked proof of the unique survivor inside the exact registered operational grammar.

`SFTValidation/Gates.lean` defines twelve Boolean acceptance obligations and can construct an inhabitant of `ClaimGate.Accepted gate` only when all twelve equal `true`. The runtime verifier then parses the complete ordered census and execution manifest, recomputes identities, cardinalities, one-to-one decision coverage, survivor count, controls, closure fields, empirical boundaries, certificate bindings and authoritative receipts, and calls that proof-bearing constructor for every current claim.

The exact scientific statements of the other claims are not all re-expressed as 2,776 separate Lean propositions in this version. They are validated as the repository's complete registered machine artifacts through Lean-executed checks. The source-manifest gate uses a read-only Python bridge solely to instantiate the already registered source factories; Lean consumes the result. This layer does not replace the protected admission engine, does not write receipts and does not substitute for source experiments or empirical replication.

Fail-closed custody event

The first whole-model run halted on six source-capture byte hashes. The discrepancy was traced to line-ending normalisation in the working tree, not to a changed scientific target or a failed theorem. The registered source bytes were restored, exact-byte Git attributes were added for those six captures, all 385 registered external source bindings were re-audited with no mismatch, and the unchanged Lean verification was rerun to `PASS`. The preserved halt demonstrates that source drift is detected rather than silently forgiven.

What the result supports

The result materially strengthens the model's case for formal coherence, exhaustive current-census coverage, unique-survivor enforcement, dependency and provenance integrity, cross-branch consistency and reproducibility by an implementation outside the admission engine. It does not by itself establish that every empirical claim is true in nature, that the registered grammar is the only conceivable grammar outside its stated boundary, or that SFT is superior to every rival theory. Those questions remain governed by the papers' declared experiments, falsification conditions and comparative tests.

Successor conclusion

Version 1.1.0 preserves the preceding paper's derivations, evidence classes, corrections, adverse and unresolved records, ownership limits and open frontier, while integrating the independent Lean 4 result and every later current claim in its declared scope. This paper contributes 80 current claim(s) within `engineering_translation` to the total dependency spine.

The new formal result establishes that the registered two-class operational root has one Lean-proved survivor and that all 2,777 current claims satisfy the proof-bearing artifact gates. The major conflation resolved by this update is between native theorem proof and whole-repository artifact verification: both are valuable, but they are not the same evidential act. No empirical status is strengthened merely because the artifact graph passes, and no historical halt or correction is erased.

The current evidence therefore establishes formal coherence, current-census completeness, unique-survivor enforcement and intact source, dependency, certificate and receipt custody for this version. Field-specific empirical conclusions remain exactly those authorised by their current records. Lawful discovery, falsification, stronger evidence and versioned extension remain open, and publication itself remains pending Maria Smith's explicit confirmation.